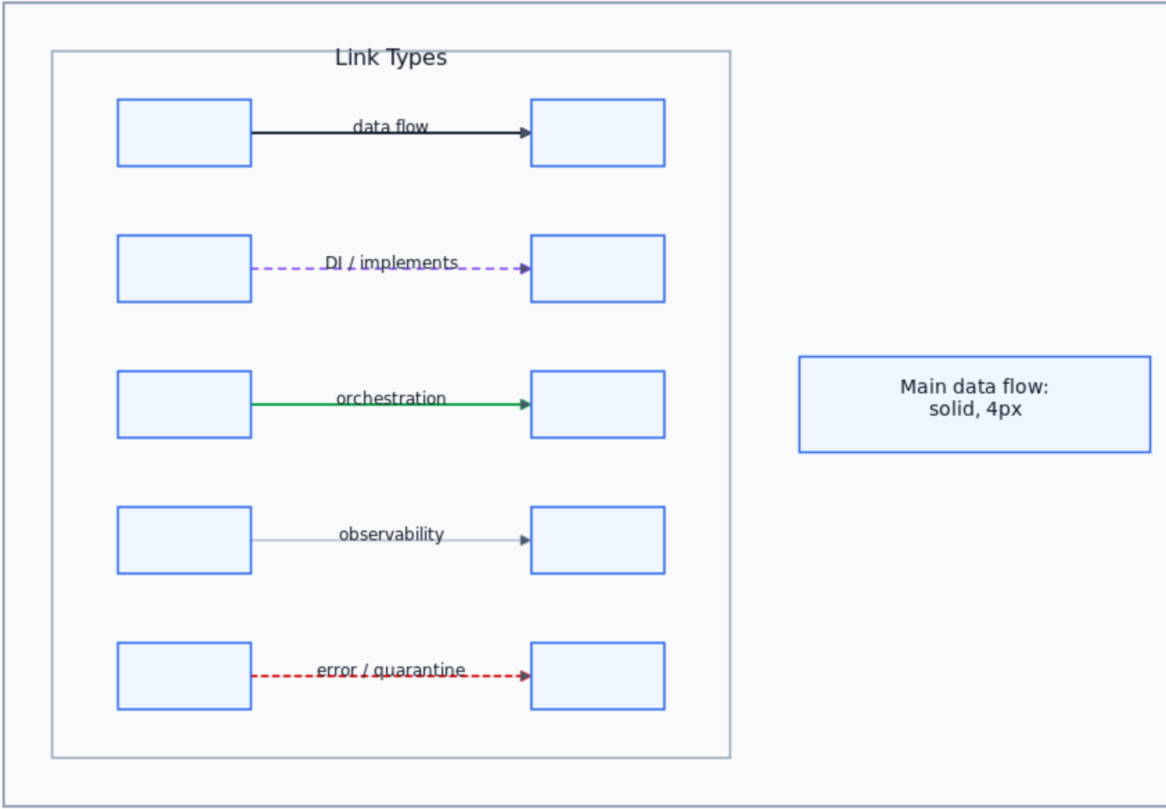

Link Types
data flow
DI / implements
orchestration
observability
error / quarantine
Main data flow:
solid, 4px

Dependency / DI:
dashed, 2px

Observability:
gray, 1px

Error

/ Quarantine: red
dashed, 2px

Codes used in
diagrams

K01 = Transform &
normalize

K02 = Validate & flatten

K03 = assay_id FK

K04 = molecule_id FK

K05 = target_id FK

K06 = injects storage

K07 = injects lock

K08 = injects checkpoint

K09 = injects quarantine

K10 = injects logger

K11 = injects metrics
K12 = injects tracing
K13 = registered in

K14 = < 5% errors

K15 = per-record error

K16 = > 20% errors

📋 Legend

K17 = INCREMENTAL

K18 = unaffected by run_type

K19 = 500, 502, 504

K20 = Schema mismatch

K21 = Missing field

K22 = Invalid format

K23 = DQ threshold

K24 = Lock timeout

K25 = Storage failure

K26 = max attempts exhausted

K27 = seed (doi, pmid, title, authors)

K28 = enriches (citations, funders, references)

K29 = enriches (abstract, mesh_terms, pub_type)

K30 = enriches (concepts, institutions, cited_by)

K31 = enriches (tldr, embedding, fields_of_study)

K32 = source merge

K33 = _deep_merge(base, override)

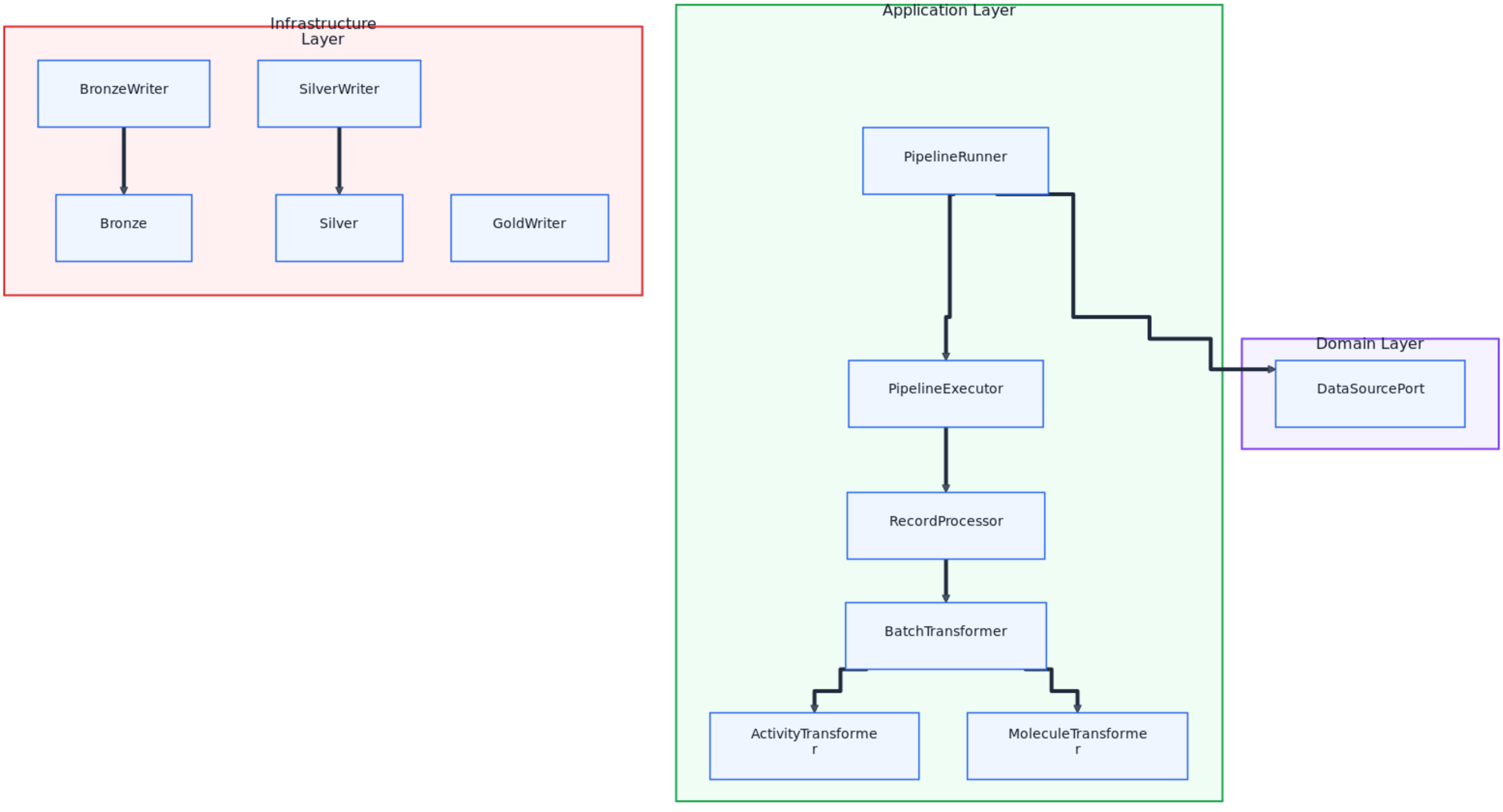
K34 = to_domain()

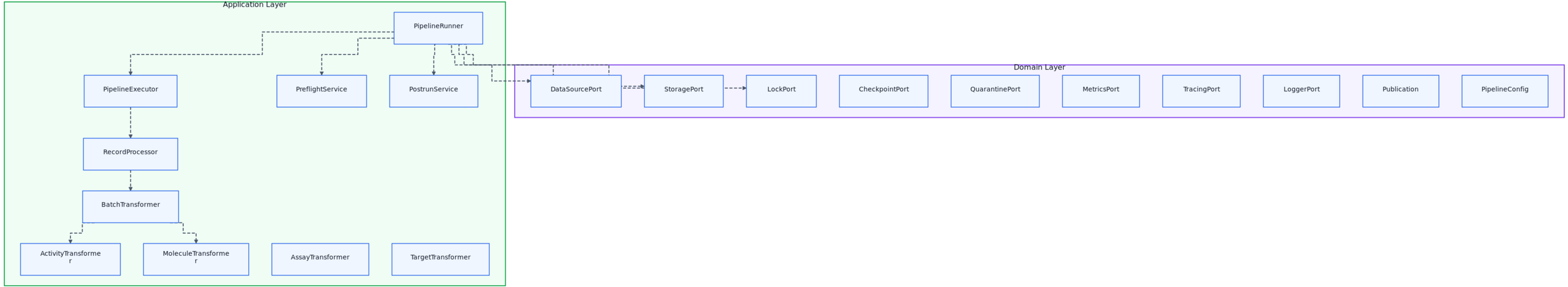
K35 =
resolve_dq_config(
)
K36 = ADR-029:
auto-resolve source_file,
dq_config_file,
filter_config_file

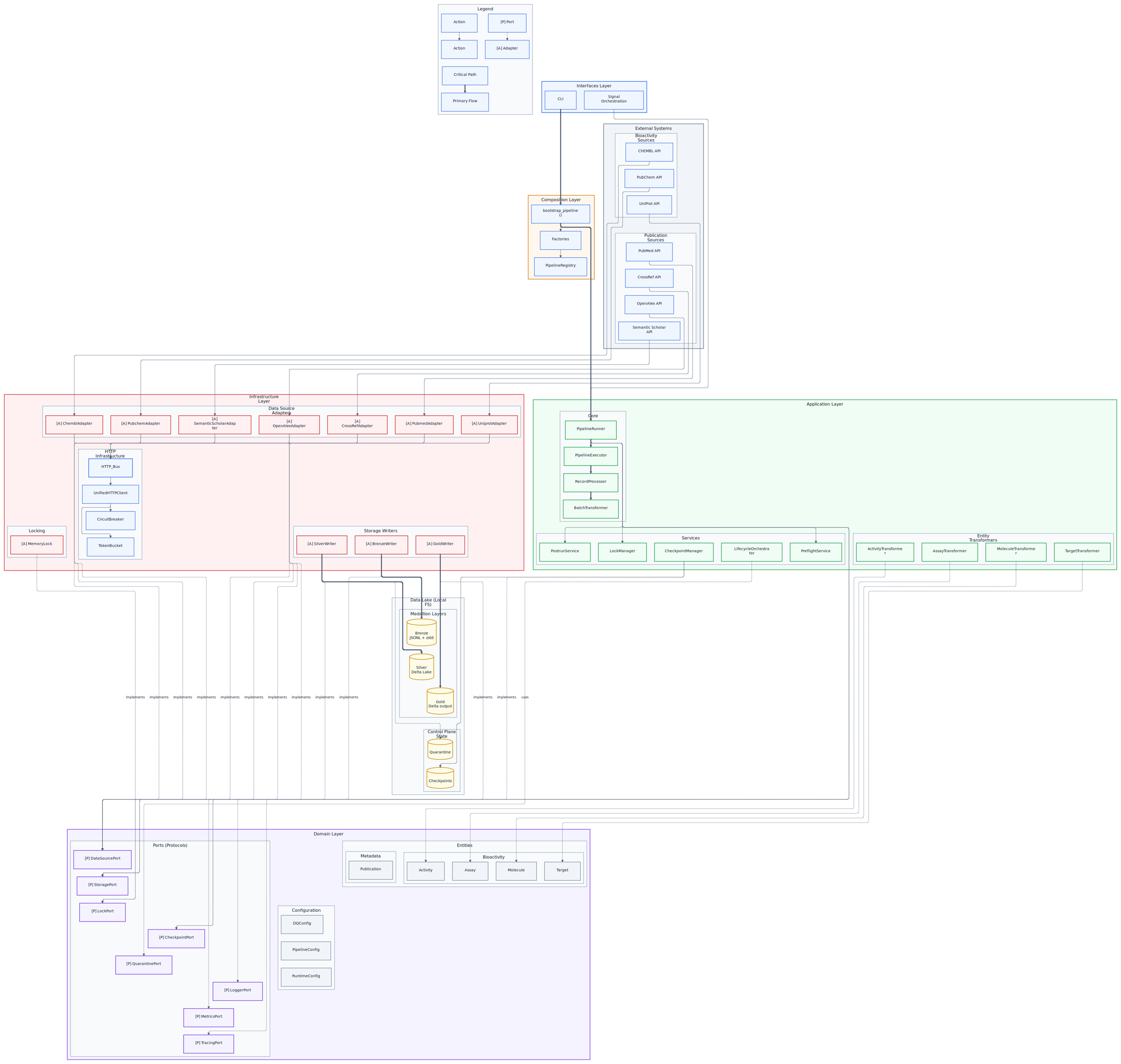
K37 = ADR-029:
auto-resolve sink paths
from provider +
entity_type

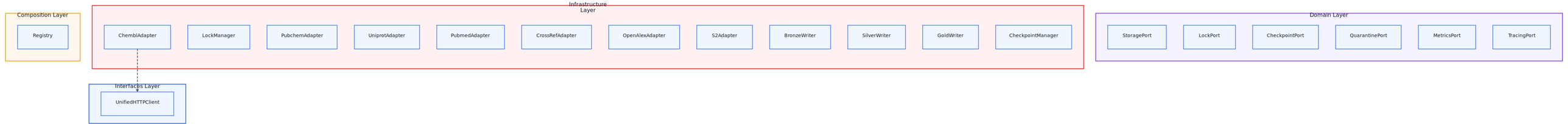
K38 = RECOVERABLE

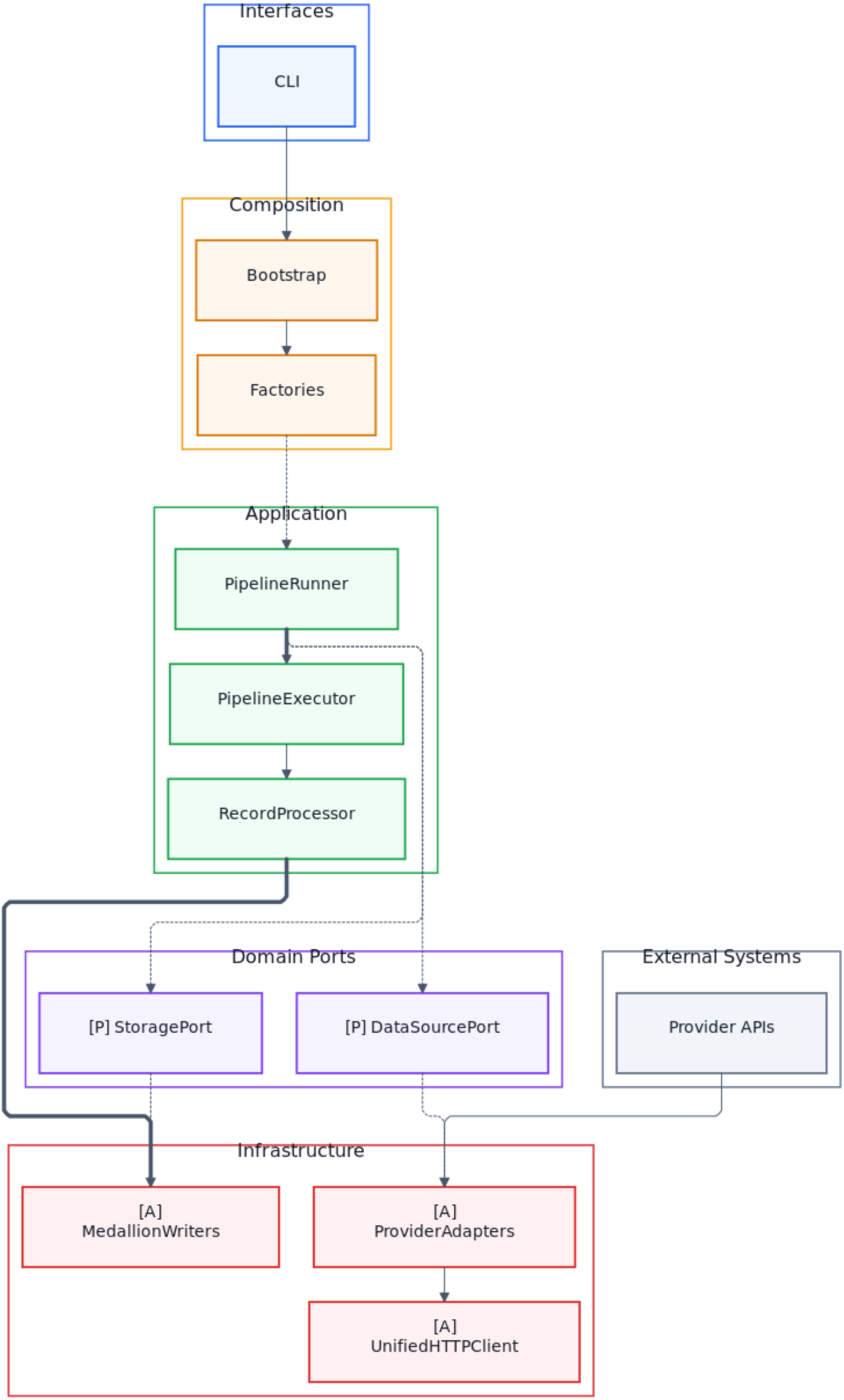
K39 = DATA_QUALITY

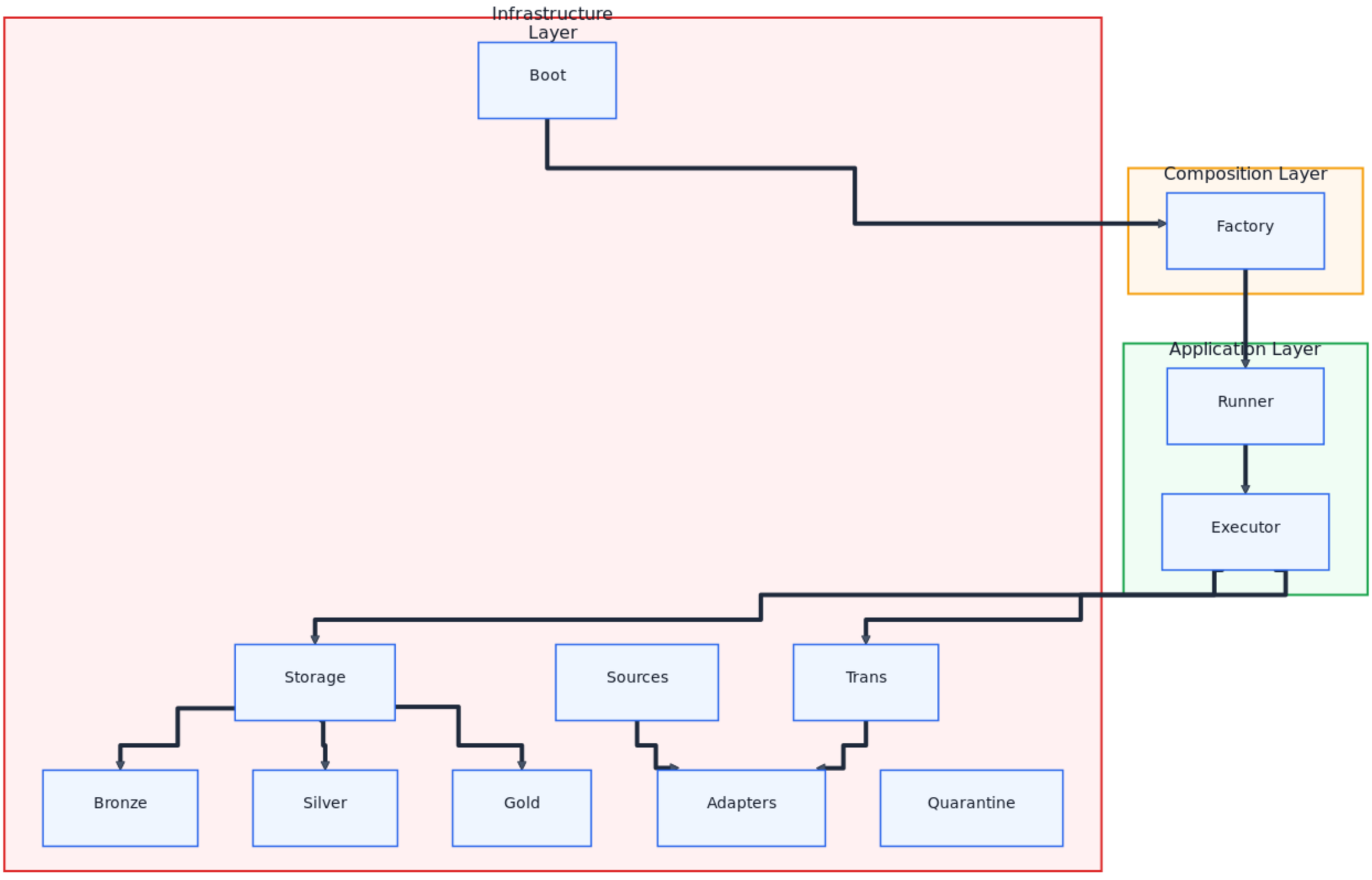

Infrastructure Layer
BronzeWriter
SilverWriter
Bronze
Silver
GoldWriter
Application Layer
PipelineRunner
PipelineExecutor
RecordProcessor
BatchTransformer
ActivityTransforme r
MoleculeTransforme r
Domain Layer
DataSourcePort

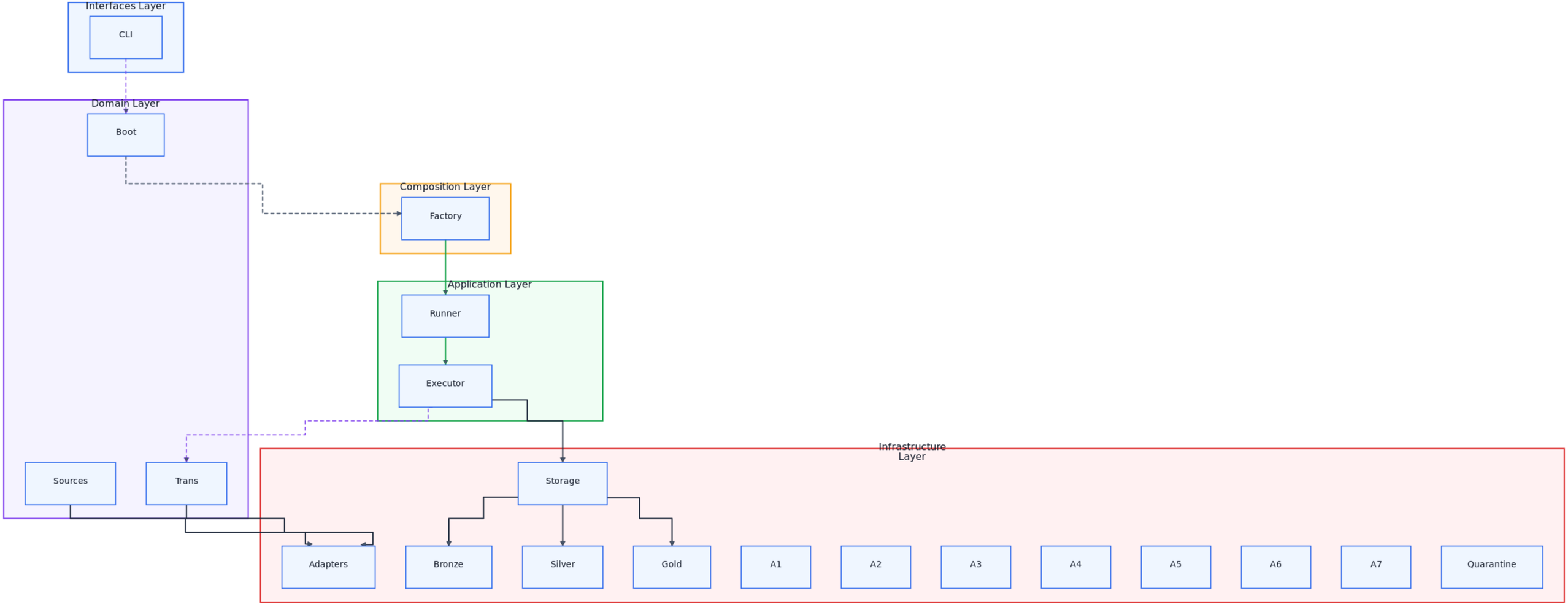

Application Layer
PipelineRunner
PipelineExecutor
PreflightService
PostrunService
RecordProcessor
BatchTransformer
ActivityTransforme
r
MoleculeTransforme
r
AssayTransformer
TargetTransformer
Domain Layer
DataSourcePort
StoragePort
LockPort
CheckpointPort
QuarantinePort
MetricsPort
TracingPort
LoggerPort
Publication
PipelineConfig

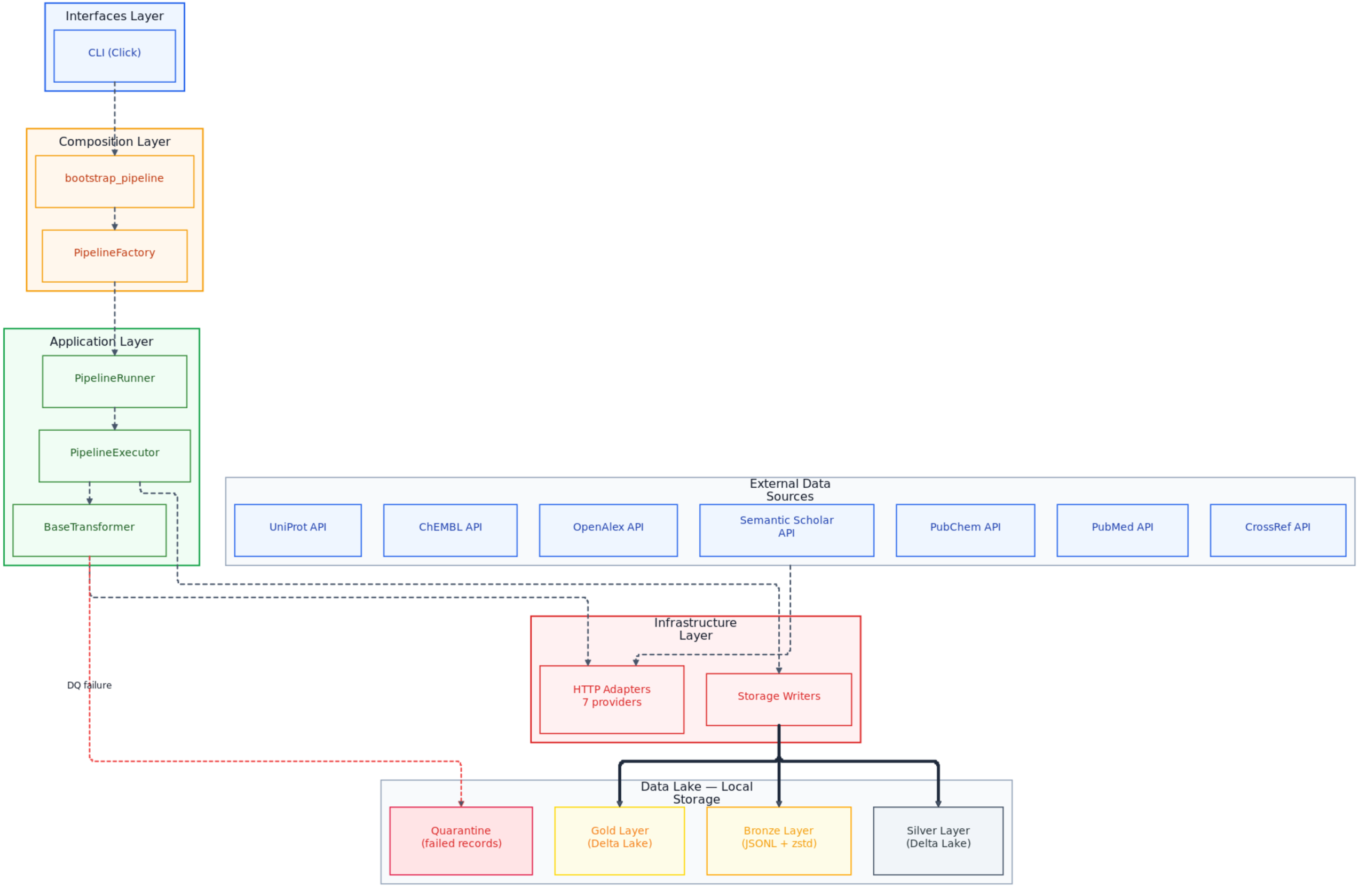

Legend
Action
[P] Port
Action
[A] Adapter
Critical Path
Primary Flow
Interfaces Layer
CLI
Signal Orchestration
External Systems
Bioactivity Sources
ChEMBL API
PubChem API
UniProt API
Publication Sources
PubMed API
CrossRef API
OpenAlex API
Semantic Scholar API
Composition Layer
bootstrap_pipeline ()
Factories
PipelineRegistry
Infrastructure Layer
Data Source Adapters
[A] ChemblAdapter
[A] PubchemAdapter
[A] SemanticScholarAdapter
[A] OpenAlexAdapter
[A] CrossRefAdapter
[A] PubmedAdapter
[A] UniprotAdapter
HTTP Infrastructure
HTTP_Bus
UnifiedHTTPClient
CircuitBreaker
TokenBucket
Locking
[A] MemoryLock
Storage Writers
[A] SilverWriter
[A] BronzeWriter
[A] GoldWriter
Application Layer
Core
PipelineRunner
PipelineExecutor
RecordProcessor
BatchTransformer
Services
PostrunService
LockManager
CheckpointManager
LifecycleOrchestrator
PreflightService
Entity Transformers
ActivityTransformer
AssayTransformer
MoleculeTransformer
TargetTransformer
Data Lake (Local FS)
Medallion Layers
Bronze JSONL + zstd
Silver Delta Lake
Gold Delta output
Control Plane State
Quarantine
Checkpoints
implements
implements
implements
implements
implements
implements
implements
implements
implements
implements
implements
implements
uses
Domain Layer
Ports (Protocols)
[P] DataSourcePort
[P] StoragePort
[P] LockPort
[P] CheckpointPort
[P] QuarantinePort
[P] LoggerPort
[P] MetricsPort
[P] TracingPort
Entities
Metadata
Publication
Bioactivity
Activity
Assay
Molecule
Target
Configuration
DQConfig
PipelineConfig
RuntimeConfig

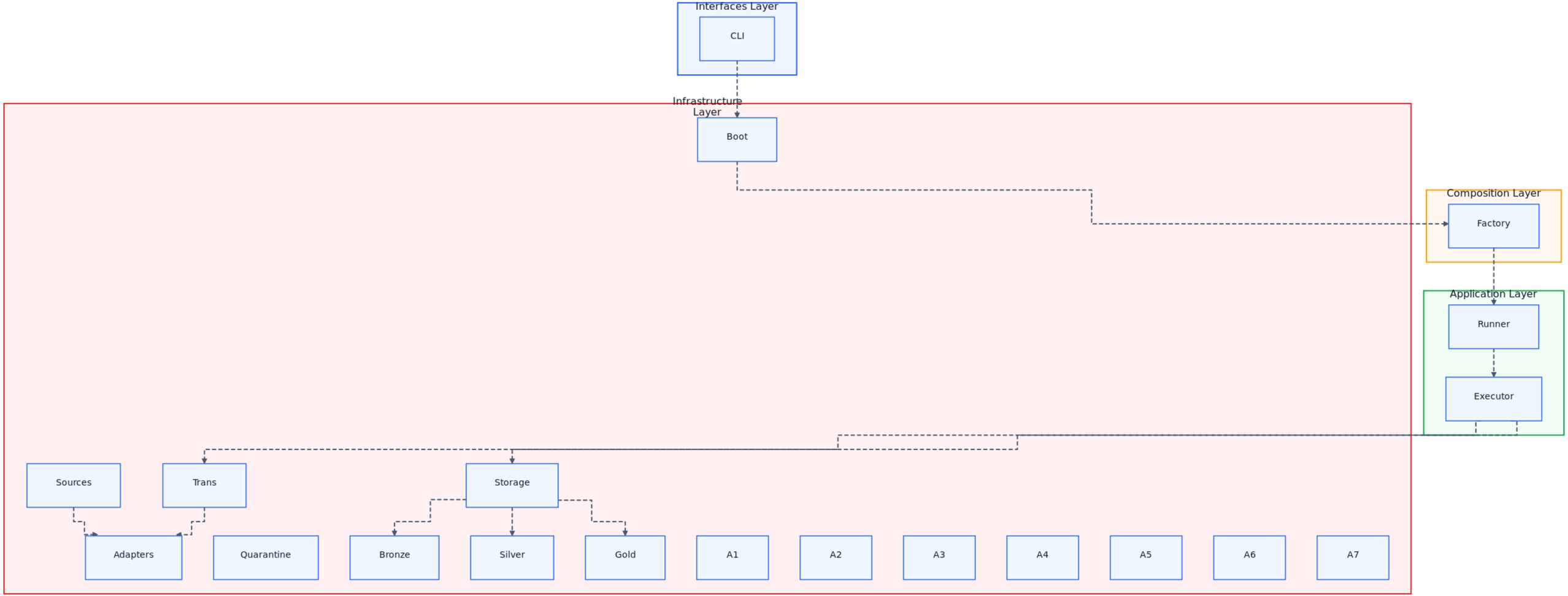

Composition Layer
Registry
Infrastructure Layer
ChemblAdapter
LockManager
PubchemAdapter
UniprotAdapter
PubmedAdapter
CrossRefAdapter
OpenAlexAdapter
S2Adapter
BronzeWriter
SilverWriter
GoldWriter
CheckpointManager
Domain Layer
StoragePort
LockPort
CheckpointPort
QuarantinePort
MetricsPort
TracingPort
Interfaces Layer
UnifiedHTTPClient

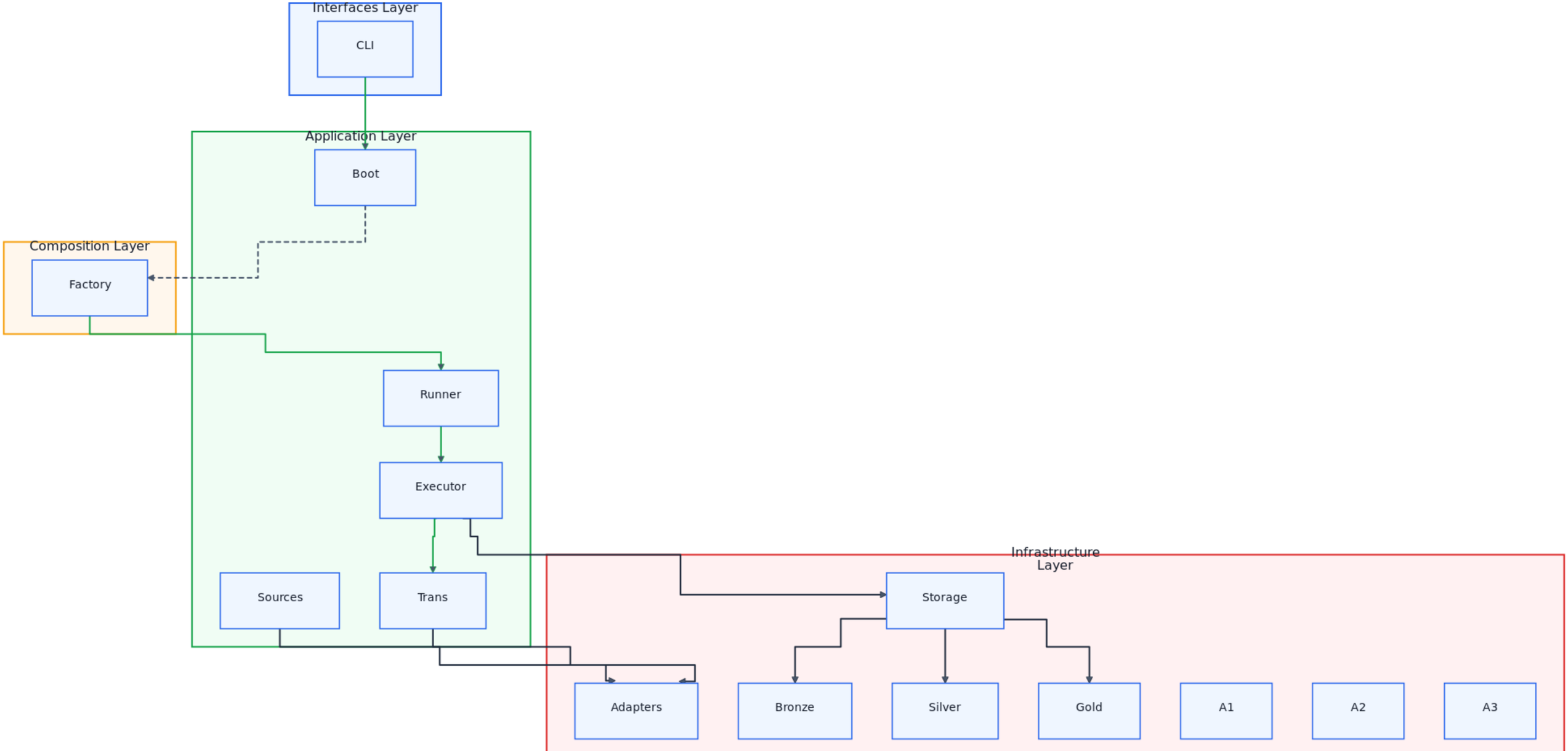

Interfaces
CLI
Composition
Bootstrap
Factories
Application
PipelineRunner
PipelineExecutor
RecordProcessor
Domain Ports
[P] StoragePort
[P] DataSourcePort
External Systems
Provider APIs
Infrastructure
[A]
MedallionWriters
[A]
ProviderAdapters
[A]
UnifiedHTTPClient

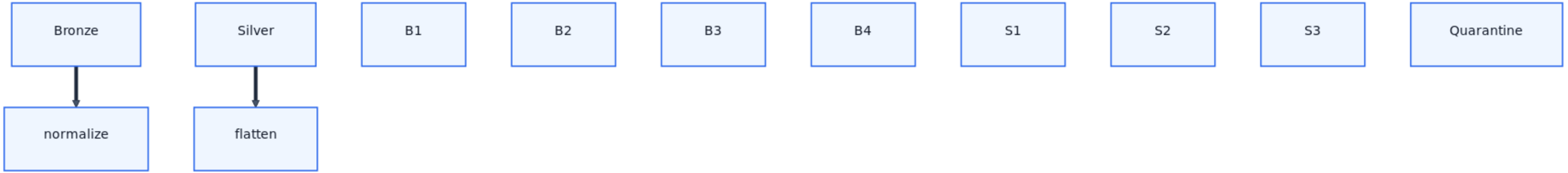

Infrastructure Layer
Boot
Composition Layer
Factory
Application Layer
Runner
Executor
Storage
Sources
Trans
Bronze
Silver
Gold
Adapters
Quarantine

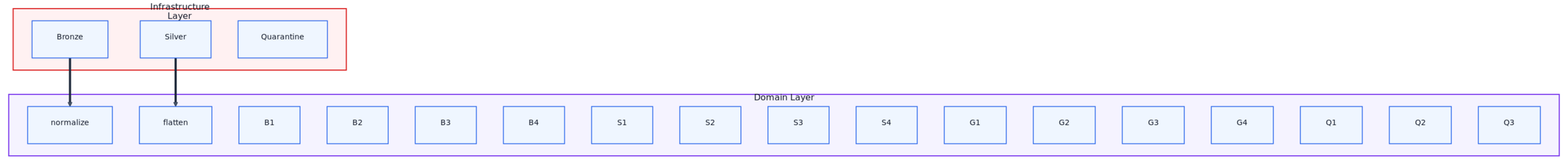

Interfaces Layer
CLI
Domain Layer
Boot
Composition Layer
Factory
Application Layer
Runner
Executor
Sources
Trans
Infrastructure Layer
Storage
Adapters
Bronze
Silver
Gold
A1
A2
A3
A4
A5
A6
A7
Quarantine

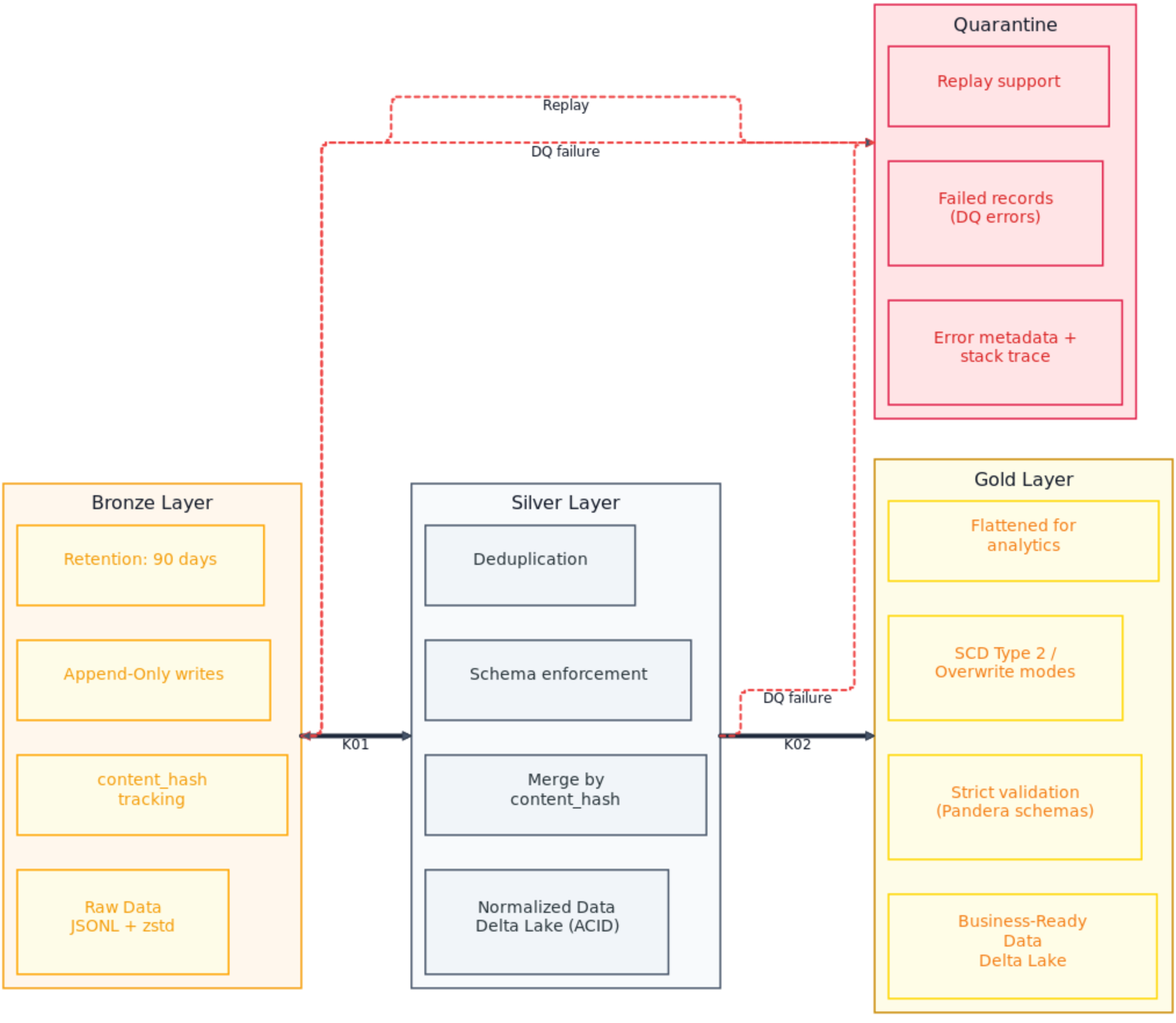

Interfaces Layer
CLI (Click)
Composition Layer
bootstrap_pipeline
PipelineFactory
Application Layer
PipelineRunner
PipelineExecutor
BaseTransformer
External Data Sources
UniProt API
ChEMBL API
OpenAlex API
Semantic Scholar API
PubChem API
PubMed API
CrossRef API
Infrastructure Layer
HTTP Adapters 7 providers
Storage Writers
DQ failure
Data Lake — Local Storage
Quarantine (failed records)
Gold Layer (Delta Lake)
Bronze Layer (JSONL + zstd)
Silver Layer (Delta Lake)

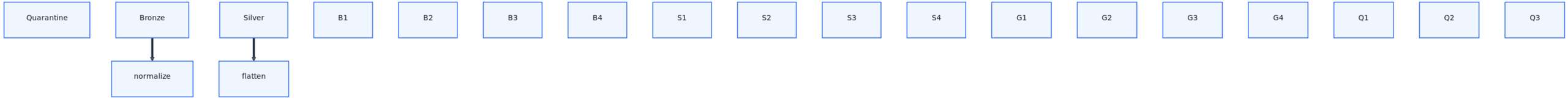

Interfaces Layer
CLI
Infrastructure Layer
Boot
Composition Layer
Factory
Application Layer
Runner
Executor
Sources
Trans
Storage
Adapters
Quarantine
Bronze
Silver
Gold
A1
A2
A3
A4
A5
A6
A7

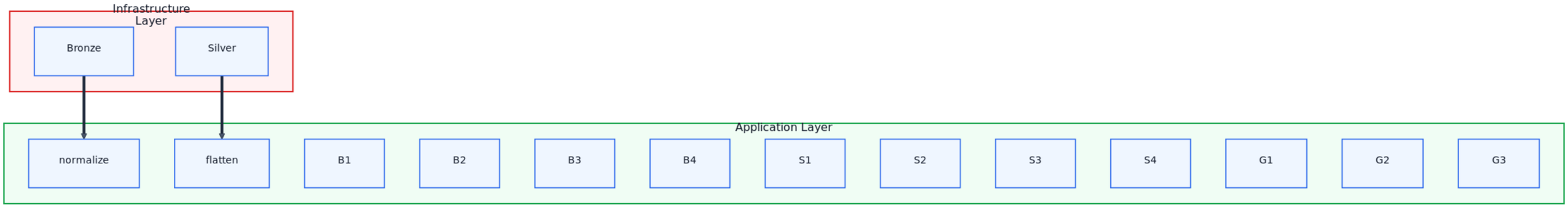

Interfaces Layer
CLI
Application Layer
Boot
Composition Layer
Factory
Runner
Executor
Sources
Trans
Infrastructure Layer
Storage
Adapters
Bronze
Silver
Gold
A1
A2
A3

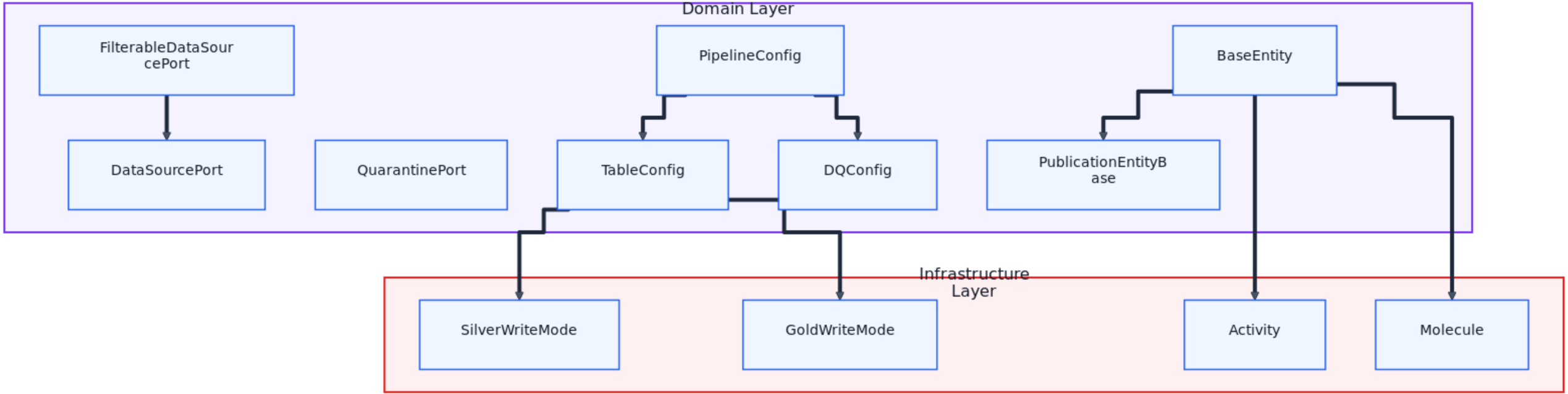

Bronze
Silver
B1
B2
B3
B4
S1
S2
S3
Quarantine
normalize
flatten

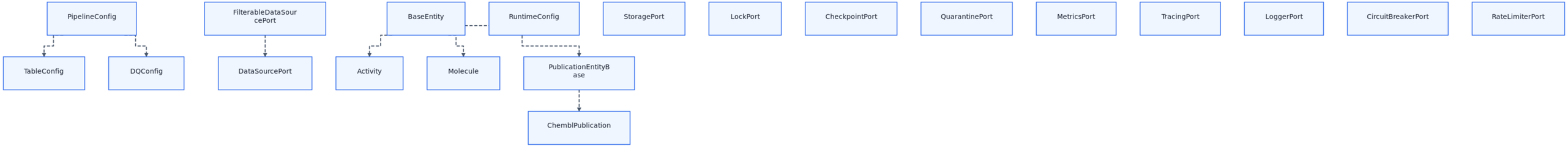

Infrastructure Layer
Bronze
Silver
Quarantine
Domain Layer
normalize
flatten
B1
B2
B3
B4
S1
S2
S3
S4
G1
G2
G3
G4
Q1
Q2
Q3

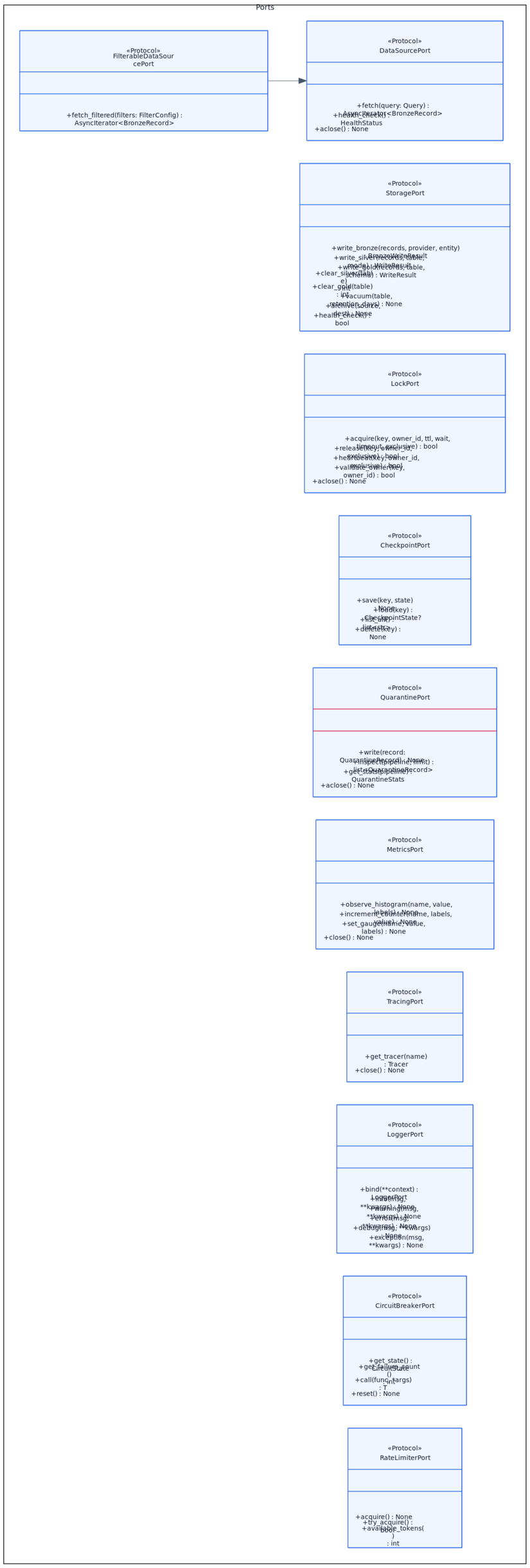

Quarantine
Replay support
Failed records
(DQ errors)
Error metadata +
stack trace
Replay
DQ failure
Bronze Layer
Retention: 90 days
Append-Only writes
content_hash
tracking
Raw Data
JSONL + zstd
K01
Silver Layer
Deduplication
Schema enforcement
Merge by
content_hash
Normalized Data
Delta Lake (ACID)
DQ failure
K02
Gold Layer
Flattened for
analytics
SCD Type 2 /
Overwrite modes
Strict validation
(Pandera schemas)
Business-Ready
Data
Delta Lake

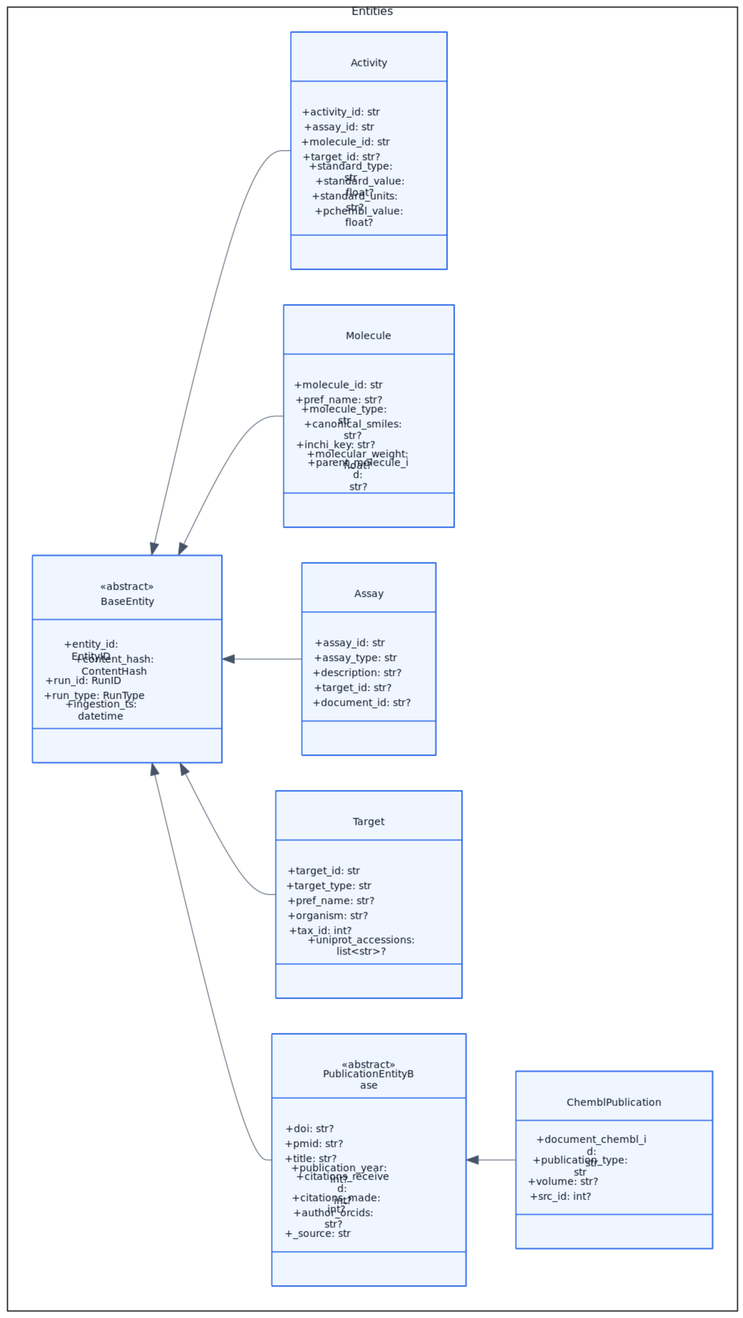

Quarantine
Bronze
normalize
Silver
flatten
B1
B2
B3
B4
S1
S2
S3
S4
G1
G2
G3
G4
Q1
Q2
Q3

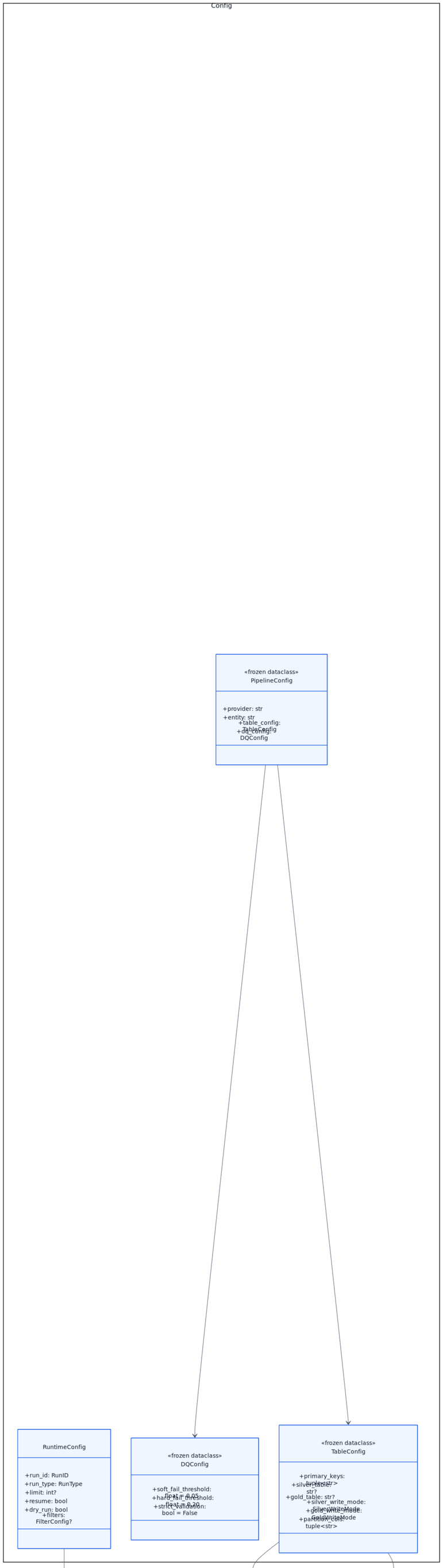

Infrastructure Layer
Bronze
Silver
Application Layer
normalize
flatten
B1
B2
B3
B4
S1
S2
S3
S4
G1
G2
G3

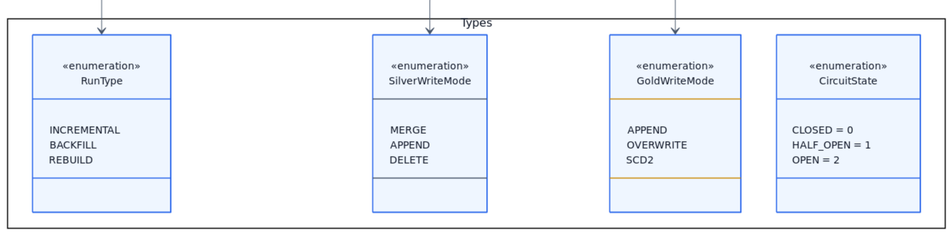

Domain Layer
FilterableDataSource Port
DataSourcePort
QuarantinePort
PipelineConfig
TableConfig
DQConfig
BaseEntity
PublicationEntityBase
Infrastructure Layer
SilverWriteMode
GoldWriteMode
Activity
Molecule

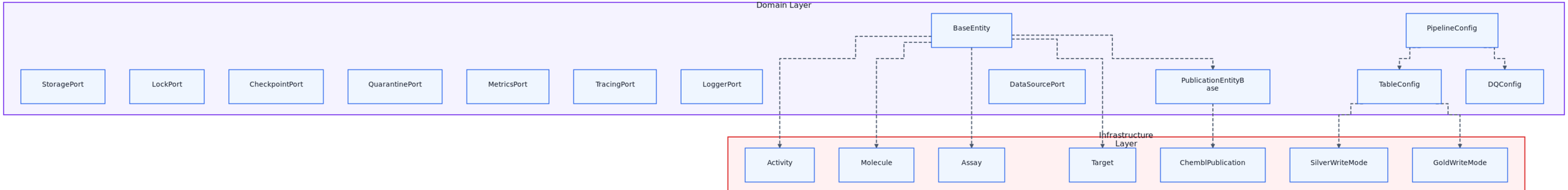

PipelineConfig
FilterableDataSourcePort
BaseEntity
RuntimeConfig
StoragePort
LockPort
CheckpointPort
QuarantinePort
MetricsPort
TracingPort
LoggerPort
CircuitBreakerPort
RateLimiterPort
TableConfig
DQConfig
DataSourcePort
Activity
Molecule
PublicationEntityBase
ChemblPublication

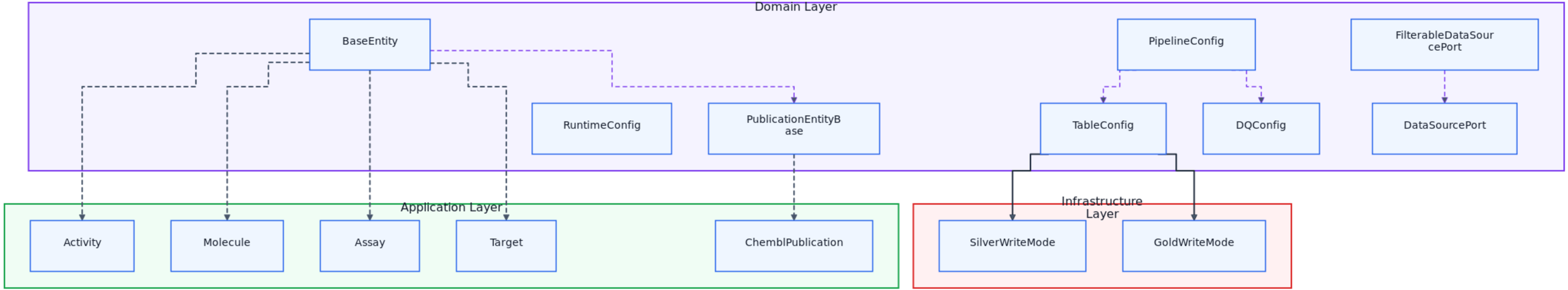

Ports
«Protocol»
FilterableDataSourcePort
+fetch_filtered(filters: FilterConfig) : AsyncIterator<BronzeRecord>
«Protocol»
DataSourcePort
+fetch(query: Query) : AsyncIterator<BronzeRecord>
+health_check() : HealthStatus
+aclose() : None
«Protocol»
StoragePort
+write_bronze(records, provider, entity) : BronzeWriteResult
+write_silver(records, table, mode) : WriteResult
+write_gold(records, table, schema) : WriteResult
+clear_silver(table)
+clear_gold(table)
+vacuum(table, retention_days) : None
+delete_source(source, dest) : None
+health_check() : bool
«Protocol»
LockPort
+acquire(key, owner_id, ttl, wait, timeout, exclusive) : bool
+release(key, owner_id, exclusive) : bool
+heartbeat(key, owner_id, exclusive) : bool
+validate_owner(key, owner_id) : bool
+aclose() : None
«Protocol»
CheckpointPort
+save(key, state) : None
+load(key) : CheckpointState?
+list_keys() : list<str>
+delete(key) : None
«Protocol»
QuarantinePort
+write(record: QuarantineRecord) : None
+write_batch(records) : int
+get_stats(pipeline) : QuarantineStats
+aclose() : None
«Protocol»
MetricsPort
+observe_histogram(name, value, labels)
+increment(name, labels, value) : None
+set_gauge(name, value, labels) : None
+close() : None
«Protocol»
TracingPort
+get_tracer(name) : Tracer
+close() : None
«Protocol»
LoggerPort
+bind(**context) : LoggerPort
+debug(msg, **kwargs) : None
+info(msg, **kwargs) : None
+warning(msg, **kwargs) : None
+error(msg, **kwargs) : None
+exception(msg, **kwargs) : None
«Protocol»
CircuitBreakerPort
+get_state() : CircuitState
+get_failure_count() : int
+call(func, *args) : T
+reset() : None
«Protocol»
RateLimiterPort
+acquire() : None
+try_acquire() : bool
+available_tokens() : int

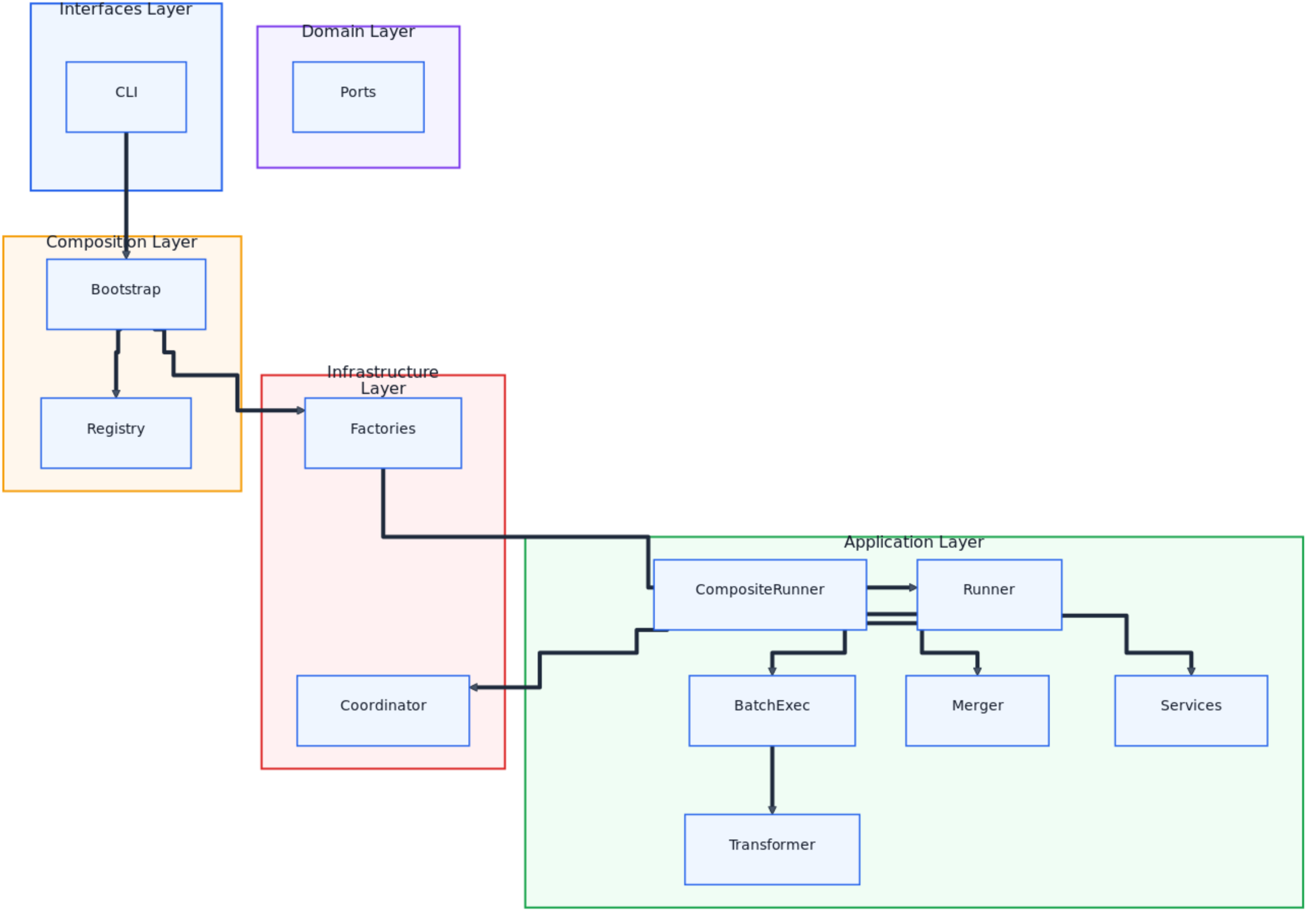

Entities
Activity
+activity_id: str
+assay_id: str
+molecule_id: str
+target_id: str?
+standard_type: str
+standard_value: float?
+standard_units: str?
+pchembl_value: float?
Molecule
+molecule_id: str
+pref_name: str?
+molecule_type: str?
+canonical_smiles: str?
+inchi_key: str?
+molecular_weight: float?
+parent_molecule_id: str?
«abstract»
BaseEntity
+entity_id: EntityID
+content_hash: ContentHash
+run_id: RunID
+run_type: RunType
+ingestion_ts: datetime
Assay
+assay_id: str
+assay_type: str
+description: str?
+target_id: str?
+document_id: str?
Target
+target_id: str
+target_type: str
+pref_name: str?
+organism: str?
+tax_id: int?
+uniprot_accessions: list<str>?
«abstract»
PublicationEntityBase
+doi: str?
+pmid: str?
+title: str?
+publication_year: int?
+citations_received: int?
+citations_made: int?
+author_orcids: str?
+_source: str
ChemblPublication
+document_chembl_id: str
+publication_type: str
+volume: str?
+src_id: int?

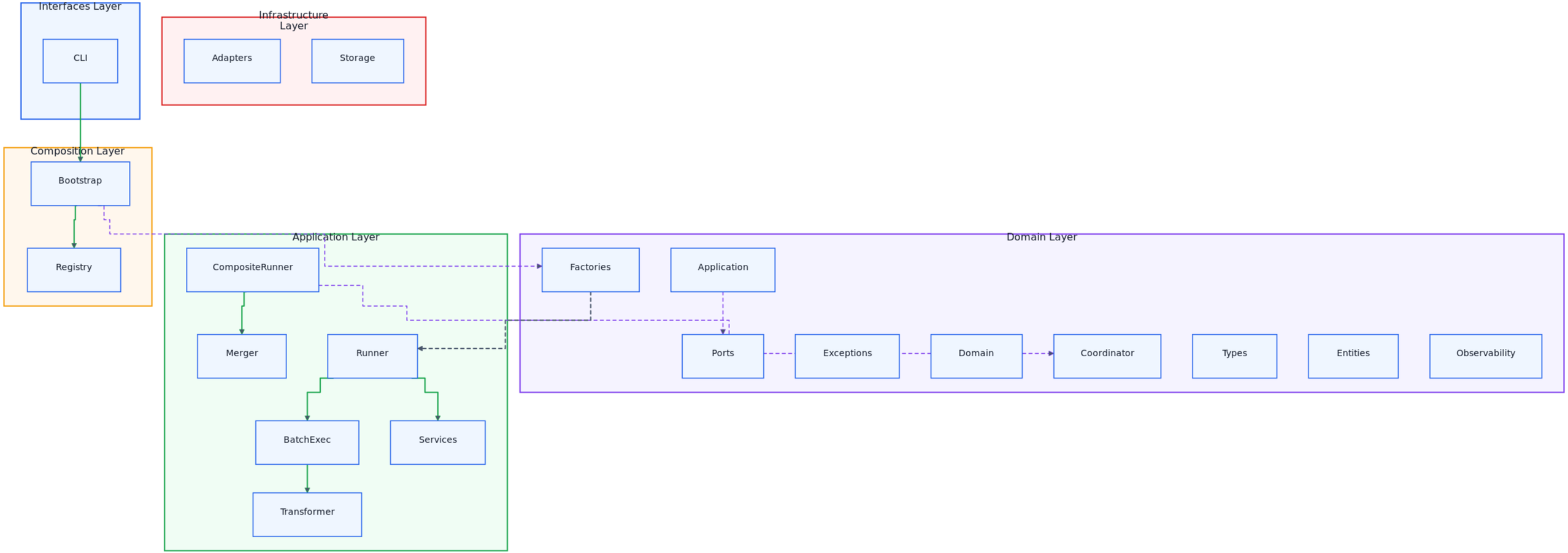

Config
«frozen dataclass»
PipelineConfig
+provider: str
+entity: str
+table_config: TableConfig
+dq_config: DQConfig
RuntimeConfig
+run_id: RunID
+run_type: RunType
+limit: int?
+resume: bool
+dry_run: bool
+filters: FilterConfig?
«frozen dataclass»
DQConfig
+soft_fail_threshold: float = 0.05
+hard_fail_threshold: float = 0.20
+strict_validation: bool = False
«frozen dataclass»
TableConfig
+primary_keys: tuple<str>
+silver_table: str?
+gold_table: str?
+silver_write_mode: SilverWriteMode
+gold_write_mode: GoldWriteMode
+partition_cols: tuple<str>

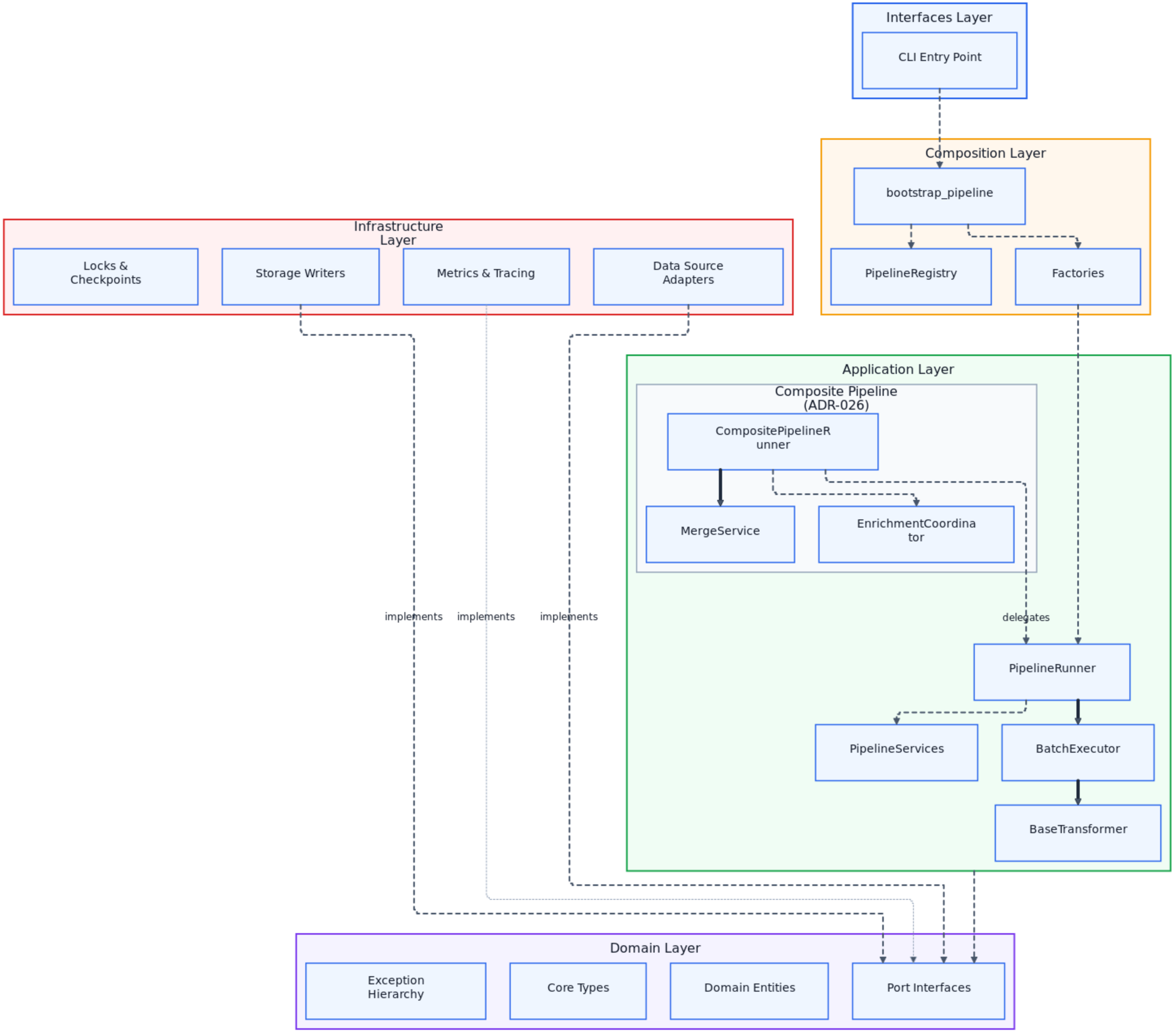

Types
«enumeration»
RunType
INCREMENTAL
BACKFILL
REBUILD
«enumeration»
SilverWriteMode
MERGE
APPEND
DELETE
«enumeration»
GoldWriteMode
APPEND
OVERWRITE
SCD2
«enumeration»
CircuitState
CLOSED = 0
HALF_OPEN = 1
OPEN = 2

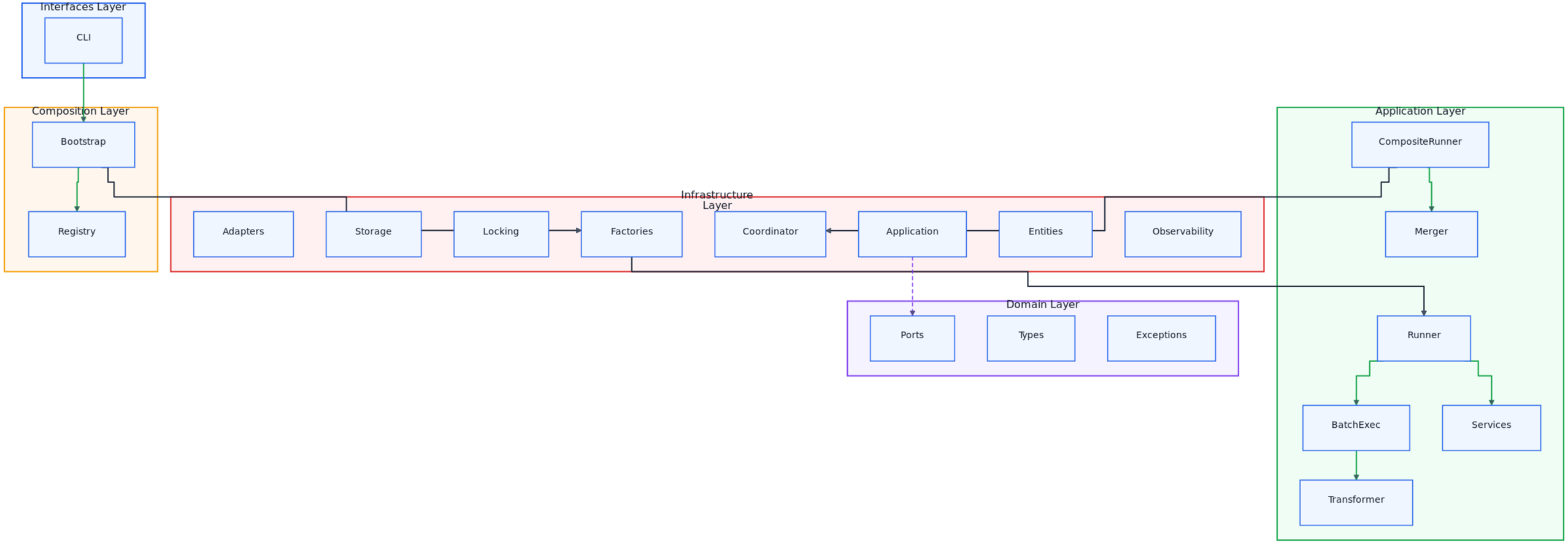

Domain Layer
BaseEntity
PipelineConfig
StoragePort
LockPort
CheckpointPort
QuarantinePort
MetricsPort
TracingPort
LoggerPort
DataSourcePort
PublicationEntityBase
TableConfig
DQConfig
Infrastructure Layer
Activity
Molecule
Assay
Target
ChemblPublication
SilverWriteMode
GoldWriteMode

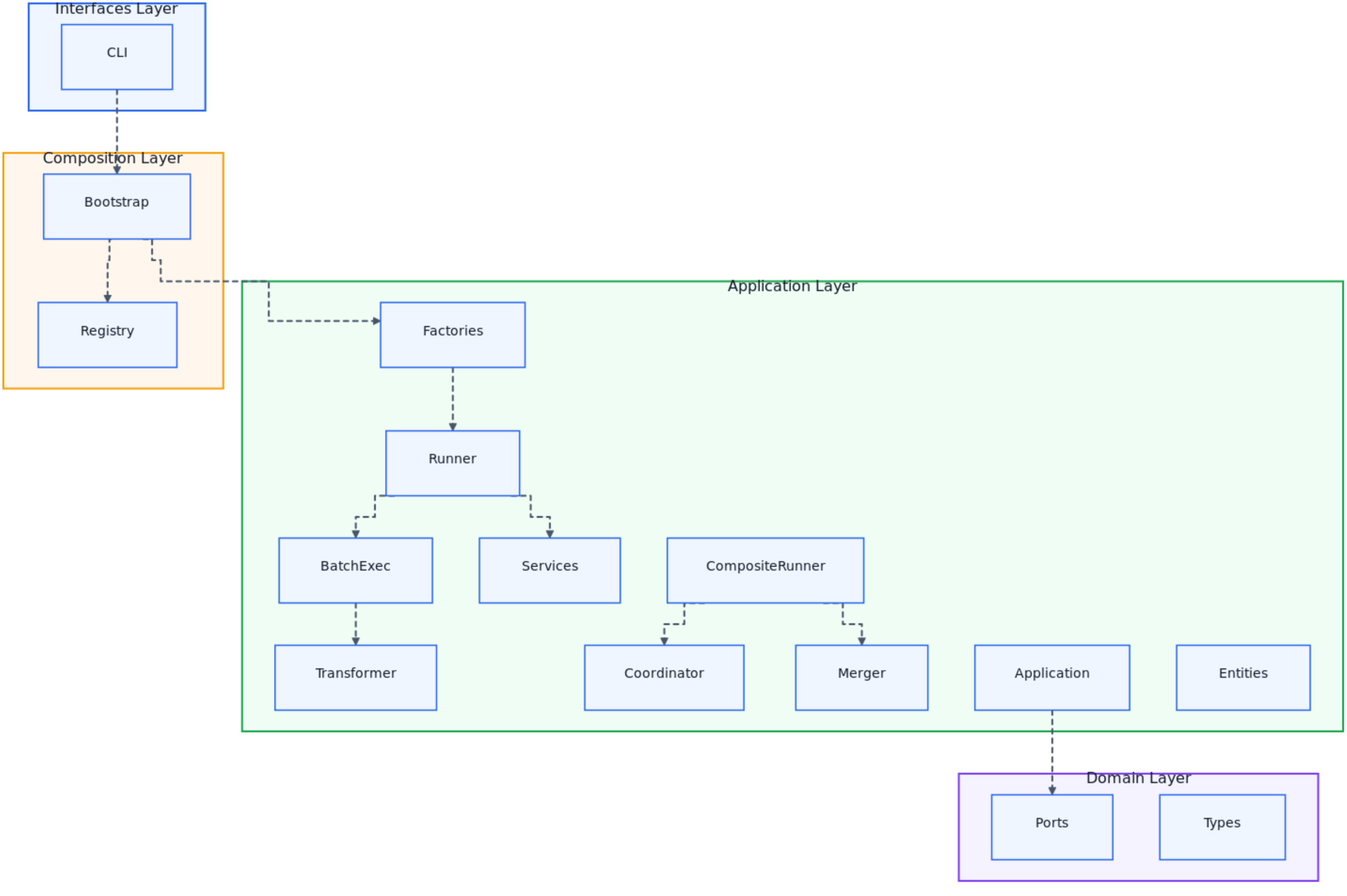

Domain Layer
BaseEntity
RuntimeConfig
PublicationEntityBase
PipelineConfig
TableConfig
DQConfig
FilterableDataSourcePort
DataSourcePort
Application Layer
Activity
Molecule
Assay
Target
ChemblPublication
Infrastructure Layer
SilverWriteMode
GoldWriteMode

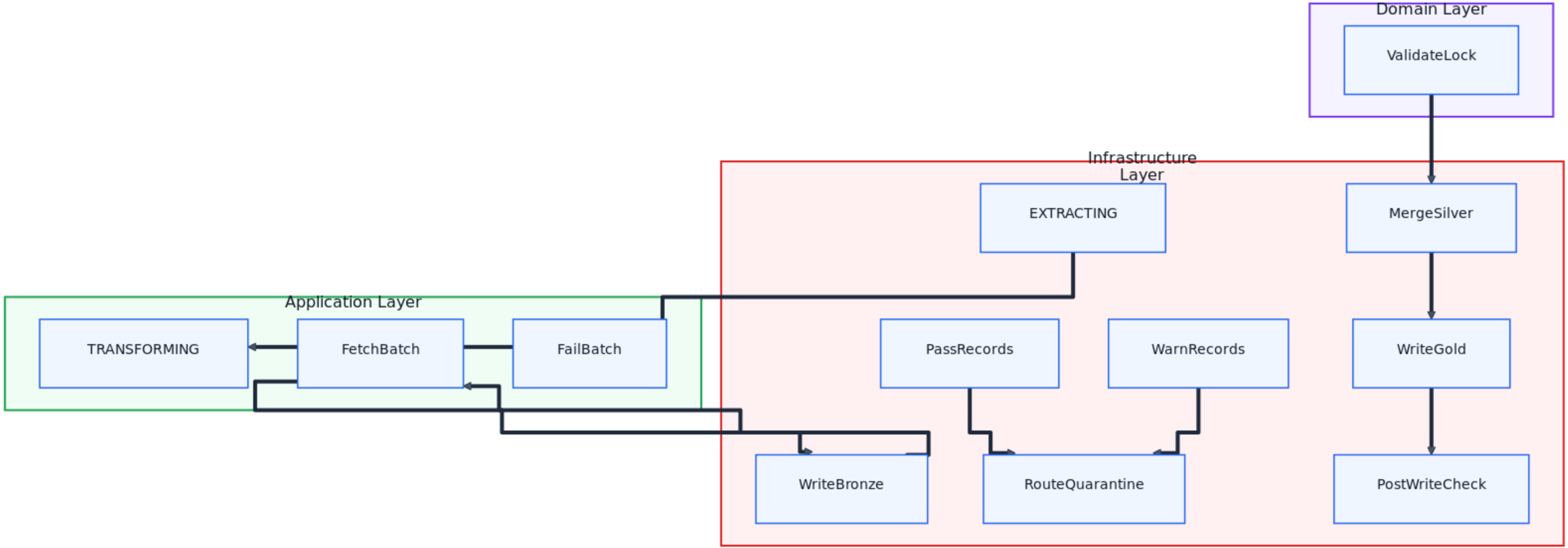

Interfaces Layer
CLI
Domain Layer
Ports
Composition Layer
Bootstrap
Registry
Infrastructure Layer
Factories
Application Layer
CompositeRunner
Runner
Coordinator
BatchExec
Merger
Services
Transformer

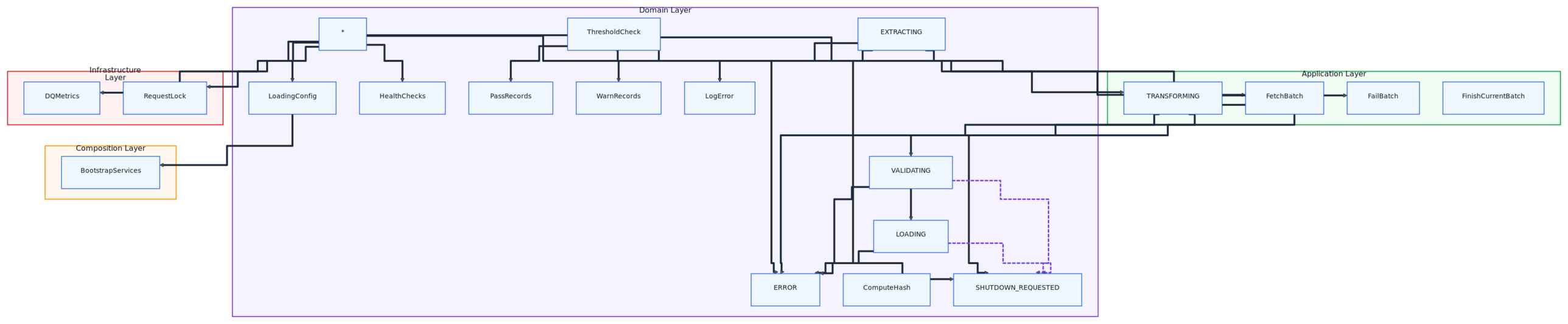

Interfaces Layer
CLI
Infrastructure Layer
Adapters
Storage
Composition Layer
Bootstrap
Registry
Application Layer
CompositeRunner
Merger
Runner
BatchExec
Services
Transformer
Domain Layer
Factories
Application
Ports
Exceptions
Domain
Coordinator
Types
Entities
Observability

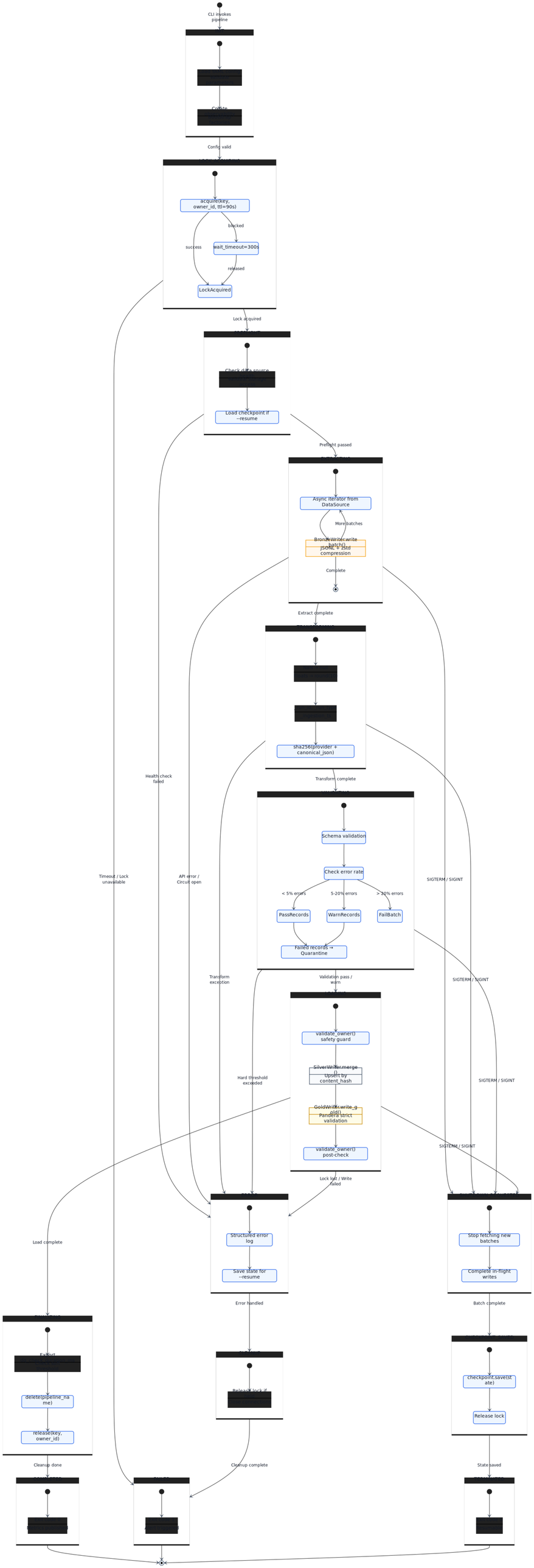

Interfaces Layer
CLI Entry Point
Composition Layer
bootstrap_pipeline
PipelineRegistry
Factories
Infrastructure Layer
Locks & Checkpoints
Storage Writers
Metrics & Tracing
Data Source Adapters
Application Layer
Composite Pipeline (ADR-026)
CompositePipelineRunner
MergeService
EnrichmentCoordinator
PipelineRunner
PipelineServices
BatchExecutor
BaseTransformer
implements
implements
implements
delegates
Domain Layer
Exception Hierarchy
Core Types
Domain Entities
Port Interfaces

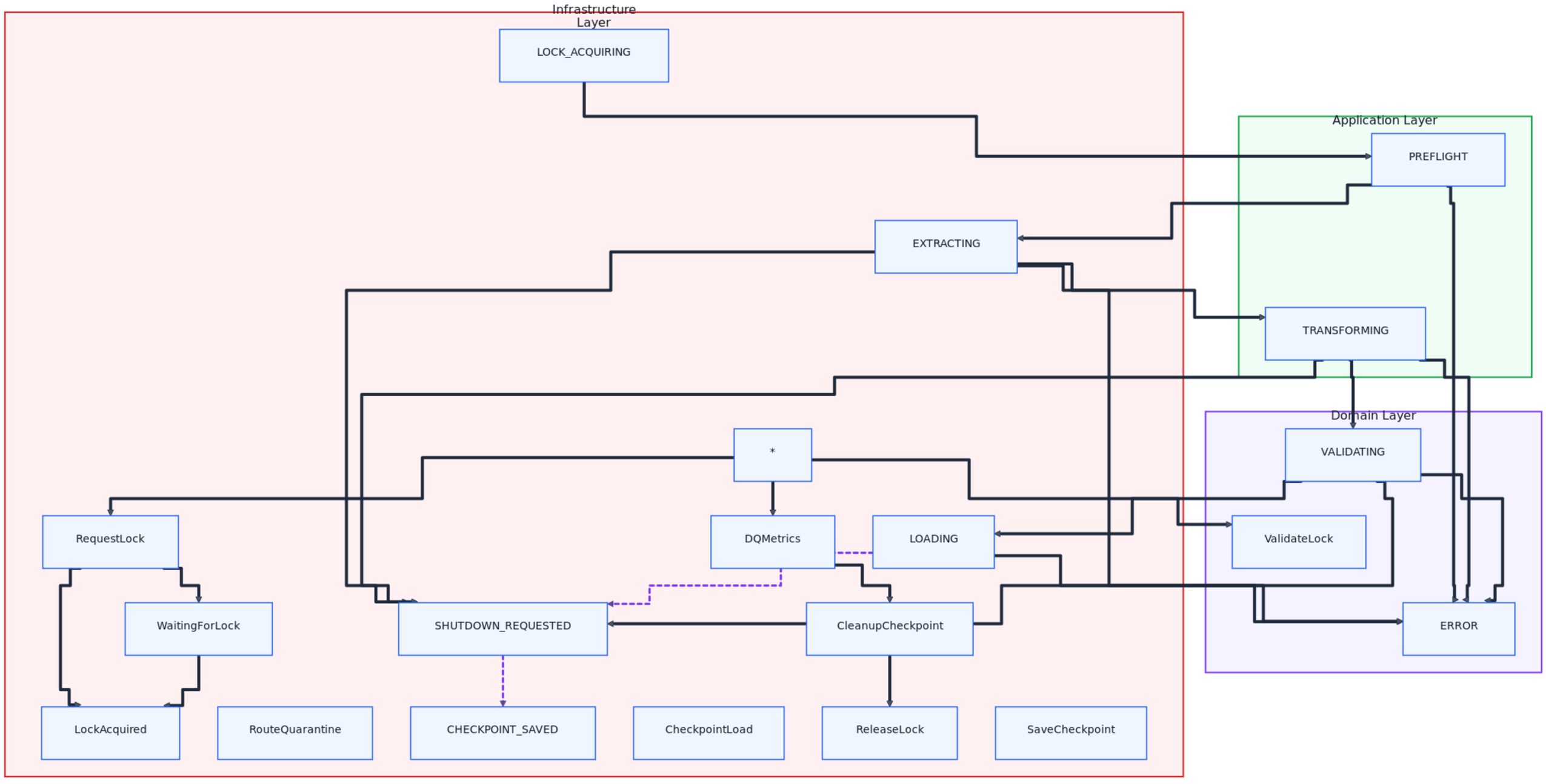

Interfaces Layer
CLI
Composition Layer
Bootstrap
Registry
Infrastructure Layer
Adapters
Storage
Locking
Factories
Coordinator
Application
Entities
Observability
Domain Layer
Ports
Types
Exceptions
Application Layer
CompositeRunner
Merger
Runner
BatchExec
Services
Transformer

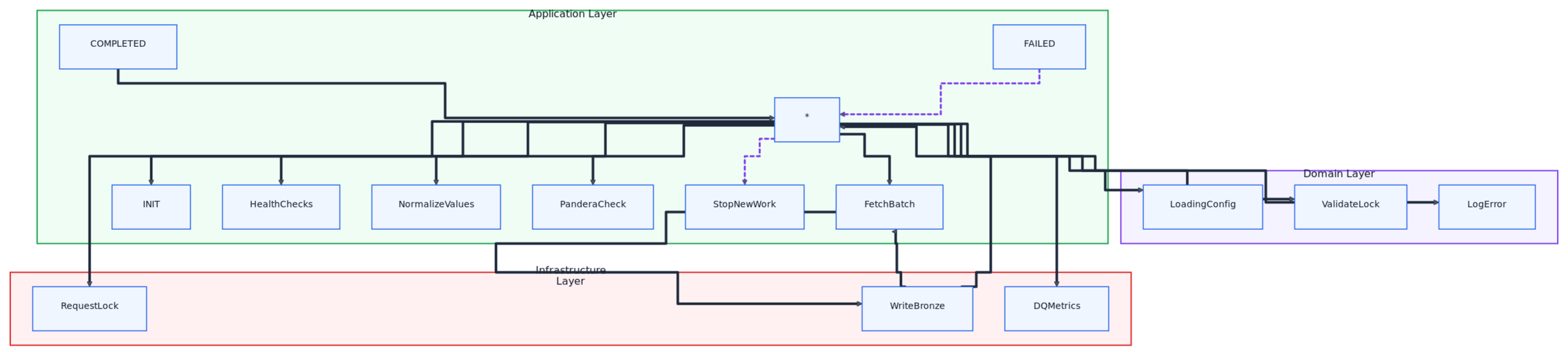

Interfaces Layer
CLI
Composition Layer
Bootstrap
Registry
Application Layer
Factories
Runner
BatchExec
Services
CompositeRunner
Transformer
Coordinator
Merger
Application
Entities
Domain Layer
Ports
Types

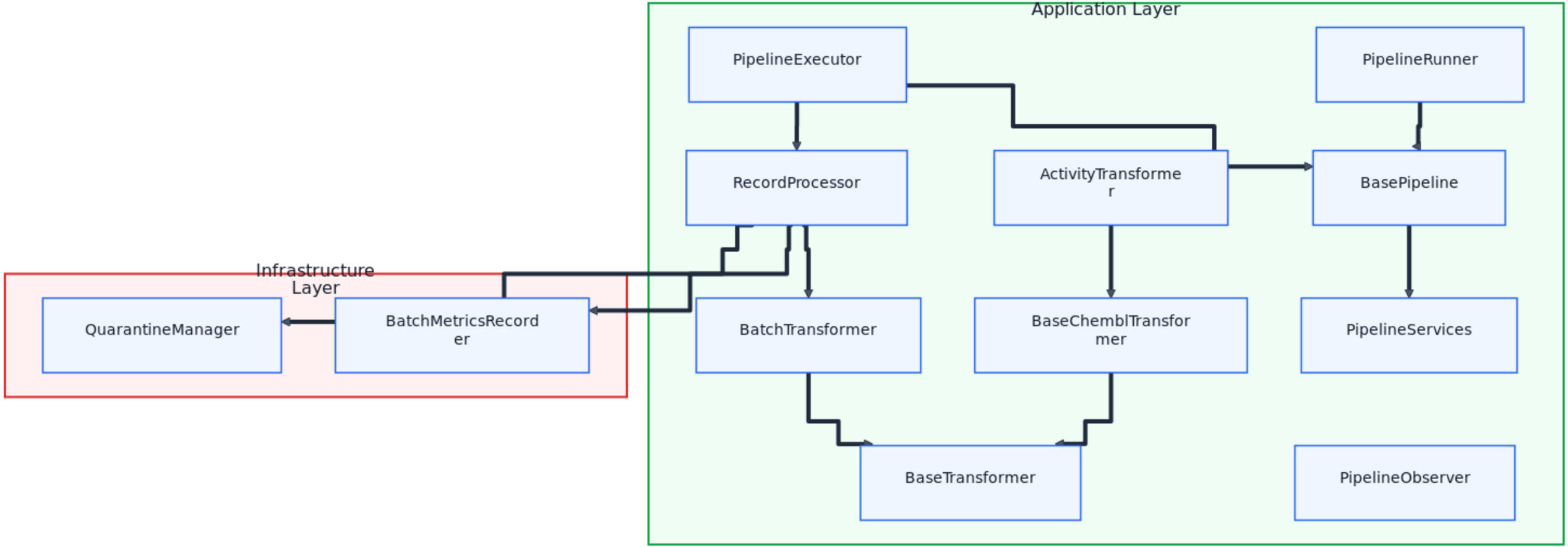

Domain Layer
ValidateLock
Infrastructure Layer
EXTRACTING
MergeSilver
Application Layer
TRANSFORMING
FetchBatch
FailBatch
PassRecords
WarnRecords
WriteGold
WriteBronze
RouteQuarantine
PostWriteCheck

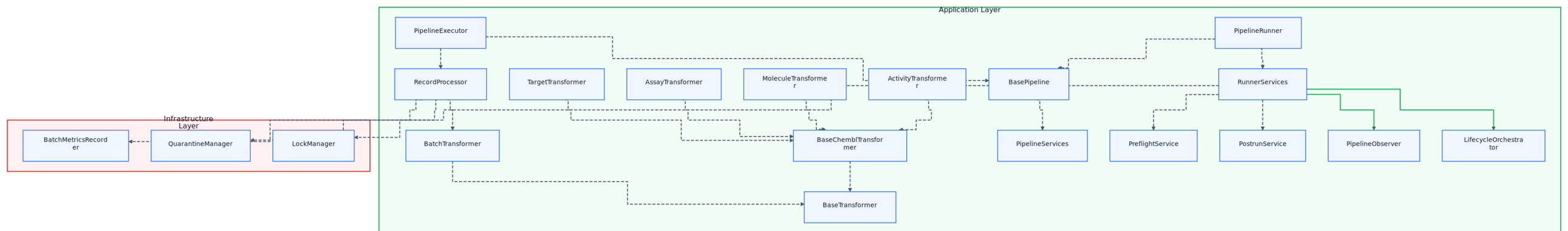

Domain Layer
*
ThresholdCheck
EXTRACTING
Infrastructure Layer
DQMetrics
RequestLock
LoadingConfig
HealthChecks
PassRecords
WarnRecords
LogError
Application Layer
TRANSFORMING
FetchBatch
FailBatch
FinishCurrentBatch
Composition Layer
BootstrapServices
VALIDATING
LOADING
ERROR
ComputeHash
SHUTDOWN_REQUESTED


CLI invokes pipeline
Config valid
acquire(key, owner_id, ttl=90s)
blocked
success
wait_timeout=300s
released
LockAcquired
Lock acquired
Check data source
Load checkpoint if --resume
Preflight passed
Async iterator from DataSource
More batches
BronzeWriter.write batch()
JSONL + zstd compression
Complete
Extract complete
sha256(provider + canonical_json)
Transform complete
Schema validation
Check error rate
< 5% errors
5-20% errors
> 20% errors
PassRecords
WarnRecords
FailBatch
Failed records → Quarantine
Validation pass / warn
validate_owner() safety guard
SilverWriter.merge()
upsert by content_hash
GoldWriter.write_gold()
Pandera strict validation
validate_owner() post-check
Lock lost / Write failed
Structured error log
Save state for --resume
Error handled
Stop fetching new batches
Complete in-flight writes
Batch complete
checkpoint.save(state)
Release lock
State saved
Export
delete(pipeline_name)
release(key, owner_id)
Cleanup done
Release lock if held
Cleanup complete
Load complete
Health check failed
Timeout / Lock unavailable
API error / Circuit open
Transform exception
Hard threshold exceeded
Lock lost / Write failed
SIGTERM / SIGINT
SIGTERM / SIGINT
SIGTERM / SIGINT
SIGTERM / SIGINT

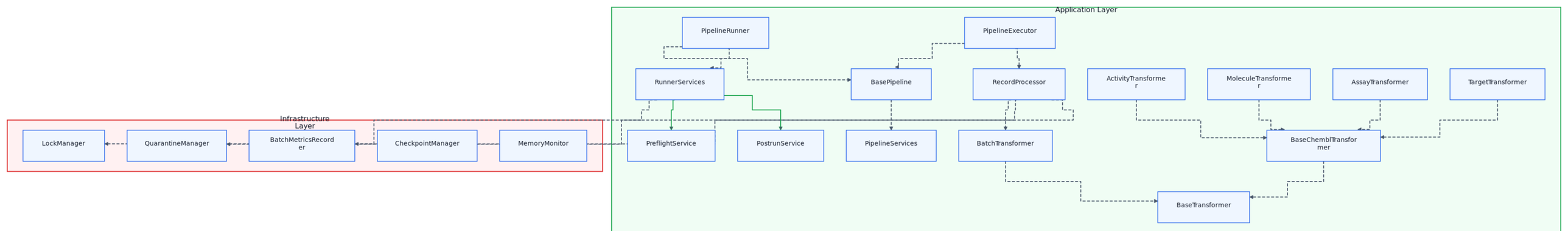

Infrastructure Layer
LOCK_ACQUIRING
Application Layer
PREFLIGHT
EXTRACTING
TRANSFORMING
Domain Layer
*
VALIDATING
RequestLock
DQMetrics
LOADING
ValidateLock
WaitingForLock
SHUTDOWN_REQUESTED
CleanupCheckpoint
ERROR
LockAcquired
RouteQuarantine
CHECKPOINT_SAVED
CheckpointLoad
ReleaseLock
SaveCheckpoint

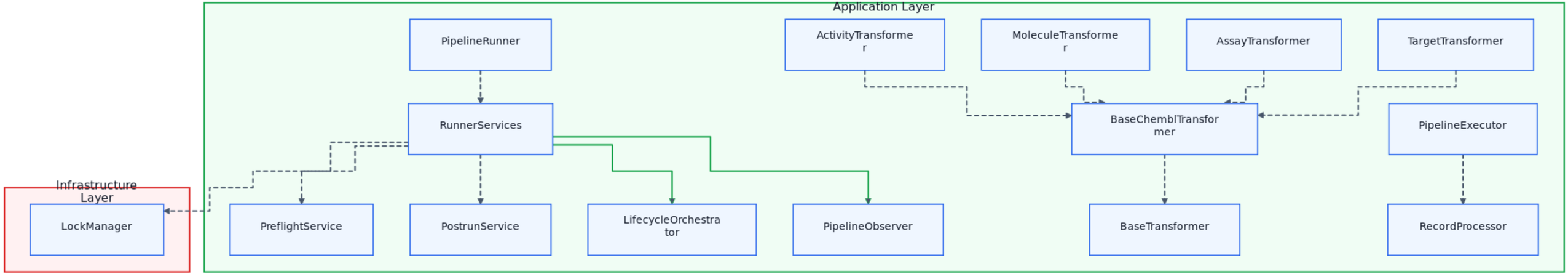

Application Layer
COMPLETED
FAILED
*
INIT
HealthChecks
NormalizeValues
PanderaCheck
StopNewWork
FetchBatch
Domain Layer
LoadingConfig
ValidateLock
LogError
Infrastructure Layer
RequestLock
WriteBronze
DQMetrics


Application Layer
PipelineExecutor
PipelineRunner
RecordProcessor
ActivityTransforme
r
BasePipeline
Infrastructure
Layer
QuarantineManager
BatchMetricsRecord
er
BatchTransformer
BaseChemblTransfor
mer
PipelineServices
BaseTransformer
PipelineObserver


Application Layer
PipelineExecutor
PipelineRunner
RecordProcessor
TargetTransformer
AssayTransformer
MoleculeTransforme
r
ActivityTransforme
r
BasePipeline
RunnerServices
Infrastructure
Layer
BatchMetricsRecord
er
QuarantineManager
LockManager
BatchTransformer
BaseChemblTransfor
mer
PipelineServices
PreflightService
PostrunService
PipelineObserver
LifecycleOrchestra
tor
BaseTransformer

Transformers
ActivityTransforme
r
+entity: str =
"activity"
#_transform_impl(record) :
SilverRecord
MoleculeTransforme
r
+entity: str =
"molecule"
#_transform_impl(record) :
SilverRecord
AssayTransformer
+entity: str =
"assay"
#_transform_impl(record) :
SilverRecord
TargetTransformer
+entity: str =
"target"
#_transform_impl(record) :
SilverRecord
«abstract»
BaseChemblTransfor
mer
#_normalize_chembl_id
(id)
#_normalize_units(value,
units) : tuple
«abstract»
BaseTransformer
#config:
ProcessingConfig
#stages:
TransformStages
#logger:
LoggerPort
+transform(record: BronzeRecord) :
SilverRecord | None
+transform_batch(records) :
list[SilverRecord]
#_transform_impl(record) :
SilverRecord | None
#compute_hash(provider, data)
#serialize_json(obj)
#content_hash
#extract_silver_record(entity
_id, record, field_record
) : SilverRecord
#_extract_specific(record,
**kwargs) : dict
#_create_entity(**kwargs)
: BaseEntity
Services
LockManager
-_lock: LockPort
-_logger:
LoggerPort
-_ttl_default_interval
int = 30
+acquire_lock(resource, owner_id)
+release_lock(resource,
owner_id) : bool
+extend_lock(resource,
owner_id) : bool
CheckpointManager
-_checkpoint:
CheckpointPort
-_logger:
LoggerPort
+load_checkpoint(key) :
Checkpoint | None
+save_checkpoint(key,
checkpoint) : None
+delete_checkpoint(key)
: None
QuarantineManager
-_quarantine:
QuarantinePort
-_logger:
LoggerPort
+quarantine_record(record,
errors) : str
PreflightService
-_data_source:
DataSourcePort
-_storage:
StoragePort
-_logger:
LoggerPort
+run_checks() :
PreflightResult
-_check_source_health() :
CheckResult
-_check_storage_health
() : CheckResult
PostrunService
-_storage:
StoragePort
-_logger:
LoggerPort
-_config:
DQConfig
+run_checks(execution_metrics) :
PostrunResult
-_check_row_counts(
execution_metrics) :
CheckResult
LifecycleOrchestra
tor
-_lifecycle_service:
MedallionLifecycleService
-_policy:
MedallionPolicy
-_logger:
LoggerPort
+clear_for_run(run_type,
tables) : ClearResult
PipelineObserver
-_tracer:
TracerPort
-_logger:
LoggerPort
+observe_batch(batch_id,
record_count, duration,
error=None) : None
MemoryMonitor
-_threshold_pct:
float = 0.8
-_check_interval_ms:
int = 1000
+get_memory_stats() :
MemoryStats
-_get_rss_estimate() :
MemoryStats
BatchMetricsRecord
er
-_metrics:
MetricsPort
+record_fetch(count,
duration_ms) : None
+record_write(count, bytes,
duration, layer) : None
Core
PipelineRunner
-_pipeline:
BasePipeline
-_services:
RunnerServices
-_lock_key: str
+run() : RunResult
-_acquire_lock() :
bool
-_run_preflight() :
PreflightResult
-_run_postrun() :
PostrunResult
-_release_lock() :
None
PipelineExecutor
-_pipeline:
BasePipeline
-_executor:
RecordProcessor
-_checkpoint_interval:
int = 1000
+execute() :
ExecutionResult
-_load_and_process() :
AsyncIterator[BatchResult]
-_log_progress(batch_idx:
int) : None
«frozen dataclass»
RunnerServices
+lock_manager:
LockManager
+checkpoint_manager:
CheckpointManager
+preflight_service:
PreflightService
+postrun_service:
PostrunService
+lifecycle_orchestrator:
LifecycleOrchestrator
+lock_context_holder?
RecordProcessor
-_transformer:
BaseTransformer
-_batch_transformer:
BatchTransformer
-_quarantine:
QuarantineManager
-_metrics:
BatchMetricsRecorder
+process_batch(records) :
BatchResult
-_process_stream(records) :
AsyncIterator[BatchResult]
«abstract»
BasePipeline
#config:
PipelineConfig
#runner:
PipelineRunner
#services:
PipelineServices
+create(config, runtime,
**overrides) : BasePipeline
+pipeline_id() : str
+provider() : str
+run_id() : RunID
BatchTransformer
-_transformer:
BaseTransformer
-_logger:
LoggerPort
+transform_batch(records) :
TransformResult
-_transform_stream(records) :
AsyncIterator[TransformResult]
«frozen dataclass»
PipelineServices
+data_source:
DataSourcePort
+storage:
StoragePort
+lock: LockPort
+checkpoint:
CheckpointPort
+quarantine:
QuarantinePort
+logger:
LoggerPort
+tracer:
TracerPort
+dq_monitor:
DQMonitorPort?


Application Layer
PipelineRunner
PipelineExecutor
RunnerServices
BasePipeline
RecordProcessor
ActivityTransforme
r
MoleculeTransforme
r
AssayTransformer
TargetTransformer
Infrastructure
Layer
LockManager
QuarantineManager
BatchMetricsRecord
er
CheckpointManager
MemoryMonitor
PreflightService
PostrunService
PipelineServices
BatchTransformer
BaseChemblTransfor
mer
BaseTransformer


Application Layer
PipelineRunner
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TargetTransformer
RunnerServices
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mer
PipelineExecutor
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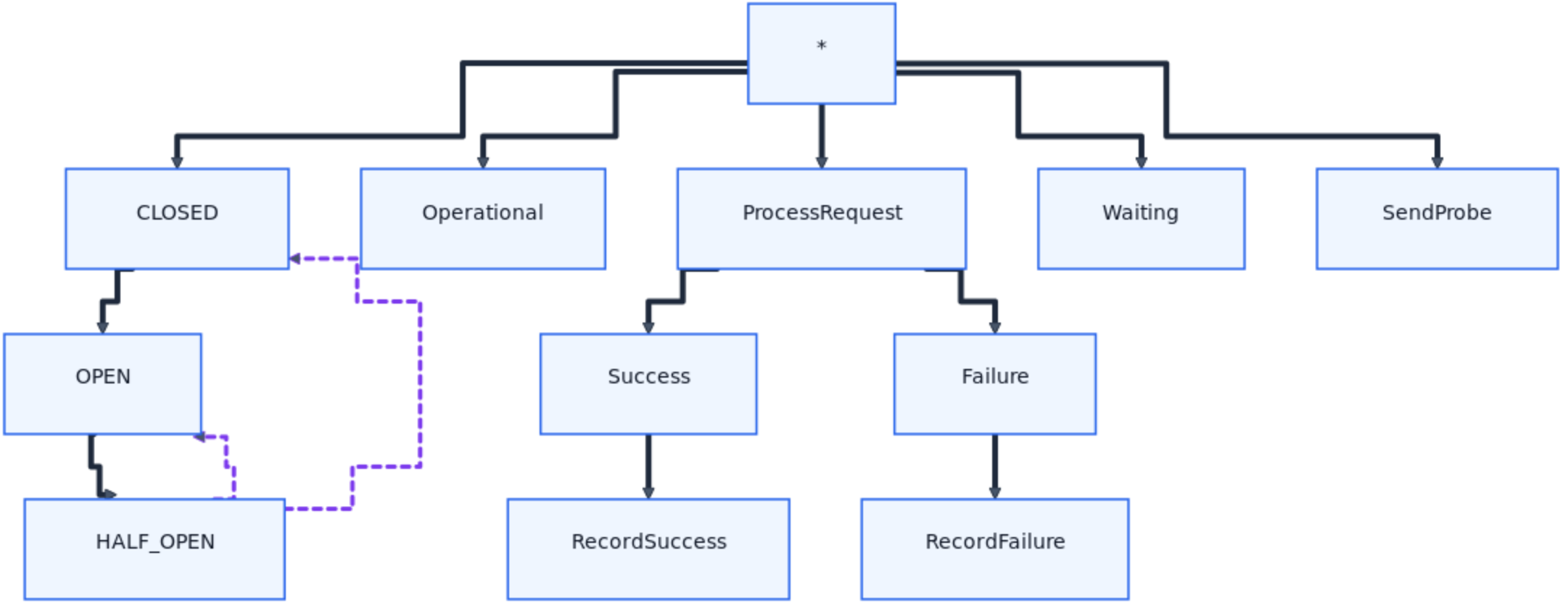
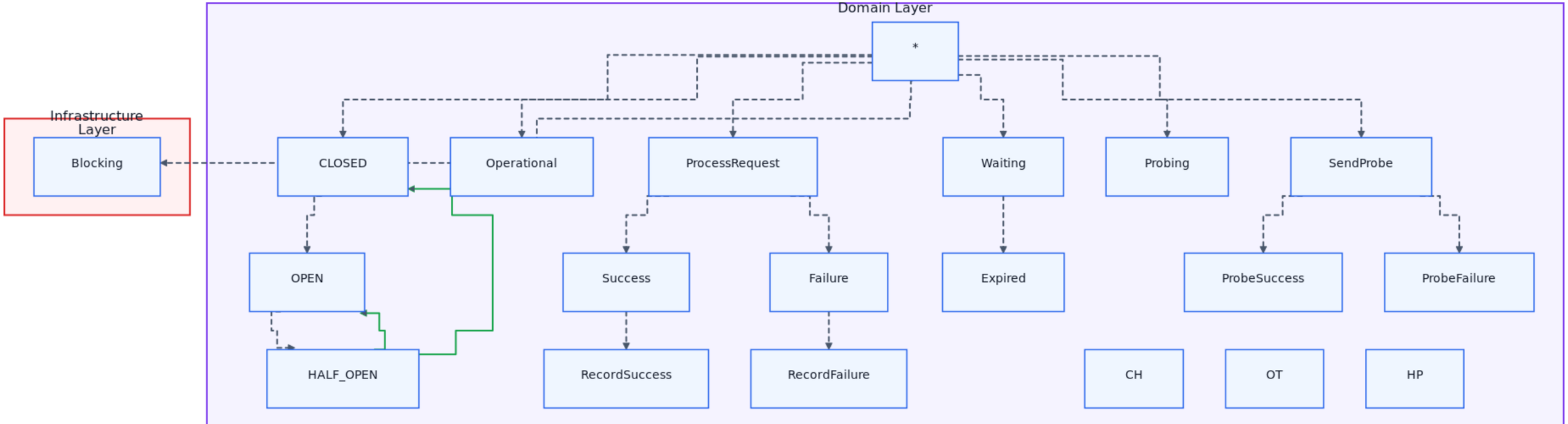
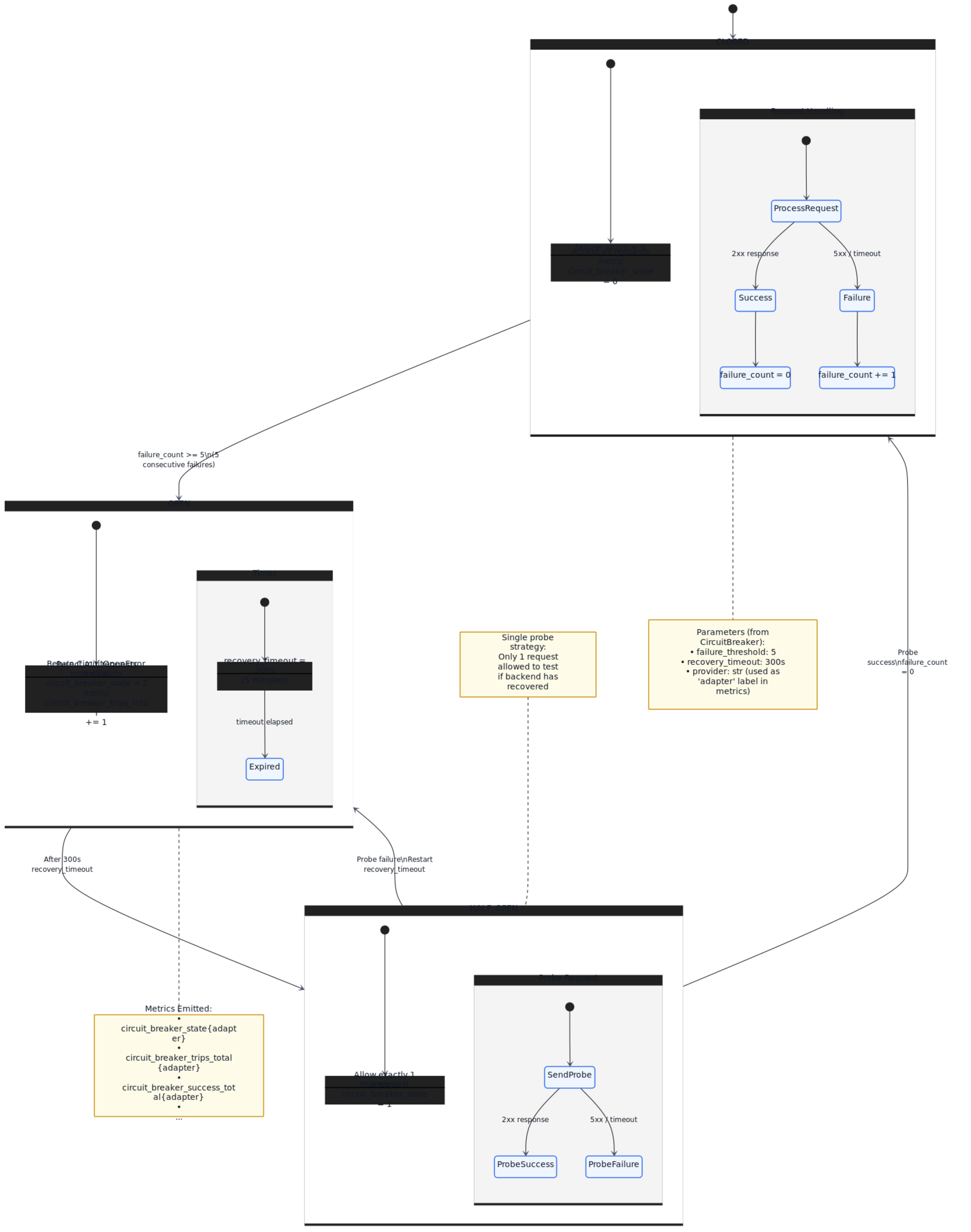
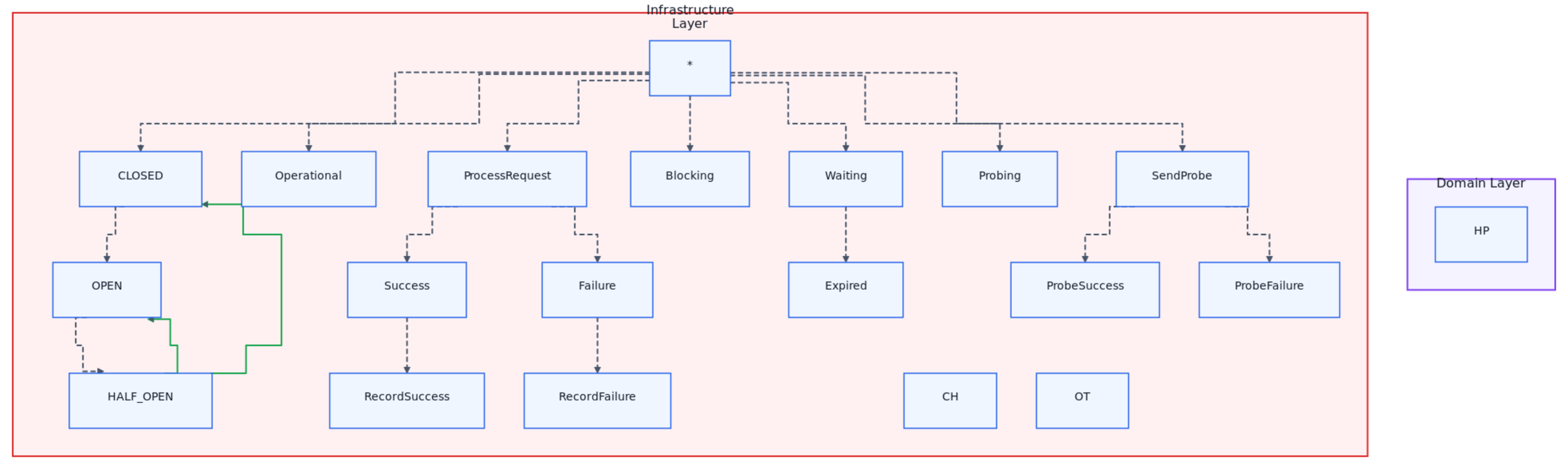
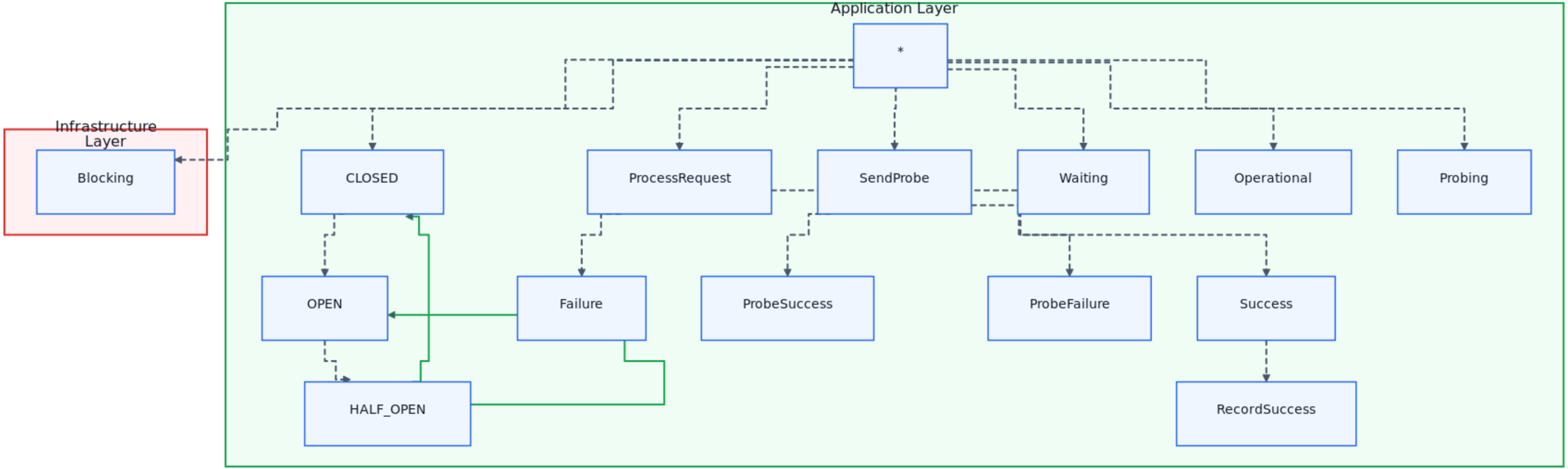
Layer
LockManager
PreflightService
PostrunService
LifecycleOrchestra
tor
PipelineObserver
BaseTransformer
RecordProcessor

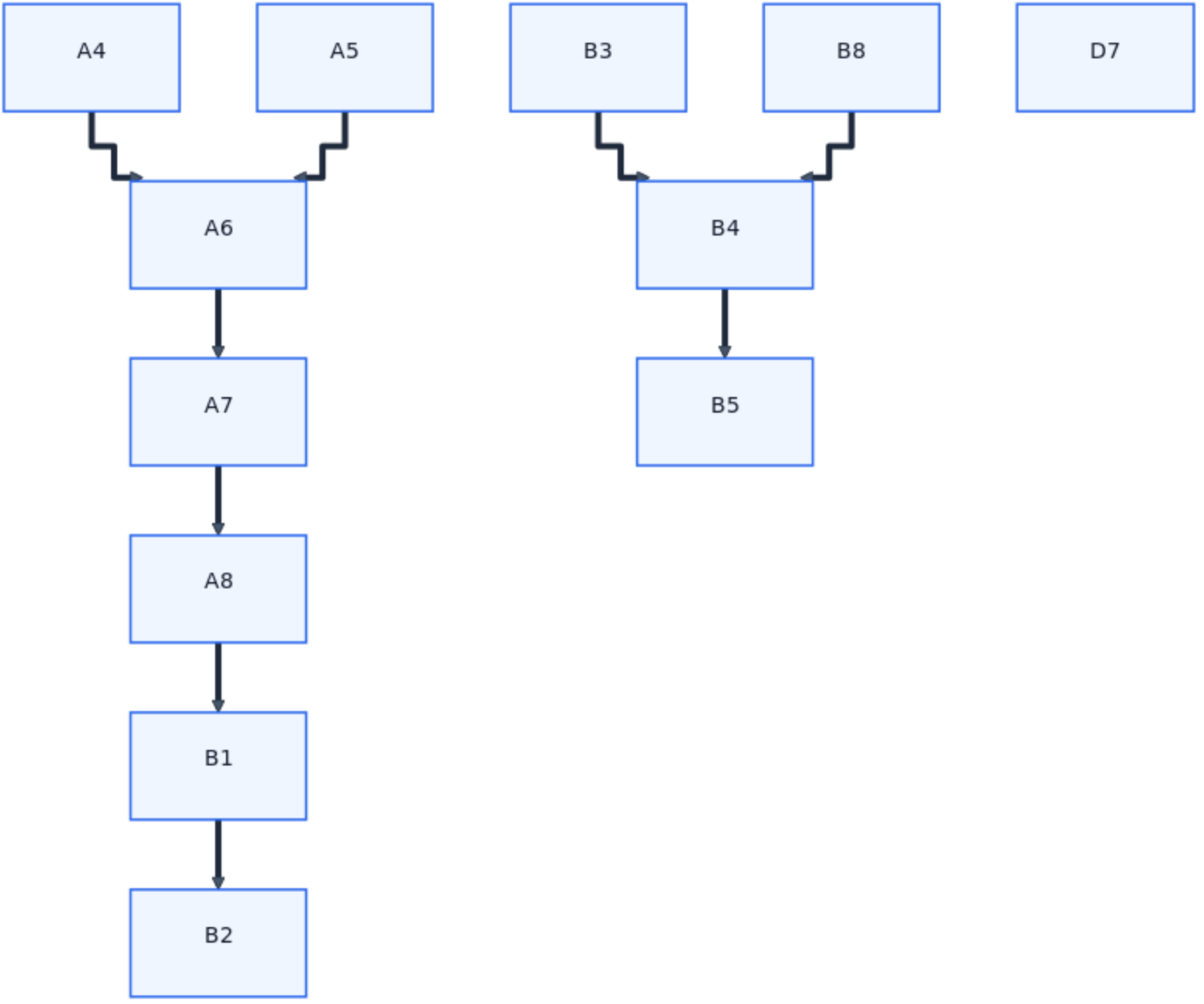

*
CLOSED
Operational
ProcessRequest
Waiting
SendProbe
OPEN
HALF_OPEN
Success
Failure
RecordSuccess
RecordFailure

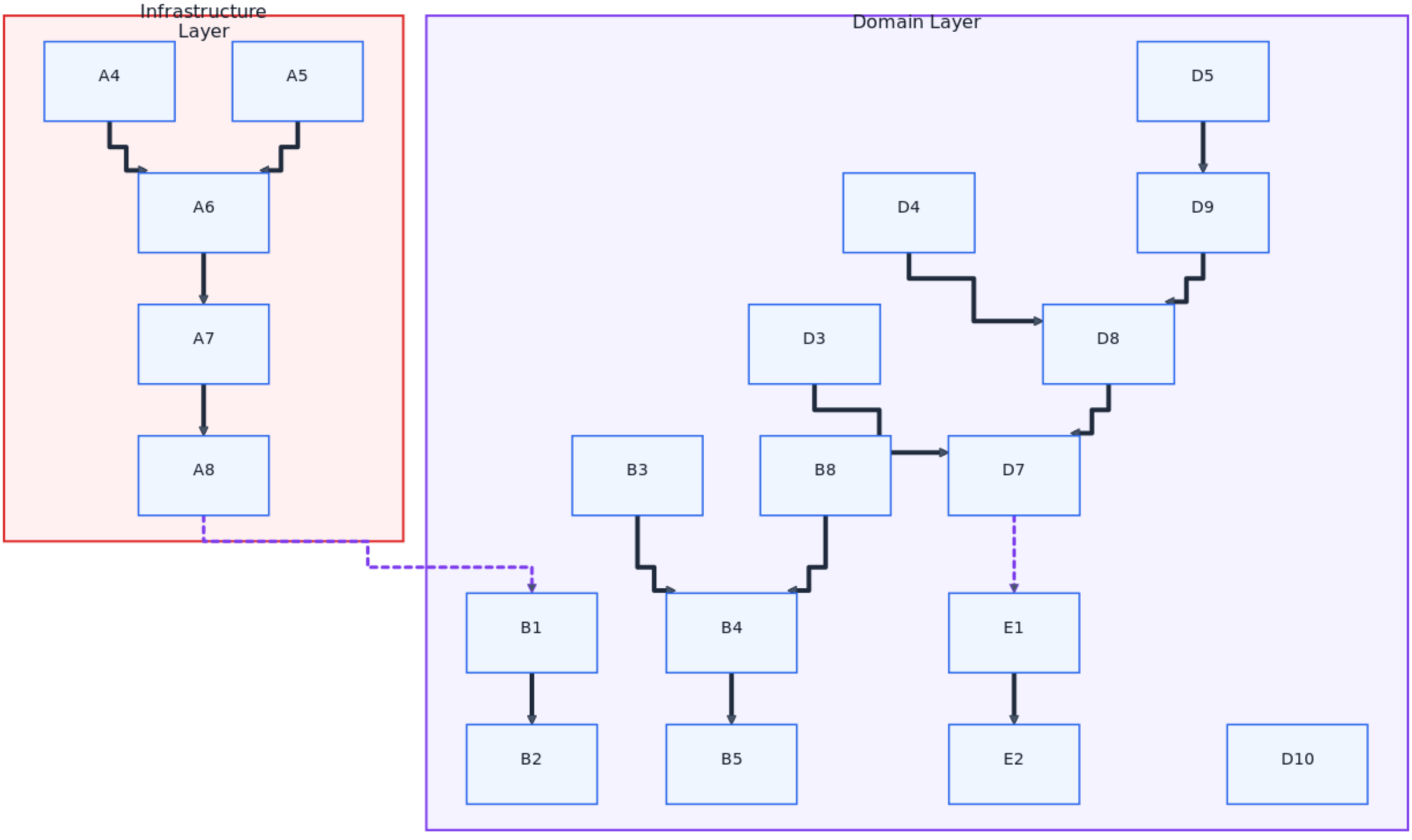

Domain Layer
*
Infrastructure Layer
Blocking
CLOSED
Operational
ProcessRequest
Waiting
Probing
SendProbe
OPEN
HALF_OPEN
Success
Failure
Expired
ProbeSuccess
ProbeFailure
RecordSuccess
RecordFailure
CH
OT
HP

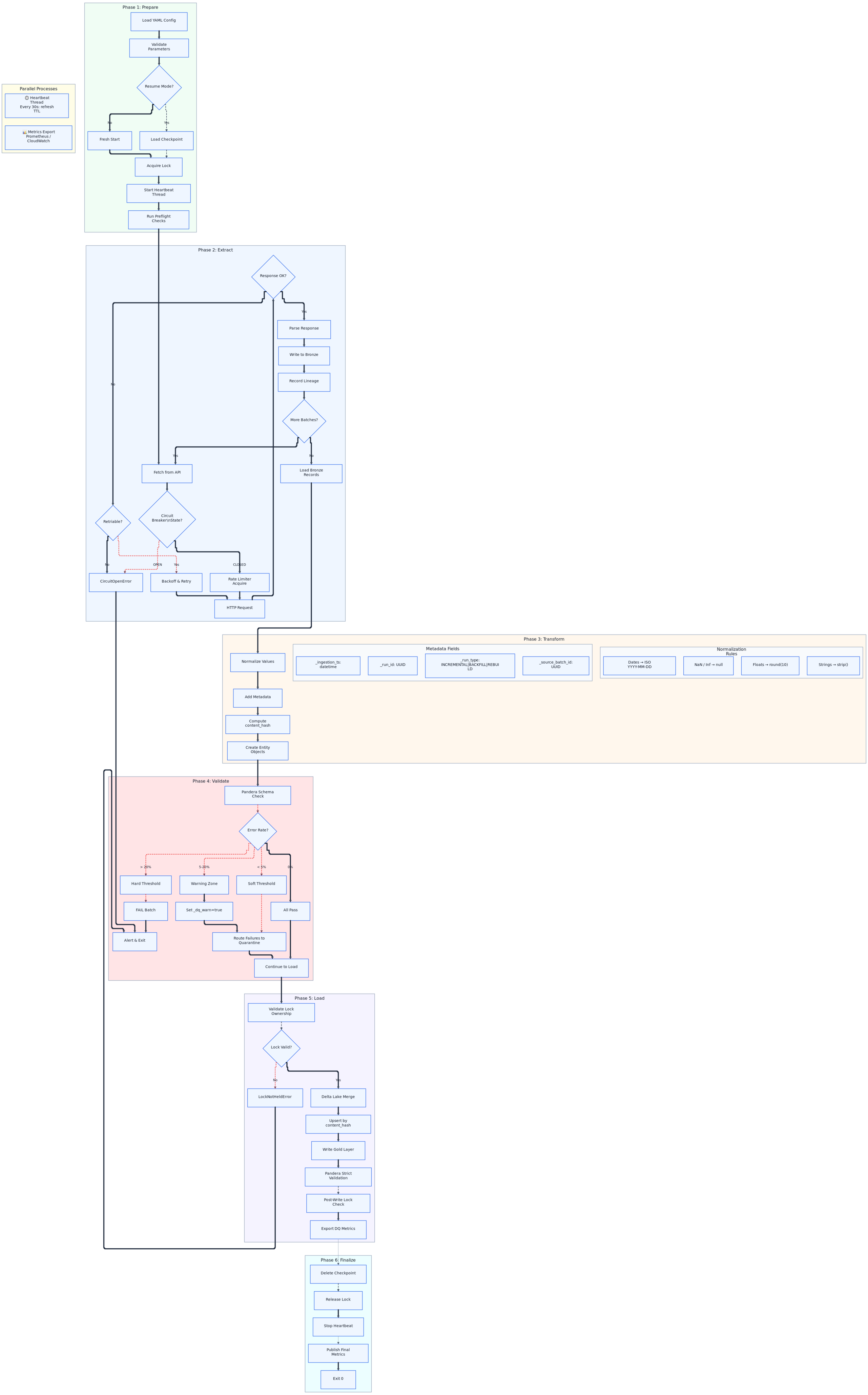

CLOSED
Request Handling
ProcessRequest
2xx response
5xx / timeout
Success
Failure
failure_count = 0
failure_count += 1
failure_count >= 5\n(5 consecutive failures)
OPEN
Timer
recovery_timeout = 300s (5 minutes)
timeout elapsed
Expired
+= 1
Single probe strategy: Only 1 request allowed to test if backend has recovered
Parameters (from CircuitBreaker):
• failure_threshold: 5
• recovery_timeout: 300s
• provider: str (used as 'adapter' label in metrics)
Probe success\nfailure_count = 0
After 300s recovery_timeout
Probe failure\nRestart recovery_timeout
HALF_OPEN
Probe Request
SendProbe
2xx response
5xx / timeout
ProbeSuccess
ProbeFailure
Metrics Emitted:
• circuit_breaker_state{adapter}
• circuit_breaker_trips_total{adapter}
• circuit_breaker_success_total{adapter}
• ...

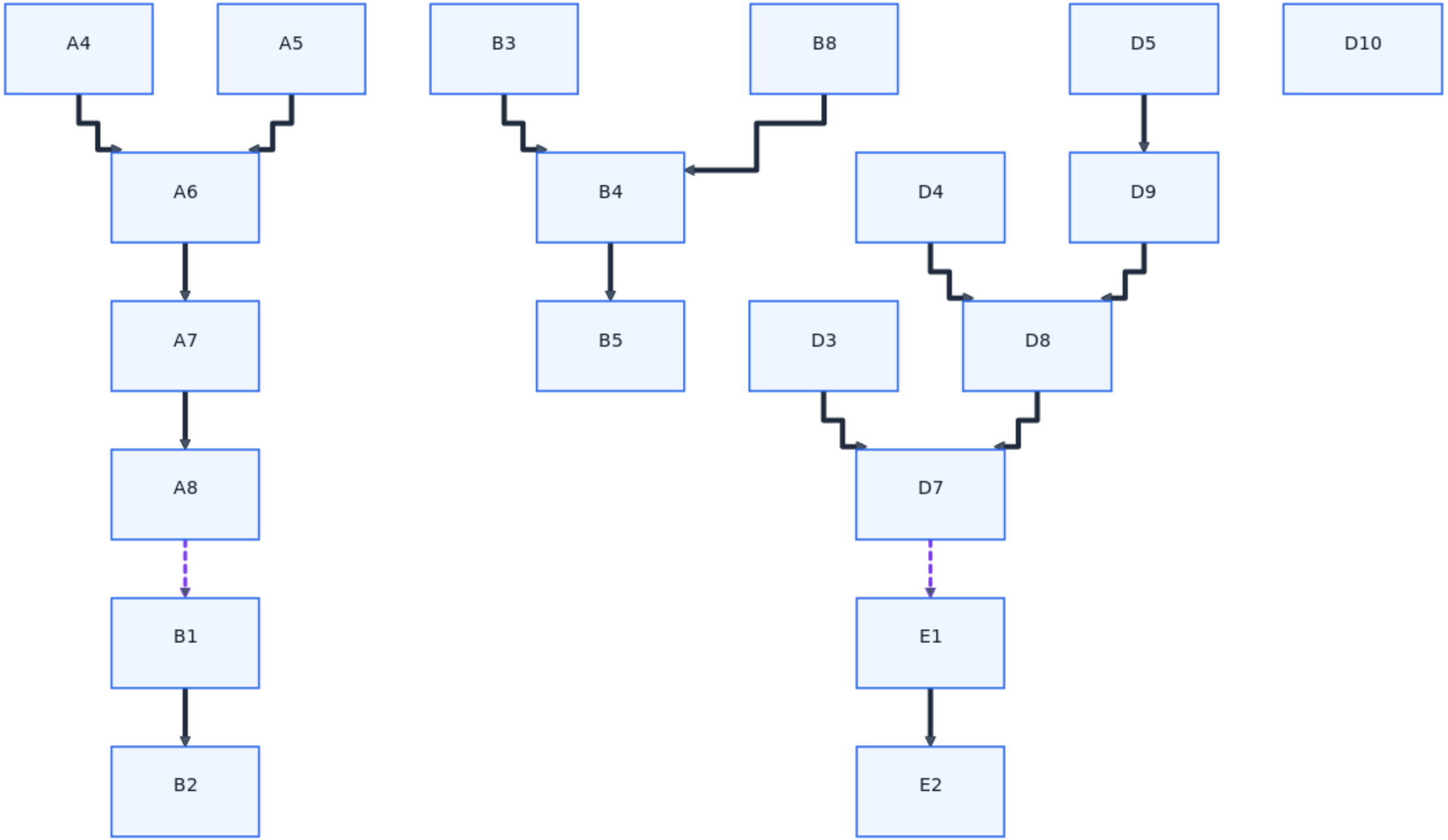

Infrastructure Layer
*
CLOSED
Operational
ProcessRequest
Blocking
Waiting
Probing
SendProbe
OPEN
HALF_OPEN
Success
Failure
Expired
ProbeSuccess
ProbeFailure
RecordSuccess
RecordFailure
CH
OT
Domain Layer
HP

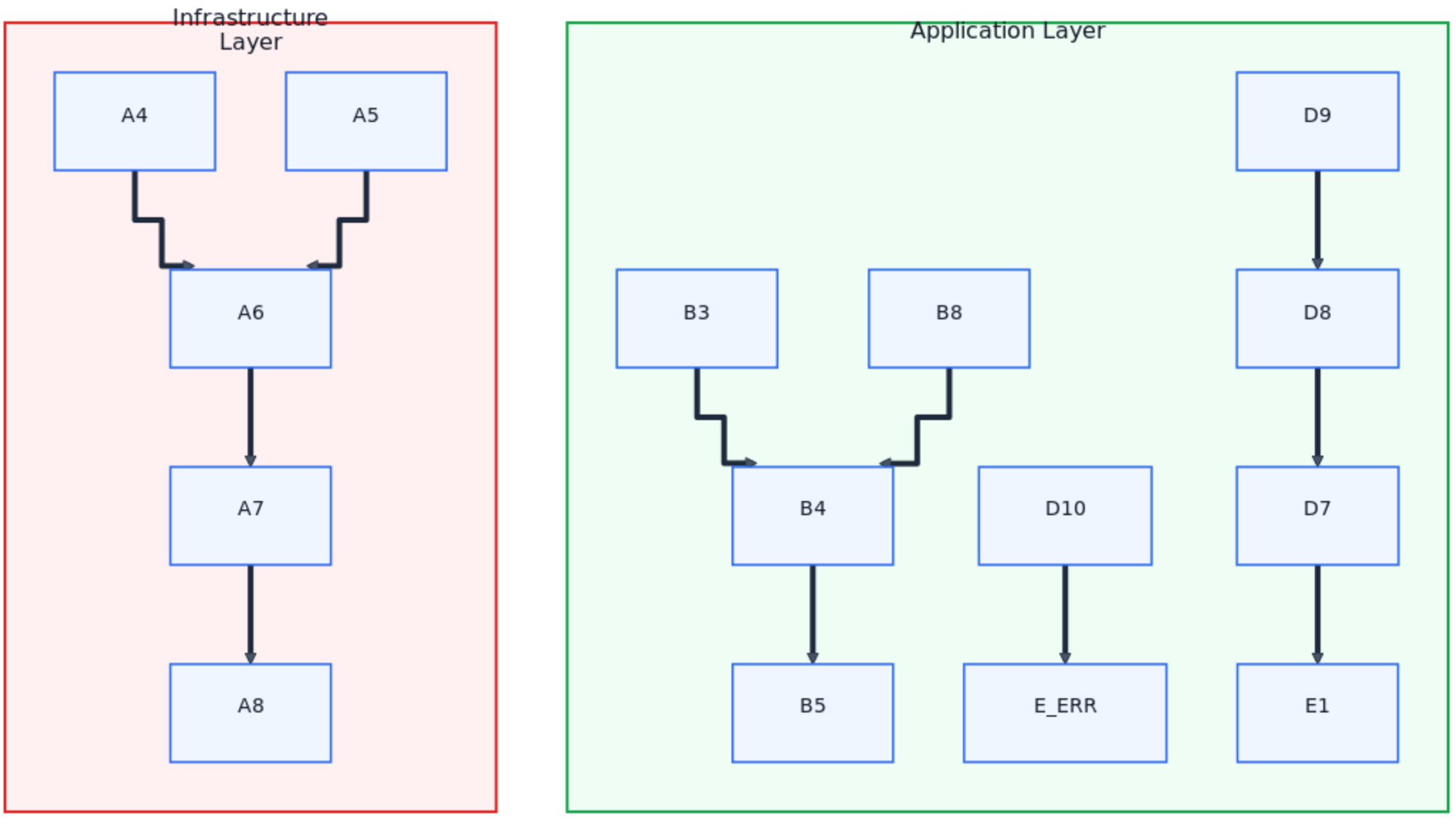

Application Layer
*
Infrastructure Layer
Blocking
CLOSED
ProcessRequest
SendProbe
Waiting
Operational
Probing
OPEN
Failure
ProbeSuccess
ProbeFailure
Success
HALF_OPEN
RecordSuccess

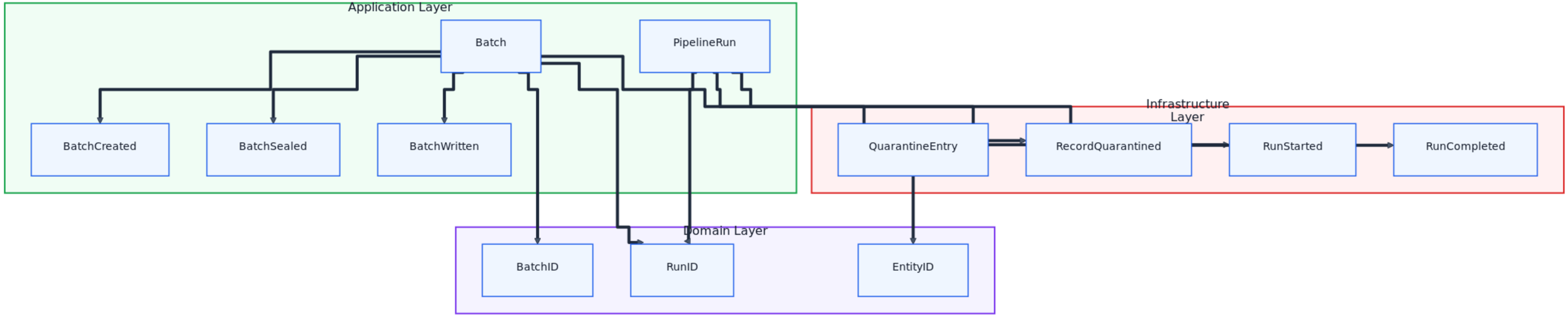

A4
A5
B3
B8
D7
A6
B4
A7
B5
A8
B1
B2

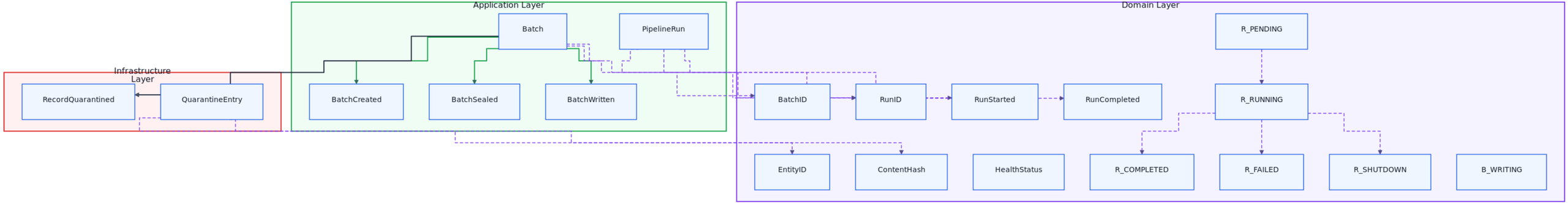

Infrastructure Layer
A4
A5
A6
A7
A8
Domain Layer
D5
D4
D9
D3
D8
B3
B8
D7
B1
B4
E1
B2
B5
E2
D10

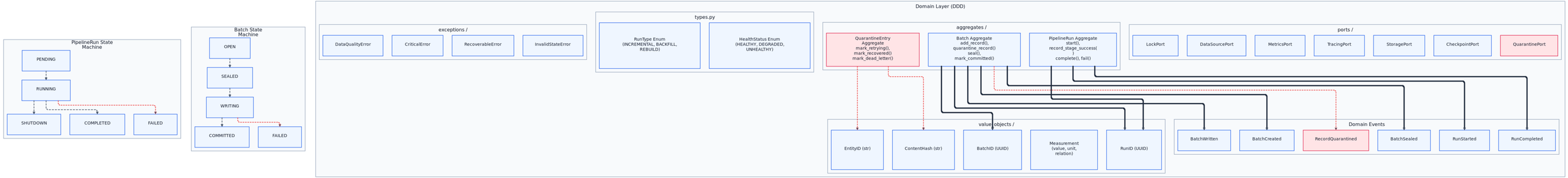

Parallel Processes
⏱️ Heartbeat Thread
Every 30s: refresh TTL
📊 Metrics Export
Prometheus / CloudWatch
Phase 1: Prepare
Load YAML Config
Validate Parameters
Resume Mode?
No
Yes
Fresh Start
Load Checkpoint
Acquire Lock
Start Heartbeat Thread
Run Preflight Checks
Phase 2: Extract
Response OK?
Yes
No
Parse Response
Write to Bronze
Record Lineage
More Batches?
Yes
No
Fetch from API
Load Bronze Records
Retriable?
Circuit Breaker\nState?
No
OPEN
Yes
CLOSED
CircuitOpenError
Backoff & Retry
Rate Limiter Acquire
HTTP Request
Phase 3: Transform
Normalize Values
Metadata Fields
_ingestion_ts: datetime
_run_id: UUID
_run_type: INCREMENTAL|BACKFILL|REBUILD
_source_batch_id: UUID
Normalization Rules
Dates → ISO YYYY-MM-DD
NaN / Inf → null
Floats → round(10)
Strings → strip()
Add Metadata
Compute content_hash
Create Entity Objects
Phase 4: Validate
Pandera Schema Check
Error Rate?
> 20%
5-20%
< 5%
0%
Hard Threshold
Warning Zone
Soft Threshold
All Pass
FAIL Batch
Set _dq_warn=true
Alert & Exit
Route Failures to Quarantine
Continue to Load
Phase 5: Load
Validate Lock Ownership
Lock Valid?
No
Yes
LockNotHeldError
Delta Lake Merge
Upsert by content_hash
Write Gold Layer
Pandera Strict Validation
Post-Write Lock Check
Export DQ Metrics
Phase 6: Finalize
Delete Checkpoint
Release Lock
Stop Heartbeat
Publish Final Metrics
Exit 0

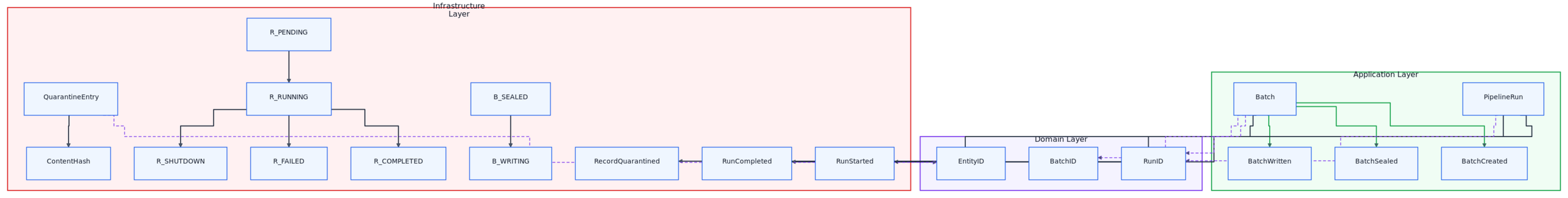

A4
A5
B3
B8
D5
D10
A6
B4
D4
D9
A7
B5
D3
D8
A8
D7
B1
E1
B2
E2

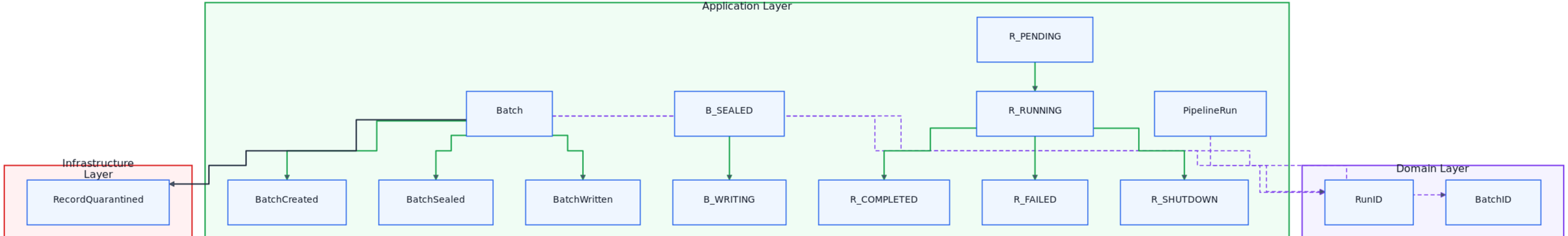

Infrastructure Layer
A4
A5
A6
A7
A8
Application Layer
D9
B3
B8
D8
B4
D10
D7
B5
E_ERR
E1

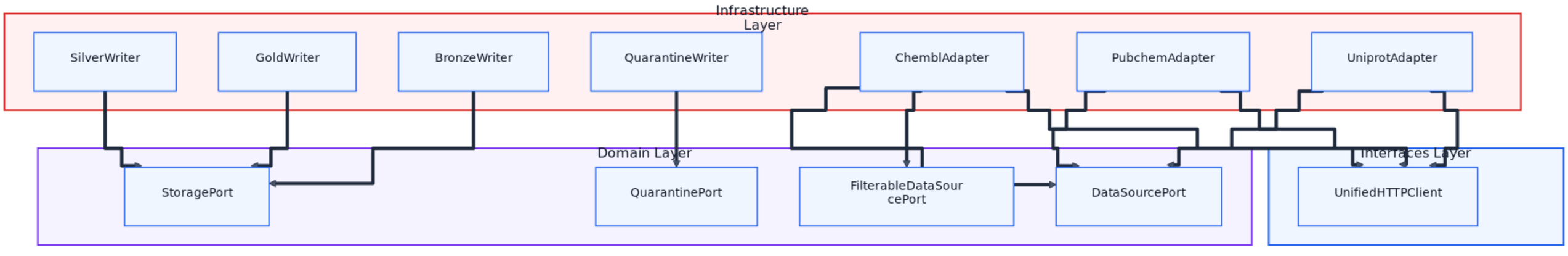

Application Layer
Batch
PipelineRun
BatchCreated
BatchSealed
BatchWritten
Infrastructure Layer
QuarantineEntry
RecordQuarantined
RunStarted
RunCompleted
Domain Layer
BatchID
RunID
EntityID

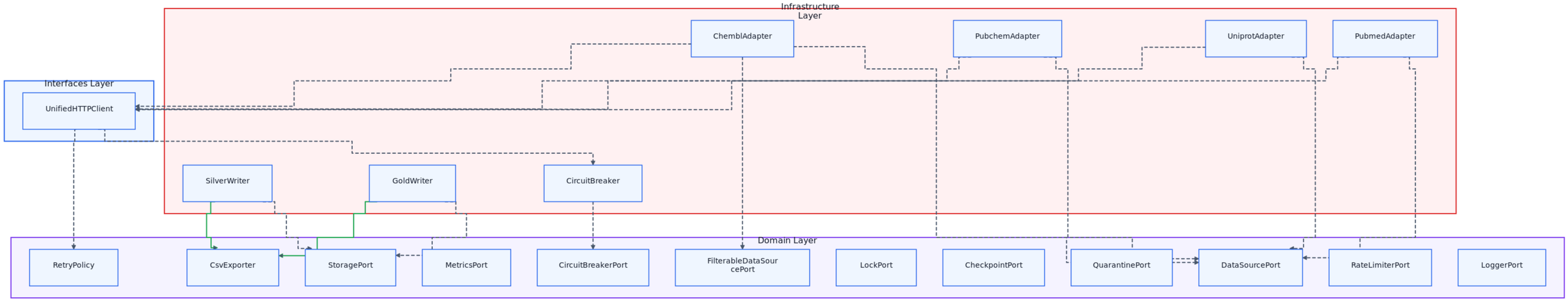

Application Layer
Batch
PipelineRun
Infrastructure Layer
RecordQuarantined
QuarantineEntry
BatchCreated
BatchSealed
BatchWritten
Domain Layer
BatchID
RunID
RunStarted
RunCompleted
R_PENDING
R_RUNNING
EntityID
ContentHash
HealthStatus
R_COMPLETED
R_FAILED
R_SHUTDOWN
B_WRITING

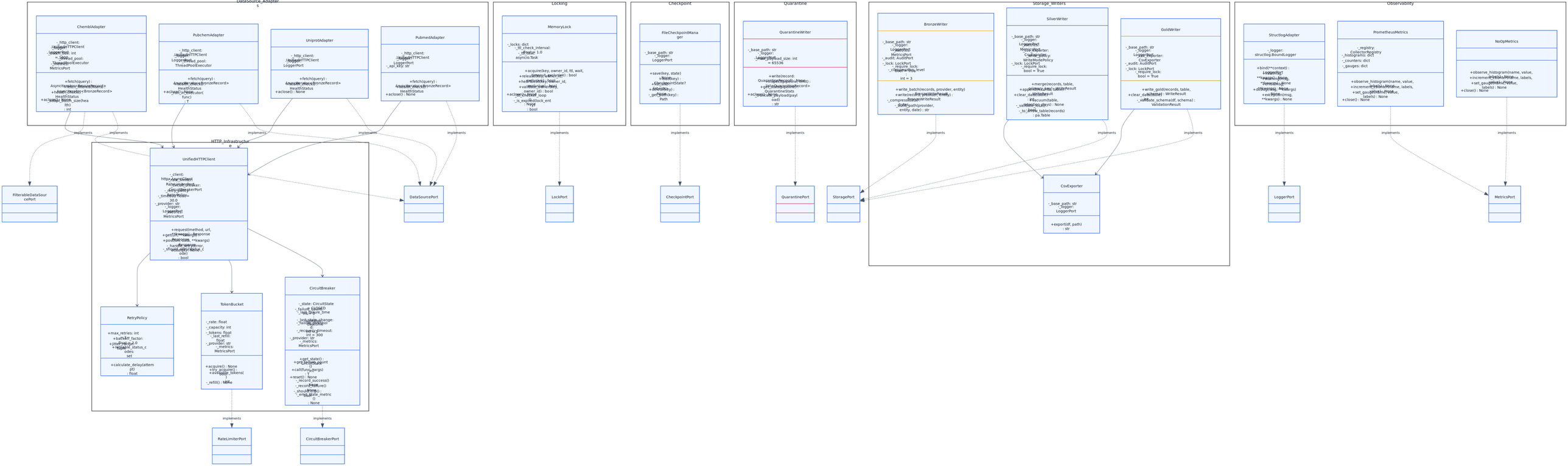

PipelineRun State Machine
PENDING
RUNNING
SHUTDOWN
COMPLETED
FAILED
Batch State Machine
OPEN
SEALED
WRITING
COMMITTED
FAILED
Domain Layer (DDD)
exceptions /
DataQualityError
CriticalError
RecoverableError
InvalidStateError
types.py
RunType Enum (INCREMENTAL, BACKFILL, REBUILD)
HealthStatus Enum (HEALTHY, DEGRADED, UNHEALTHY)
aggregates /
QuarantineEntry Aggregate mark_retrying(), mark_recovered() mark_dead_letter()
Batch Aggregate add_record(), quarantine_record() seal(), mark_committed()
PipelineRun Aggregate start(), record_stage_success( ) complete(), fail()
ports /
LockPort
DataSourcePort
MetricsPort
TracingPort
StoragePort
CheckpointPort
QuarantinePort
value objects /
EntityID (str)
ContentHash (str)
BatchID (UUID)
Measurement (value, unit, relation)
RunID (UUID)
Domain Events
BatchWritten
BatchCreated
RecordQuarantined
BatchSealed
RunStarted
RunCompleted

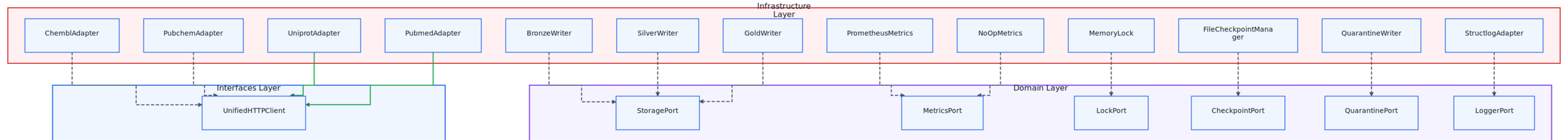

Infrastructure Layer
R_PENDING
QuarantineEntry
R_RUNNING
B_SEALED
ContentHash
R_SHUTDOWN
R_FAILED
R_COMPLETED
B_WRITING
RecordQuarantined
RunCompleted
RunStarted
Domain Layer
EntityID
BatchID
RunID
Application Layer
Batch
PipelineRun
BatchWritten
BatchSealed
BatchCreated

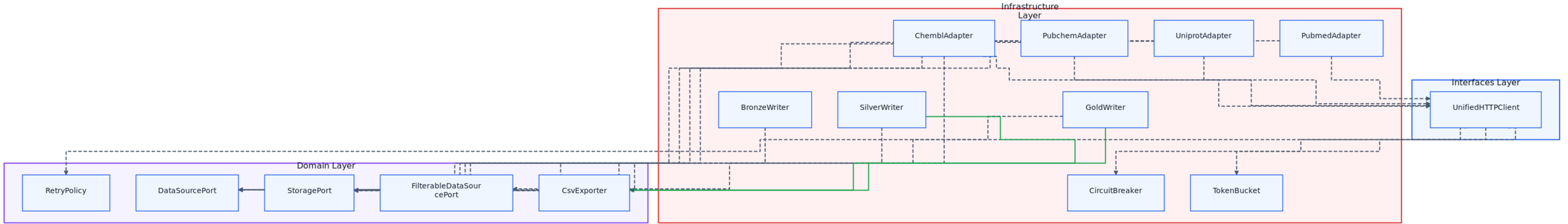

Application Layer
R_PENDING
Batch
B_SEALED
R_RUNNING
PipelineRun
Infrastructure Layer
RecordQuarantined
BatchCreated
BatchSealed
BatchWritten
B_WRITING
R_COMPLETED
R_FAILED
R_SHUTDOWN
Domain Layer
RunID
BatchID


Infrastructure Layer
SilverWriter
GoldWriter
BronzeWriter
QuarantineWriter
ChemblAdapter
PubchemAdapter
UniprotAdapter
Domain Layer
StoragePort
QuarantinePort
FilterableDataSour cePort
DataSourcePort
Interfaces Layer
UnifiedHTTPClient


Infrastructure Layer
ChemblAdapter
PubchemAdapter
UniprotAdapter
PubmedAdapter
Interfaces Layer
UnifiedHTTPClient
SilverWriter
GoldWriter
CircuitBreaker
Domain Layer
RetryPolicy
CsvExporter
StoragePort
MetricsPort
CircuitBreakerPort
FilterableDataSourcePort
LockPort
CheckpointPort
QuarantinePort
DataSourcePort
RateLimiterPort
LoggerPort


DataSource_Adapters
ChemblAdapter
- _http_client: UnifiedHTTPClient
- _logger: LoggerPort
- _page_size: int = 1000
- _thread_pool: ThreadPoolExecutor
- _metrics: MetricsPort
+fetch(query) : AsyncIterator[BronzeRecord]
+fetch_by_id(id) : BronzeRecord
+health_check() : HealthStatus
+aclose() : None
PubchemAdapter
- _http_client: UnifiedHTTPClient
- _logger: LoggerPort
- _thread_pool: ThreadPoolExecutor
+fetch(query) : AsyncIterator[BronzeRecord]
+health_check() : HealthStatus
+aclose() : None
UniprotAdapter
- _http_client: UnifiedHTTPClient
- _logger: LoggerPort
+fetch(query) : AsyncIterator[BronzeRecord]
+health_check() : HealthStatus
+aclose() : None
PubmedAdapter
- _http_client: UnifiedHTTPClient
- _logger: LoggerPort
- _api_key: str
+fetch(query) : AsyncIterator[BronzeRecord]
+health_check() : HealthStatus
+aclose() : None
Locking
MemoryLock
- _locks: dict
- _ttl_check_interval: float = 1.0
- _cleanup_task: asyncio.Task
+acquire(key, owner_id, ttl, wait, timeout) : bool
+release(key, owner_id) : bool
+heartbeat(key, owner_id, ttl) : bool
+validate_owner(key, owner_id) : bool
+aclose() : None
- _cleanup_loop()
- _is_expired(lock_entry) : bool
Checkpoint
FileCheckpointManager
- _base_path: str
- _logger: LoggerPort
+save(key, state) : None
+load(key) : CheckpointState
+delete(key) : None
- _get_path(key) : Path
Quarantine
QuarantineWriter
- _base_path: str
- _logger: LoggerPort
- _max_buffer_size: int = 65536
+write(record: QuarantineRecord, reason: str) : None
+get_stats() : QuarantineStats
+aclose() : None
- _flush_buffer(payload) : str
Storage_Writers
BronzeWriter
- _base_path: str
- _logger: LoggerPort
- _metrics: MetricsPort
- _audit: AuditPort
- _lock: LockPort
- _require_lock: bool = True
- _compression_level: int = 3
+write_batch(records, provider, entity) : BronzeWriteResult
+write(record, provider, entity) : BronzeWriteResult
- _build_path(provider, entity, date) : str
SilverWriter
- _base_path: str
- _logger: LoggerPort
- _metrics: MetricsPort
- _audit: AuditPort
- _lock: LockPort
- _require_lock: bool = True
+merge(records, table, primary_key) : SilverWriteResult
+append_only(records, table) : SilverWriteResult
+clear_table(table) : SilverWriteResult
- _read_existing(table, keys) : pa.Table
- _to_arrow_table(records) : pa.Table
GoldWriter
- _base_path: str
- _logger: LoggerPort
- _csv_exporter: CsvExporter
- _audit: AuditPort
- _lock: LockPort
- _require_lock: bool = True
+write_gold(records, table, schema) : WriteResult
+validate_schema(df, schema) : ValidationResult
CsvExporter
- _base_path: str
- _logger: LoggerPort
+export(df, path) : str
Observability
StructlogAdapter
- _logger: structlog.BoundLogger
+bind(**context) : LoggerPort
+unbind(*keys) : None
+debug(msg, **kwargs) : None
+info(msg, **kwargs) : None
+warning(msg, **kwargs) : None
+error(msg, **kwargs) : None
+exception(msg, **kwargs) : None
PrometheusMetrics
- _registry: CollectorRegistry
- _histograms: dict
- _counters: dict
- _gauges: dict
+observe_histogram(name, value, labels) : None
+increment_counter(name, labels, value) : None
+set_gauge(name, value, labels) : None
+close() : None
NoOpMetrics
+observe_histogram(name, value, labels) : None
+increment_counter(name, labels, value) : None
+set_gauge(name, value, labels) : None
+close() : None
HTTP_Infrastructure
UnifiedHTTPClient
- _client: httpx.AsyncClient
- _rate_limiter: RateLimiterPort
- _circuit_breaker: CircuitBreaker
- _retry_policy: RetryPolicy
- _timeout: float = 30.0
- _provider: str
- _logger: LoggerPort
- _metrics: MetricsPort
+request(method, url, **kwargs) : httpx.Response
+get(url, **kwargs) : httpx.Response
+post(url, **kwargs) : httpx.Response
- _handle_error_response(response) : None
- _should_retry(error, attempt) : bool
RetryPolicy
+max_retries: int = 3
+backoff_factor: float = 2.0
+jitter: bool = True
+retryable_status_codes: set
+calculate_delay(attempt) : float
TokenBucket
- _rate: float
- _capacity: int
- _tokens: float
- _last_refill: float
- _provider: str
- _metrics: MetricsPort
+acquire() : None
+try_acquire(tokens) : bool
- _refill() : None
CircuitBreaker
- _state: CircuitState = CLOSED
- _failure_count: int
- _last_failure_time: float
- _last_state_change: float
- _failure_threshold: int = 5
- _recovery_timeout: int = 300
- _provider: str
- _metrics: MetricsPort
+get_state() : CircuitState
+get_stats() : CircuitStats
+call(func, *args, **kwargs) : T
+reset() : None
- _record_success()
- _record_failure()
- _should_attempt() : bool
- _emit_state_metric() : None
Ports
FilterableDataSourcePort
DataSourcePort
LockPort
CheckpointPort
QuarantinePort
StoragePort
RateLimiterPort
CircuitBreakerPort
LoggerPort
MetricsPort
implements


Infrastructure Layer
ChemblAdapter
PubchemAdapter
UniprotAdapter
PubmedAdapter
BronzeWriter
SilverWriter
GoldWriter
PrometheusMetrics
NoOpMetrics
MemoryLock
FileCheckpointMana ger
QuarantineWriter
StructlogAdapter
Interfaces Layer
UnifiedHTTPClient
Domain Layer
StoragePort
MetricsPort
LockPort
CheckpointPort
QuarantinePort
LoggerPort


Infrastructure Layer
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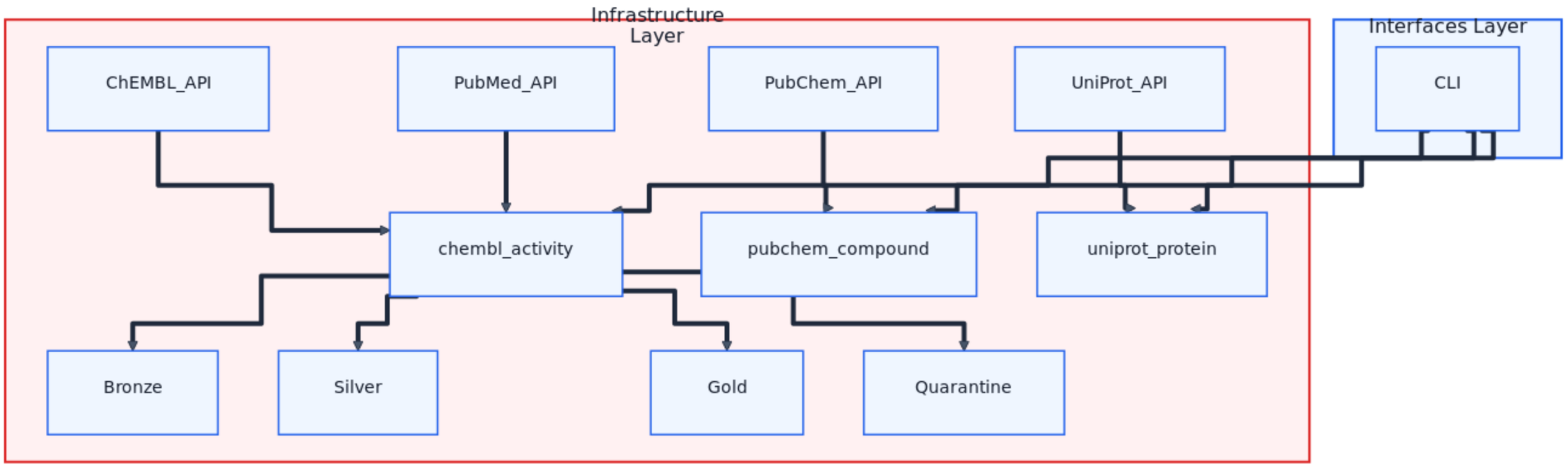
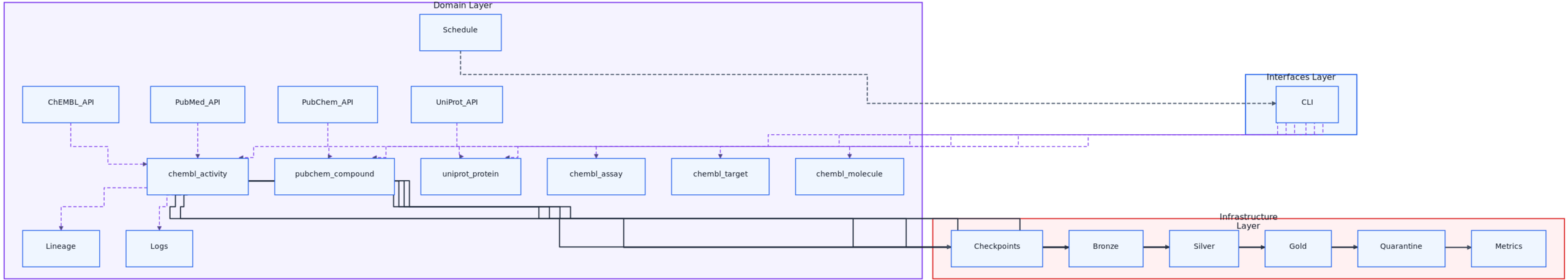
CsvExporter


Infrastructure Layer
Interfaces Layer
ChEMBL_API
PubMed_API
PubChem_API
UniProt_API
CLI
chembl_activity
pubchem_compound
uniprot_protein
Bronze
Silver
Gold
Quarantine


Domain Layer
Schedule
ChEMBL_API
PubMed_API
PubChem_API
UniProt_API
chembl_activity
pubchem_compound
uniprot_protein
chembl_assay
chembl_target
chembl_molecule
Lineage
Logs
Interfaces Layer
CLI
Infrastructure Layer
Checkpoints
Bronze
Silver
Gold
Quarantine
Metrics

External APIs

🔵 ChEMBL API
ebi.ac.uk / chembl

🔵 PubMed API
eutils.ncbi.nlm.nih.gov

🔵 PubChem API
pubchem.ncbi.nlm.nih.gov

🔵 UniProt API
uniprot.org

Local Machine (Single Instance)

CLI Execution

⏰ Local scheduler
(cron / systemd)

🖥️ CLI / Manual run
PipelineRunner

Local Pipeline Workers

📦 chembl_activity
local pipeline

📦 chembl_molecule
local pipeline

📦 chembl_assay
local pipeline

📦 chembl_target
local pipeline

📦 pubchem_compound
local pipeline

📦 uniprot_protein
local pipeline

Local Observability

📊 In-process metrics

📋 Local logs
stdout / file

Local filesystem (data /)

📁 lineage / Parquet

📁 quarantine / Parquet

📁 silver / Delta Lake

📁 checkpoints / JSON

📁 gold / Delta / Parquet

📁 bronze / JSONL+zstd

lock

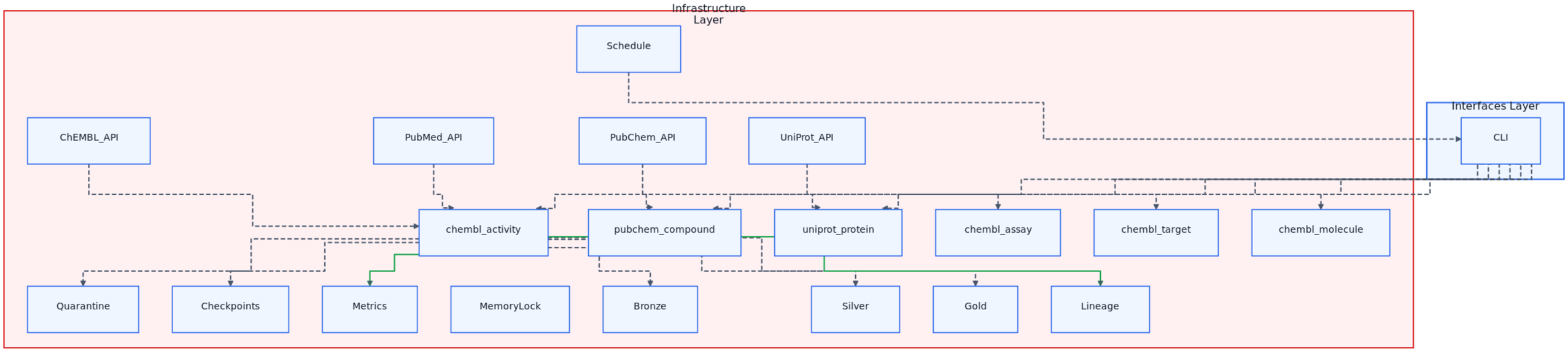
lock

lock

lock

In-Process Locking

🔒 MemoryLock
TTL 90s / heartbeat 30s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Gold
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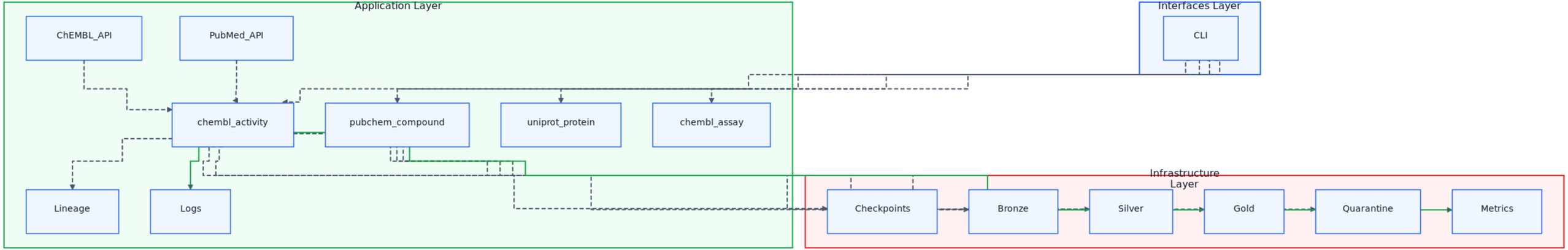
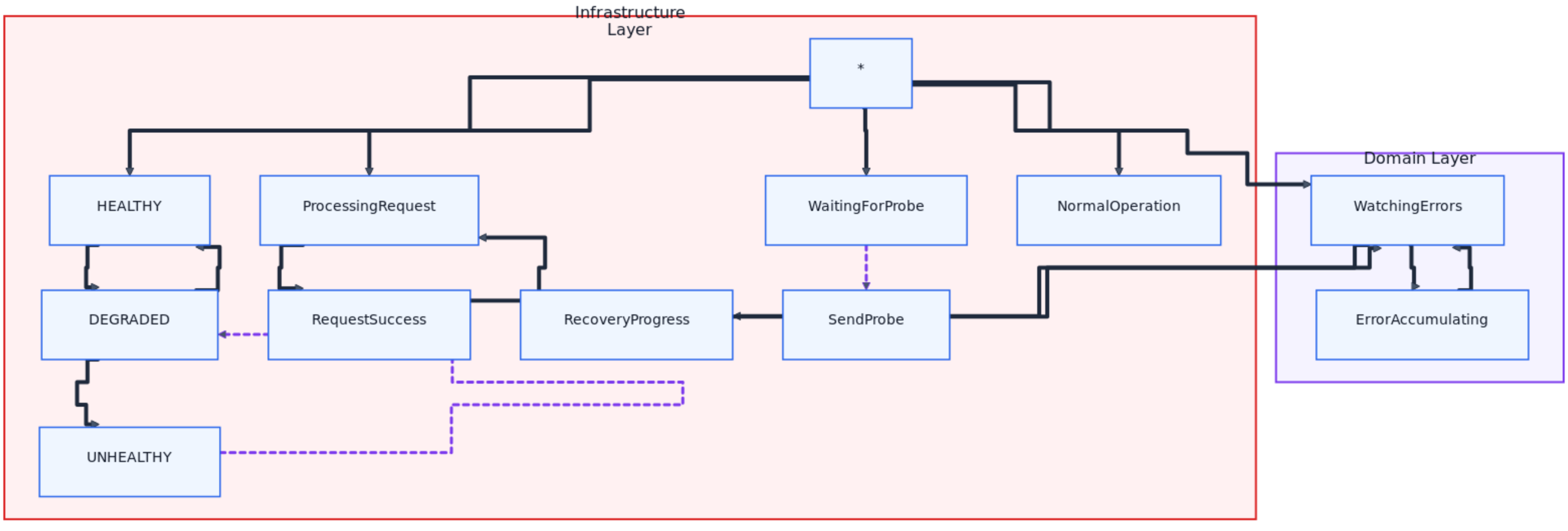
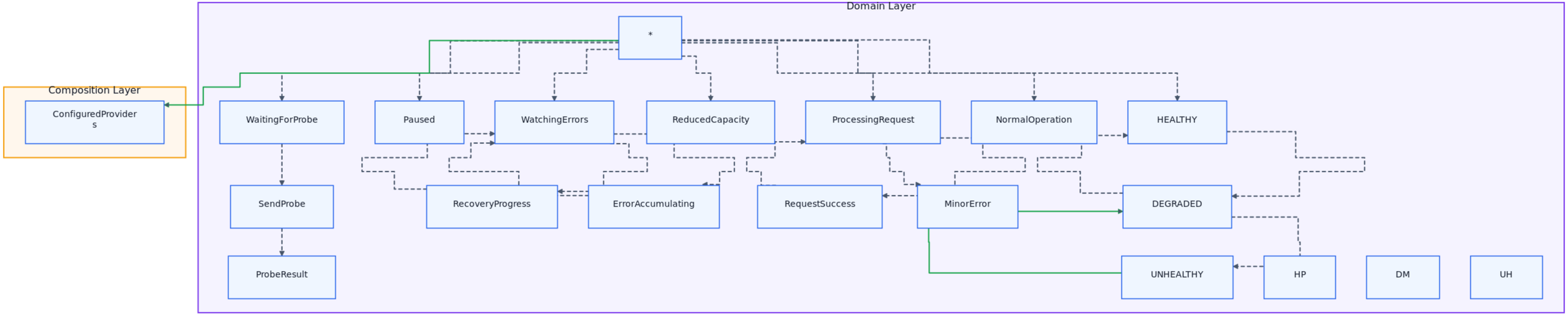

Application Layer
ChEMBL_API
PubMed_API
chembl_activity
pubchem_compound
uniprot_protein
chembl_assay
Lineage
Logs
Interfaces Layer
CLI
Infrastructure Layer
Checkpoints
Bronze
Silver
Gold
Quarantine
Metrics


Infrastructure Layer
*
HEALTHY
ProcessingRequest
WaitingForProbe
NormalOperation
DEGRADED
RequestSuccess
RecoveryProgress
SendProbe
UNHEALTHY
Domain Layer
WatchingErrors
ErrorAccumulating


Domain Layer
Composition Layer
ConfiguredProviders
*
WaitingForProbe
Paused
WatchingErrors
ReducedCapacity
ProcessingRequest
NormalOperation
HEALTHY
SendProbe
RecoveryProgress
ErrorAccumulating
RequestSuccess
MinorError
DEGRADED
ProbeResult
UNHEALTHY
HP
DM
UH

ProviderSpecificBehavior

ConfiguredProviders

ChEMBL: No rate limit, / status.json, 1000 records
PubChem: 5 req / sec, compound probe, 100 compounds
UniProt: 100 req / sec, search probe, 500 proteins
PubMed: 3 req / sec, efetch probe, 200 articles

Provider: ChEMBL
Base URL: ebi.ac.uk / chembl
Health Endpoint: / chembl / api / data / status.json
Rate Limit: None

Initial state

HEALTHY

✅ All systems operational
batch_size = default (1000)
retry_count = 0
metric
provider_health = 2

Request Processing

ProcessingRequest

2xx response

next request

1-2 consecutive errors

RequestSuccess

MinorError

1-2 consecutive errors\nOR latency > 2x baseline

0 errors for 5 minutes

DEGRADED

⚠️ Reduced performance
batch_size = default ÷ 2 (500)
circuit_breaker = monitoring
metric
provider_health = 1

Recovery Monitoring

WatchingErrors

error received

count < threshold

success received

< 5 minutes stable

ErrorAccumulating

RecoveryProgress

Alert Triggers:
• P2 Alert: pipeline paused
• Slack notification
• CloudWatch alarm
Recovery:
• Probe every 5 minutes
• Requires successful health_check

≥3 consecutive errors\nOR health_check fails

1 successful health_check

Adaptive Behavior:
• Reduce batch size by 50%
• Double request timeout
• Increase retry delays
• Log warning for each error

UNHEALTHY

❌ Alert P2 triggered
metric
provider_health = 0

Health Probe

Wait recovery_timeout (300s)

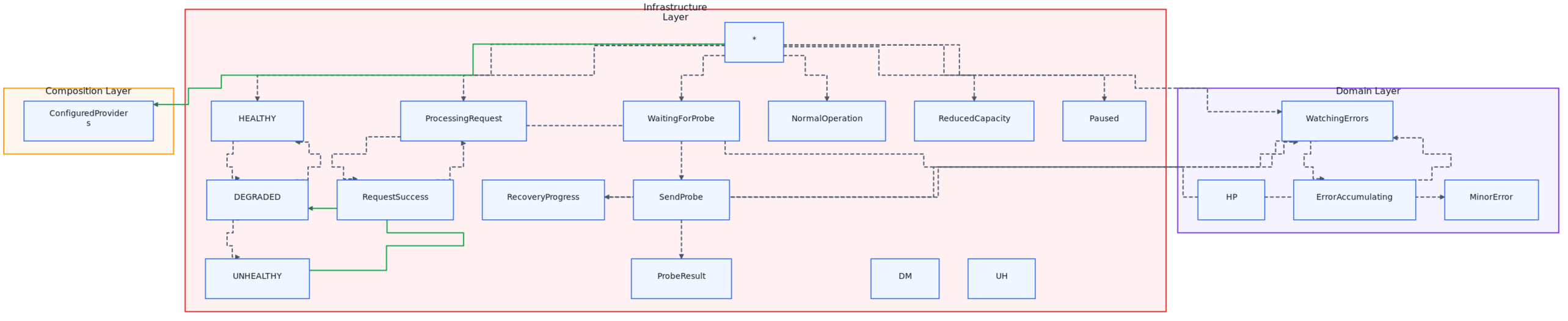
timeout elapsed

SendProbe

health_check()

ProbeResult

health_check fails\n(restart timer)


Infrastructure Layer
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Composition Layer
ConfiguredProviders
HEALTHY
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RequestSuccess
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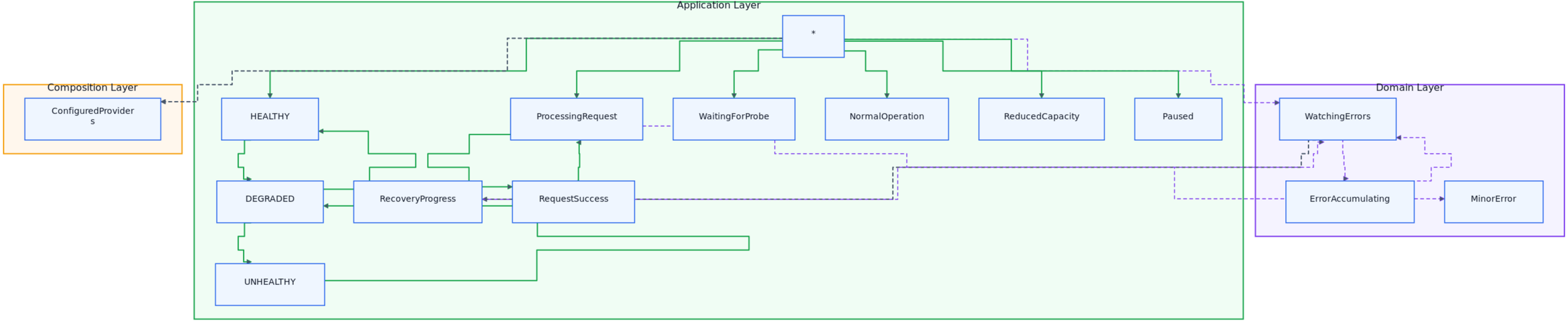
Application Layer
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RecoveryProgress
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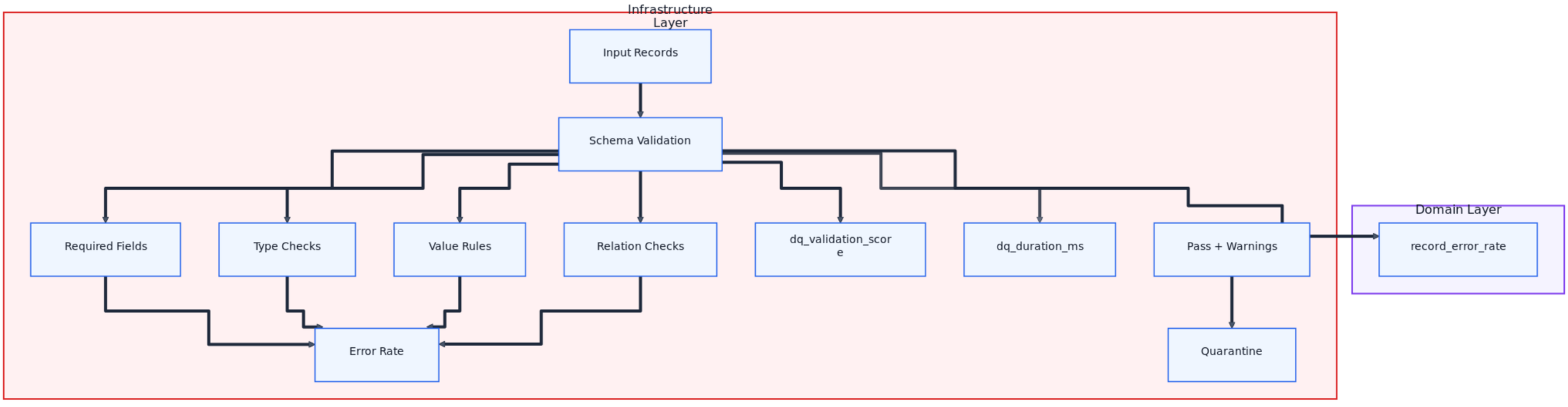
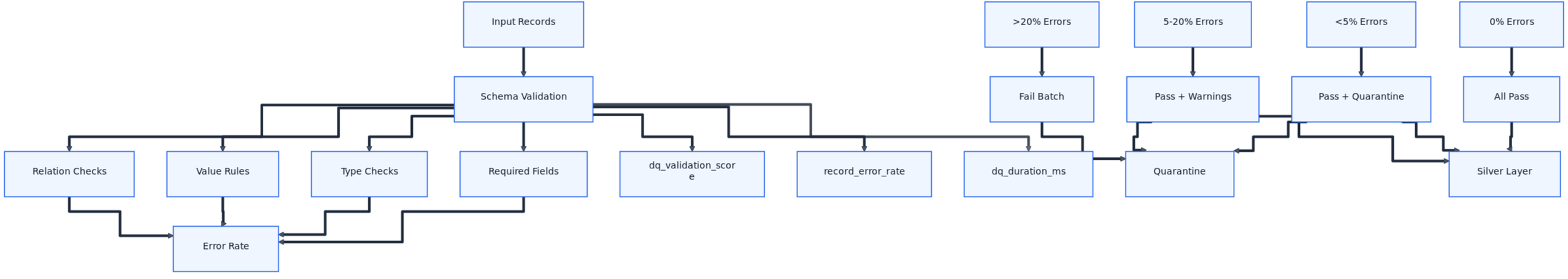
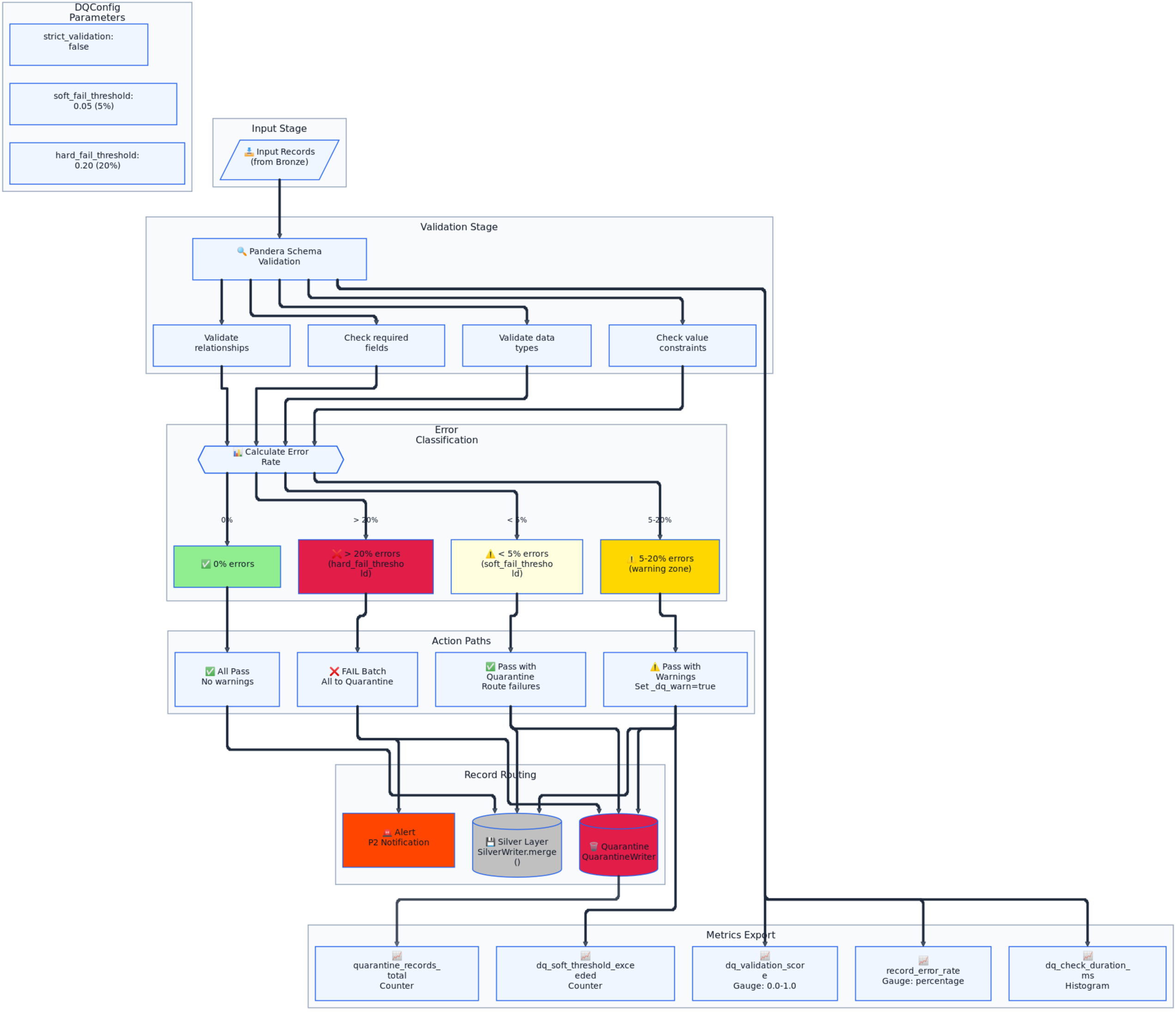
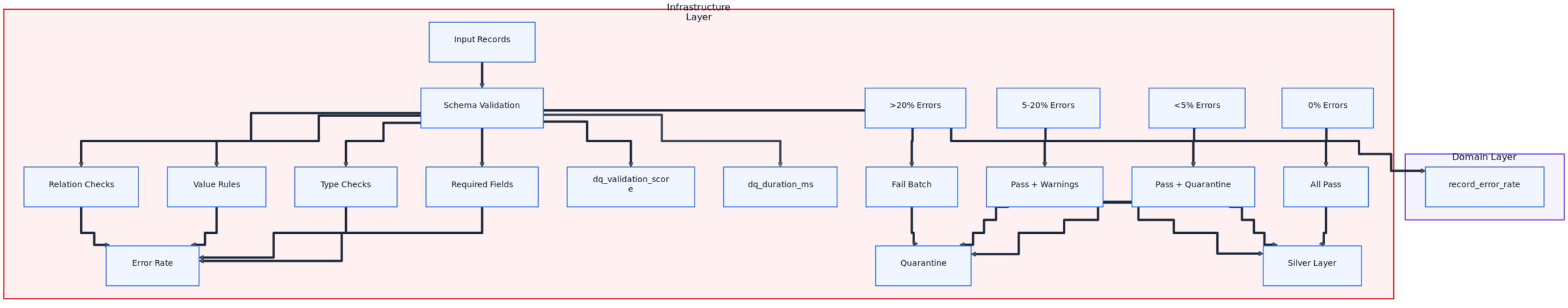
MinorError
UNHEALTHY


Infrastructure Layer
Input Records
Schema Validation
Required Fields
Type Checks
Value Rules
Relation Checks
dq_validation_score
dq_duration_ms
Pass + Warnings
Error Rate
Quarantine
Domain Layer
record_error_rate


Input Records
Schema Validation
Relation Checks
Value Rules
Type Checks
Required Fields
dq_validation_score
record_error_rate
dq_duration_ms
Error Rate
>20% Errors
5-20% Errors
<5% Errors
0% Errors
Fail Batch
Pass + Warnings
Pass + Quarantine
All Pass
Quarantine
Silver Layer


DQConfig Parameters
strict_validation: false
soft_fail_threshold: 0.05 (5%)
hard_fail_threshold: 0.20 (20%)
Input Stage
Input Records (from Bronze)
Validation Stage
Pandera Schema Validation
Validate relationships
Check required fields
Validate data types
Check value constraints
Error Classification
Calculate Error Rate
0%
> 20%
< 5%
5-20%
0% errors
> 20% errors (hard_fail_threshold)
< 5% errors (soft_fail_threshold)
5-20% errors (warning zone)
Action Paths
All Pass No warnings
FAIL Batch All to Quarantine
Pass with Quarantine Route failures
Pass with Warnings Set _dq_warn=true
Record Routing
Alert P2 Notification
Silver Layer SilverWriter.merge()
Quarantine QuarantineWriter
Metrics Export
quarantine_records_total Counter
dq_soft_threshold_exceeded Counter
dq_validation_score Gauge: 0.0-1.0
record_error_rate Gauge: percentage
dq_check_duration_ms Histogram


Infrastructure Layer
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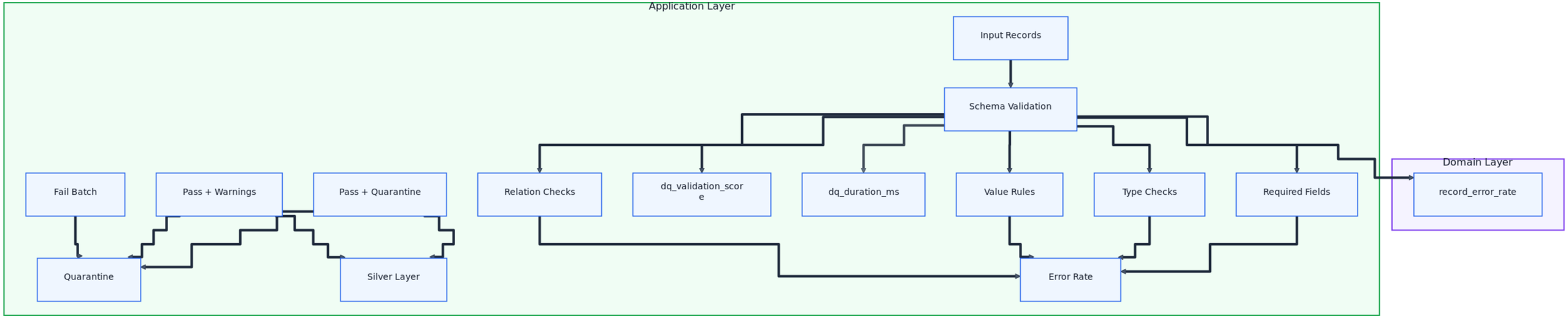
Relation Checks
Value Rules
Type Checks
Required Fields
dq_validation_score
dq_duration_ms
Error Rate
>20% Errors
5-20% Errors
<5% Errors
0% Errors
Fail Batch
Pass + Warnings
Pass + Quarantine
All Pass
Quarantine
Silver Layer
Domain Layer
record_error_rate


Application Layer
Input Records
Schema Validation
Relation Checks
dq_validation_score
dq_duration_ms
Value Rules
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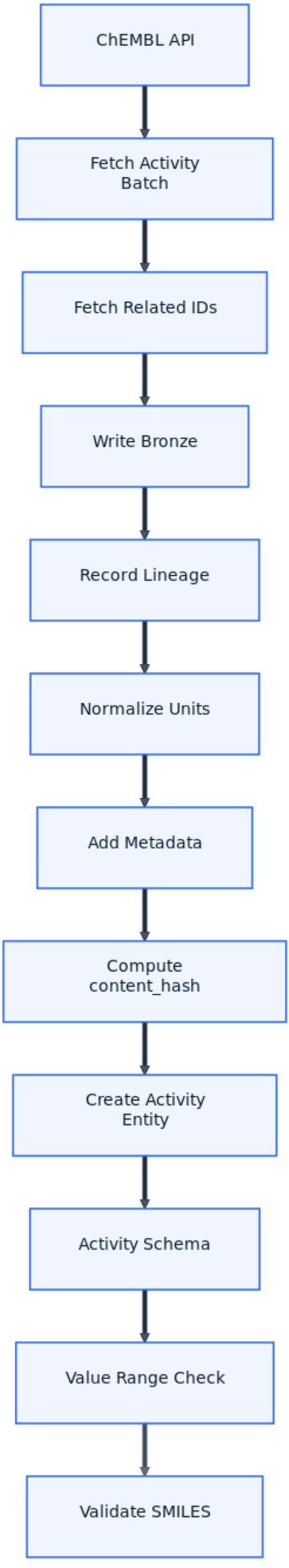
Domain Layer
record_error_rate
Error Rate
Fail Batch
Pass + Warnings
Pass + Quarantine
Quarantine
Silver Layer


ChEMBL API
Fetch Activity Batch
Fetch Related IDs
Write Bronze
Record Lineage
Normalize Units
Add Metadata
Compute content_hash
Create Activity Entity
Activity Schema
Value Range Check
Validate SMILES

ChEMBL API

Fetch Activity Batch

Fetch Related IDs

Write Bronze

Record Lineage

Normalize Units

Add Metadata

Compute content_hash

Create Activity Entity

Activity Schema

Value Range Check

Validate SMILES

Error Rate Gate

Silver Activity

Gold by Target

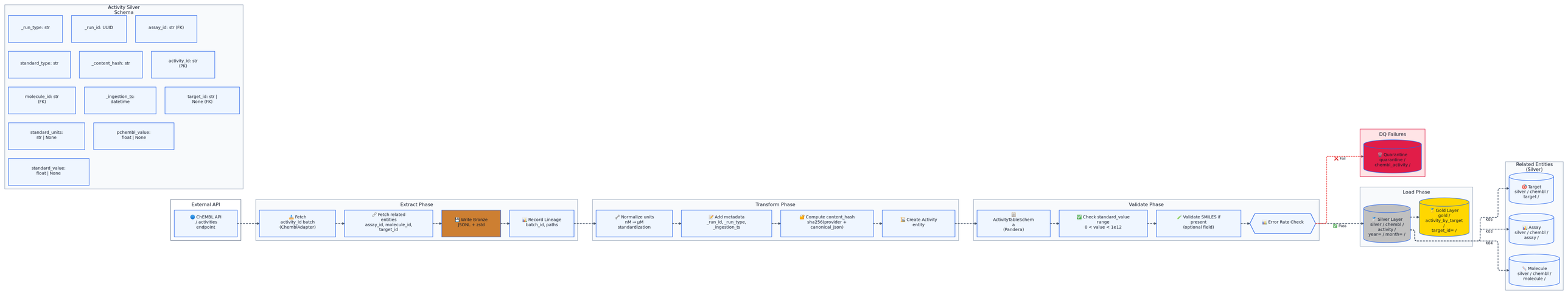
Assay

Molecule

Target

Quarantine

activity_id (PK)


Activity Silver Schema
_run_type: str
_run_id: UUID
assay_id: str (FK)
standard_type: str
_content_hash: str
activity_id: str (PK)
molecule_id: str (FK)
_ingestion_ts: datetime
target_id: str | None (FK)
standard_units: str | None
pchembl_value: float | None
standard_value: float | None
External API
ChEMBL API / activities endpoint
Extract Phase
Fetch activity_id batch (ChemblAdapter)
Fetch related entities assay_id, molecule_id, target_id
Write Bronze JSONL + zstd
Record Lineage batch_id, paths
Transform Phase
Normalize units nM → µM standardization
Add metadata _run_id, _run_type, _ingestion_ts
Compute content_hash sha256(provider + canonical_json)
Create Activity entity
Validate Phase
ActivityTableSchema (Pandera)
Check standard_value range 0 < value < 1e12
Validate SMILES if present (optional field)
Error Rate Check
Fail
Pass
DQ Failures
Quarantine quarantine / chembl_activity /
Load Phase
Silver Layer silver / chembl / activity / year= / month= /
Gold Layer gold / activity_by_target / target_id= /
Related Entities (Silver)
Target silver / chembl / target /
Assay silver / chembl / assay /
Molecule silver / chembl / molecule /
K01
K03
K04

Local FS

ChEMBL API

Fetch Activity Batch

Fetch Related IDs

Write Bronze

Record Lineage

Normalize Units

Add Metadata

Compute content_hash

Create Activity Entity

Activity Schema

Value Range Check

Validate SMILES

Error Rate Gate

Silver Activity

Gold by Target

Assay

Molecule

Target

Quarantine

ChEMBL API

Fetch Activity Batch

Fetch Related IDs

Write Bronze

Record Lineage

Normalize Units

Add Metadata

Compute content_hash

Create Activity Entity

Activity Schema

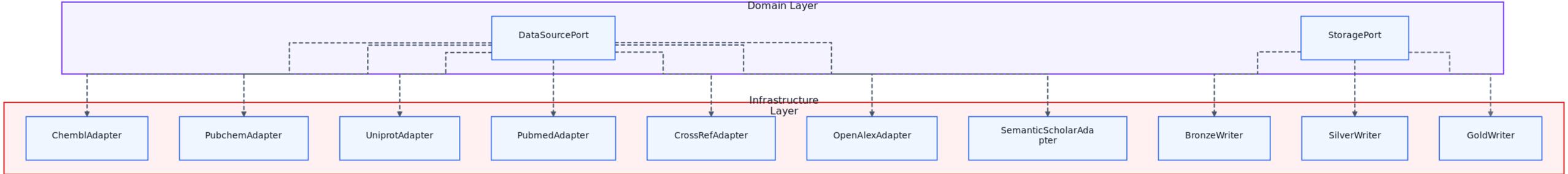
Value Range Check

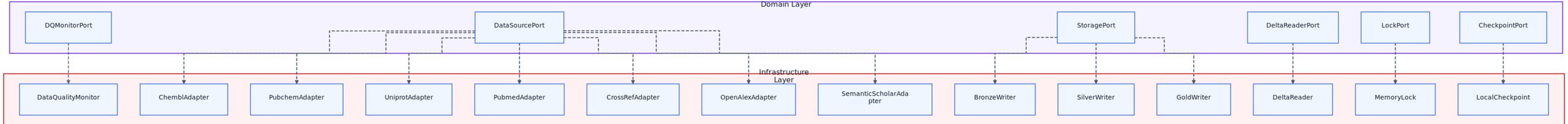
Validate SMILES

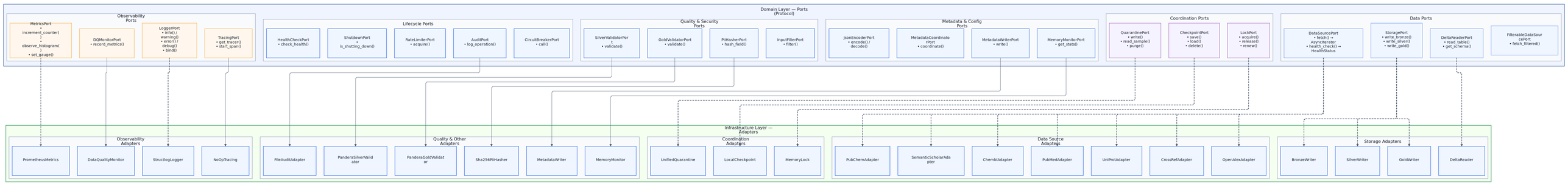
Error Rate Gate

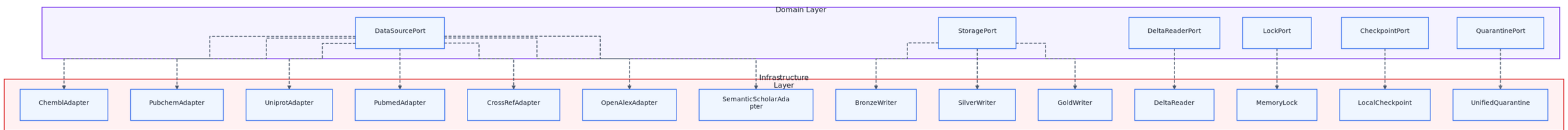
Silver Activity

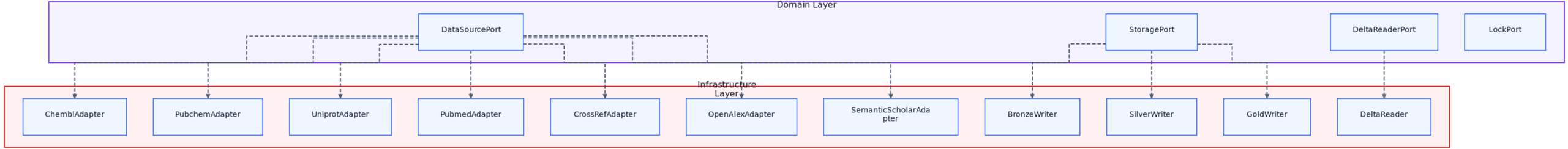
Gold by Target

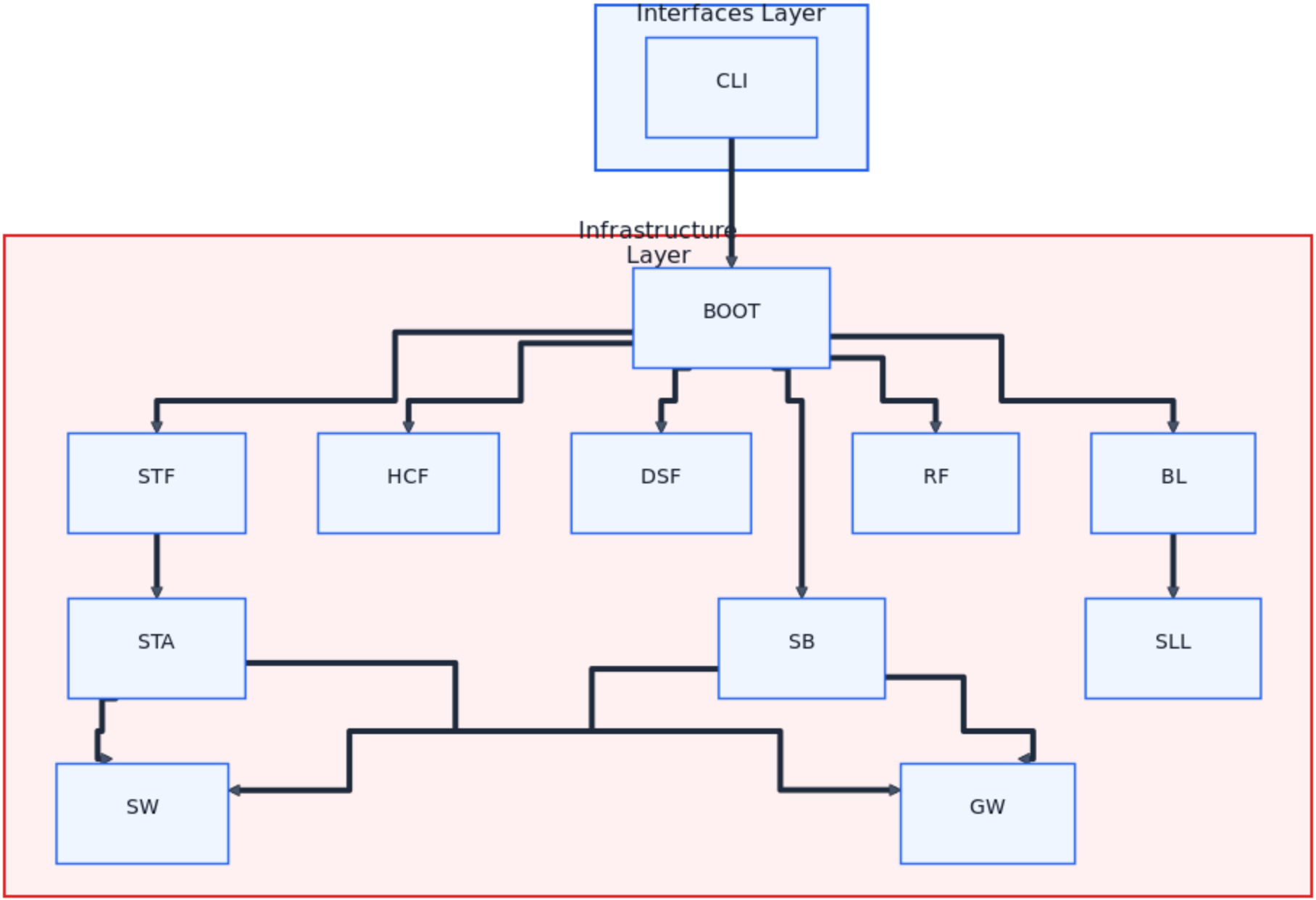

Domain Layer
DataSourcePort
StoragePort
Infrastructure Layer
ChemblAdapter
PubchemAdapter
UniprotAdapter
PubmedAdapter
CrossRefAdapter
OpenAlexAdapter
SemanticScholarAdapter
BronzeWriter
SilverWriter
GoldWriter

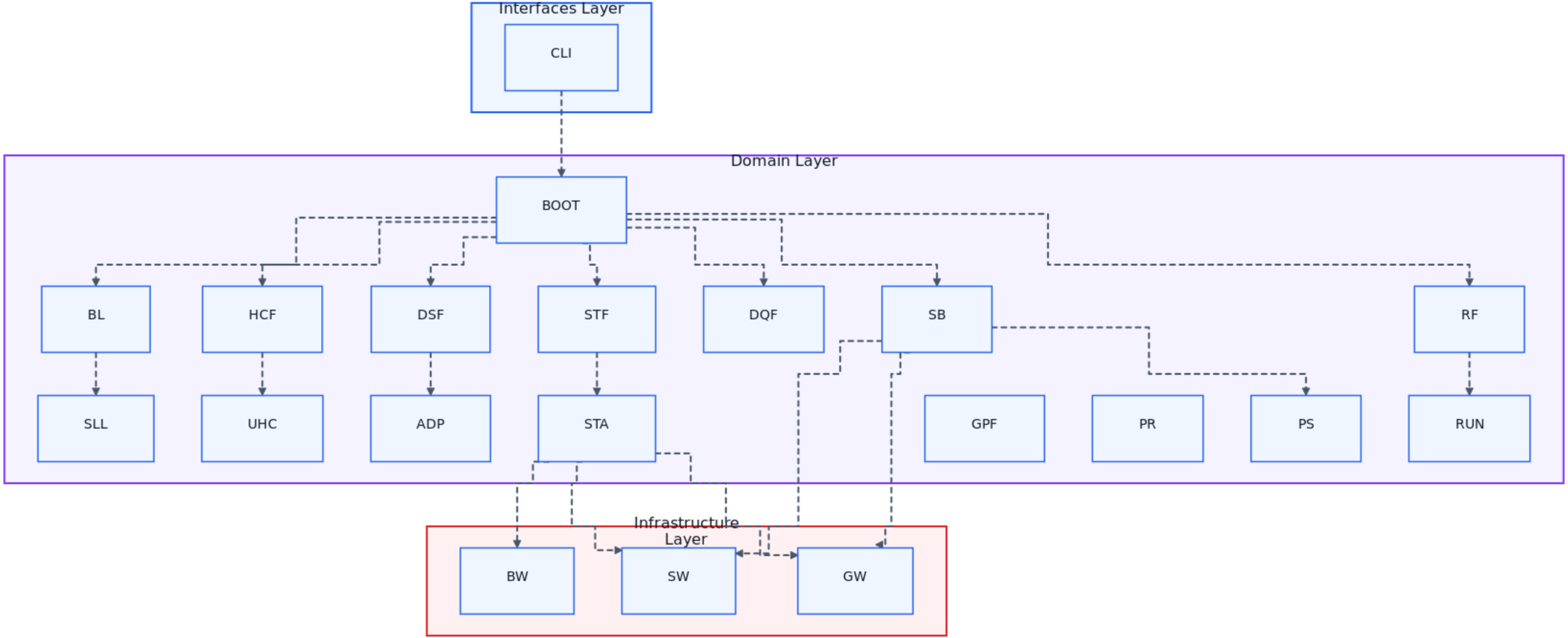

Domain Layer
DQMonitorPort
DataSourcePort
StoragePort
DeltaReaderPort
LockPort
CheckpointPort
Infrastructure Layer
DataQualityMonitor
ChemblAdapter
PubchemAdapter
UniprotAdapter
PubmedAdapter
CrossRefAdapter
OpenAlexAdapter
SemanticScholarAdapter
BronzeWriter
SilverWriter
GoldWriter
DeltaReader
MemoryLock
LocalCheckpoint


Domain Layer — Ports (Protocol)
Observability Ports
MetricsPort
• increment_counter( )
• observe_histogram( )
• set_gauge()
DQMonitorPort
• record_metrics()
LoggerPort
• info() / warning()
• error() / debug()
• bind()
TracingPort
• get_tracer()
• start_span()
Lifecycle Ports
HealthCheckPort
• check_health()
ShutdownPort
• is_shutting_down()
RateLimiterPort
• acquire()
AuditPort
• log_operation()
CircuitBreakerPort
• call()
Quality & Security Ports
SilverValidatorPort
• validate()
GoldValidatorPort
• validate()
PiiHasherPort
• hash_field()
InputFilterPort
• filter()
Metadata & Config Ports
JsonEncoderPort
• encode() / decode()
MetadataCoordinatorPort
• coordinate()
MetadataWriterPort
• write()
MemoryMonitorPort
• get_stats()
Coordination Ports
QuarantinePort
• write()
• read_sample()
• purge()
CheckpointPort
• save()
• load()
• delete()
LockPort
• acquire()
• release()
• renew()
Data Ports
DataSourcePort
• fetch() → AsyncIterator
• health_check() → HealthStatus
StoragePort
• write_bronze()
• write_silver()
• write_gold()
DeltaReaderPort
• read_table()
• get_schema()
FilterableDataSourcePort
• fetch_filtered()
Infrastructure Layer — Adapters
Observability Adapters
PrometheusMetrics
DataQualityMonitor
StructlogLogger
NoOpTracing
Quality & Other Adapters
FileAuditAdapter
PanderaSilverValidator
PanderaGoldValidator
Sha256PiiHasher
MetadataWriter
MemoryMonitor
Coordination Adapters
UnifiedQuarantine
LocalCheckpoint
MemoryLock
Data Source Adapters
PubChemAdapter
SemanticScholarAdapter
ChemblAdapter
PubMedAdapter
UniProtAdapter
CrossRefAdapter
OpenAlexAdapter
Storage Adapters
BronzeWriter
SilverWriter
GoldWriter
DeltaReader


Domain Layer
DataSourcePort
StoragePort
DeltaReaderPort
LockPort
CheckpointPort
QuarantinePort
Infrastructure Layer
ChemblAdapter
PubchemAdapter
UniprotAdapter
PubmedAdapter
CrossRefAdapter
OpenAlexAdapter
SemanticScholarAdapter
BronzeWriter
SilverWriter
GoldWriter
DeltaReader
MemoryLock
LocalCheckpoint
UnifiedQuarantine


Domain Layer
DataSourcePort
StoragePort
DeltaReaderPort
LockPort
Infrastructure Layer
ChemblAdapter
PubchemAdapter
UniprotAdapter
PubmedAdapter
CrossRefAdapter
OpenAlexAdapter
SemanticScholarAdapter
BronzeWriter
SilverWriter
GoldWriter
DeltaReader


Interfaces Layer
CLI
Infrastructure Layer
BOOT
STF
HCF
DSF
RF
BL
STA
SB
SLL
SW
GW


Interfaces Layer
CLI
Domain Layer
BOOT
BL
HCF
DSF
STF
DQF
SB
RF
SLL
UHC
ADP
STA
GPF
PR
PS
RUN
Infrastructure Layer
BW
SW
GW

Entry Point

CLI run command

bootstrap / runtime / assembly.py

Composition Factories

Logger & Observability

BootstrapLogger
• configure structlog

DQServicesFactory
• create()
→ DQ analyzers + monitor

Client & Data Source

HttpClientFactory
• create(provider)
→ UnifiedHTTPClient

DataSourceFactory
• create(provider, config)
→ DataSourcePort impl

Storage & Services

StorageFactory
• create(config)
→ Bronze / Silver / Gold writers

StorageAdapter
• create_bronze / silver / gold

ServicesBuilder
• wire storage / lock / obs
• build() → PipelineServices

Pipeline Construction

RunnerFactory
• create_runner()
→ PipelineRunner

GenericPipelineFactory
• pipeline / transformer / schema bindings

injects

uses

uses

injects

uses

uses

injects

injects

injects

injects

injects

injects

injects

registers

SW

GW

ML

LC

UQ

PMM

NOT

PLREG

Created Instances

Infra Instances

[A] UnifiedHTTPClient
+ TokenBucket
+ CircuitBreaker

[A] Provider Adapter (ChemblAdapter, etc.)

BronzeWriter

StructlogLogger

Application Instances

PipelineServices (frozen bundle)

PipelineRunner (fully assembled)

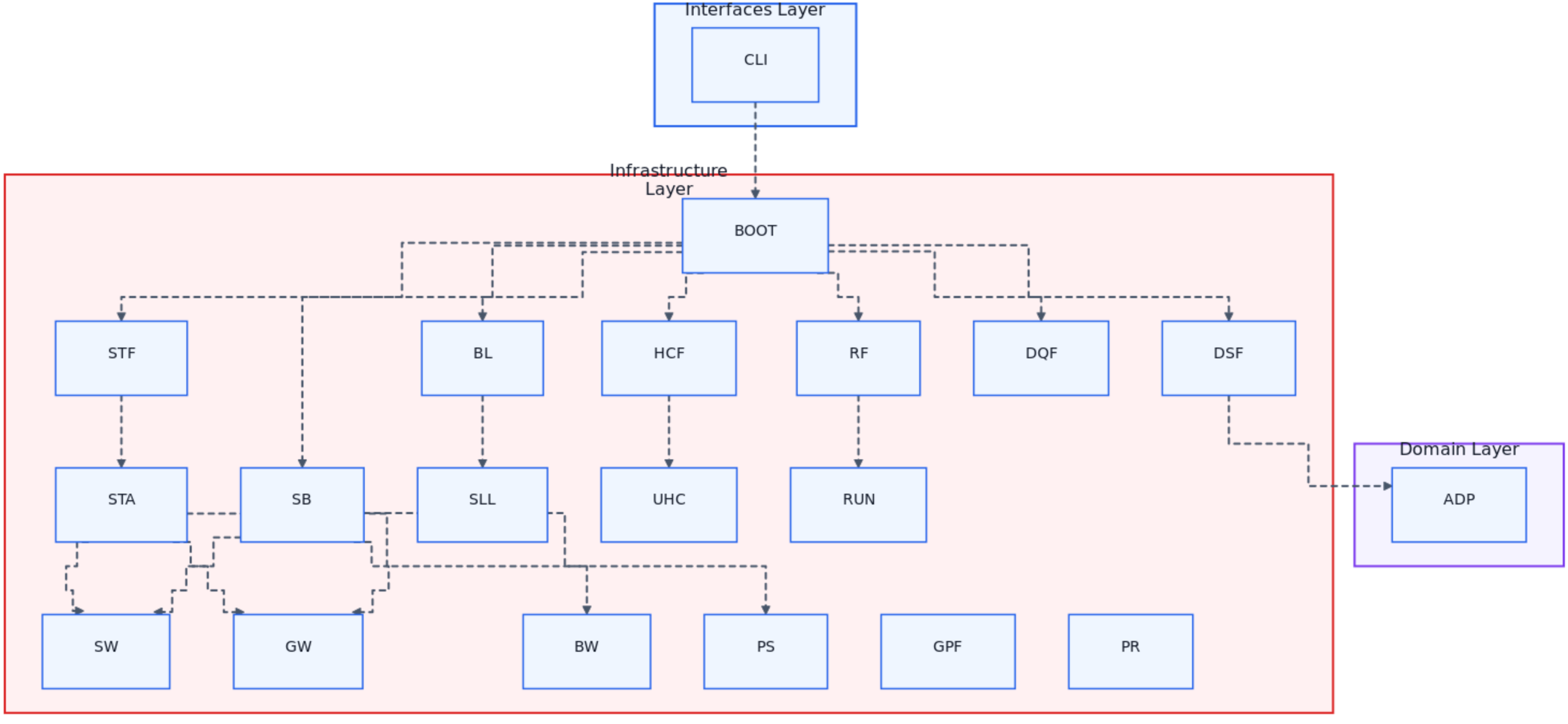
Registration

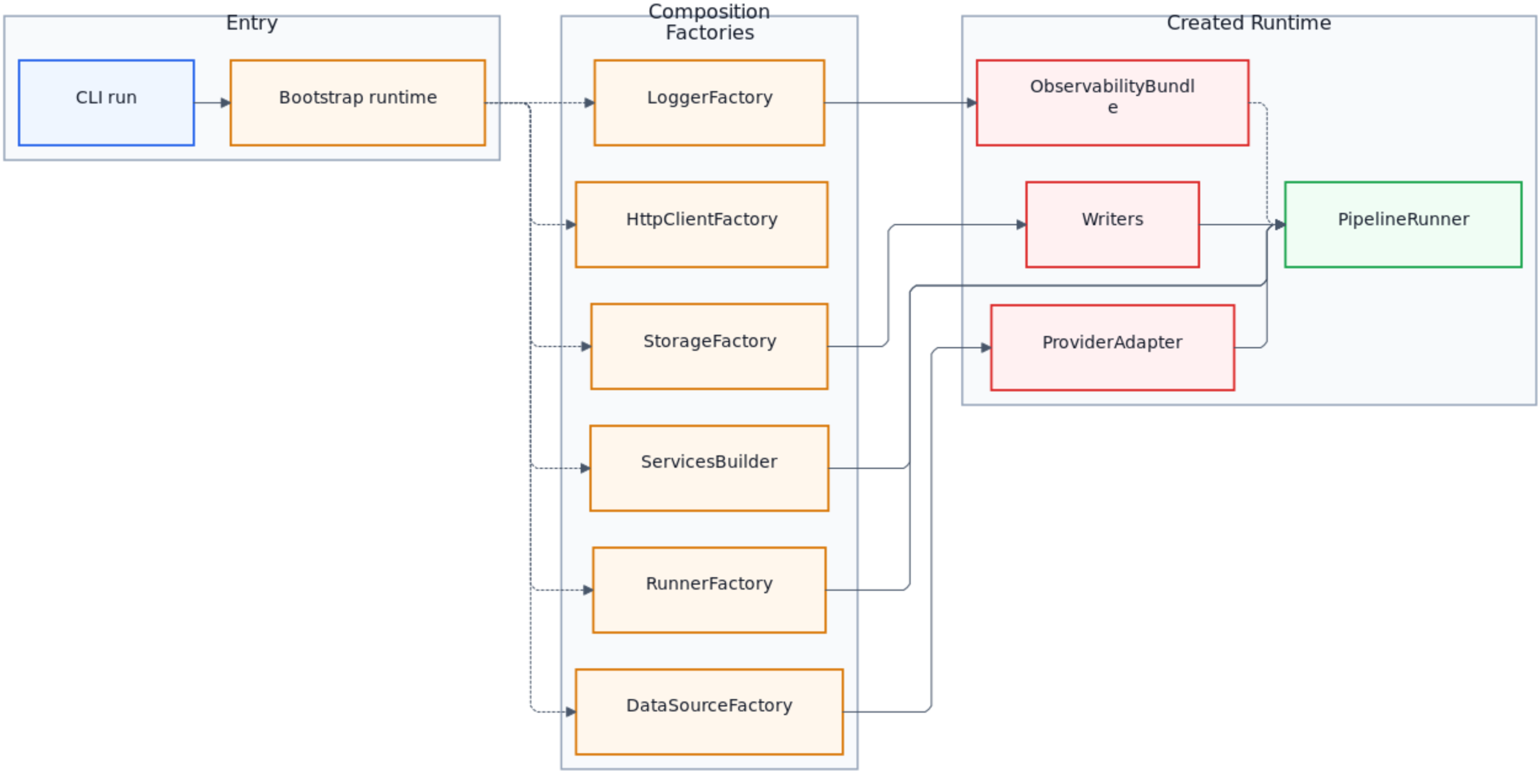
ProviderRegistry
• 8 providers registered

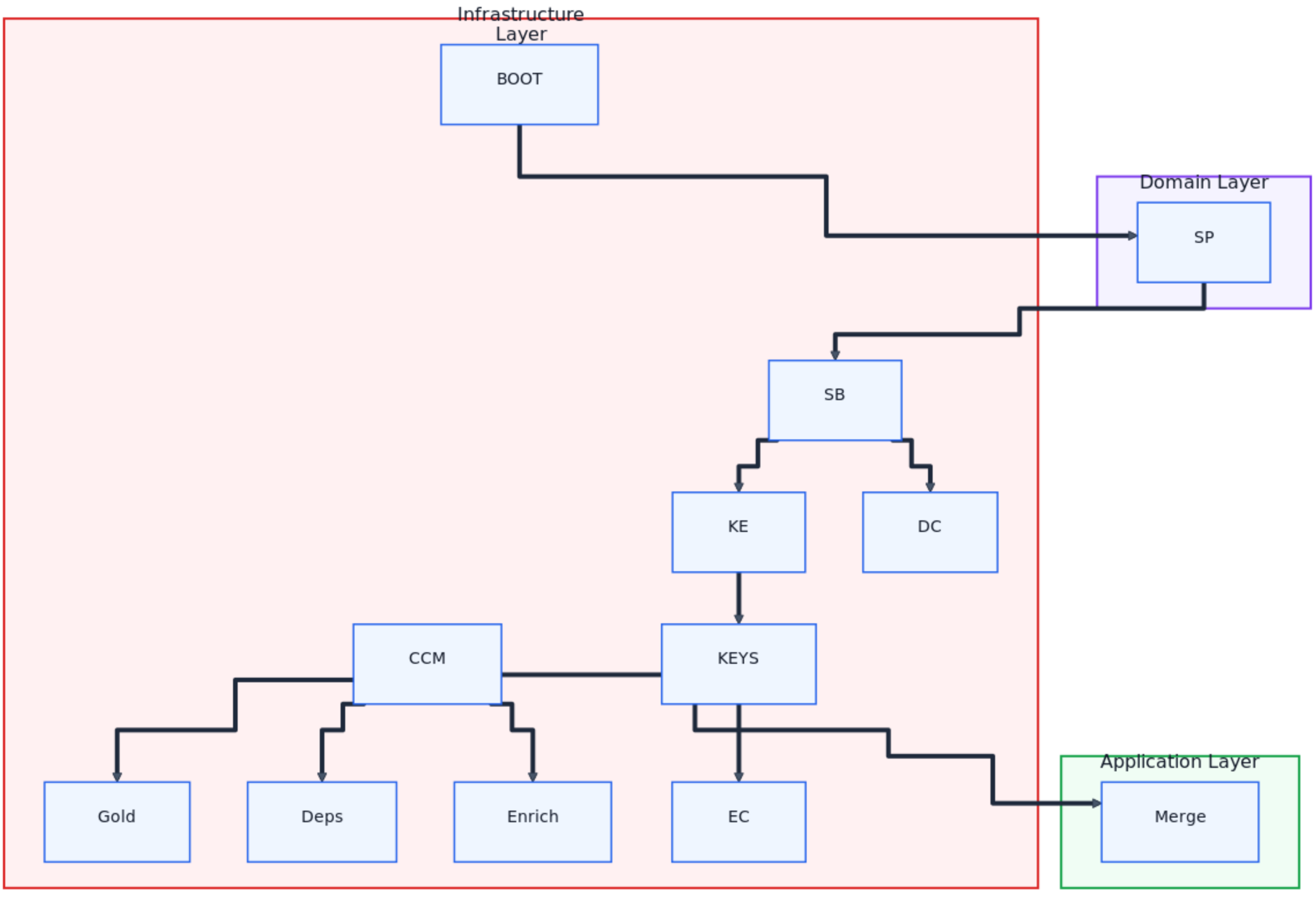
ObservabilityBundle
• logger + metrics + tracing

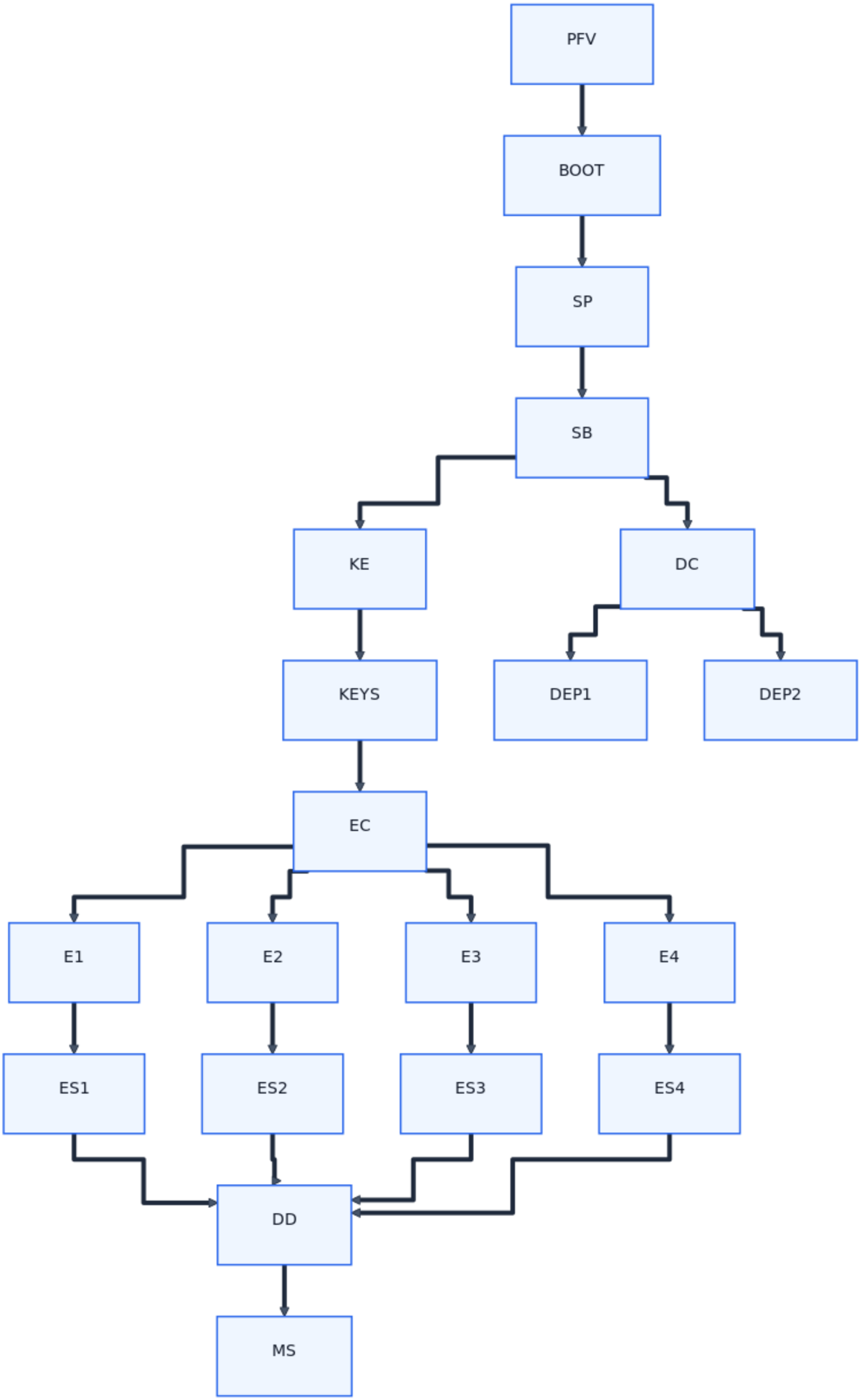
maps

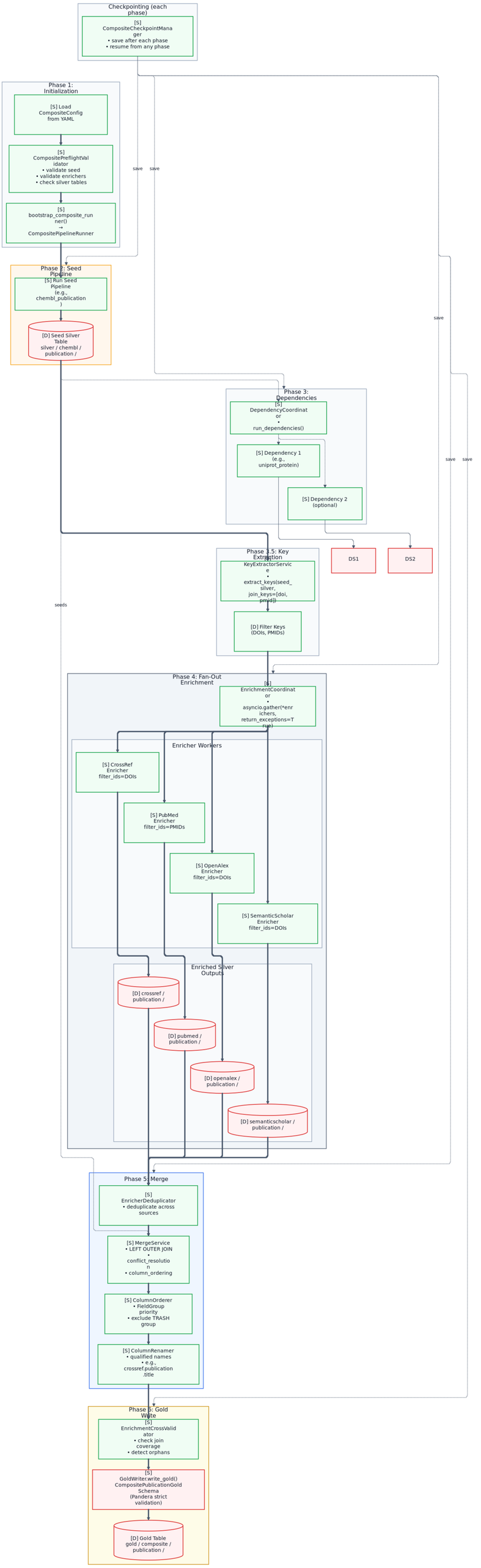
DSR


Interfaces Layer
CLI
Infrastructure Layer
BOOT
STF
BL
HCF
RF
DQF
DSF
STA
SB
SLL
UHC
RUN
Domain Layer
ADP
SW
GW
BW
PS
GPF
PR


Entry
CLI run
Bootstrap runtime
Composition Factories
LoggerFactory
HttpClientFactory
StorageFactory
ServicesBuilder
RunnerFactory
DataSourceFactory
Created Runtime
ObservabilityBundle
Writers
PipelineRunner
ProviderAdapter


Infrastructure Layer
BOOT
Domain Layer
SP
SB
KE
DC
CCM
KEYS
Gold
Deps
Enrich
EC
Application Layer
Merge


PFV
BOOT
SP
SB
KE
DC
KEYS
DEP1
DEP2
EC
E1
E2
E3
E4
ES1
ES2
ES3
ES4
DD
MS


Checkpointing (each phase)
[S] CompositeCheckpointManager
• save after each phase
• resume from any phase
Phase 1: Initialization
[S] Load CompositeConfig from YAML
[S] CompositePreflightValidator
• validate seed
• validate enrichers
• check silver tables
[S] bootstrap_composite_runner()
→ CompositePipelineRunner
save
save
Phase 2: Seed Pipeline
[S] Run Seed Pipeline (e.g., chembl_publication)
[D] Seed Silver Table silver / chembl / publication /
save
Phase 3: Dependencies
[S] DependencyCoordinator
• run_dependencies()
[S] Dependency 1 (e.g., uniprot_protein)
[S] Dependency 2 (optional)
save
save
DS1
DS2
Phase 3.5: Key Extraction
[S] KeyExtractorService
• extract_keys(seed_silver, join_keys=[doi, pmid])
[D] Filter Keys (DOIs, PMIDs)
seeds
Phase 4: Fan-Out Enrichment
[S] EnrichmentCoordinator
• asyncio.gather(*enrichers, return_exceptions=True)
Enricher Workers
[S] CrossRef Enricher filter_ids=DOIs
[S] PubMed Enricher filter_ids=PMIDs
[S] OpenAlex Enricher filter_ids=DOIs
[S] SemanticScholar Enricher filter_ids=DOIs
Enriched Silver Outputs
[D] crossref / publication /
[D] pubmed / publication /
[D] openalex / publication /
[D] semanticscholar / publication /
Phase 5: Merge
[S] EnricherDeduplicator
• deduplicate across sources
[S] MergeService
• LEFT OUTER JOIN
• conflict_resolution
• column_ordering
[S] ColumnOrderer
• FieldGroup priority
• exclude TRASH group
[S] ColumnRenamer
• qualified names
• e.g., crossref.publication.title
Phase 6: Gold Write
[S] EnrichmentCrossValidator
• check join coverage
• detect orphans
[S] GoldWriter.write_gold() CompositePublicationGold Schema (Pandera strict validation)
[D] Gold Table gold / composite / publication /

Infrastructure Layer

PFV

BOOT

Domain Layer

SP

SB

KE

DC

KEYS

DEP1

DEP2

EC

E1

E2

E3

E4

ES1

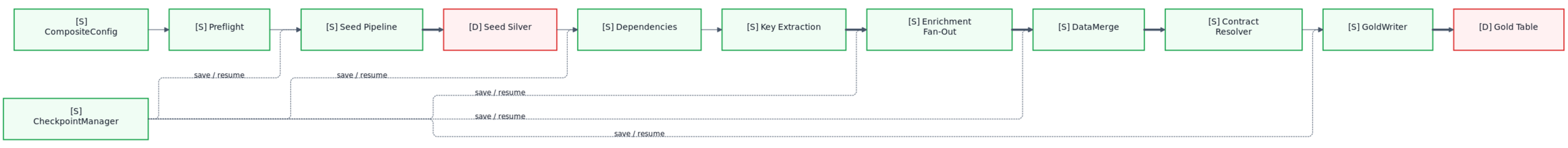
ES2

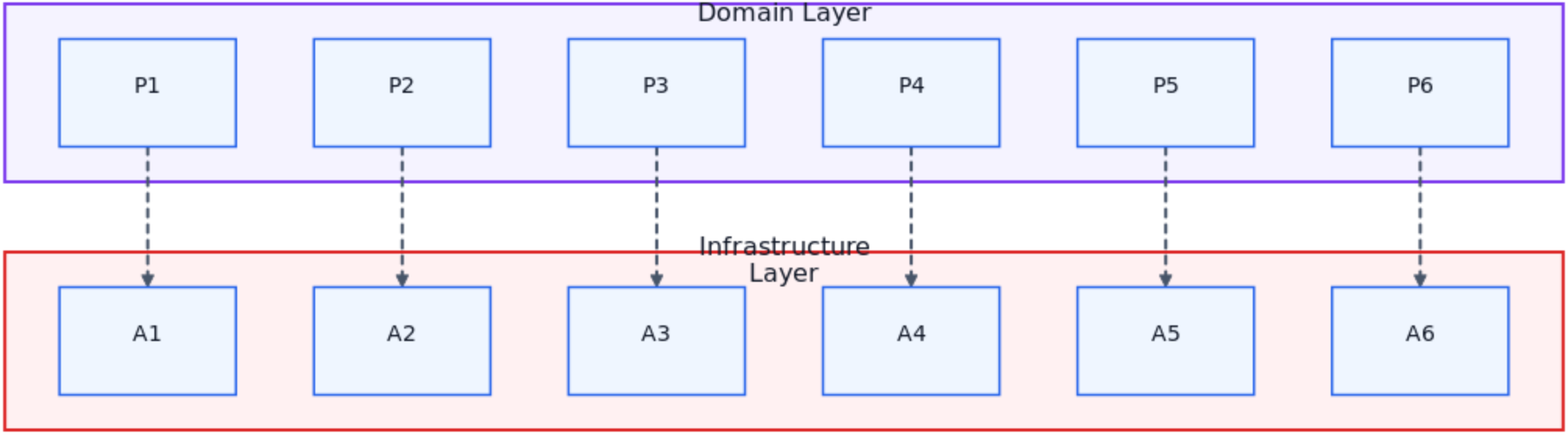
ES3

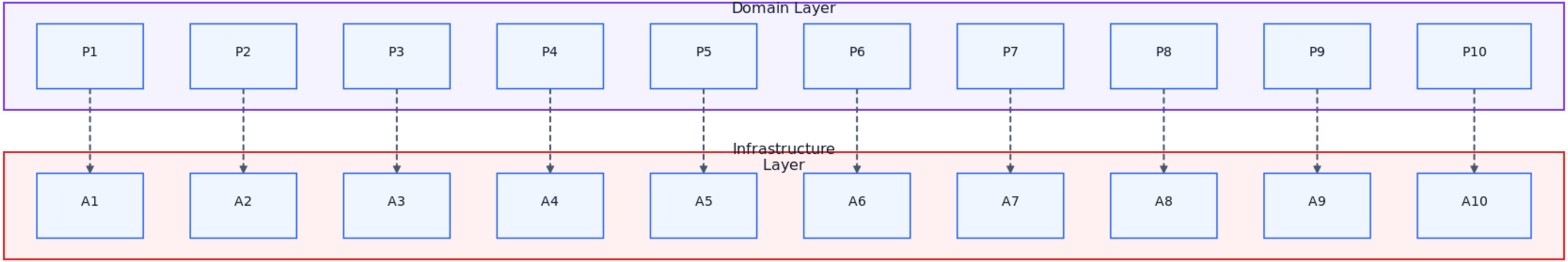
ES4

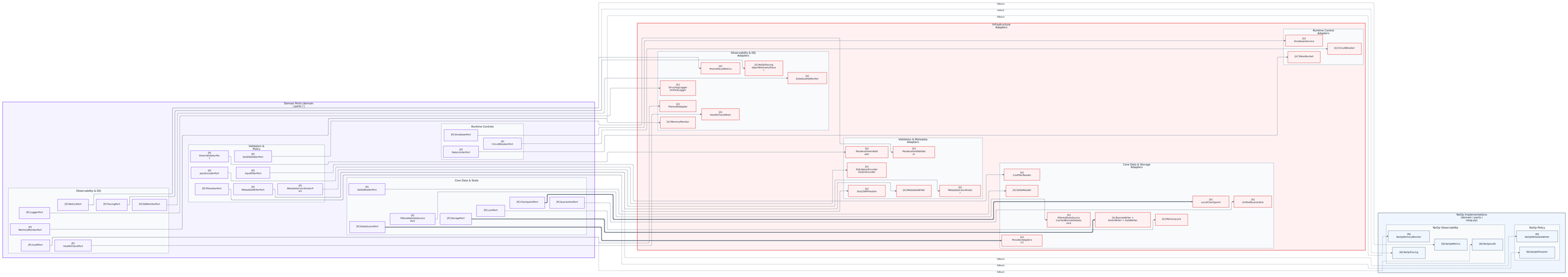
DD

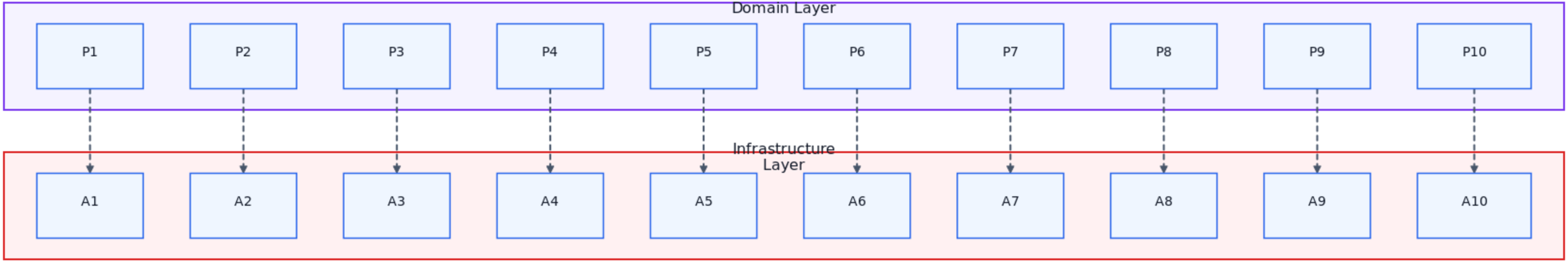
MS

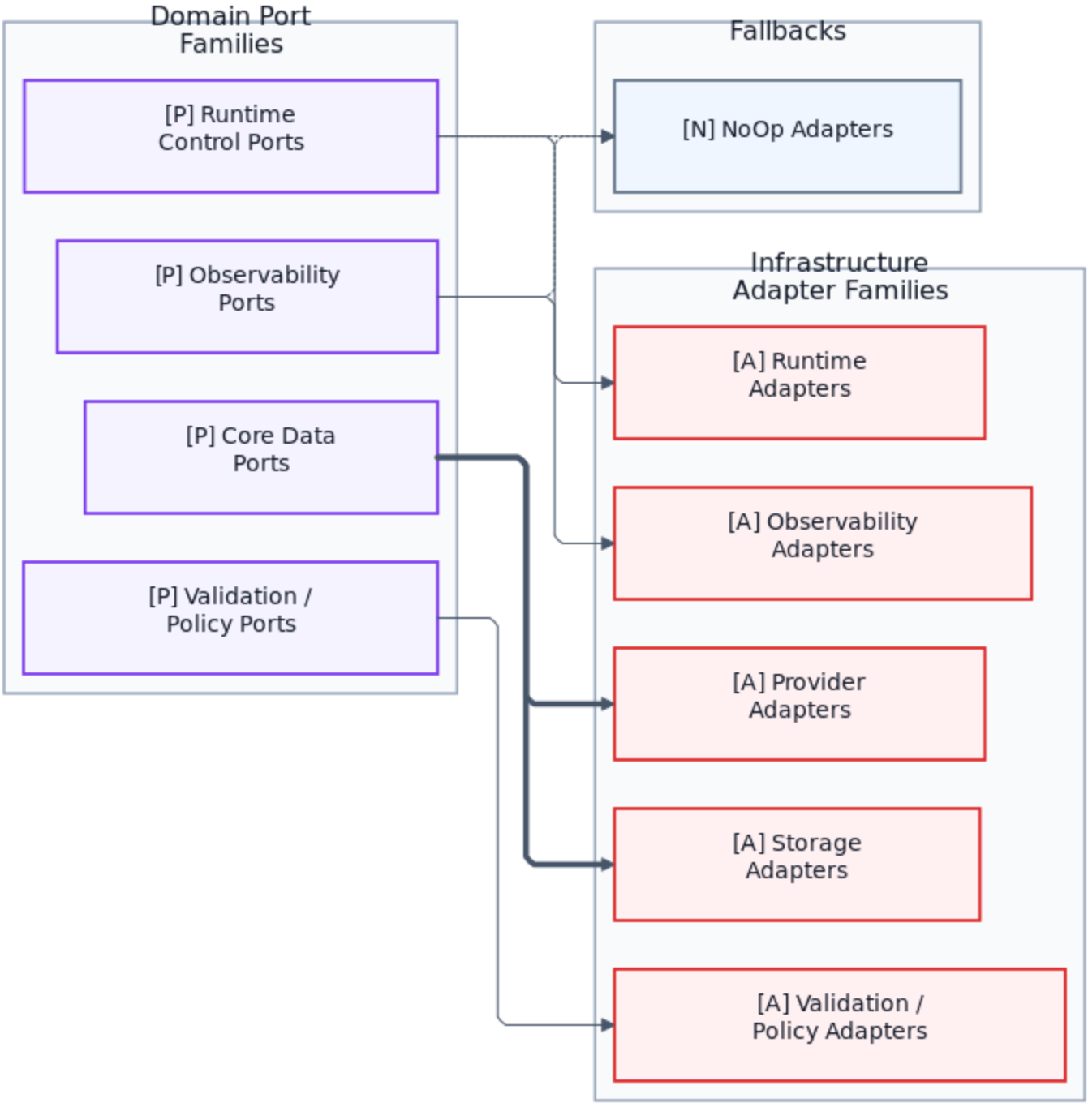

[S]
CompositeConfig
[S] Preflight
[S] Seed Pipeline
[D] Seed Silver
[S] Dependencies
[S] Key Extraction
[S] Enrichment
Fan-Out
[S] DataMerge
[S] Contract
Resolver
[S] GoldWriter
[D] Gold Table
[S]
CheckpointManager
save / resume
save / resume
save / resume
save / resume
save / resume

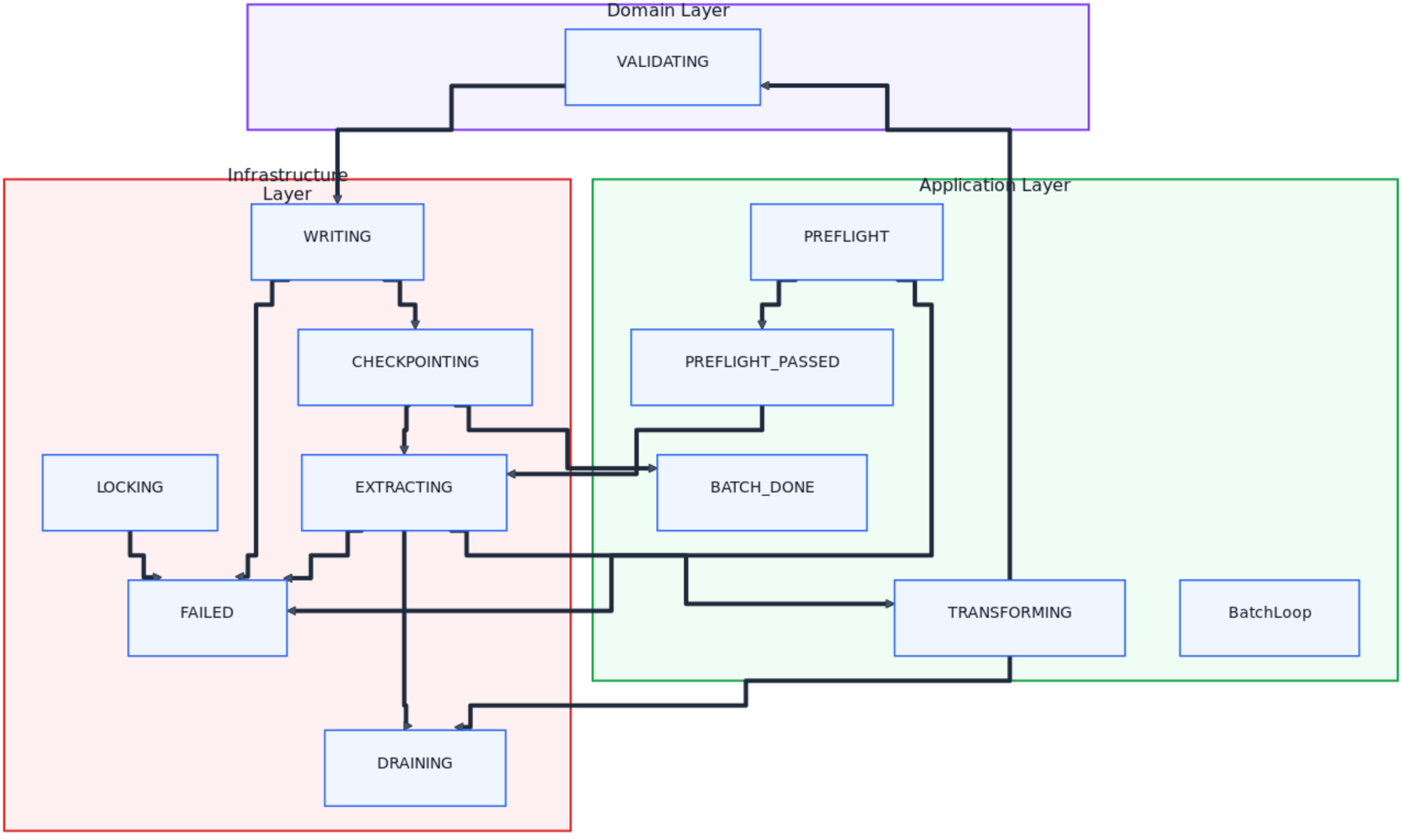

Domain Layer
P1
P2
P3
P4
P5
P6
Infrastructure Layer
A1
A2
A3
A4
A5
A6

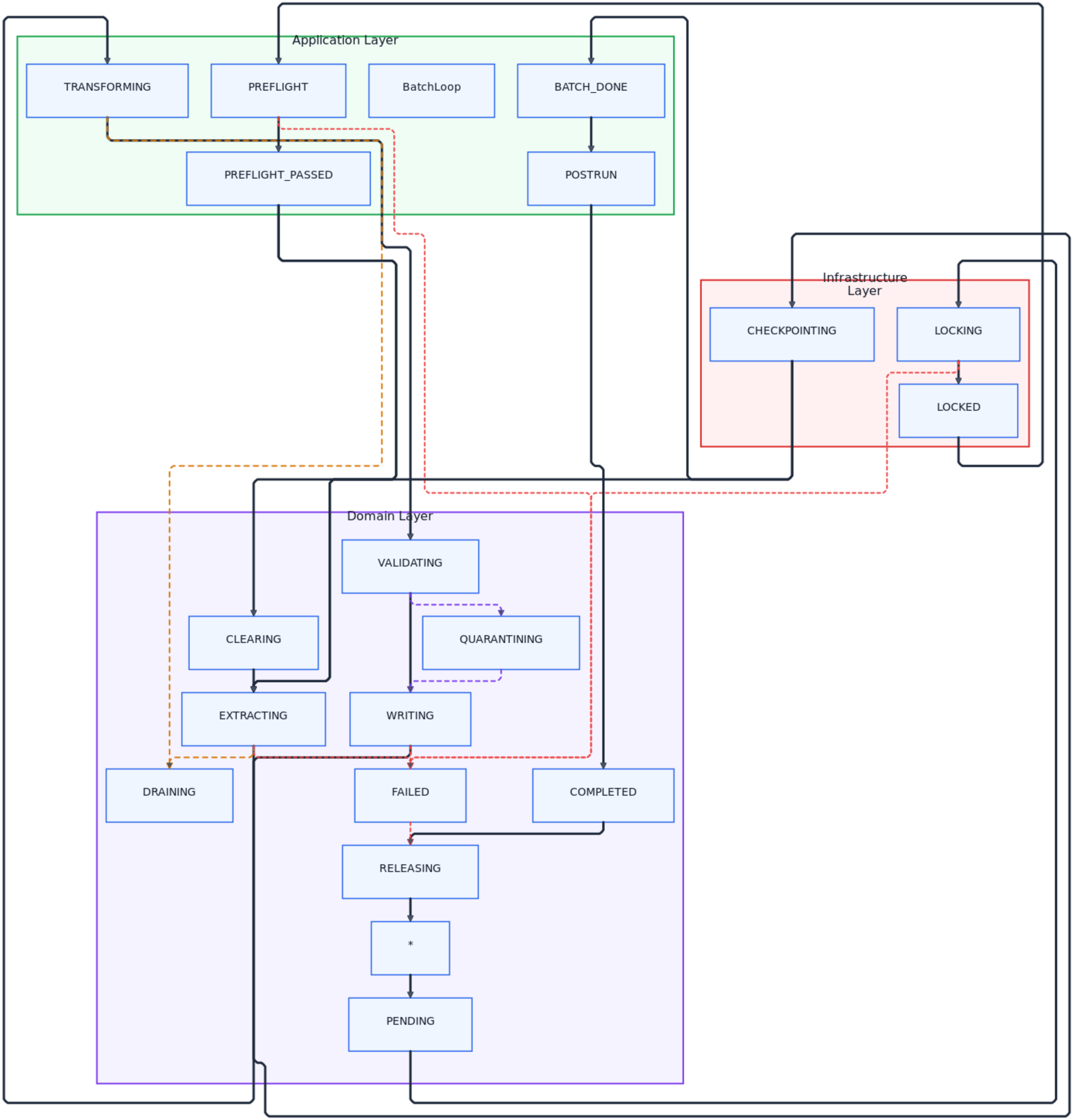

Domain Layer
P1
P2
P3
P4
P5
P6
P7
P8
P9
P10
Infrastructure Layer
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10

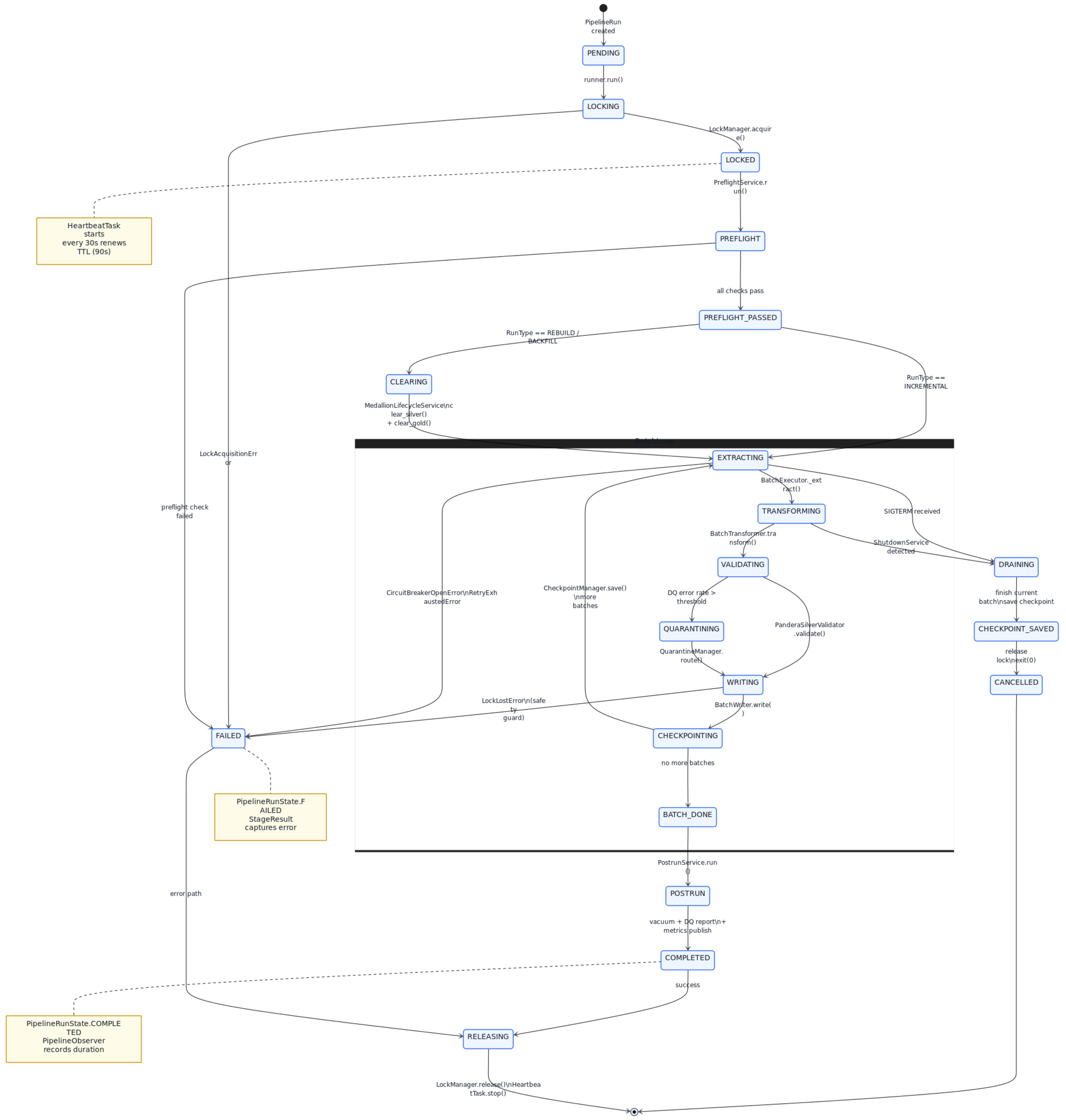

fallback
default
fallback
Domain Ports (domain / ports /)
Observability & DQ
[P] LoggerPort
[P] MetricsPort
[P] TracingPort
[P] DQMonitorPort
[P] MemoryMonitorPort
[P] AuditPort
[P] HealthCheckPort
Validation & Policy
[P] SilverValidationPort
[P] GoldValidationPort
[P] JsonEncoderPort
[P] InputMaxPort
[P] PiiHasherPort
[P] MetadataWriterPort
[P] MetadataCoordinatorPort
Runtime Controls
[P] ShutdownPort
[P] CircuitBreakerPort
[P] RateLimiterPort
Core Data & State
[P] DeltaReaderPort
[P] FilteredDataSource Port
[P] DataSourcePort
[P] StoragePort
[P] LockPort
[P] CheckpointPort
[P] QuarantinePort
Infrastructure Adapters
Observability & DQ Adapters
[A] PrometheusMetrics
[A] NoOpTracing OpenTelemetryTracer
[A] DataQualityMonitor
[A] StructlogLogger UnifiedLogger
[A] FileAuditAdapter
[A] HealthCheckMain
[A] MemoryMonitor
Runtime Control Adapters
[A] ShutdownService
[A] CircuitBreaker
[A] TokenBucket
Validation & Metadata Adapters
[A] PanderaSilverValidator
[A] PanderaGoldValidator
[A] SHA256Hasher
[A] MetadataWriter
[A] MetadataCoordinator
Core Data & Storage Adapters
[A] CsvFileReader
[A] DeltaReader
[A] FilteredDataSource CachedBronzeDataSource
[A] BronzeWriter + SilverWriter + GoldWriter
[A] MemoryLock
[A] LocalCheckpoint
[A] UnifiedQuarantine
[A] ProviderAdapters (5)
NoOp Implementations (development / noop-pyi)
NoOp Observability
[N] NoOpMemoryMonitor
[N] NoOpTracing
[N] NoOpMetrics
[N] NoOpAudit
NoOp Policy
[N] NoOpMetadataWriter
[N] NoOpPiiHasher
fallback
fallback
fallback

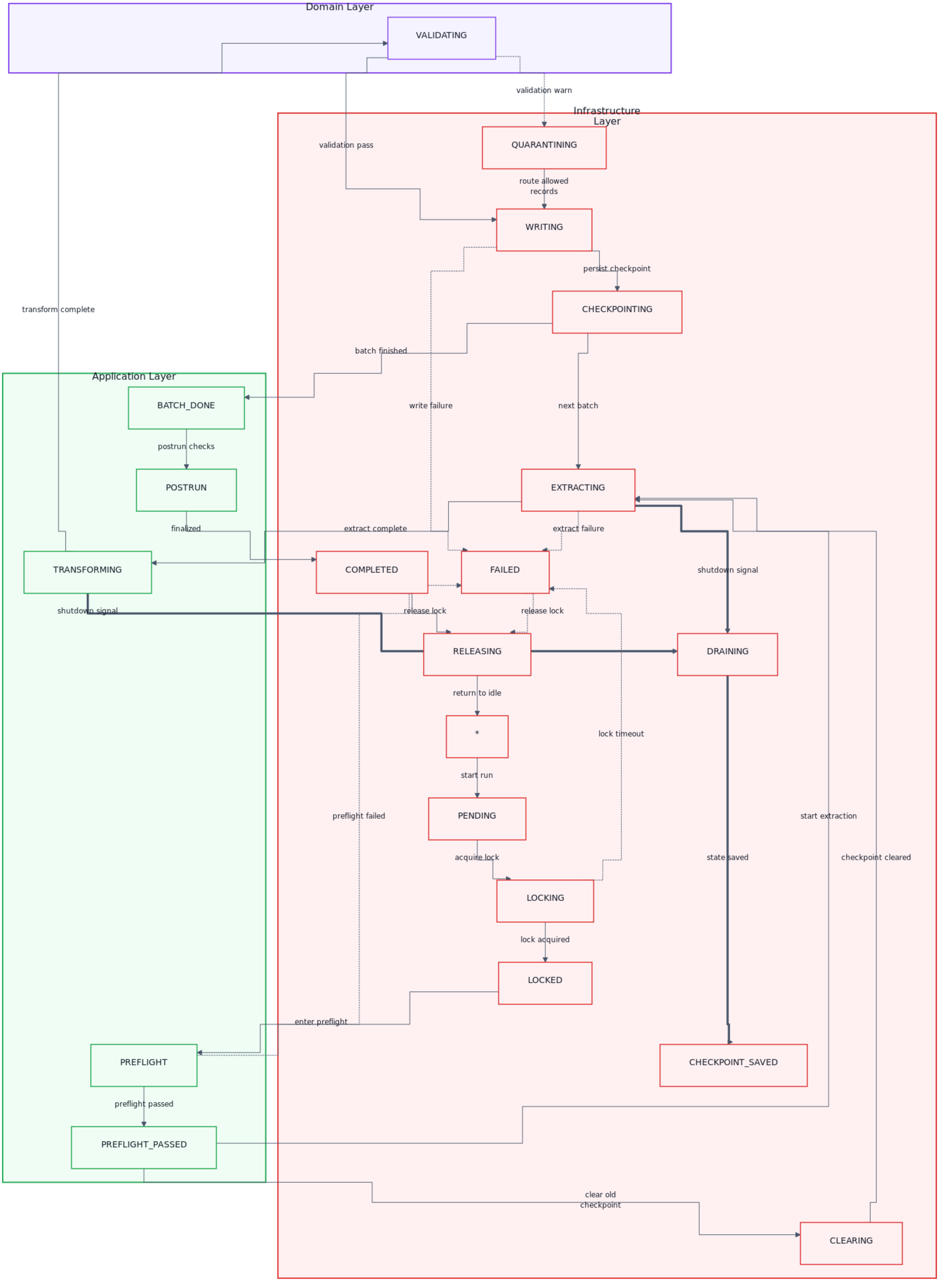

Domain Layer
P1
P2
P3
P4
P5
P6
P7
P8
P9
P10
Infrastructure Layer
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10

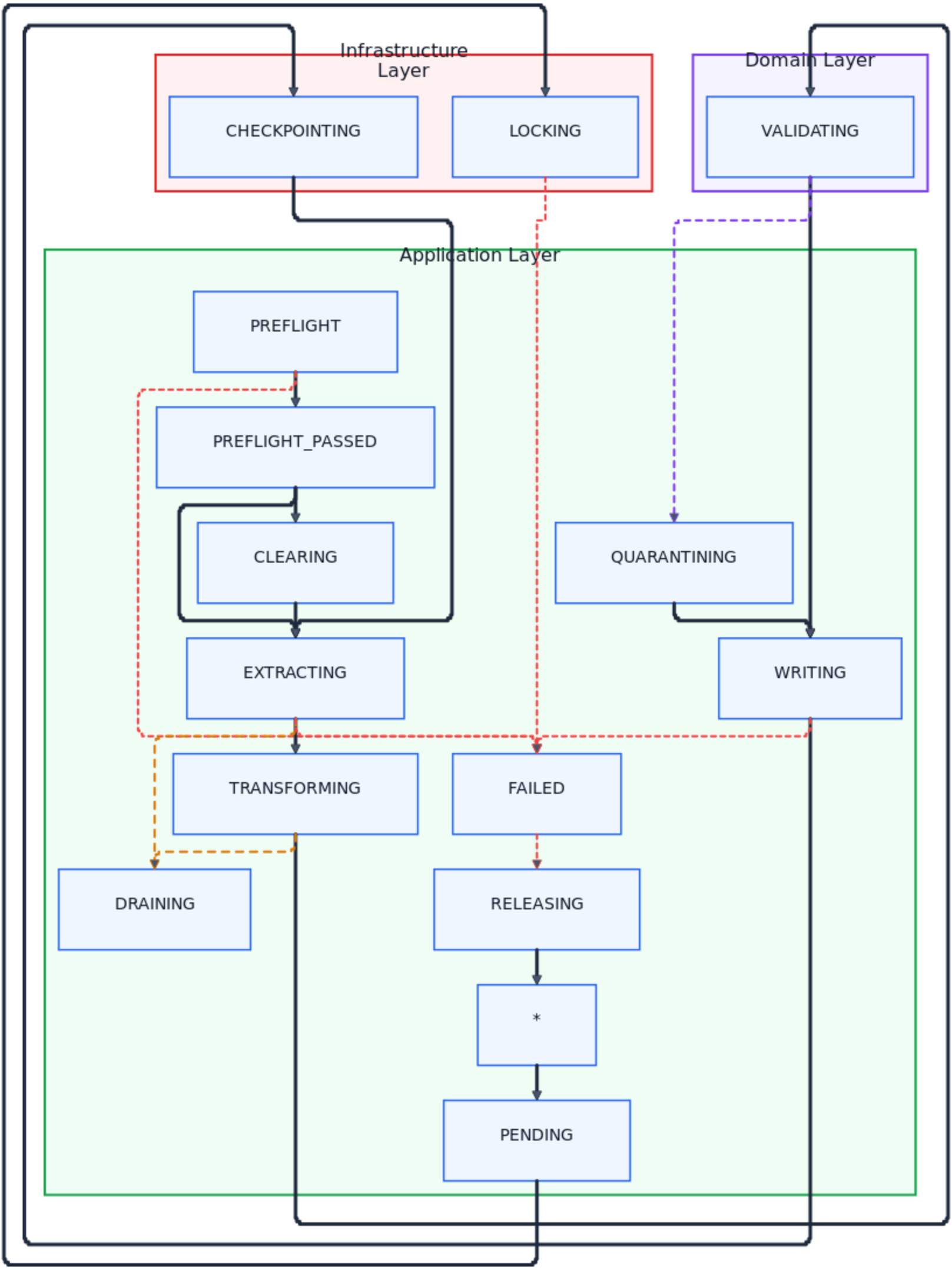

Domain Port Families
[P] Runtime Control Ports
[P] Observability Ports
[P] Core Data Ports
[P] Validation / Policy Ports
Fallbacks
[N] NoOp Adapters
Infrastructure Adapter Families
[A] Runtime Adapters
[A] Observability Adapters
[A] Provider Adapters
[A] Storage Adapters
[A] Validation / Policy Adapters

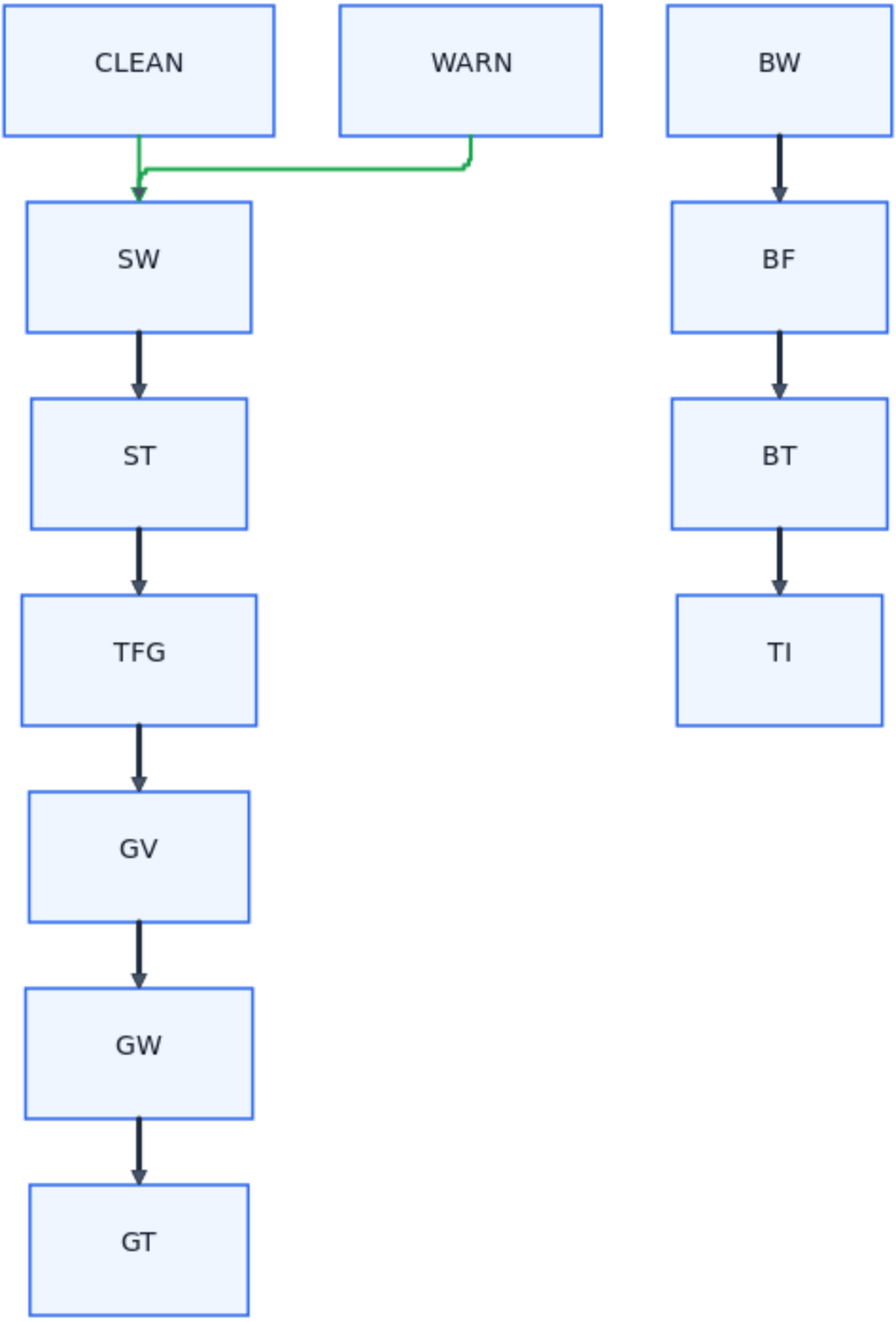

Domain Layer
VALIDATING
Infrastructure Layer
Application Layer
WRITING
PREFLIGHT
CHECKPOINTING
PREFLIGHT_PASSED
LOCKING
EXTRACTING
BATCH_DONE
FAILED
TRANSFORMING
BatchLoop
DRAINING

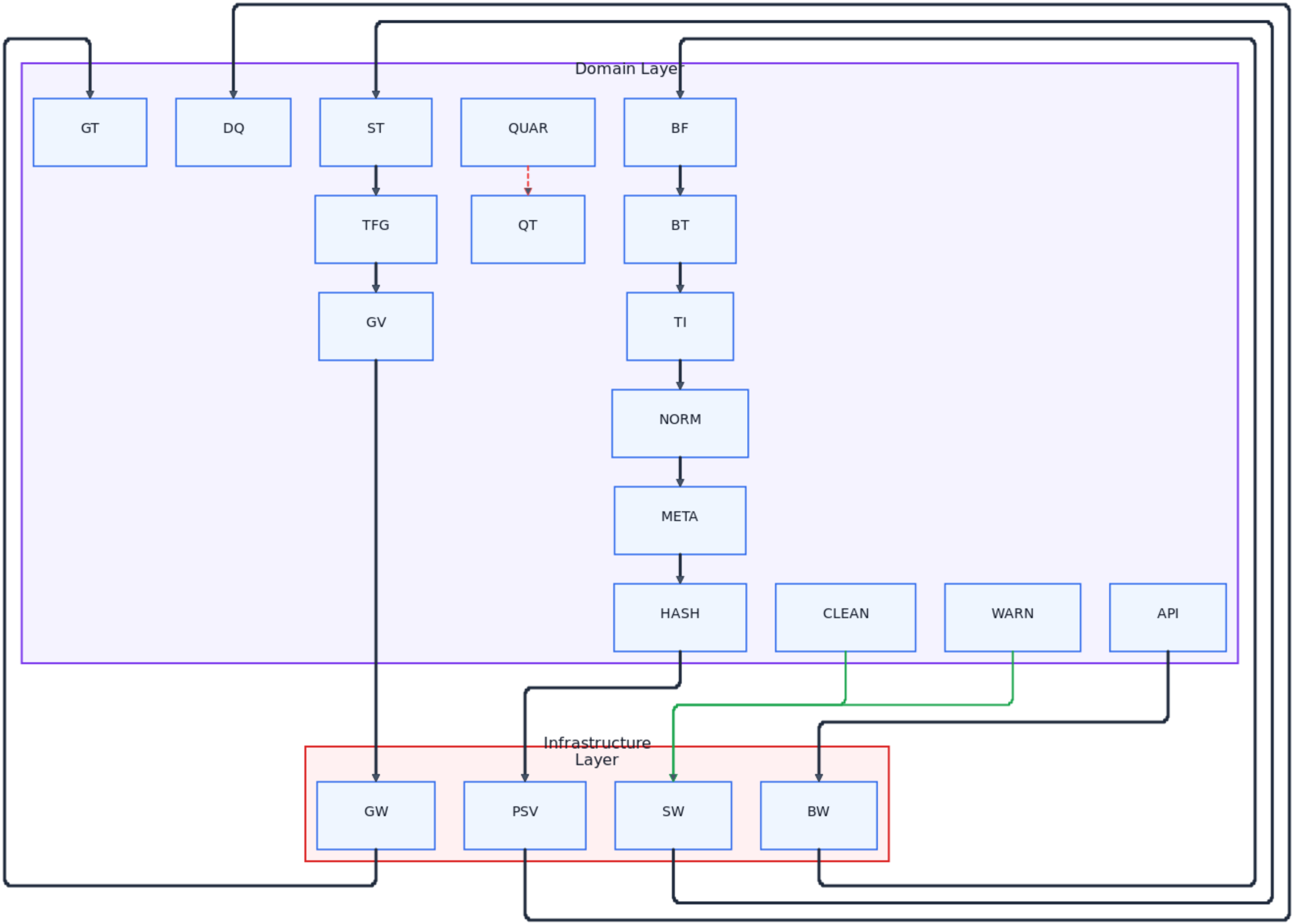

Application Layer
TRANSFORMING
PREFLIGHT
BatchLoop
BATCH_DONE
PREFLIGHT_PASSED
POSTRUN
Infrastructure Layer
CHECKPOINTING
LOCKING
LOCKED
Domain Layer
VALIDATING
CLEARING
QUARANTINING
EXTRACTING
WRITING
DRAINING
FAILED
COMPLETED
RELEASING
*
PENDING

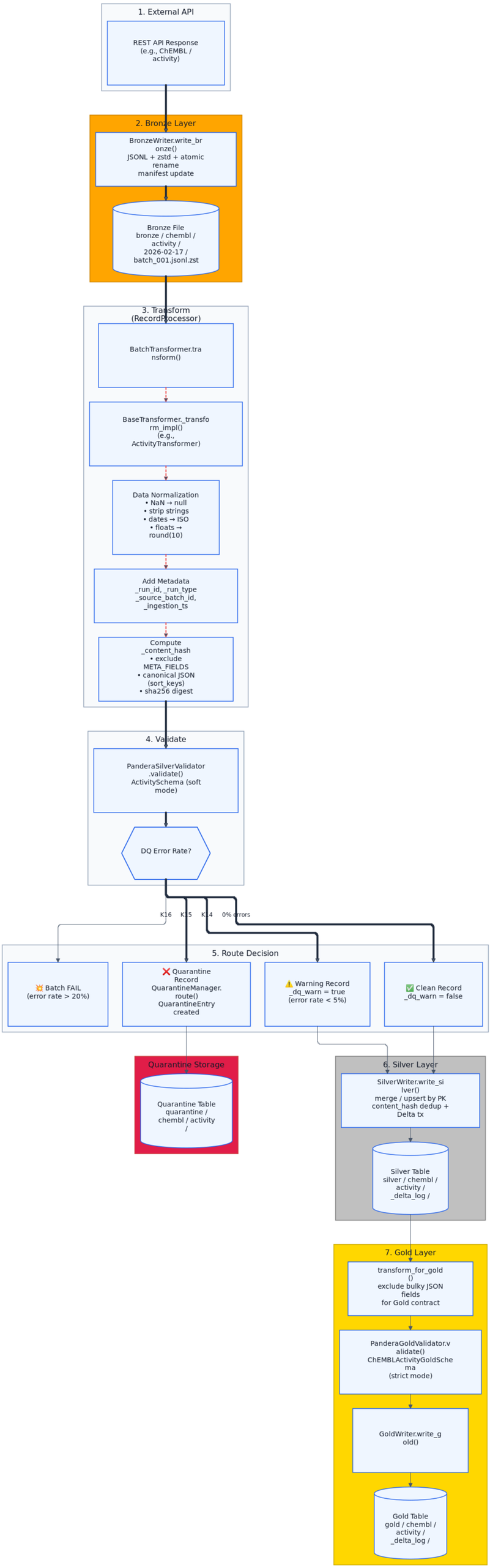

PipelineRun created
PENDING
runner.run()
LOCKING
LockManager.acquire()
LOCKED
PreflightService.run()
HeartbeatTask starts every 30s renews TTL (90s)
PREFLIGHT
all checks pass
PREFLIGHT_PASSED
RunType == REBUILD / BACKFILL
RunType == INCREMENTAL
CLEARING
MedallionLifecycleService\nclear_silver() + clear_gold()
LockAcquisitionError
preflight check failed
EXTRACTING
BatchExecutor._extract()
TRANSFORMING
SIGTERM received
BatchTransformer.transform()
ShutdownService detected
VALIDATING
DRAINING
CircuitBreakerOpenError\nRetryExhaustedError
CheckpointManager.save()\nmore batches
DQ error rate > threshold
finish current batch\nsave checkpoint
QUARANTINING
PanderaSilverValidator.validate()
CHECKPOINT_SAVED
QuarantineManager.route()
release lock\nexit(0)
WRITING
CANCELLED
LockLostError\n(safety guard)
BatchWriter.write()
FAILED
CHECKPOINTING
no more batches
PipelineRunState.FAILED StageResult captures error
BATCH_DONE
PostrunService.run()
error path
POSTRUN
vacuum + DQ report\n+ metrics publish
COMPLETED
success
PipelineRunState.COMPLETED PipelineObserver records duration
RELEASING
LockManager.release()\nHeartbeatTask.stop()

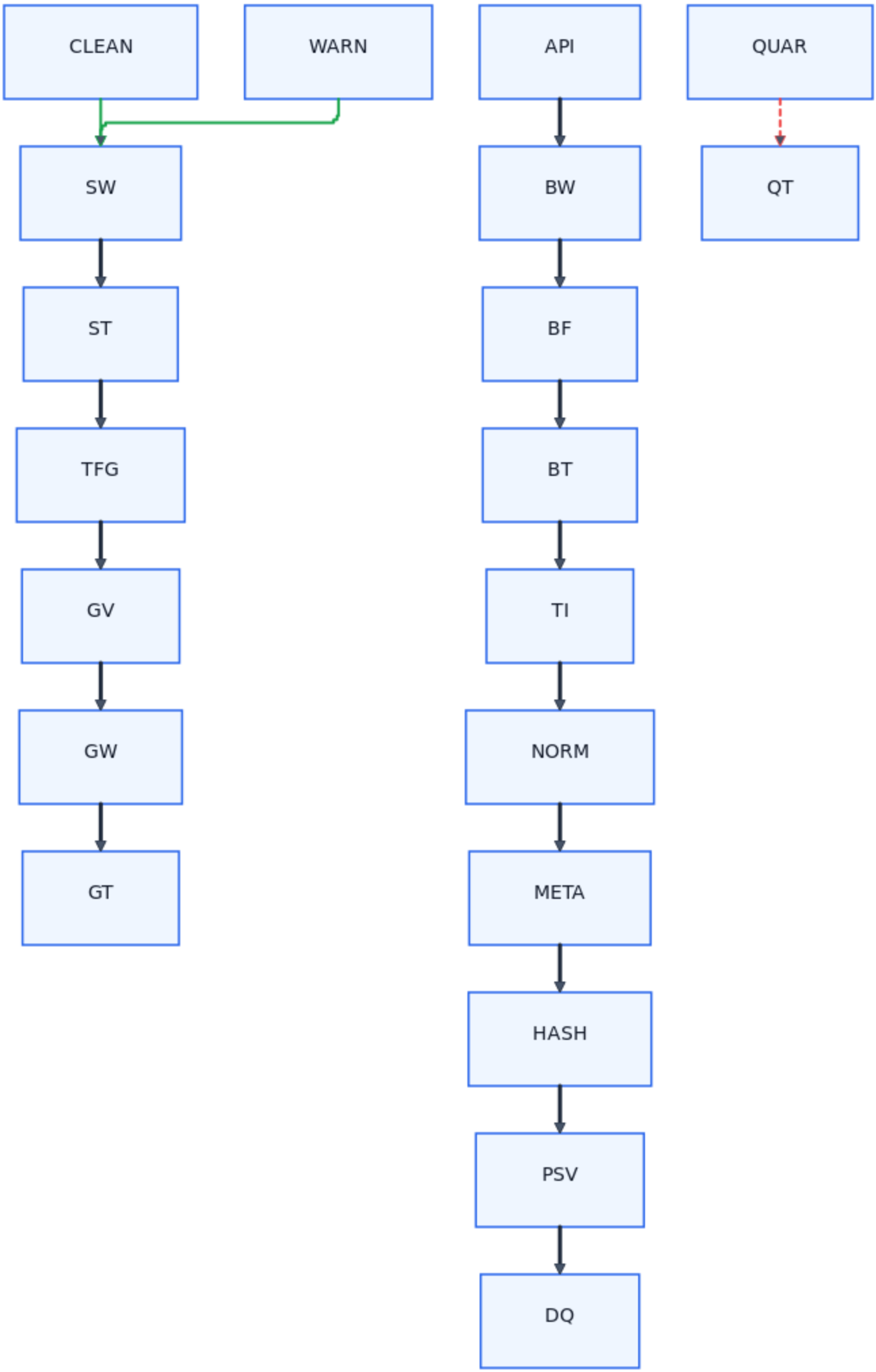

Domain Layer
VALIDATING
validation warn
Infrastructure Layer
validation pass
QUARANTINING
route allowed records
WRITING
persist checkpoint
CHECKPOINTING
transform complete
batch finished
Application Layer
BATCH_DONE
write failure
next batch
postrun checks
POSTRUN
EXTRACTING
finalized
extract complete
extract failure
TRANSFORMING
COMPLETED
FAILED
shutdown signal
shutdown signal
release lock
release lock
RELEASING
DRAINING
return to idle
*
lock timeout
start run
preflight failed
PENDING
start extraction
acquire lock
state saved
checkpoint cleared
LOCKING
lock acquired
LOCKED
enter preflight
PREFLIGHT
CHECKPOINT_SAVED
preflight passed
PREFLIGHT_PASSED
clear old checkpoint
CLEARING

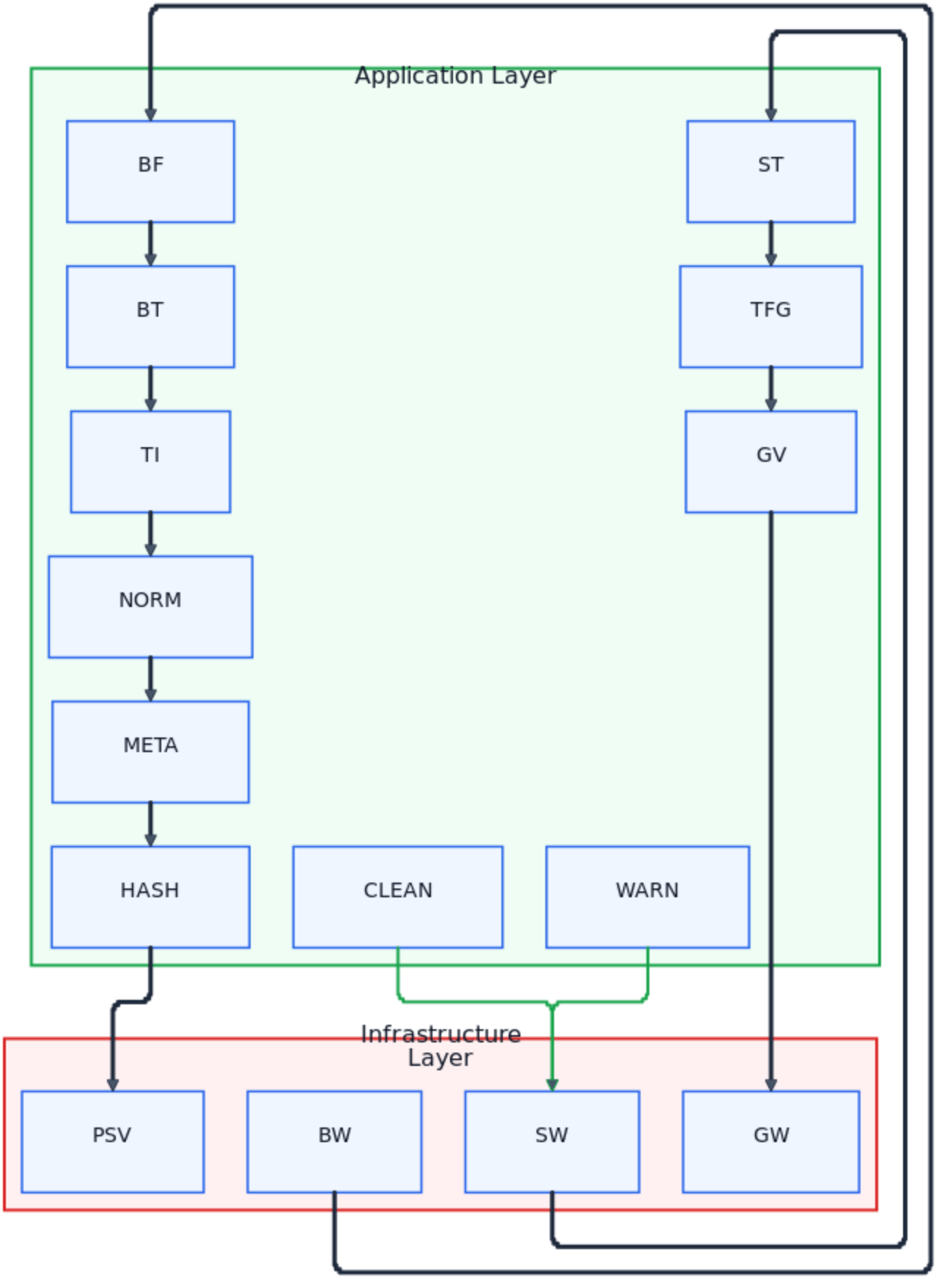

Infrastructure Layer
CHECKPOINTING
LOCKING
Domain Layer
VALIDATING
Application Layer
PREFLIGHT
PREFLIGHT_PASSED
CLEARING
QUARANTINING
EXTRACTING
WRITING
TRANSFORMING
FAILED
DRAINING
RELEASING
*
PENDING

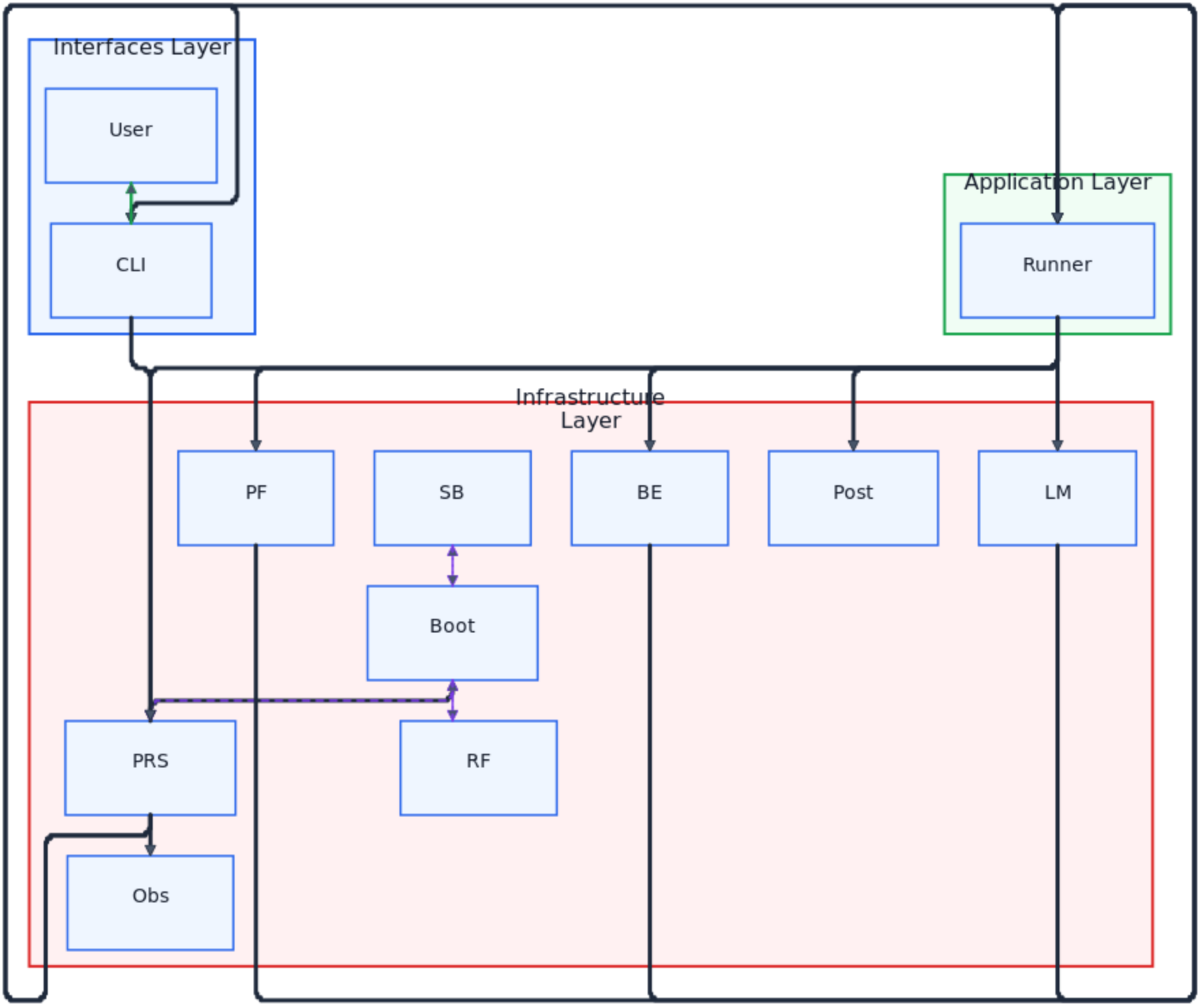

CLEAN
WARN
BW
SW
BF
ST
BT
TFG
TI
GV
GW
GT

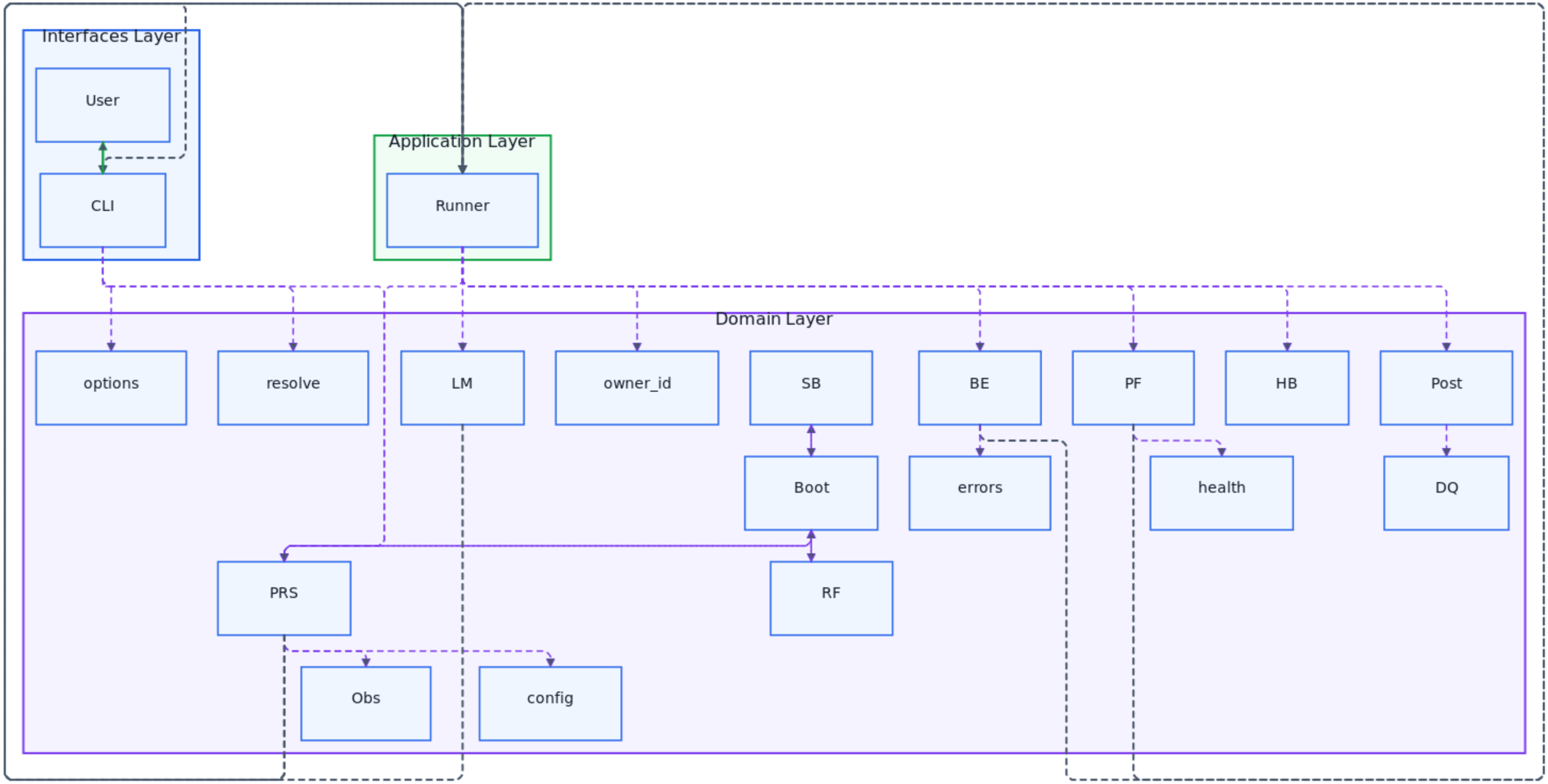

Domain Layer
GT
DQ
ST
QUAR
BF
TFG
QT
BT
GV
TI
NORM
META
HASH
CLEAN
WARN
API
Infrastructure Layer
GW
PSV
SW
BW

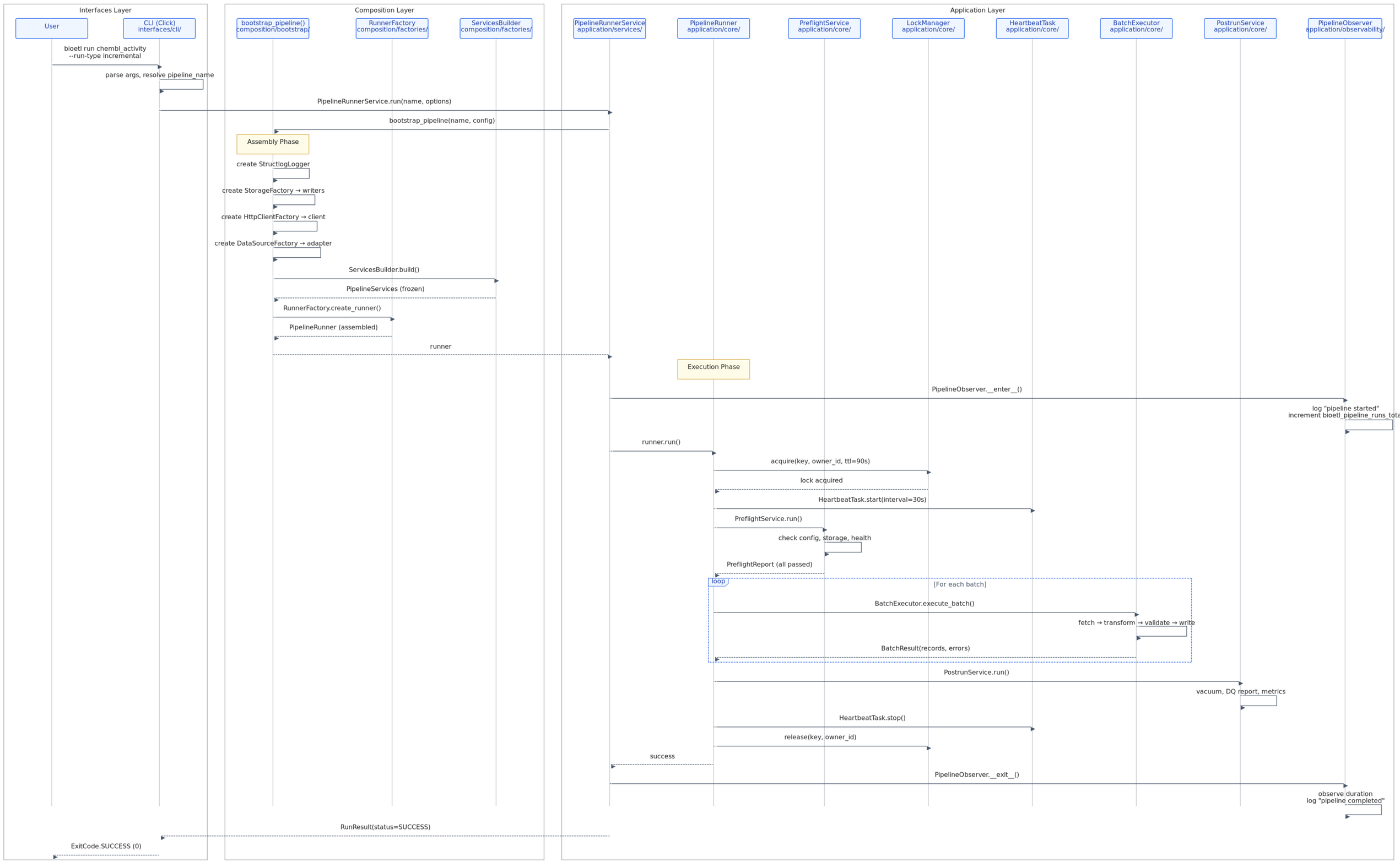

1. External API
REST API Response (e.g., ChEMBL / activity)
2. Bronze Layer
BronzeWriter.write_bronze() JSONL + zstd + atomic rename manifest update
Bronze File bronze / chembl / activity / 2026-02-17 / batch_001.jsonl.zst
3. Transform (RecordProcessor)
BatchTransformer.transform()
BaseTransformer._transform_impl() (e.g., ActivityTransformer)
Data Normalization • NaN → null • strip strings • dates → ISO • floats → round(10)
Add Metadata _run_id, _run_type _source_batch_id, _ingestion_ts
Compute _content_hash • exclude META_FIELDS • canonical JSON (sort_keys) • sha256 digest
4. Validate
PanderaSilverValidator.validate() ActivitySchema (soft mode)
DQ Error Rate?
K16
K15
K14
0% errors
5. Route Decision
💥 Batch FAIL (error rate > 20%)
❌ Quarantine Record QuarantineManager.route() QuarantineEntry created
⚠️ Warning Record _dq_warn = true (error rate < 5%)
✅ Clean Record _dq_warn = false
Quarantine Storage
Quarantine Table quarantine / chembl / activity /
6. Silver Layer
SilverWriter.write_silver() merge / upsert by PK content_hash dedup + Delta tx
Silver Table silver / chembl / activity / _delta_log /
7. Gold Layer
transform_for_gold() exclude bulky JSON fields for Gold contract
PanderaGoldValidator.validate() ChEMBLActivityGoldSchema (strict mode)
GoldWriter.write_gold()
Gold Table gold / chembl / activity / _delta_log /

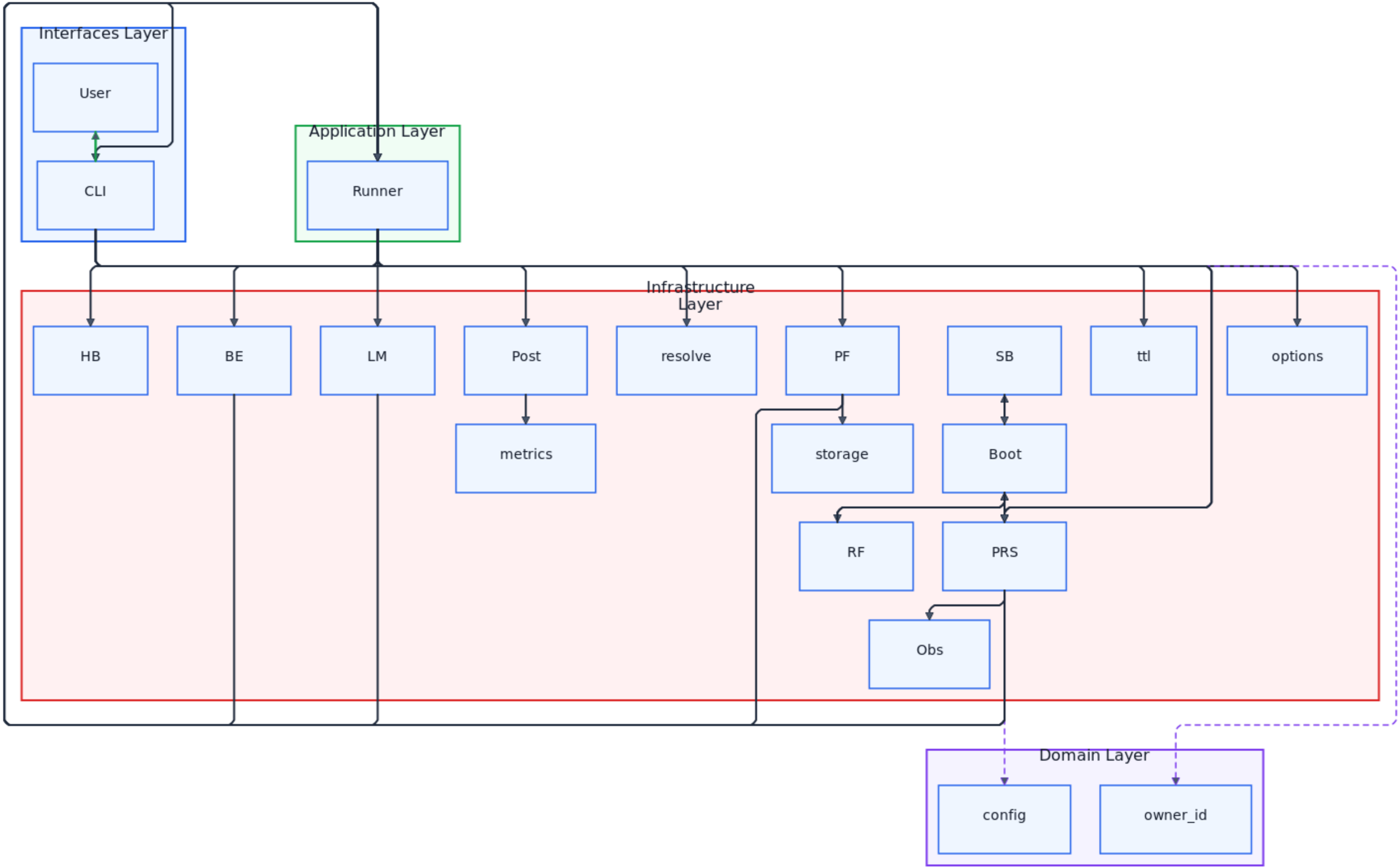

CLEAN
WARN
API
QUAR
SW
BW
QT
ST
BF
TFG
BT
GV
TI
GW
NORM
GT
META
HASH
PSV
DQ

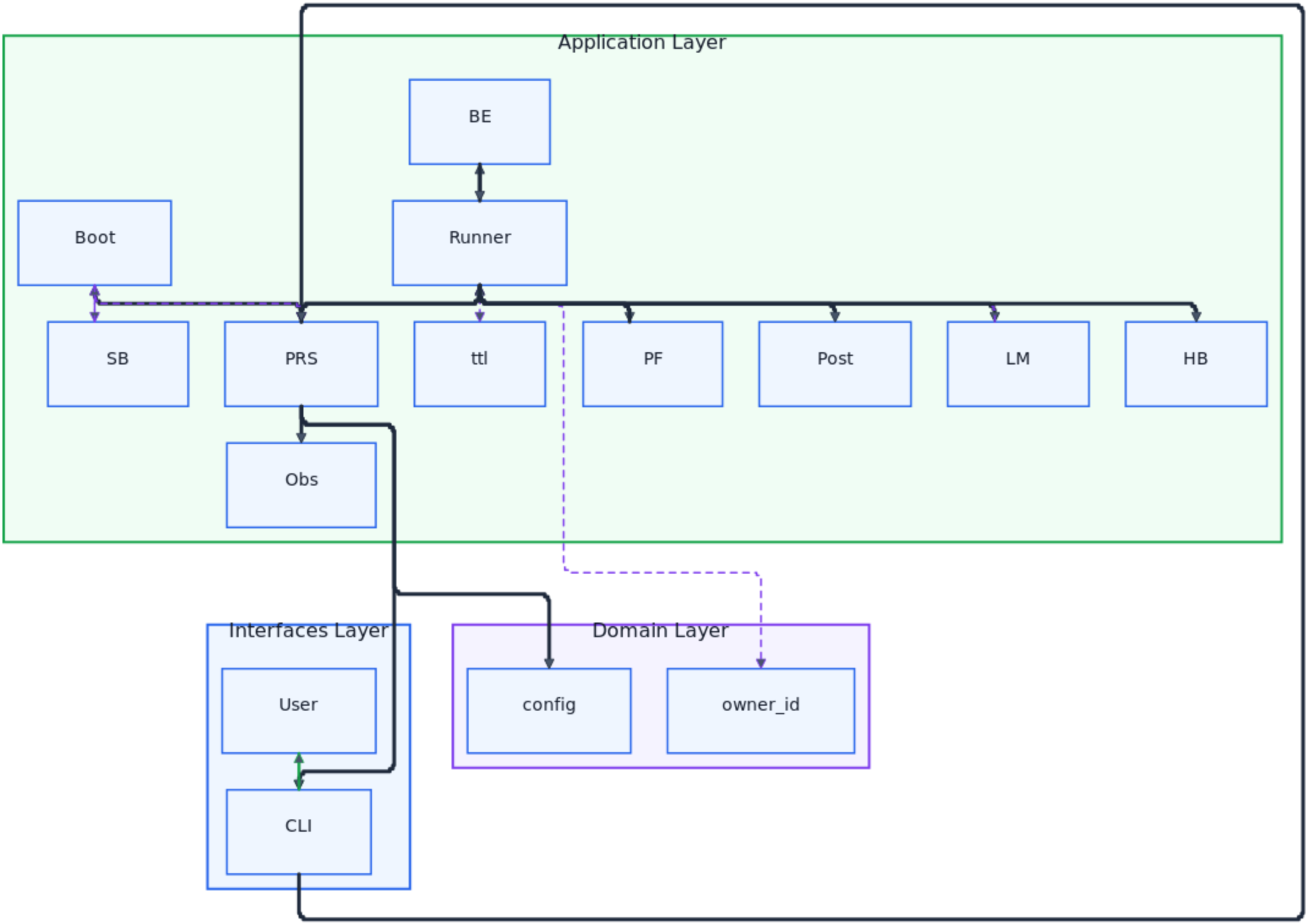

Application Layer
BF
BT
TI
NORM
META
HASH
ST
TFG
GV
CLEAN
WARN
Infrastructure Layer
PSV
BW
SW
GW

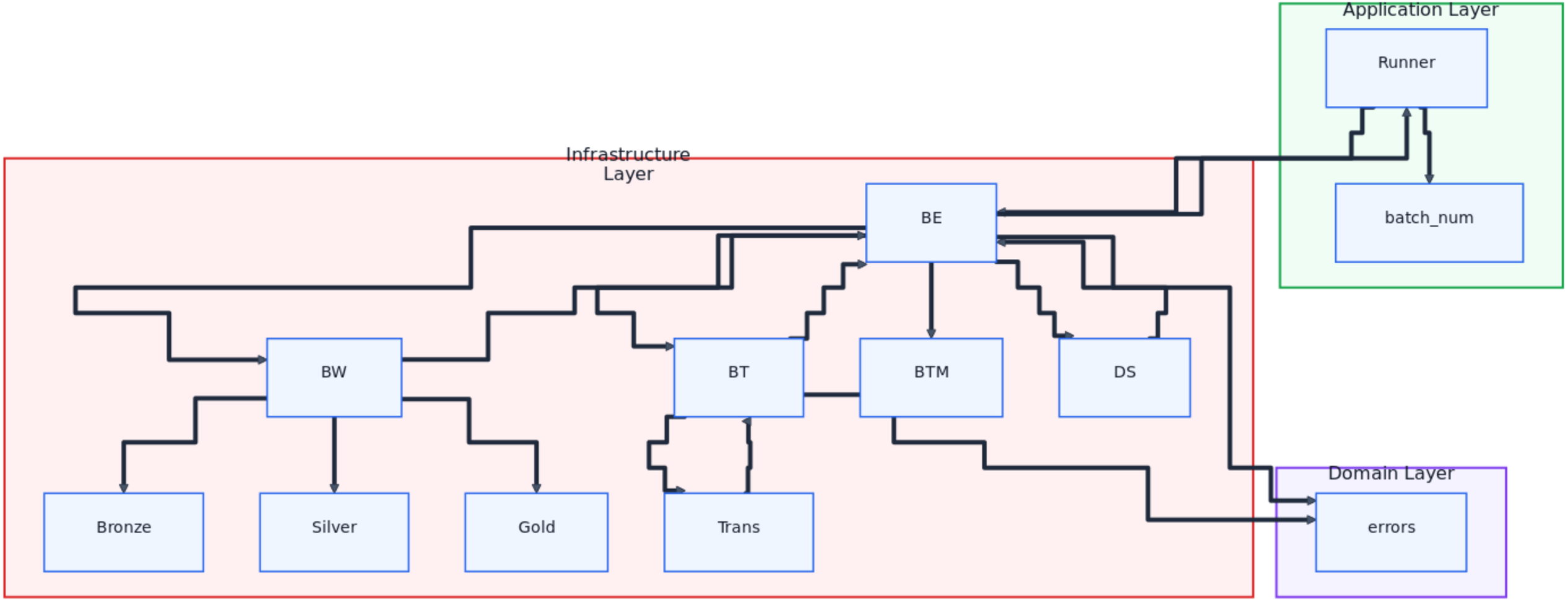

Interfaces Layer
User
CLI
Application Layer
Runner
Infrastructure Layer
PF
SB
BE
Post
LM
Boot
PRS
RF
Obs

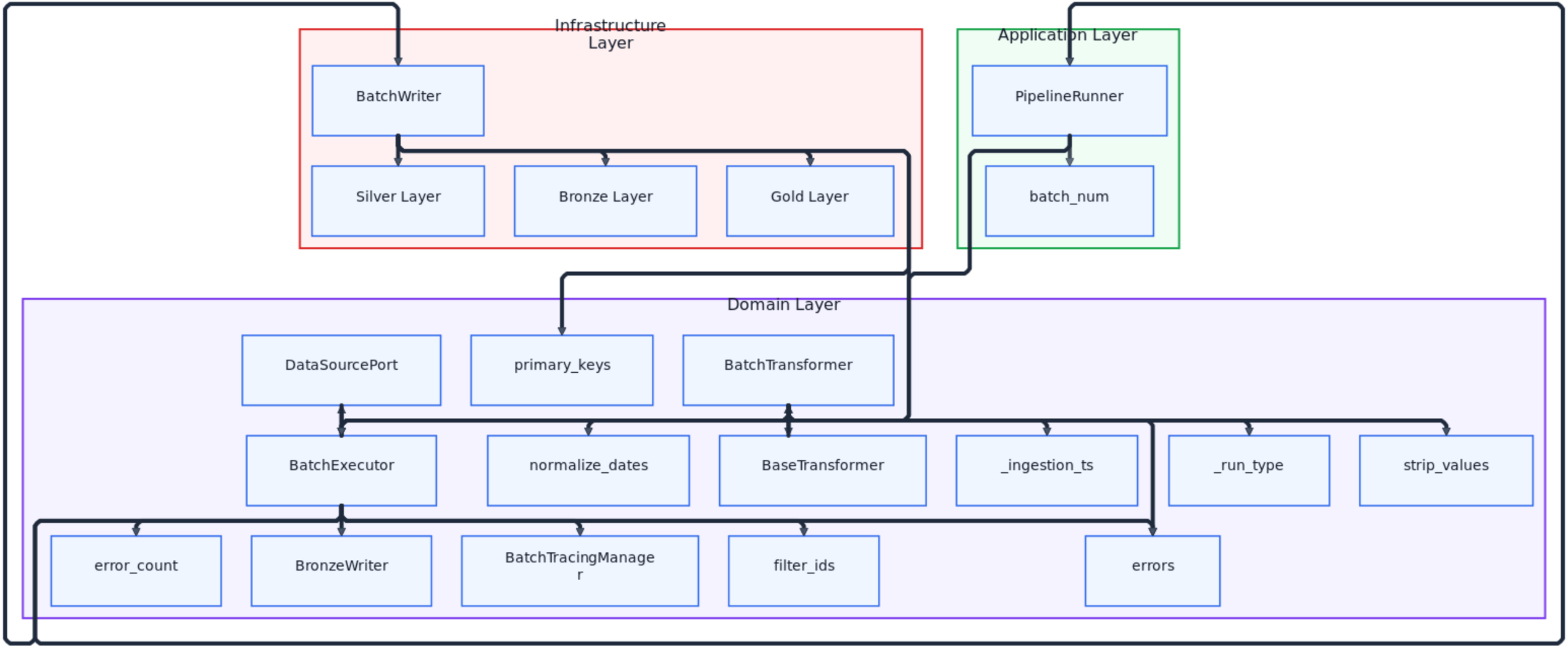

Interfaces Layer
User
CLI
Application Layer
Runner
Domain Layer
options
resolve
LM
owner_id
SB
BE
PF
HB
Post
Boot
errors
health
DQ
PRS
RF
Obs
config

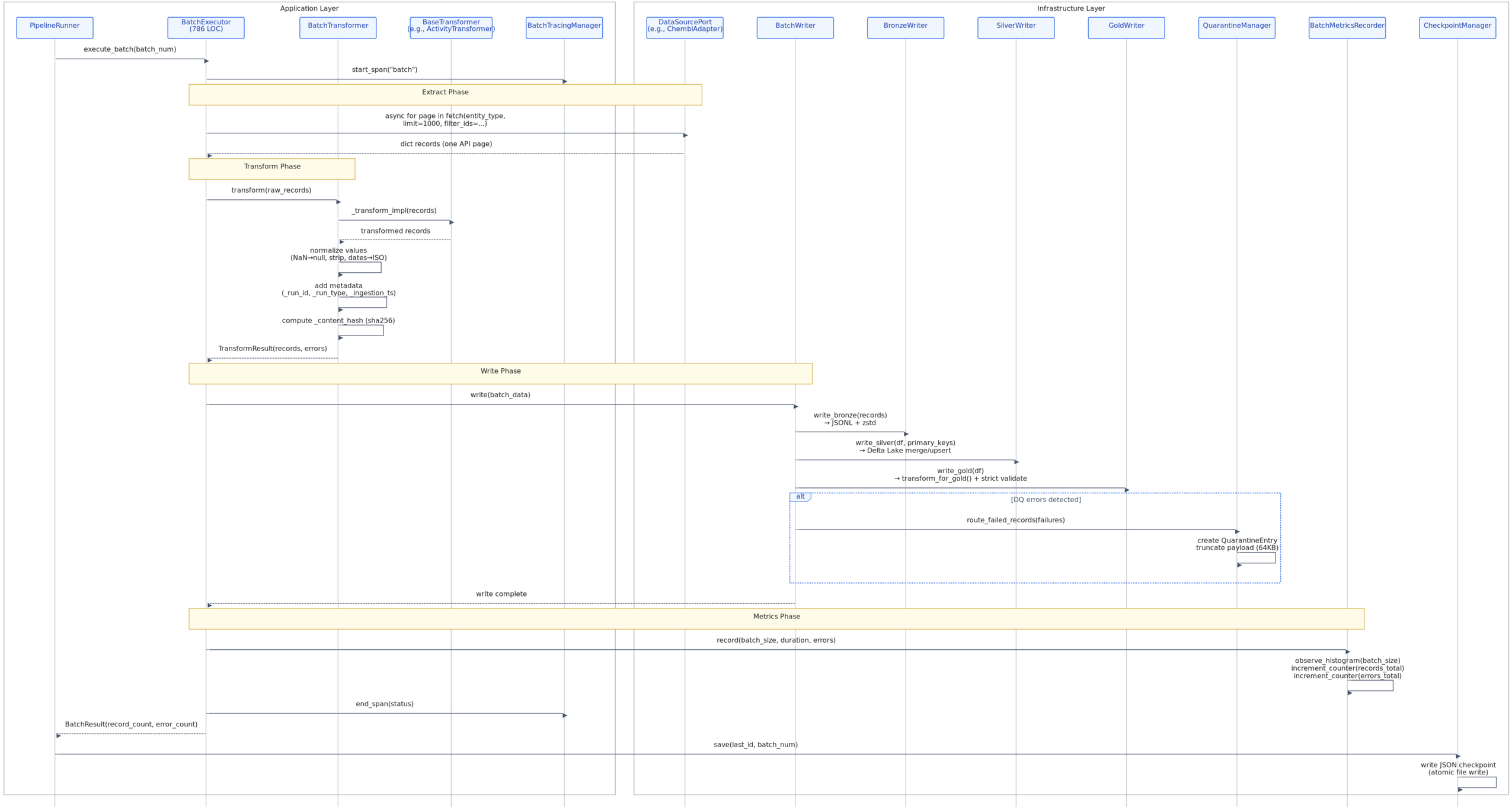

Interfaces Layer
Composition Layer
Application Layer
User
CLI (Click) interfaces/cli/
bootstrap_pipeline() composition/bootstrap/
RunnerFactory composition/factories/
ServicesBuilder composition/factories/
PipelineRunnerService application/services/
PipelineRunner application/core/
PreflightService application/core/
LockManager application/core/
HeartbeatTask application/core/
BatchExecutor application/core/
PostrunService application/core/
PipelineObserver application/observability/
bioetl run chembl_activity --run-type incremental
parse args, resolve pipeline_name
PipelineRunnerService.run(name, options)
bootstrap_pipeline(name, config)
Assembly Phase
create StructlogLogger
create StorageFactory → writers
create HttpClientFactory → client
create DataSourceFactory → adapter
ServicesBuilder.build()
PipelineServices (frozen)
RunnerFactory.create_runner()
PipelineRunner (assembled)
runner
Execution Phase
PipelineObserver.__enter__()
log "pipeline started"
increment bioetl_pipeline_runs_total
runner.run()
acquire(key, owner_id, ttl=90s)
lock acquired
HeartbeatTask.start(interval=30s)
PreflightService.run()
check config, storage, health
PreflightReport (all passed)
loop
[For each batch]
BatchExecutor.execute_batch()
fetch → transform → validate → write
BatchResult(records, errors)
PostrunService.run()
vacuum, DQ report, metrics
HeartbeatTask.stop()
release(key, owner_id)
success
PipelineObserver.__exit__()
observe duration
log "pipeline completed"
RunResult(status=SUCCESS)
ExitCode.SUCCESS (0)

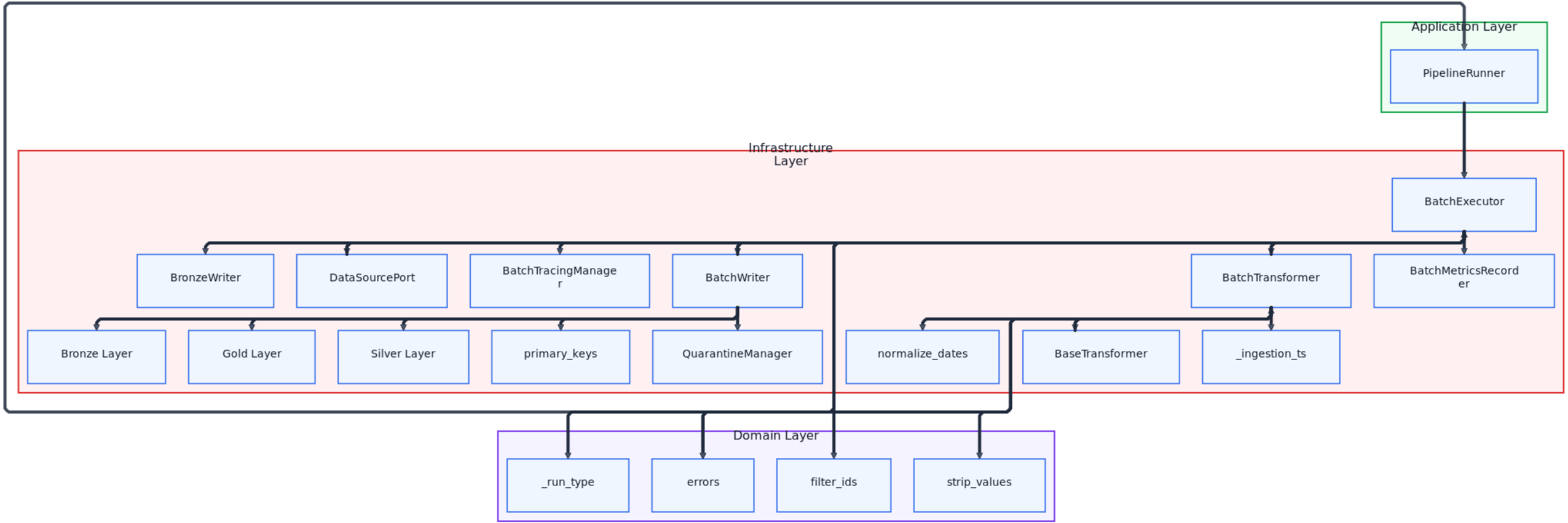

Interfaces Layer
User
CLI
Application Layer
Runner
Infrastructure Layer
HB
BE
LM
Post
resolve
PF
SB
ttl
options
metrics
storage
Boot
RF
PRS
Obs
Domain Layer
config
owner_id

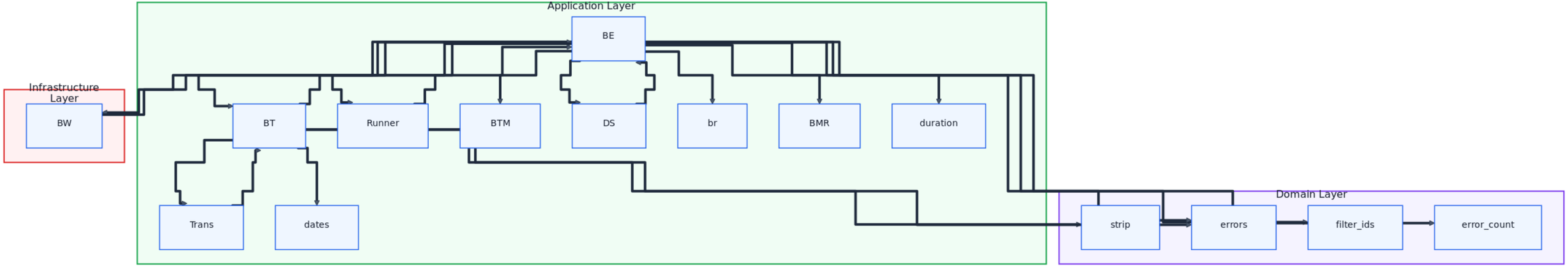

Application Layer
BE
Boot
Runner
SB
PRS
ttl
PF
Post
LM
HB
Obs
Interfaces Layer
Domain Layer
User
config
owner_id
CLI

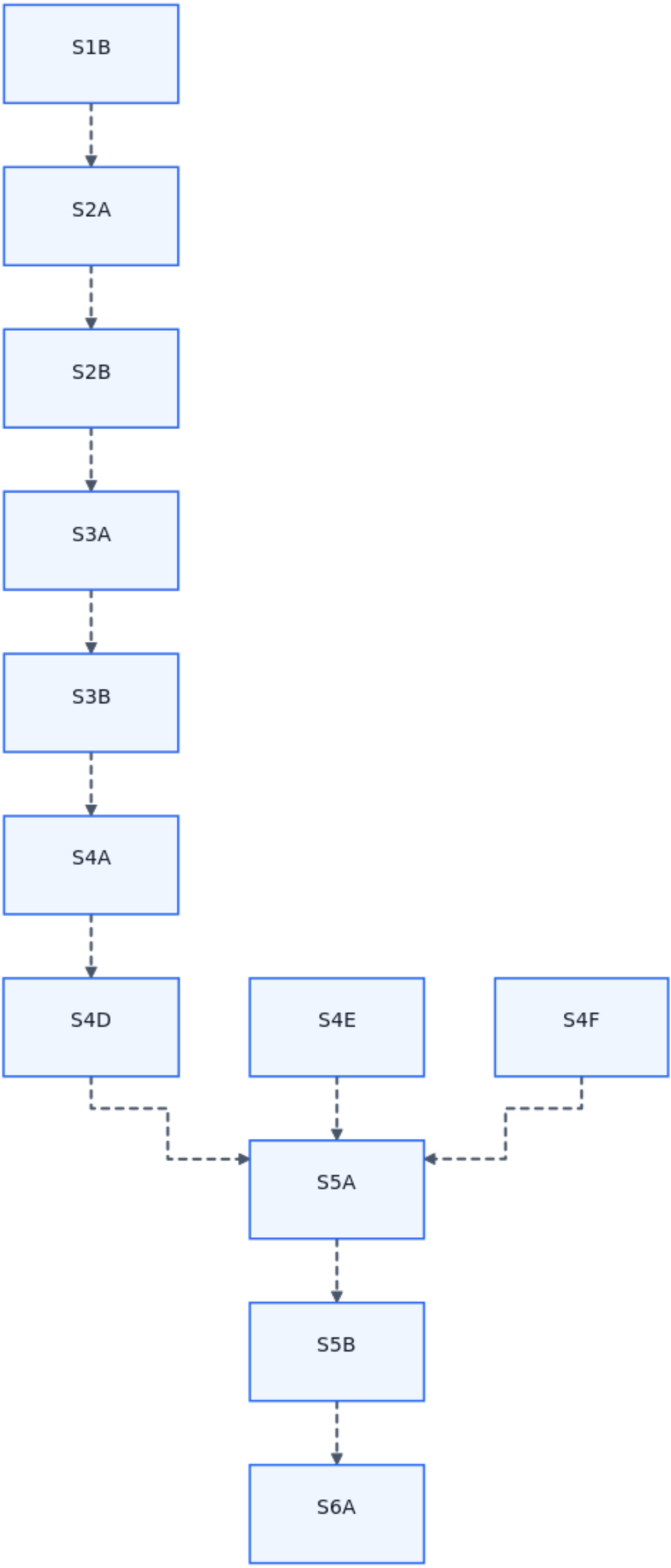

Application Layer
Runner
batch_num
Infrastructure Layer
BE
BW
BT
BTM
DS
Bronze
Silver
Gold
Trans
Domain Layer
errors

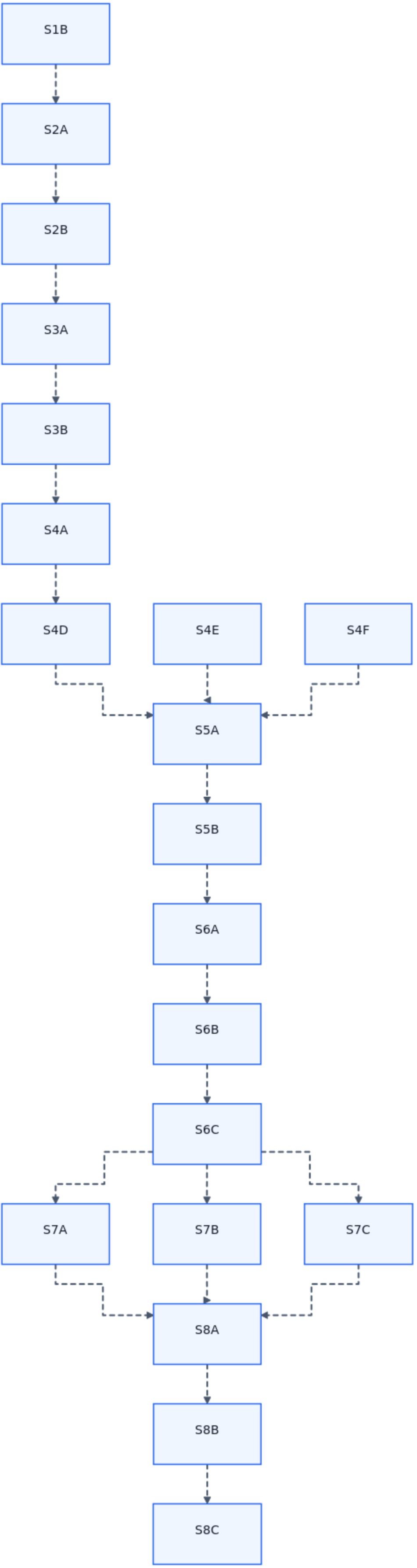

Infrastructure Layer
BatchWriter
Silver Layer
Bronze Layer
Gold Layer
Application Layer
PipelineRunner
batch_num
Domain Layer
DataSourcePort
primary_keys
BatchTransformer
BatchExecutor
normalize_dates
BaseTransformer
_ingestion_ts
_run_type
strip_values
error_count
BronzeWriter
BatchTracingManager
filter_ids
errors

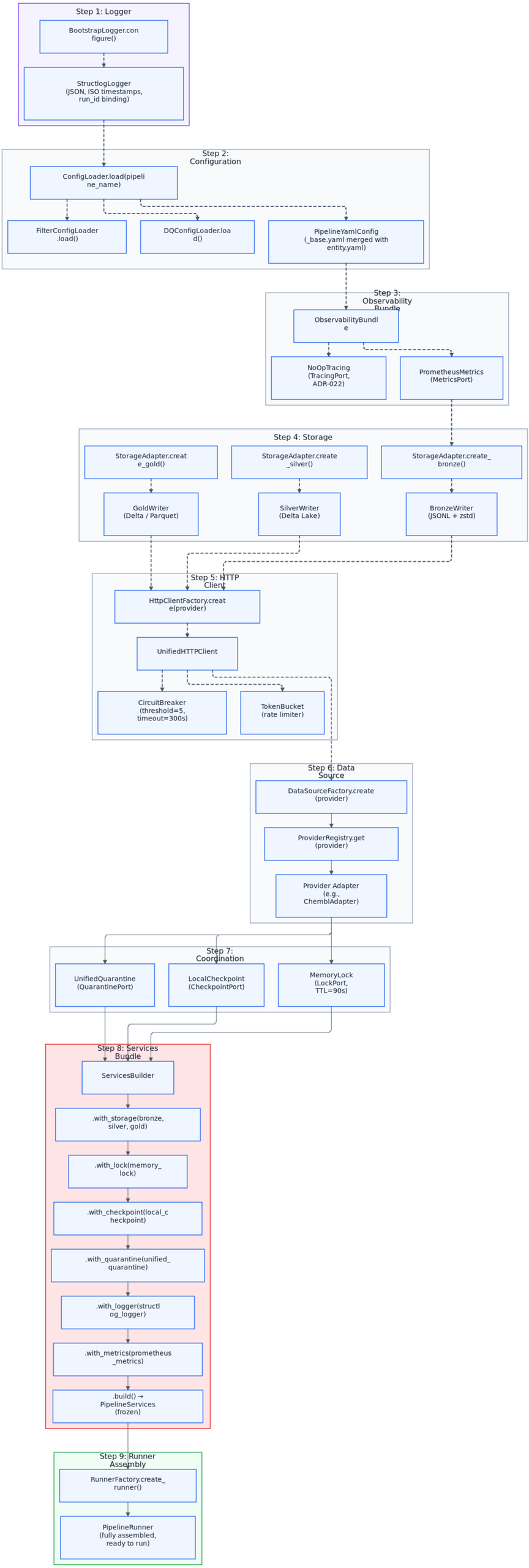

Application Layer
Infrastructure Layer
PipelineRunner
BatchExecutor (786 LOC)
BatchTransformer
BaseTransformer (e.g., ActivityTransformer)
BatchTracingManager
DataSourcePort (e.g., ChemblAdapter)
BatchWriter
BronzeWriter
SilverWriter
GoldWriter
QuarantineManager
BatchMetricsRecorder
CheckpointManager
execute_batch(batch_num)
start_span("batch")
Extract Phase
async for page in fetch(entity_type, limit=1000, filter_ids=...)
dict records (one API page)
Transform Phase
transform(raw_records)
_transform_impl(records)
transformed records
normalize values (NaN→null, strip, dates→ISO)
add metadata (_run_id, _run_type, _ingestion_ts)
compute _content_hash (sha256)
TransformResult(records, errors)
Write Phase
write(batch_data)
write_bronze(records) → JSONL + zstd
write_silver(df, primary_keys) → Delta Lake merge/upsert
write_gold(df) → transform_for_gold() + strict validate
alt
[DQ errors detected]
route_failed_records(failures)
create QuarantineEntry truncate payload (64KB)
write complete
Metrics Phase
record(batch_size, duration, errors)
observe_histogram(batch_size) increment_counter(records_total) increment_counter(errors_total)
end_span(status)
BatchResult(record_count, error_count)
save(last_id, batch_num)
write JSON checkpoint (atomic file write)

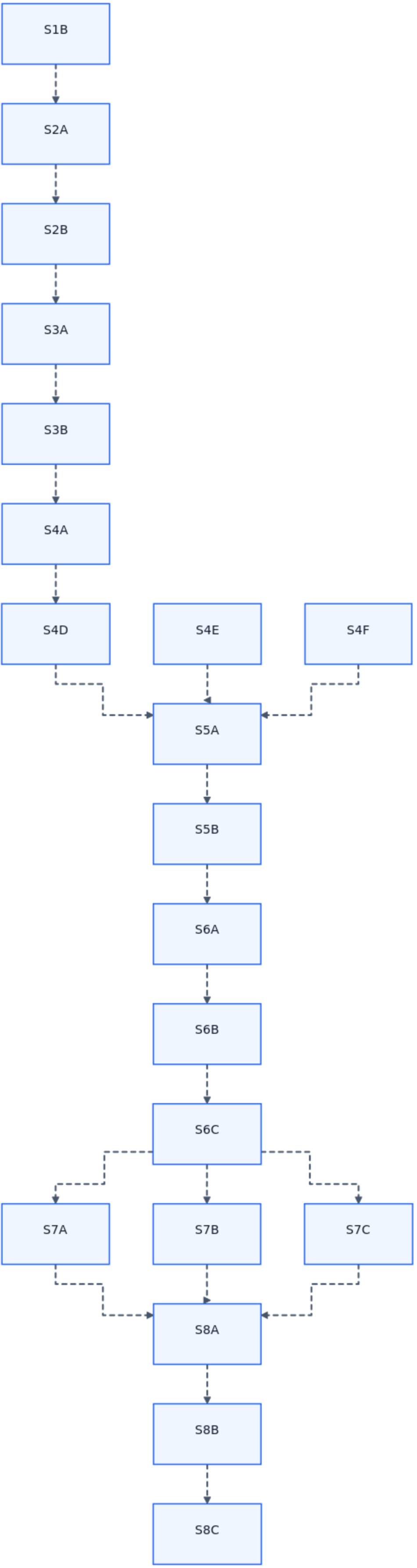

Application Layer
PipelineRunner
Infrastructure Layer
BatchExecutor
BronzeWriter
DataSourcePort
BatchTracingManager
BatchWriter
BatchTransformer
BatchMetricsRecorder
Bronze Layer
Gold Layer
Silver Layer
primary_keys
QuarantineManager
normalize_dates
BaseTransformer
_ingestion_ts
Domain Layer
_run_type
errors
filter_ids
strip_values

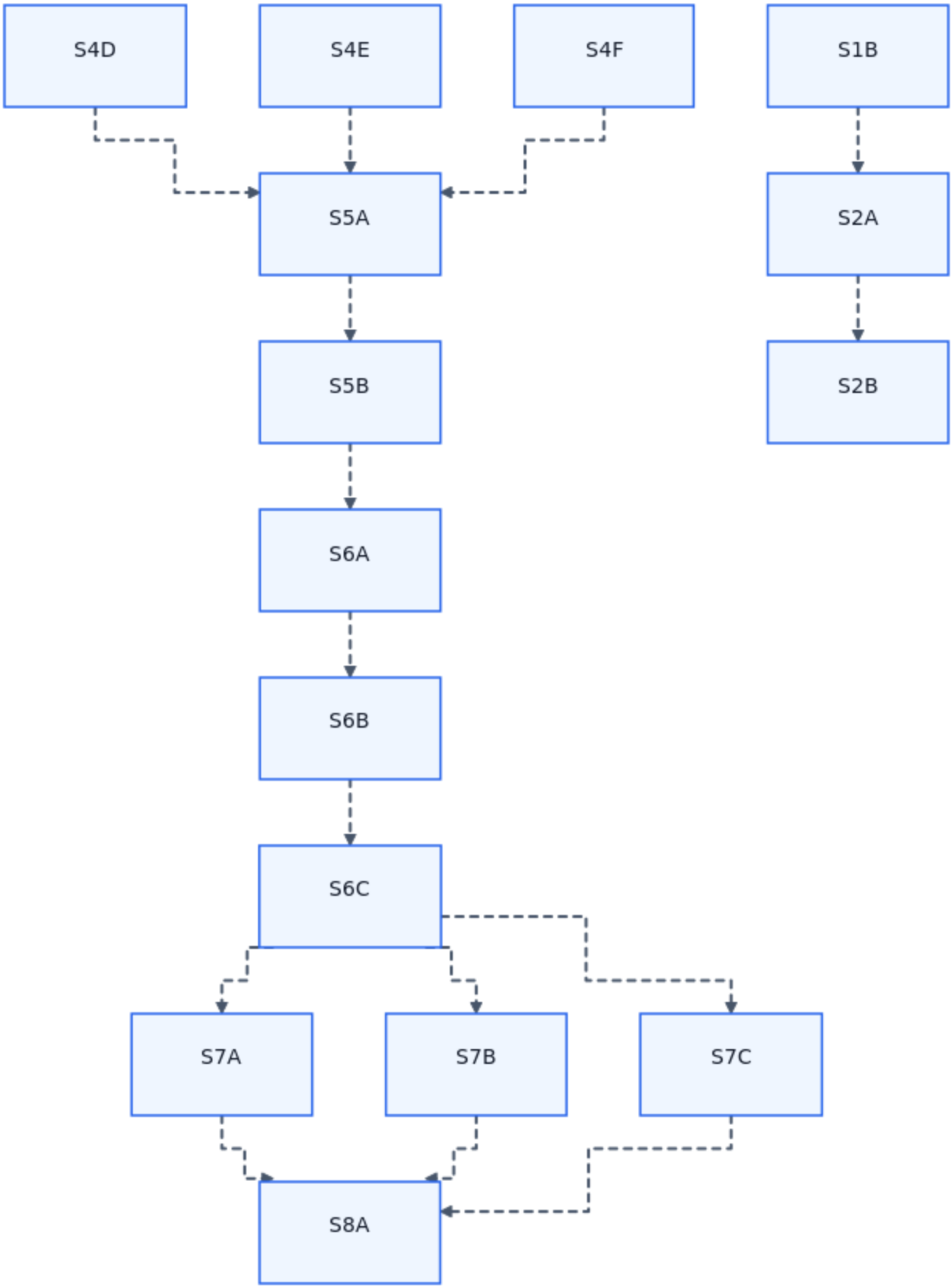

Application Layer
BE
Infrastructure Layer
BW
BT
Runner
BTM
DS
br
BMR
duration
Trans
dates
Domain Layer
strip
errors
filter_ids
error_count


S1B
S2A
S2B
S3A
S3B
S4A
S4D
S4E
S4F
S5A
S5B
S6A


S1B
S2A
S2B
S3A
S3B
S4A
S4D
S4E
S4F
S5A
S5B
S6A
S6B
S6C
S7A
S7B
S7C
S8A
S8B
S8C


Step 1: Logger
BootstrapLogger.configure()
StructlogLogger (JSON, ISO timestamps, run_id binding)
Step 2: Configuration
ConfigLoader.load(pipeline_name)
FilterConfigLoader.load()
DQConfigLoader.load()
PipelineYamlConfig (_base.yaml merged with entity.yaml)
Step 3: Observability Bundle
ObservabilityBundle
NoOpTracing (TracingPort, ADR-022)
PrometheusMetrics (MetricsPort)
Step 4: Storage
StorageAdapter.create_gold()
StorageAdapter.create_silver()
StorageAdapter.create_bronze()
GoldWriter (Delta / Parquet)
SilverWriter (Delta Lake)
BronzeWriter (JSONL + zstd)
Step 5: HTTP Client
HttpClientFactory.create(provider)
UnifiedHTTPClient
CircuitBreaker (threshold=5, timeout=300s)
TokenBucket (rate limiter)
Step 6: Data Source
DataSourceFactory.create(provider)
ProviderRegistry.get(provider)
Provider Adapter (e.g., ChemblAdapter)
Step 7: Coordination
UnifiedQuarantine (QuarantinePort)
LocalCheckpoint (CheckpointPort)
MemoryLock (LockPort, TTL=90s)
Step 8: Services Bundle
ServicesBuilder
.with_storage(bronze, silver, gold)
.with_lock(memory_lock)
.with_checkpoint(local_checkpoint)
.with_quarantine(unified_quarantine)
.with_logger(structlog_logger)
.with_metrics(prometheus_metrics)
.build() → PipelineServices (frozen)
Step 9: Runner Assembly
RunnerFactory.create_runner()
PipelineRunner (fully assembled, ready to run)


S1B
S2A
S2B
S3A
S3B
S4A
S4D
S4E
S4F
S5A
S5B
S6A
S6B
S6C
S7A
S7B
S7C
S8A
S8B
S8C


S4D
S4E
S4F
S1B
S5A
S2A
S5B
S2B
S6A
S6B
S6C
S7A
S7B
S7C
S8A

Infrastructure Layer

Bronze

Silver

Gold

QuarantineEntry

Storage Writers

Application Layer

PipelineRun

Batch

Domain Layer

DataSourcePort

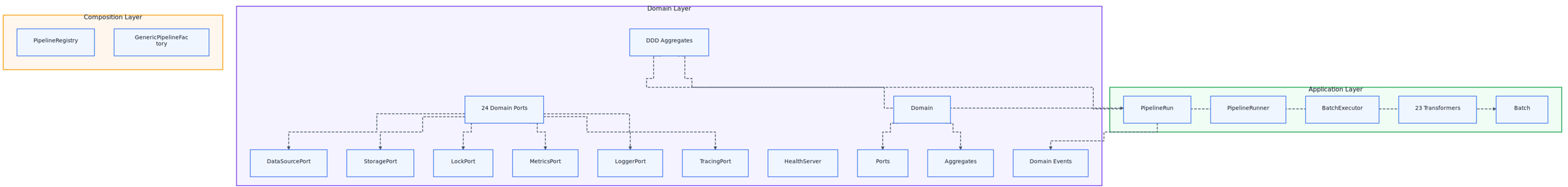
Composition Layer

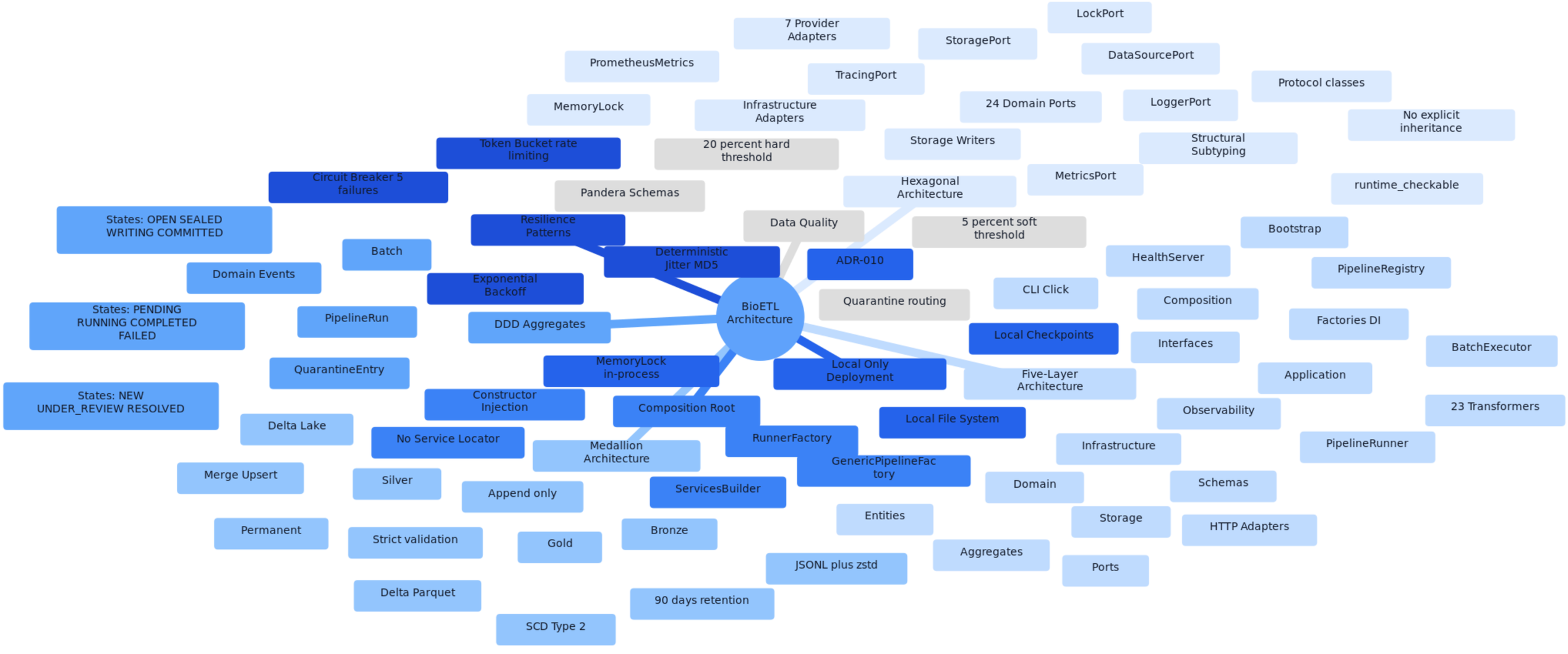
PipelineRegistry

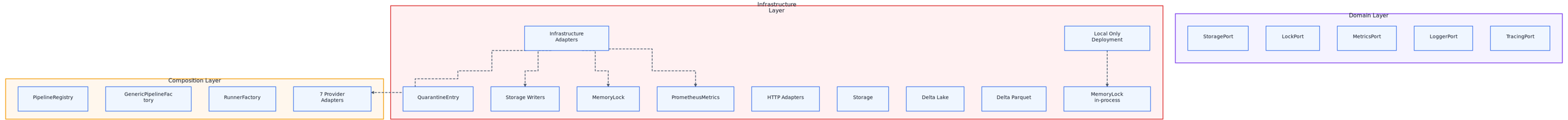
PipelineRunner

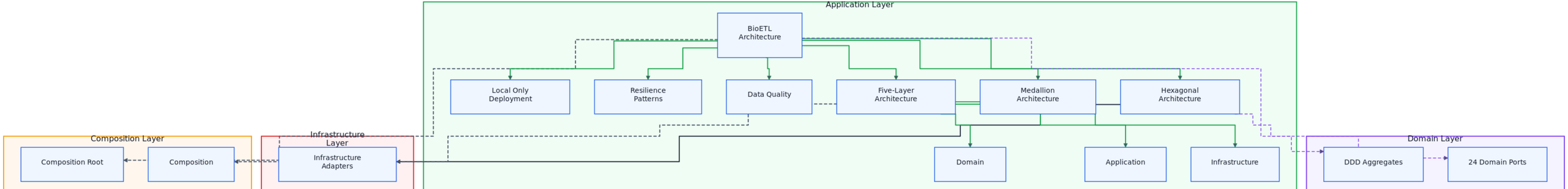
BatchExecutor

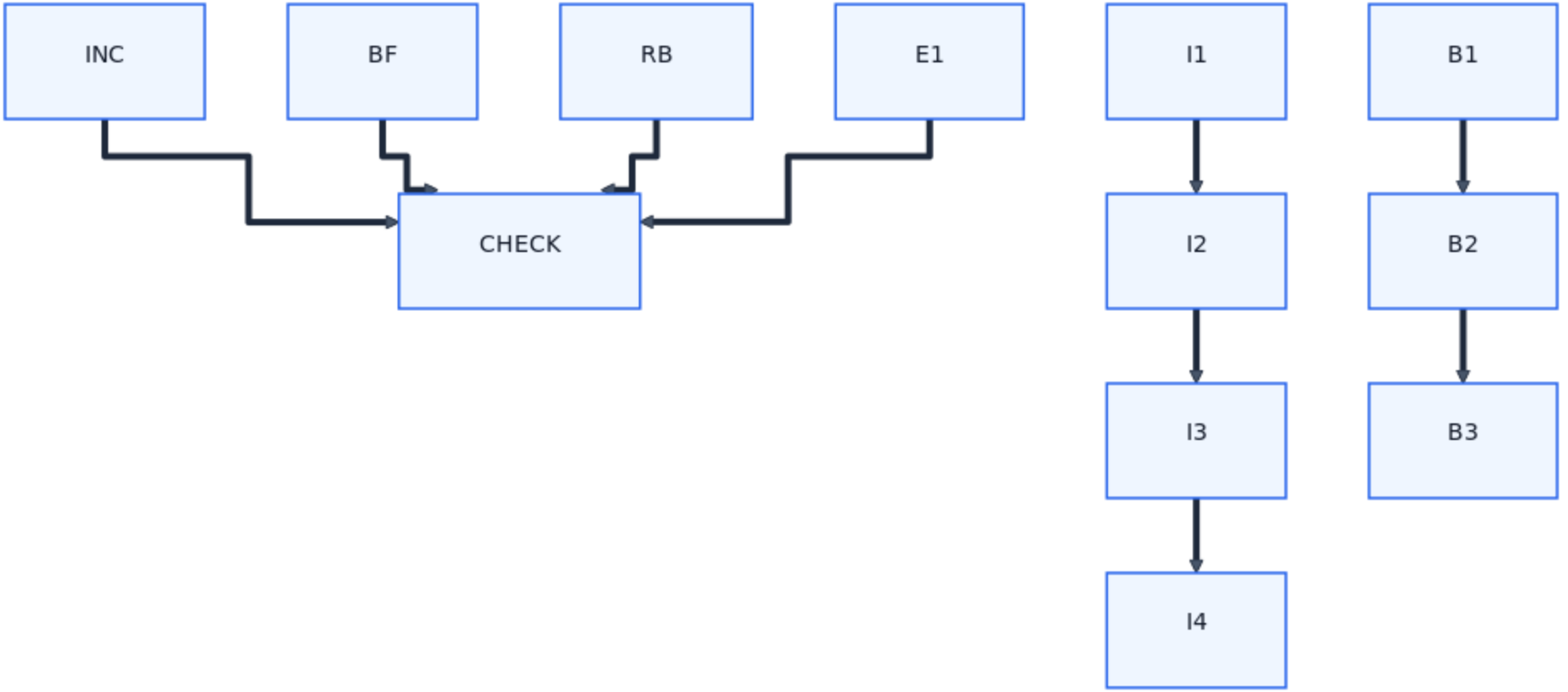
23 Transformers

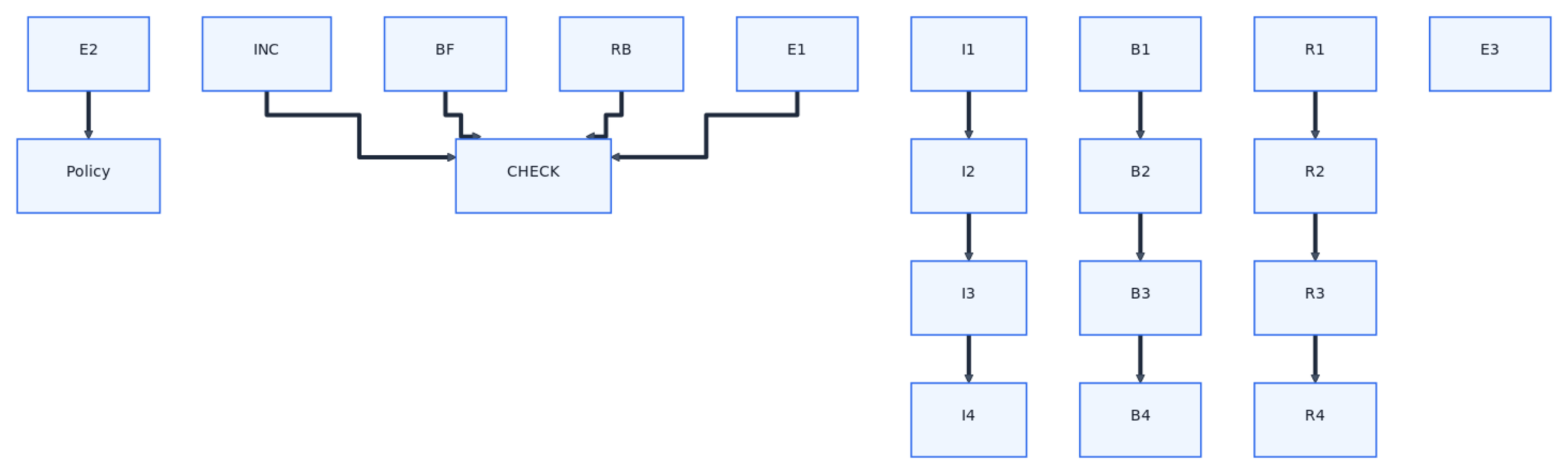

Composition Layer
PipelineRegistry
GenericPipelineFac tory
Domain Layer
DDD Aggregates
24 Domain Ports
Domain
DataSourcePort
StoragePort
LockPort
MetricsPort
LoggerPort
TracingPort
HealthServer
Ports
Aggregates
Domain Events
Application Layer
PipelineRun
PipelineRunner
BatchExecutor
23 Transformers
Batch

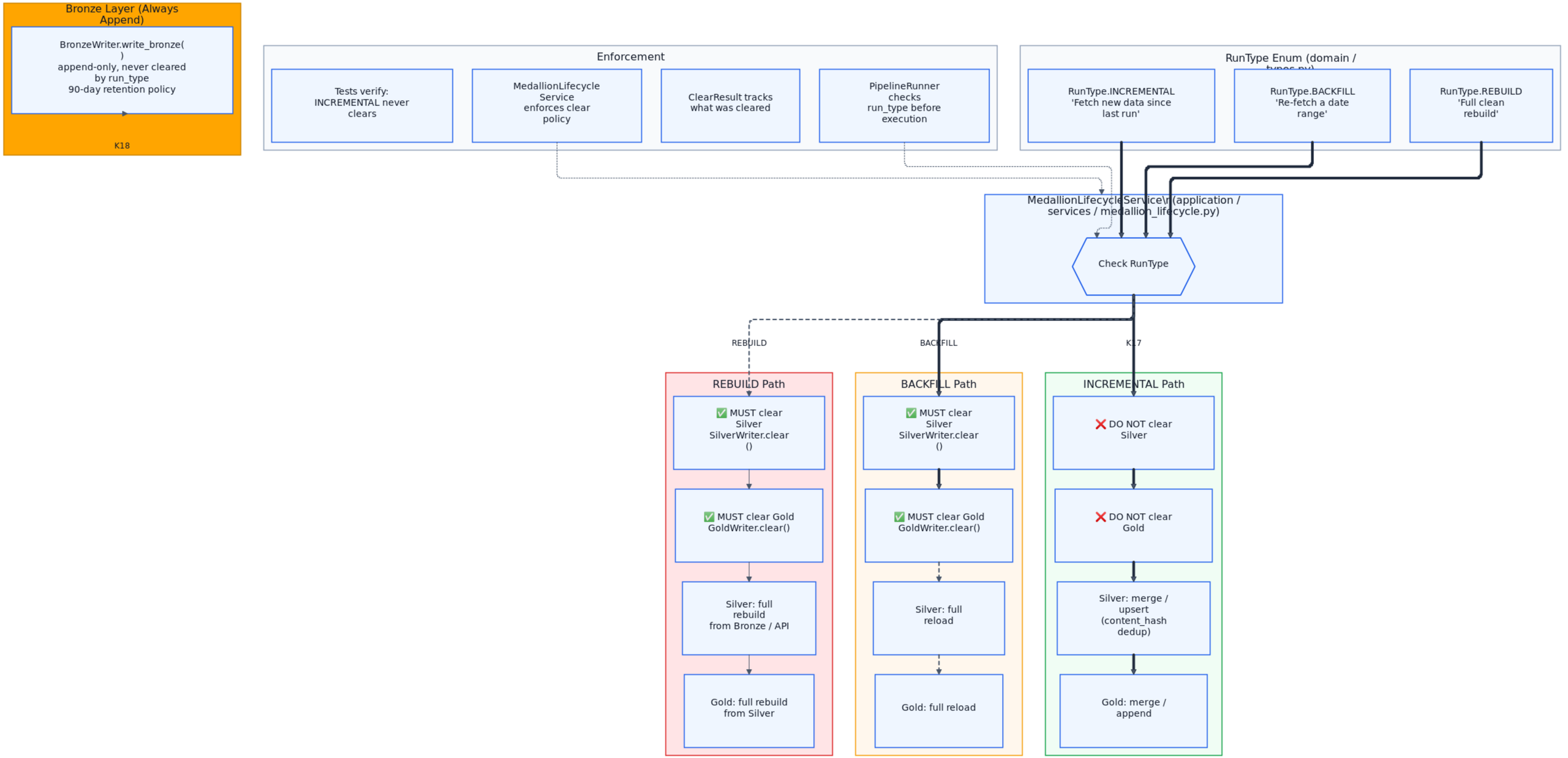

BioETL Architecture
Hexagonal Architecture
Five-Layer Architecture
Medallion Architecture
Deterministic Jitter MD5
Resilience Patterns
Token Bucket rate limiting
Circuit Breaker 5 failures
Exponential Backoff
Constructor Injection
Composition Root
ServicesBuilder
RunnerFactory
GenericPipelineFactory
ADR-010
Local Only Deployment
Local Checkpoints
MemoryLock in-process
DDD Aggregates
Quarantine routing
Data Quality
20 percent hard threshold
5 percent soft threshold
Pandera Schemas
7 Provider Adapters
Infrastructure Adapters
24 Domain Ports
Storage Writers
StoragePort
LockPort
DataSourcePort
TracingPort
PrometheusMetrics
MemoryLock
LoggerPort
Protocol classes
No explicit inheritance
Structural Subtyping
MetricsPort
runtime_checkable
Bootstrap
HealthServer
PipelineRegistry
CLI Click
Composition
Factories DI
Interfaces
BatchExecutor
Application
Observability
23 Transformers
PipelineRunner
Infrastructure
Schemas
Domain
HTTP Adapters
Storage
Ports
Aggregates
Entities
Local File System
Batch
Domain Events
PipelineRun
QuarantineEntry
Delta Lake
No Service Locator
Silver
Append only
Bronze
Gold
Strict validation
Permanent
Merge Upsert
Delta Parquet
SCD Type 2
90 days retention
JSONL plus zstd
States: OPEN SEALED WRITING COMMITTED
States: PENDING RUNNING COMPLETED FAILED
States: NEW UNDER_REVIEW RESOLVED

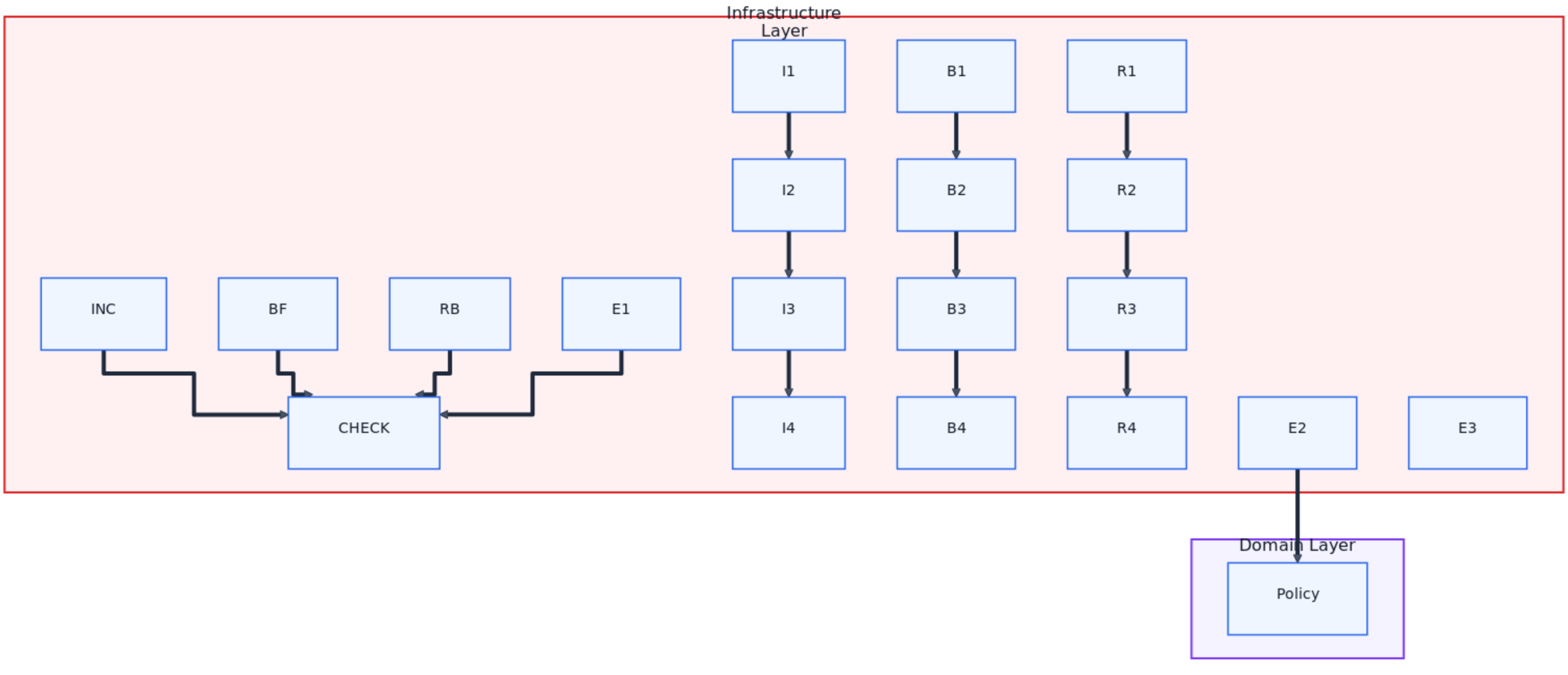

Infrastructure Layer
Infrastructure Adapters
Local Only Deployment
Domain Layer
StoragePort
LockPort
MetricsPort
LoggerPort
TracingPort
Composition Layer
PipelineRegistry
GenericPipelineFactory
RunnerFactory
7 Provider Adapters
QuarantineEntry
Storage Writers
MemoryLock
PrometheusMetrics
HTTP Adapters
Storage
Delta Lake
Delta Parquet
MemoryLock in-process

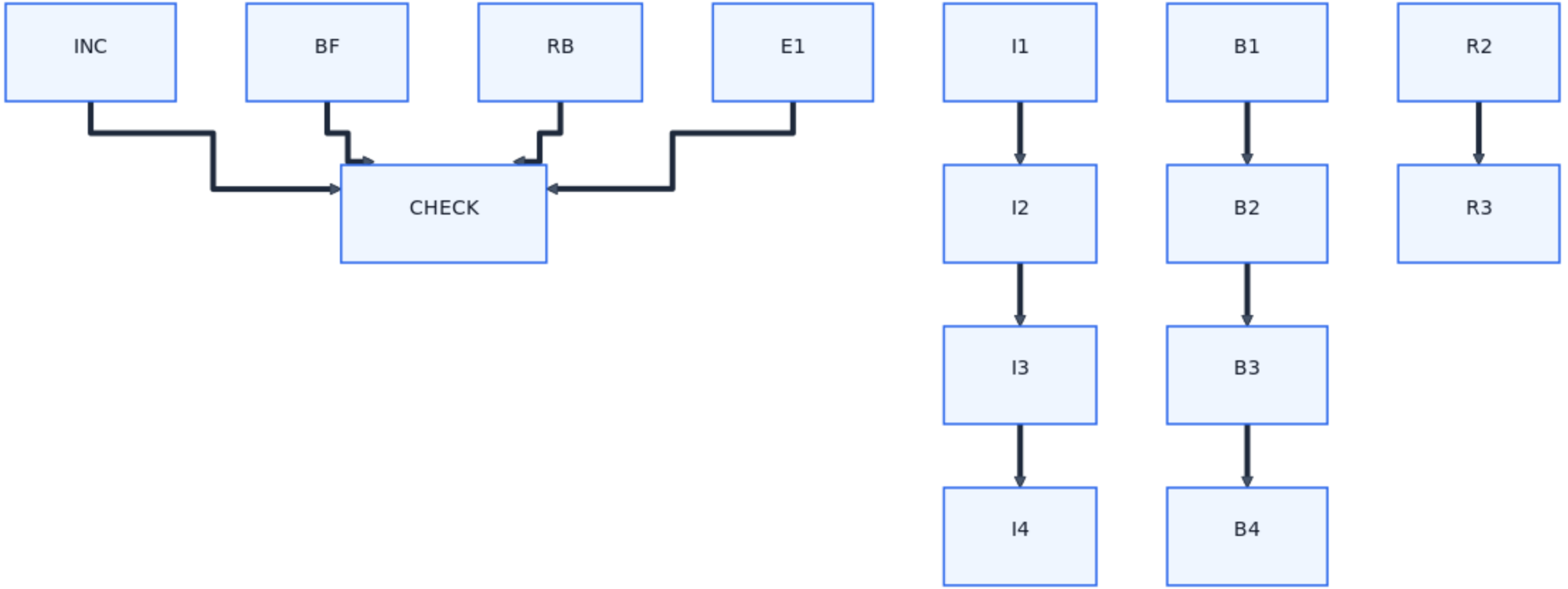

Application Layer
BioETL Architecture
Local Only Deployment
Resilience Patterns
Data Quality
Five-Layer Architecture
Medallion Architecture
Hexagonal Architecture
Composition Layer
Composition Root
Composition
Infrastructure Layer
Infrastructure Adapters
Domain
Application
Infrastructure
Domain Layer
DDD Aggregates
24 Domain Ports

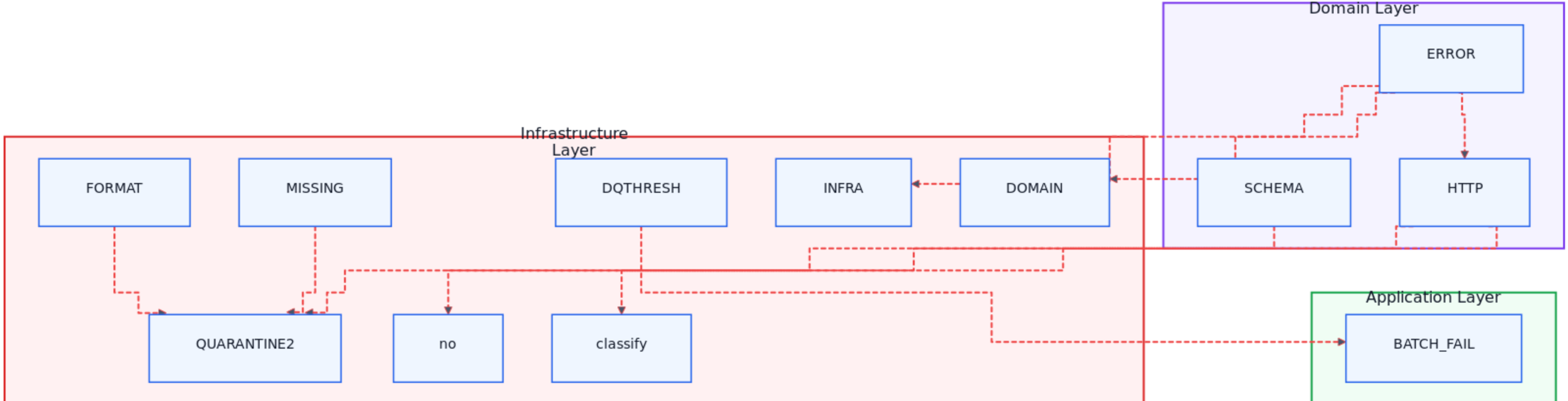

INC
BF
RB
E1
CHECK
I1
I2
I3
I4
B1
B2
B3

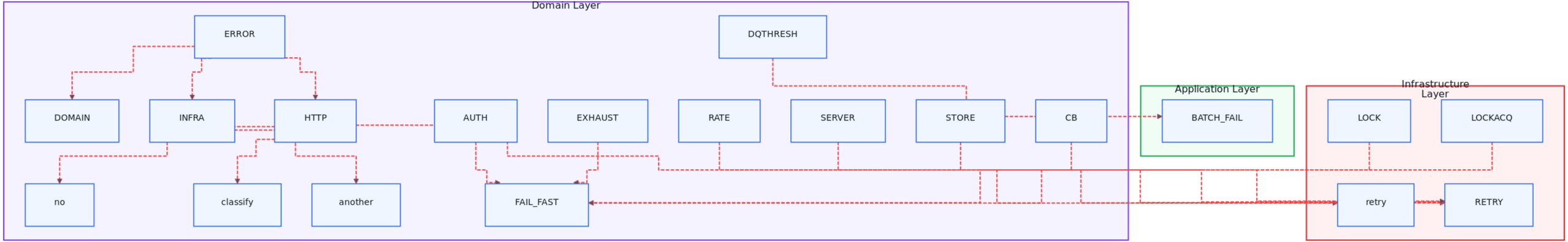

E2
INC
BF
RB
E1
I1
B1
R1
E3
Policy
CHECK
I2
B2
R2
I3
B3
R3
I4
B4
R4

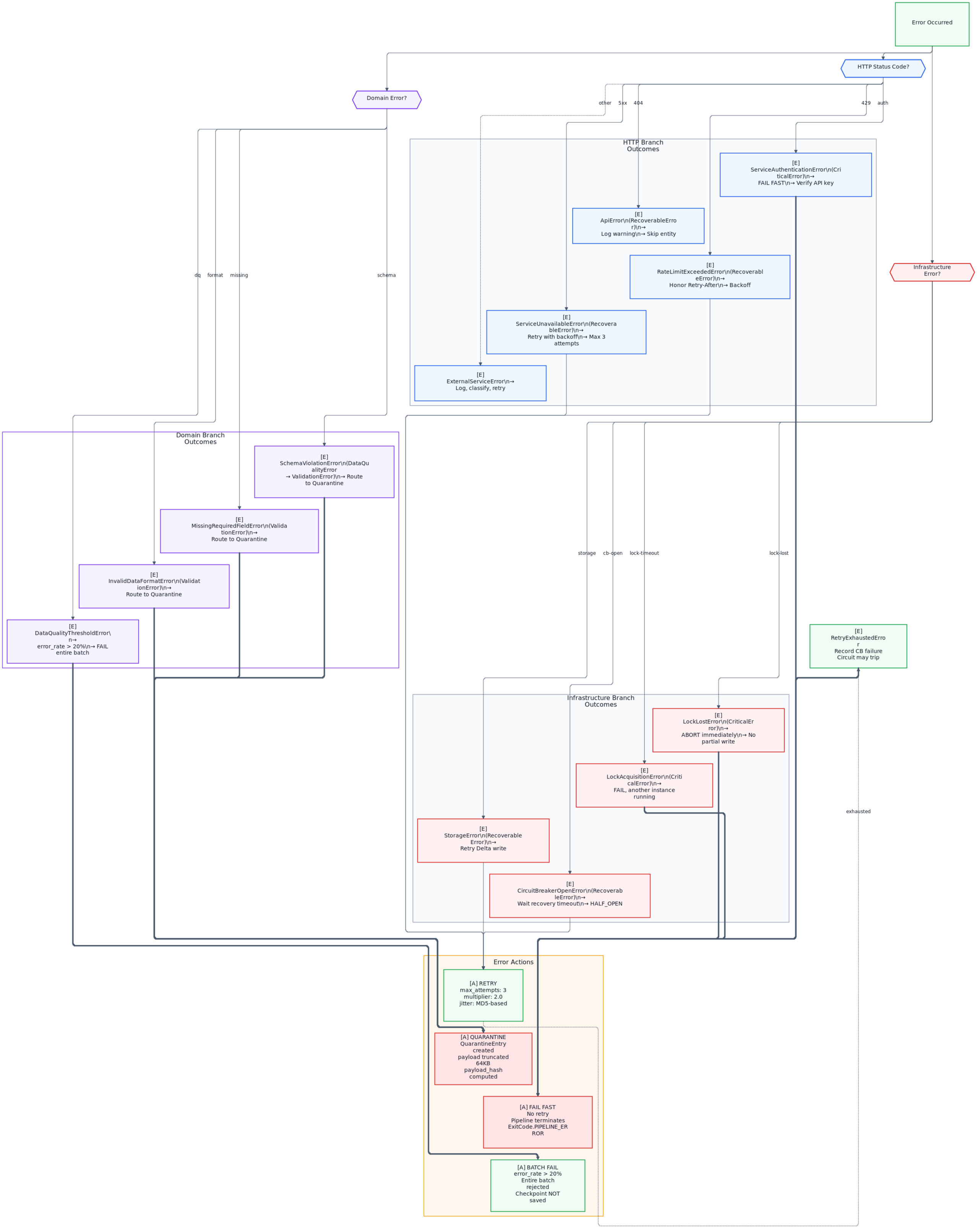

Bronze Layer (Always Append)
BronzeWriter.write_bronze( )
append-only, never cleared by run_type
90-day retention policy
K18
Enforcement
Tests verify: INCREMENTAL never clears
MedallionLifecycle Service enforces clear policy
ClearResult tracks what was cleared
PipelineRunner checks run_type before execution
RunType Enum (domain / types.py)
RunType.INCREMENTAL 'Fetch new data since last run'
RunType.BACKFILL 'Re-fetch a date range'
RunType.REBUILD 'Full clean rebuild'
MedallionLifecycleService\n(application / services / medallion_lifecycle.py)
Check RunType
REBUILD
BACKFILL
K17
REBUILD Path
✅ MUST clear Silver SilverWriter.clear ()
✅ MUST clear Gold GoldWriter.clear()
Silver: full rebuild from Bronze / API
Gold: full rebuild from Silver
BACKFILL Path
✅ MUST clear Silver SilverWriter.clear ()
✅ MUST clear Gold GoldWriter.clear()
Silver: full reload
Gold: full reload
INCREMENTAL Path
❌ DO NOT clear Silver
❌ DO NOT clear Gold
Silver: merge / upsert (content_hash dedup)
Gold: merge / append

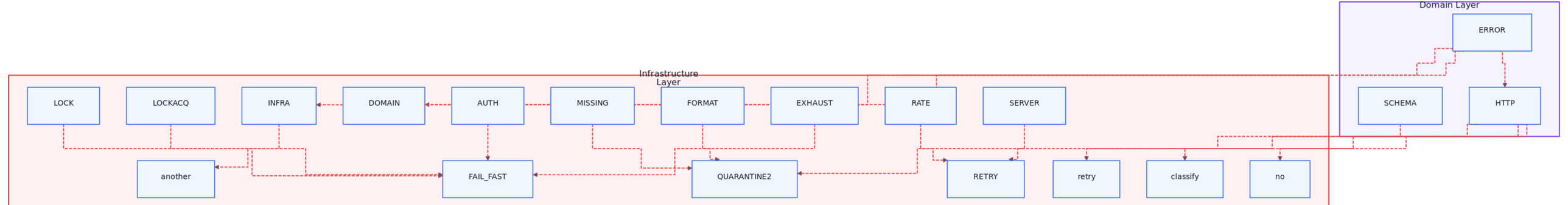

Infrastructure Layer
I1
B1
R1
I2
B2
R2
INC
BF
RB
E1
I3
B3
R3
CHECK
I4
B4
R4
E2
E3
Domain Layer
Policy

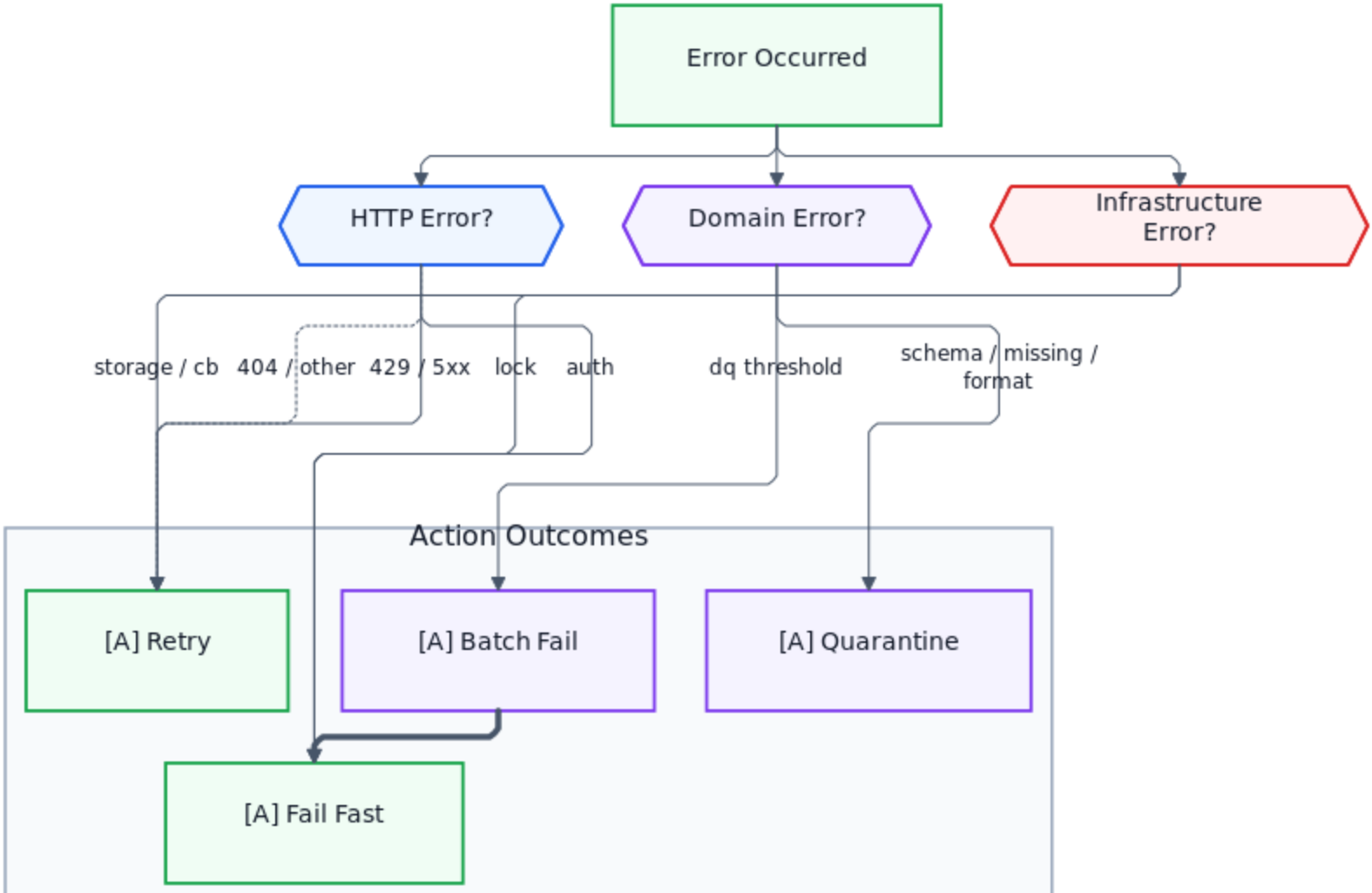

INC
BF
RB
E1
CHECK
I1
I2
I3
I4
B1
B2
B3
B4
R2
R3

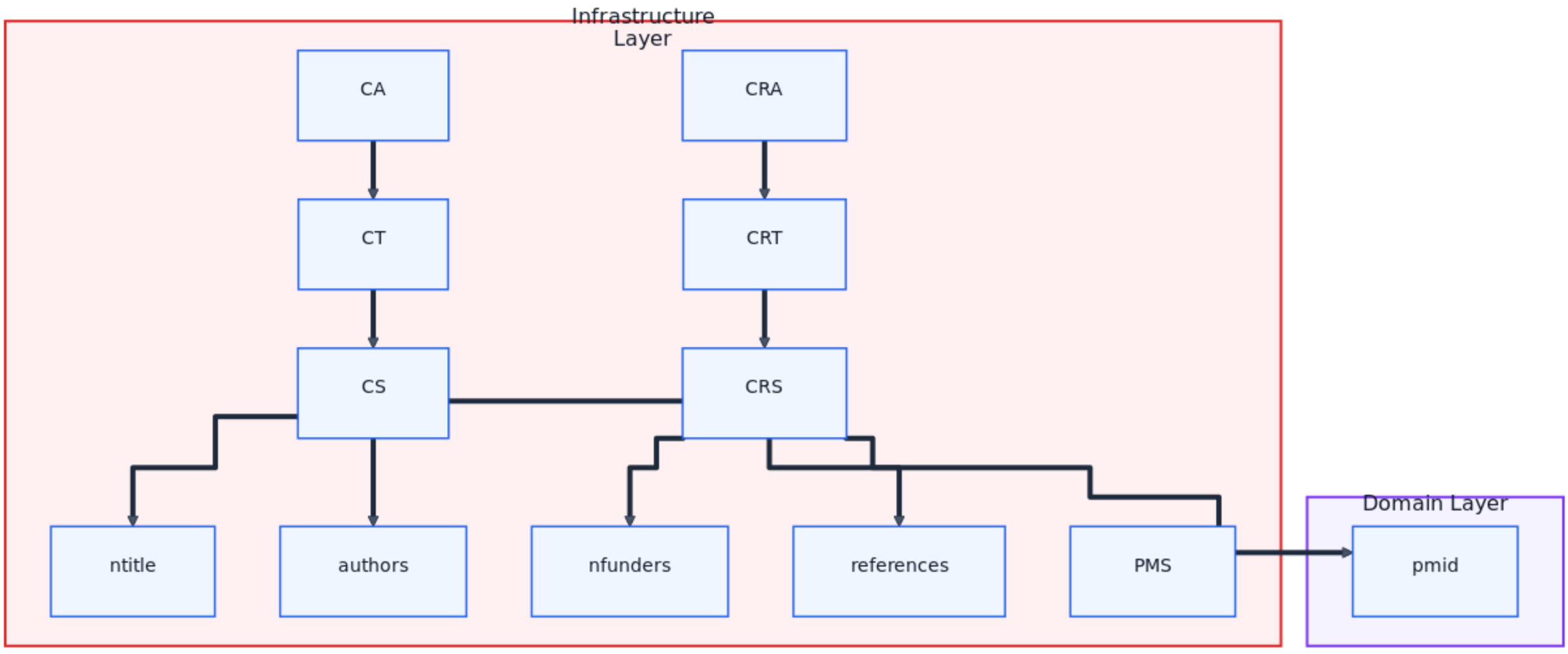

Domain Layer
ERROR
SCHEMA
HTTP
Infrastructure Layer
FORMAT
MISSING
DQTHRESH
INFRA
DOMAIN
QUARANTINE2
no
classify
Application Layer
BATCH_FAIL

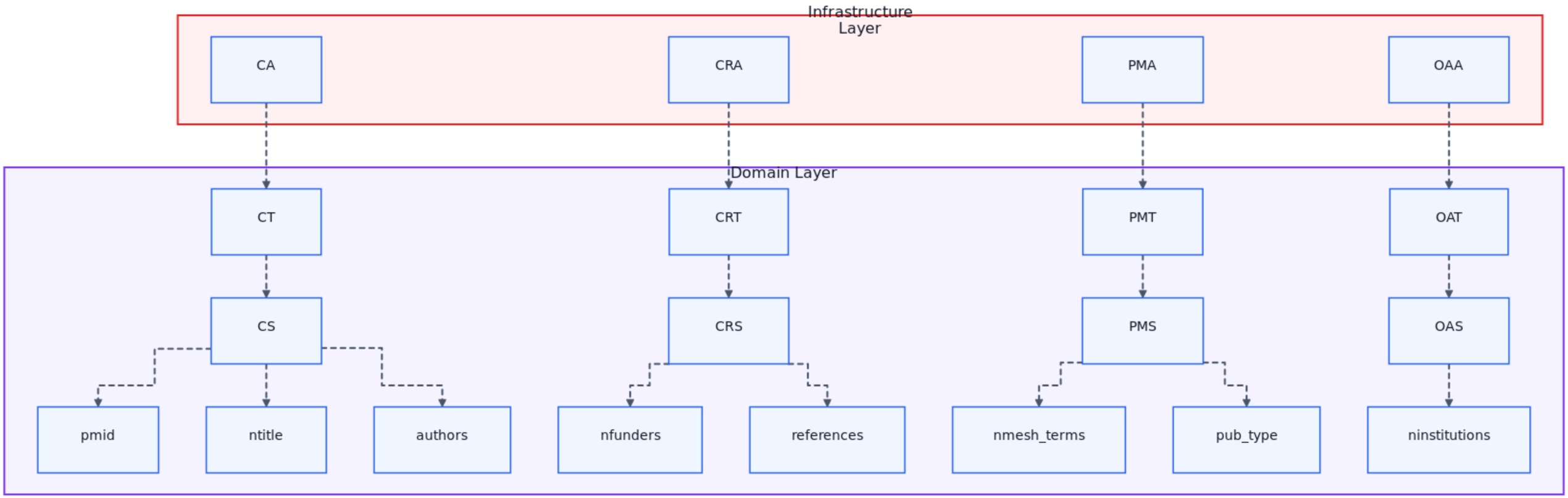

Domain Layer
ERROR
DQTHRESH
DOMAIN
INFRA
HTTP
AUTH
EXHAUST
RATE
SERVER
STORE
CB
Application Layer
BATCH_FAIL
Infrastructure Layer
LOCK
LOCKACQ
no
classify
another
FAIL_FAST
retry
RETRY

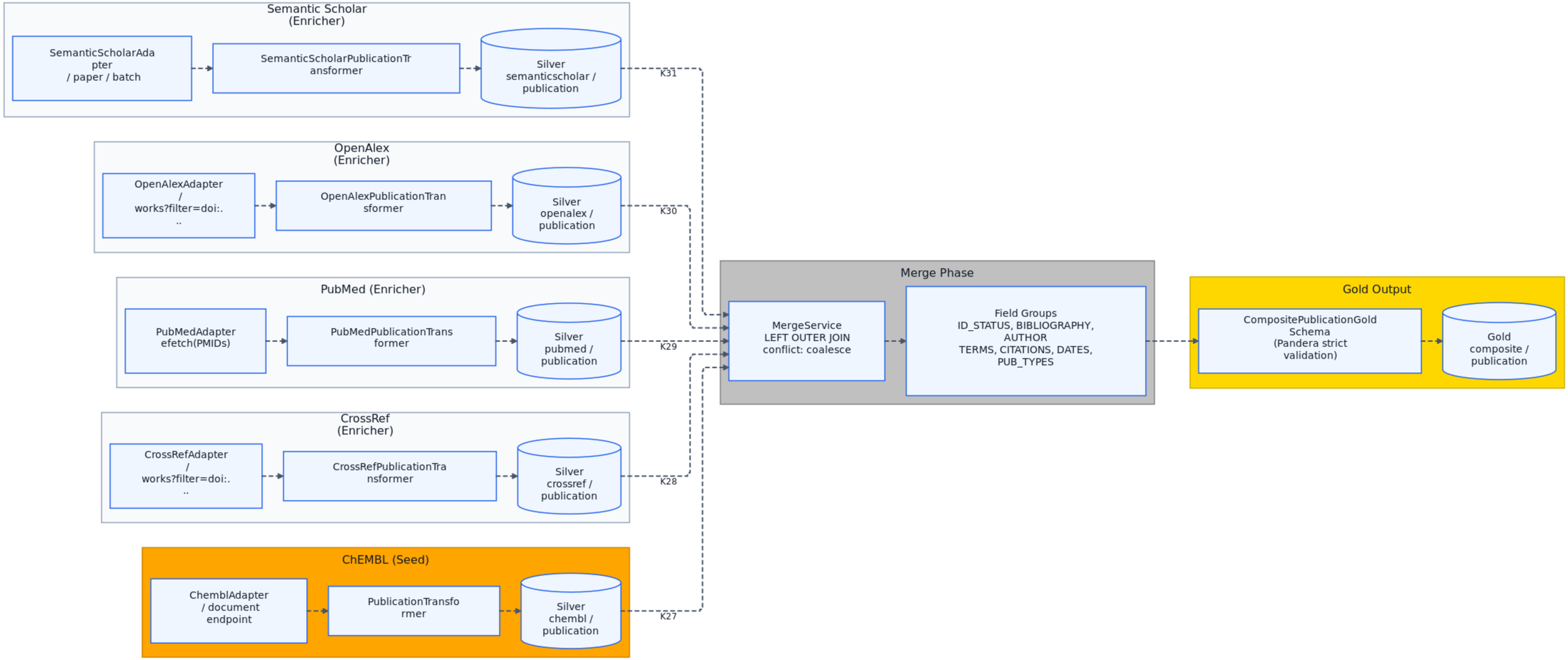

Error Occurred
HTTP Status Code?
Domain Error?
other
5xx
404
429
auth
HTTP Branch Outcomes
[E] ServiceAuthenticationError\n(CriticalError)\n→ FAIL FAST\n→ Verify API key
[E] ApiError\n(RecoverableError)\n→ Log warning\n→ Skip entity
[E] RateLimitExceededError\n(RecoverableError)\n→ Honor Retry-After\n→ Backoff
[E] ServiceUnavailableError\n(RecoverableError)\n→ Retry with backoff\n→ Max 3 attempts
[E] ExternalServiceError\n→ Log, classify, retry
Infrastructure Error?
dq
format
missing
schema
Domain Branch Outcomes
[E] SchemaViolationError\n(DataQualityError → ValidationError)\n→ Route to Quarantine
[E] MissingRequiredFieldError\n(ValidationError)\n→ Route to Quarantine
[E] InvalidDataFormatError\n(ValidationError)\n→ Route to Quarantine
[E] DataQualityThresholdError\n→ error_rate > 20%\n→ FAIL entire batch
storage
cb-open
lock-timeout
lock-lost
Infrastructure Branch Outcomes
[E] LockLostError\n(CriticalError)\n→ ABORT immediately\n→ No partial write
[E] LockAcquisitionError\n(CriticalError)\n→ FAIL, another instance running
[E] StorageError\n(RecoverableError)\n→ Retry Delta write
[E] CircuitBreakerOpenError\n(RecoverableError)\n→ Wait recovery timeout\n→ HALF_OPEN
[E] RetryExhaustedError\n→ Record CB failure Circuit may trip
exhausted
Error Actions
[A] RETRY max_attempts: 3 multiplier: 2.0 jitter: MD5-based
[A] QUARANTINE QuarantineEntry created payload truncated 64KB payload_hash computed
[A] FAIL FAST No retry Pipeline terminates ExitCode.PIPELINE_ERROR
[A] BATCH FAIL error_rate > 20% Entire batch rejected Checkpoint NOT saved

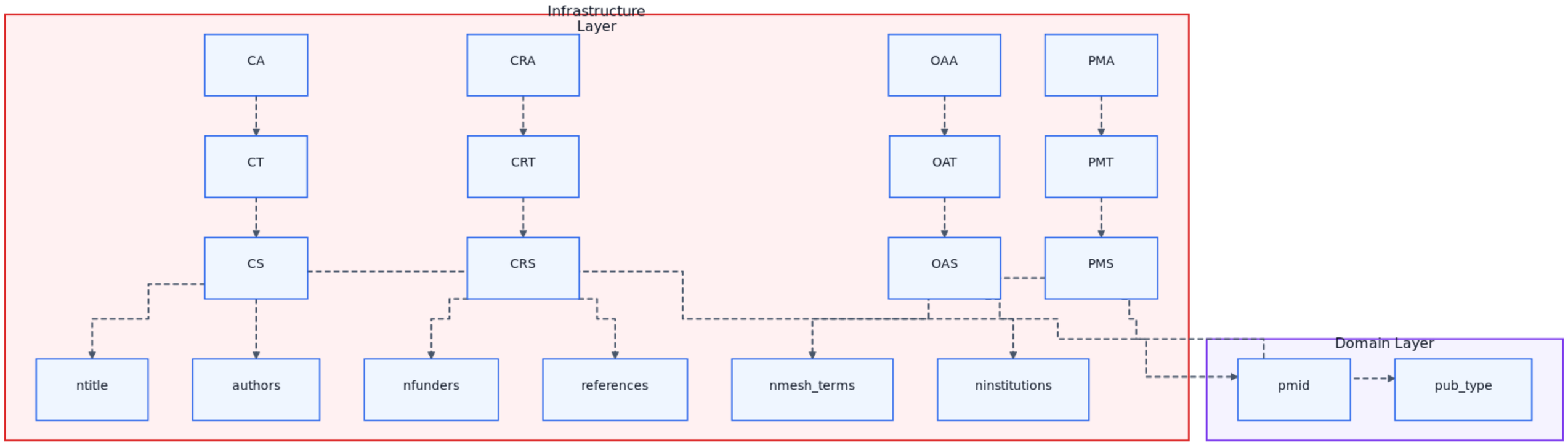

Domain Layer
ERROR
SCHEMA
HTTP
Infrastructure Layer
LOCK
LOCKACQ
INFRA
DOMAIN
AUTH
MISSING
FORMAT
EXHAUST
RATE
SERVER
another
FAIL_FAST
QUARANTINE2
RETRY
retry
classify
no

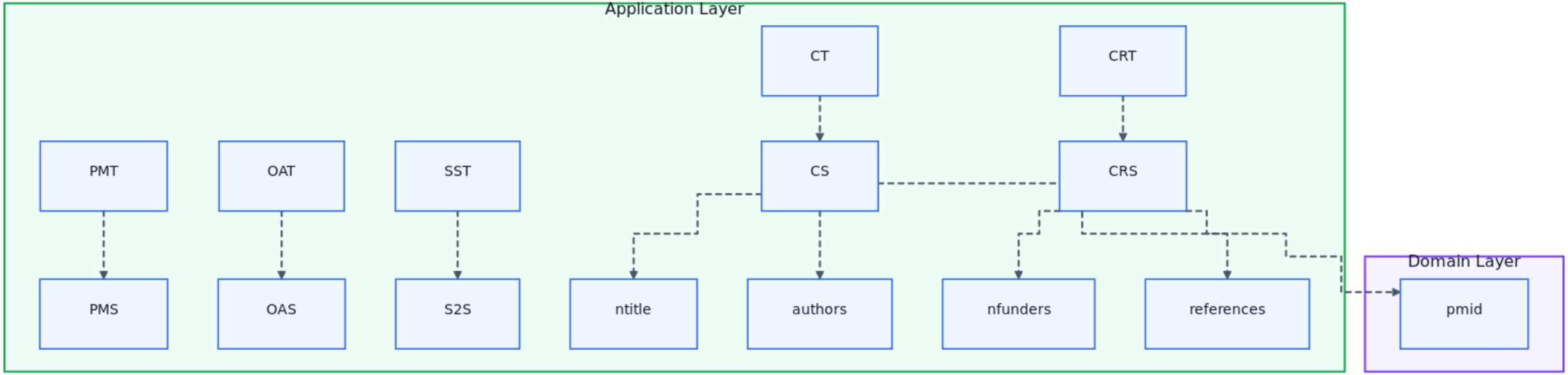

Error Occurred
HTTP Error?
Domain Error?
Infrastructure Error?
storage / cb
404 / other
429 / 5xx
lock
auth
dq threshold
schema / missing / format
Action Outcomes
[A] Retry
[A] Batch Fail
[A] Quarantine
[A] Fail Fast

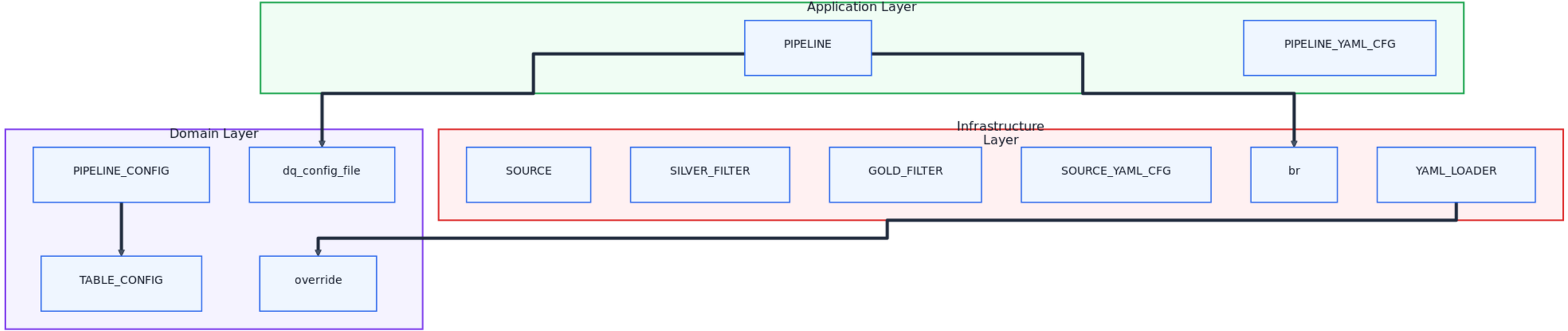

Infrastructure Layer
CA
CRA
CT
CRT
CS
CRS
ntitle
authors
nfunders
references
PMS
Domain Layer
pmid

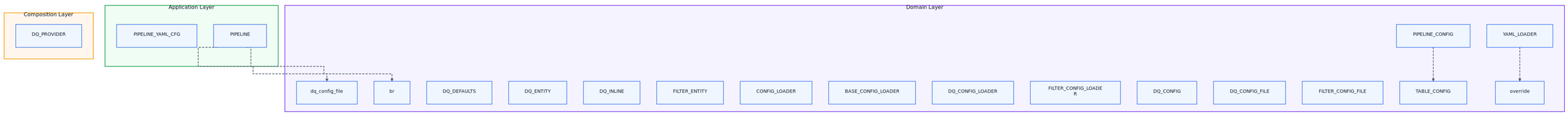

Infrastructure Layer
CA
CRA
PMA
OAA
Domain Layer
CT
CRT
PMT
OAT
CS
CRS
PMS
OAS
pmid
ntitle
authors
nfunders
references
nmesh_terms
pub_type
ninstitutions

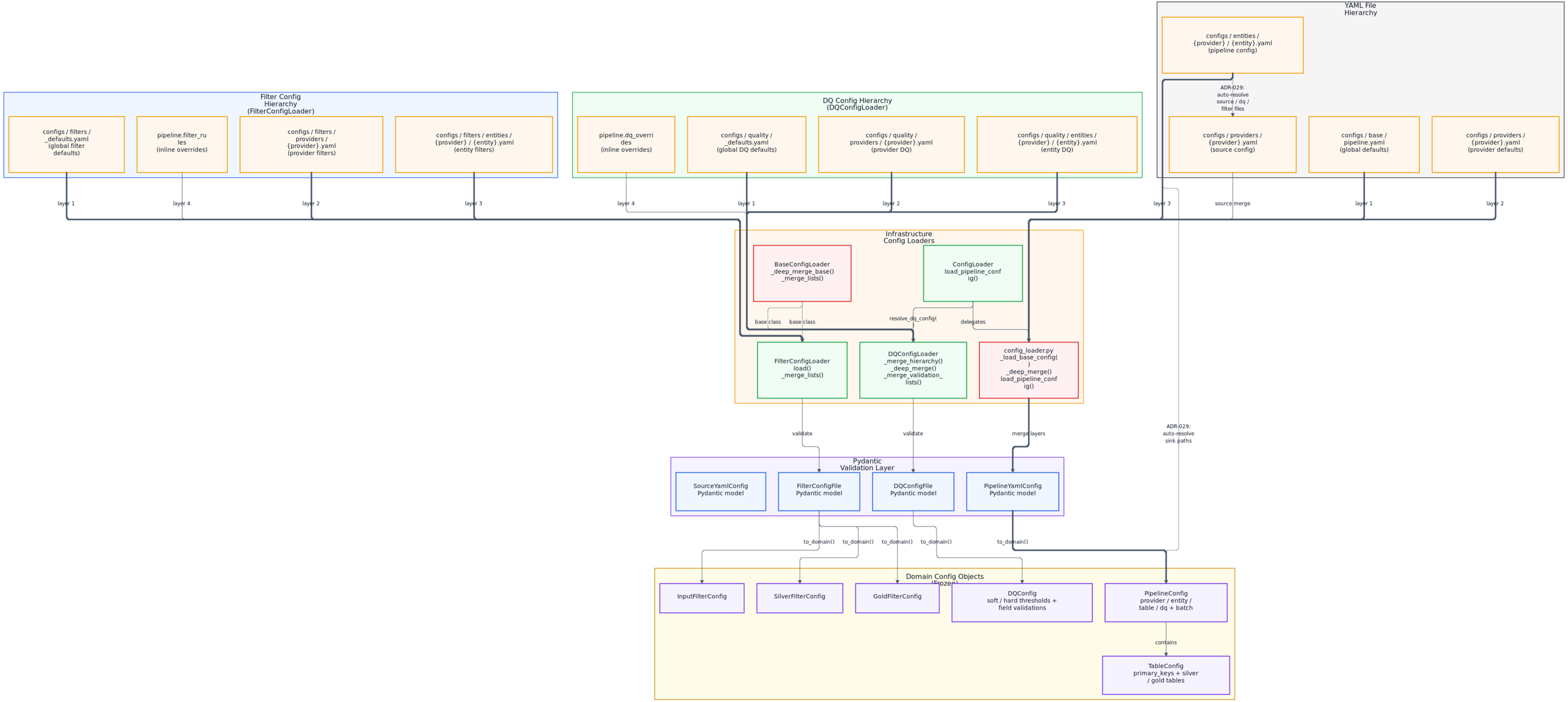

Semantic Scholar (Enricher)
SemanticScholarAdapter / paper / batch
SemanticScholarPublicationTransformer
Silver semanticscholar / publication
K31
OpenAlex (Enricher)
OpenAlexAdapter / works?filter=doi:...
OpenAlexPublicationTransformer
Silver openalex / publication
K30
PubMed (Enricher)
PubMedAdapter efetch(PMIDs)
PubMedPublicationTransformer
Silver pubmed / publication
K29
CrossRef (Enricher)
CrossRefAdapter / works?filter=doi:...
CrossRefPublicationTransformer
Silver crossref / publication
K28
ChEMBL (Seed)
ChemblAdapter / document endpoint
PublicationTransformer
Silver chembl / publication
K27
Merge Phase
MergeService LEFT OUTER JOIN conflict: coalesce
Field Groups ID_STATUS, BIBLIOGRAPHY, AUTHOR TERMS, CITATIONS, DATES, PUB_TYPES
Gold Output
CompositePublicationGold Schema (Pandera strict validation)
Gold composite / publication

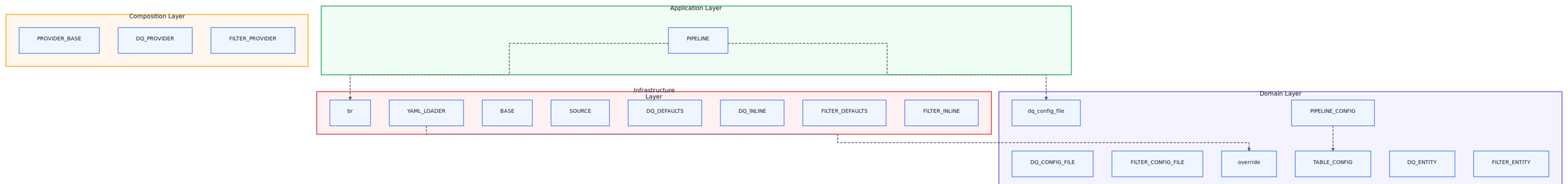

Infrastructure Layer
CA
CRA
OAA
PMA
CT
CRT
OAT
PMT
CS
CRS
OAS
PMS
ntitle
authors
nfunders
references
nmesh_terms
ninstitutions
Domain Layer
pmid
pub_type

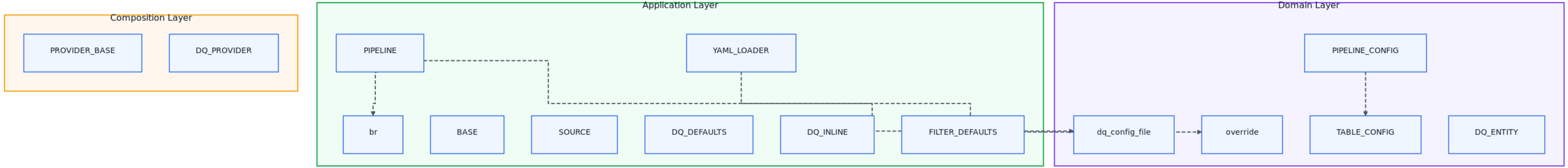

Application Layer
CT
CRT
PMT
OAT
SST
CS
CRS
PMS
OAS
S2S
ntitle
authors
nfunders
references
Domain Layer
pmid

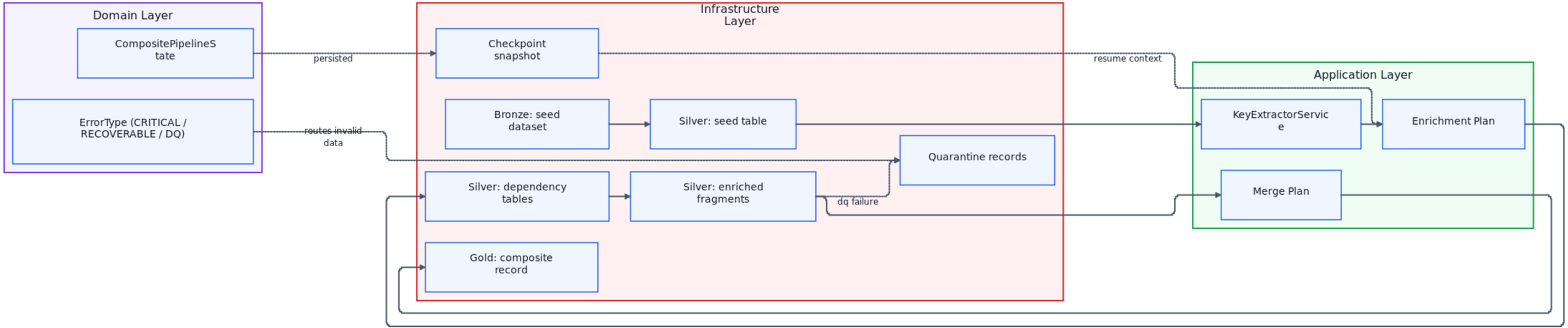

Application Layer
PIPELINE
PIPELINE_YAML_CFG
Domain Layer
PIPELINE_CONFIG
dq_config_file
TABLE_CONFIG
override
Infrastructure Layer
SOURCE
SILVER_FILTER
GOLD_FILTER
SOURCE_YAML_CFG
br
YAML_LOADER

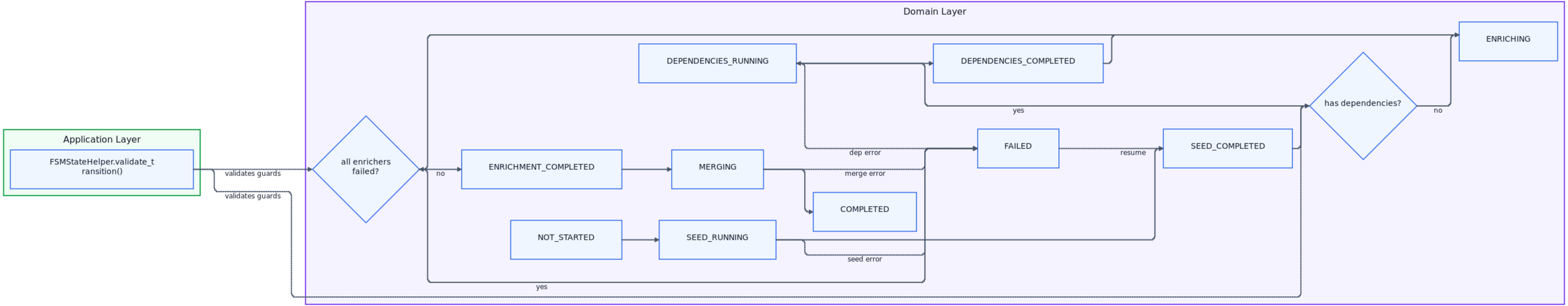

Composition Layer
DQ_PROVIDER
Application Layer
PIPELINE_YAML_CFG
PIPELINE
Domain Layer
PIPELINE_CONFIG
YAML_LOADER
dq_config_file
br
DQ_DEFAULTS
DQ_ENTITY
DQ_INLINE
FILTER_ENTITY
CONFIG_LOADER
BASE_CONFIG_LOADER
DQ_CONFIG_LOADER
FILTER_CONFIG_LOADER
DQ_CONFIG
DQ_CONFIG_FILE
FILTER_CONFIG_FILE
TABLE_CONFIG
override

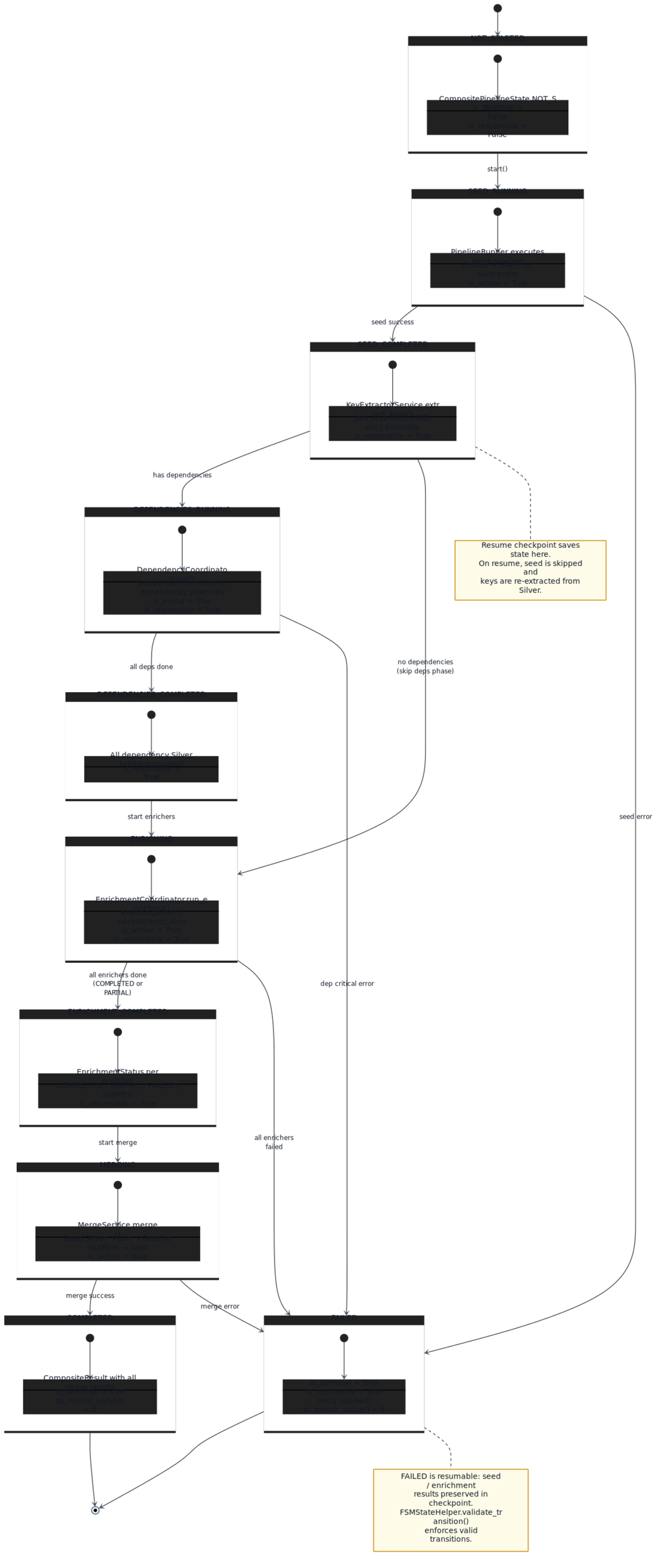

YAML File Hierarchy
configs / entities / {provider} / {entity}.yaml (pipeline config)
ADR-029: auto-resolve source / dq / filter files
configs / providers / {provider}.yaml (source config)
configs / base / pipeline.yaml (global defaults)
configs / providers / {provider}.yaml (provider defaults)
Filter Config Hierarchy (FilterConfigLoader)
configs / filters / _defaults.yaml (global filter defaults)
pipeline.filter_rules (inline overrides)
configs / filters / providers / {provider}.yaml (provider filters)
configs / filters / entities / {provider} / {entity}.yaml (entity filters)
DQ Config Hierarchy (DQConfigLoader)
pipeline.dq_overrides (inline overrides)
configs / quality / _defaults.yaml (global DQ defaults)
configs / quality / providers / {provider}.yaml (provider DQ)
configs / quality / entities / {provider} / {entity}.yaml (entity DQ)
layer 1
layer 4
layer 2
layer 3
layer 4
layer 1
layer 2
layer 3
layer 3
source merge
layer 1
layer 2
Infrastructure Config Loaders
BaseConfigLoader _deep_merge_base() _merge_lists()
ConfigLoader load_pipeline_config()
base class
base class
resolve_dq_config(
delegates
FilterConfigLoader load() _merge_lists()
DQConfigLoader _merge_hierarchy() _deep_merge() _merge_validation_lists()
config_loader.py _load_base_config() _deep_merge() load_pipeline_config()
validate
validate
merge layers
ADR-029: auto-resolve sink paths
Pydantic Validation Layer
SourceYamlConfig Pydantic model
FilterConfigFile Pydantic model
DQConfigFile Pydantic model
PipelineYamlConfig Pydantic model
to_domain()
to_domain()
to_domain()
to_domain()
to_domain()
Domain Config Objects (Frozen)
InputFilterConfig
SilverFilterConfig
GoldFilterConfig
DQConfig soft / hard thresholds + field validations
PipelineConfig provider / entity / table / dq + batch
contains
TableConfig primary_keys + silver / gold tables

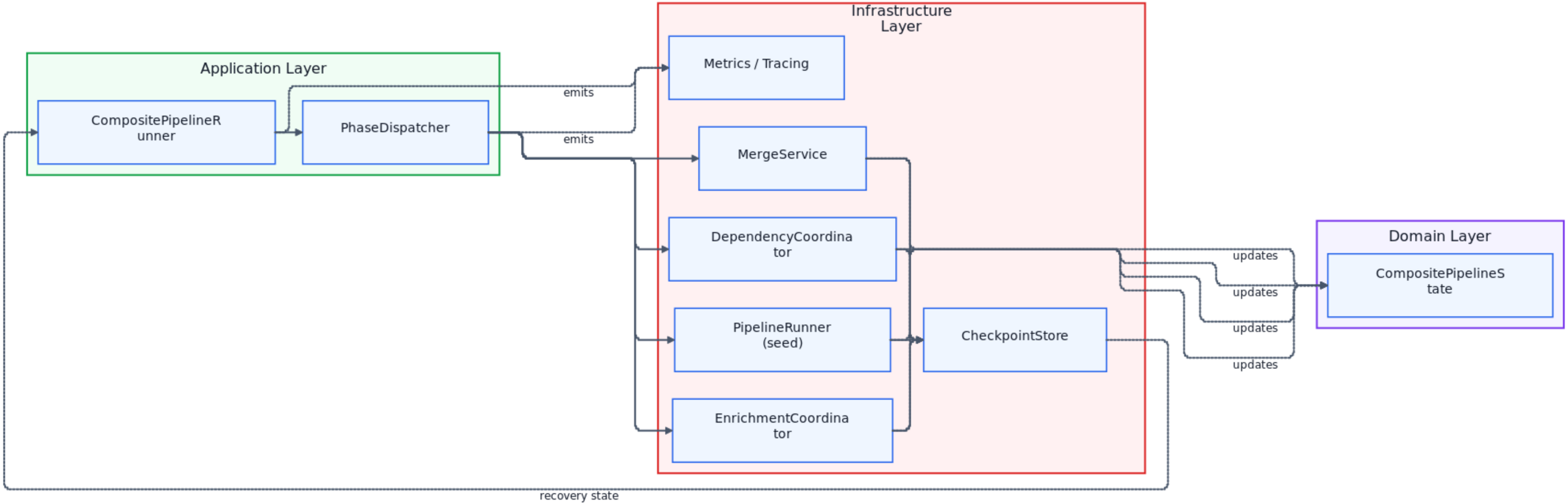

Composition Layer
PROVIDER_BASE
DQ_PROVIDER
FILTER_PROVIDER
Application Layer
PIPELINE
Infrastructure Layer
br
YAML_LOADER
BASE
SOURCE
DQ_DEFAULTS
DQ_INLINE
FILTER_DEFAULTS
FILTER_INLINE
Domain Layer
dq_config_file
PIPELINE_CONFIG
DQ_CONFIG_FILE
FILTER_CONFIG_FILE
override
TABLE_CONFIG
DQ_ENTITY
FILTER_ENTITY

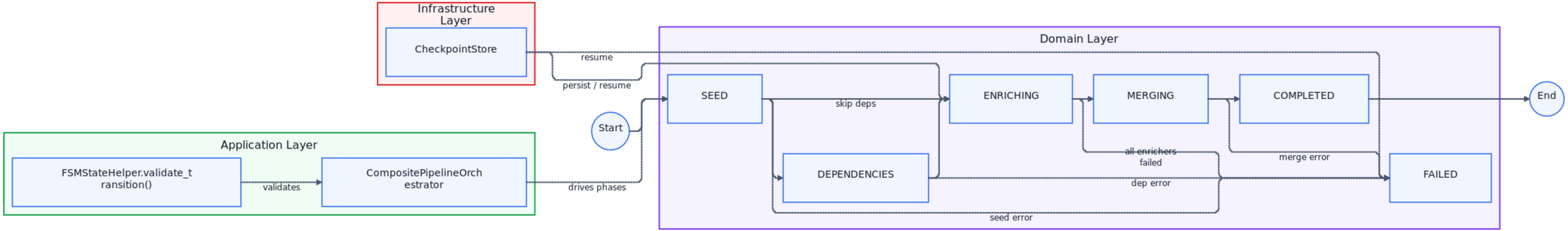

Composition Layer
PROVIDER_BASE
DQ_PROVIDER
Application Layer
PIPELINE
YAML_LOADER
br
BASE
SOURCE
DQ_DEFAULTS
DQ_INLINE
FILTER_DEFAULTS
Domain Layer
PIPELINE_CONFIG
dq_config_file
override
TABLE_CONFIG
DQ_ENTITY

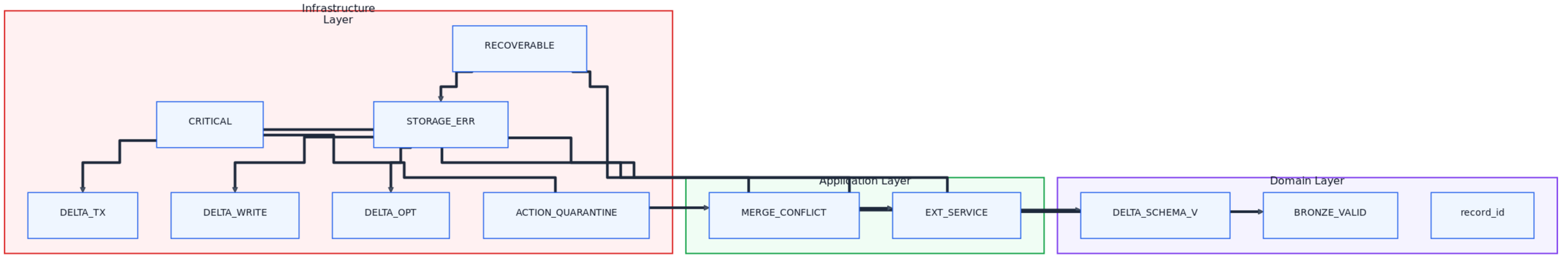

Domain Layer
CompositePipelineState
ErrorType (CRITICAL / RECOVERABLE / DQ)
persisted
routes invalid data
Infrastructure Layer
Checkpoint snapshot
Bronze: seed dataset
Silver: seed table
Quarantine records
Silver: dependency tables
Silver: enriched fragments
dq failure
Gold: composite record
resume context
Application Layer
KeyExtractorService
Enrichment Plan
Merge Plan

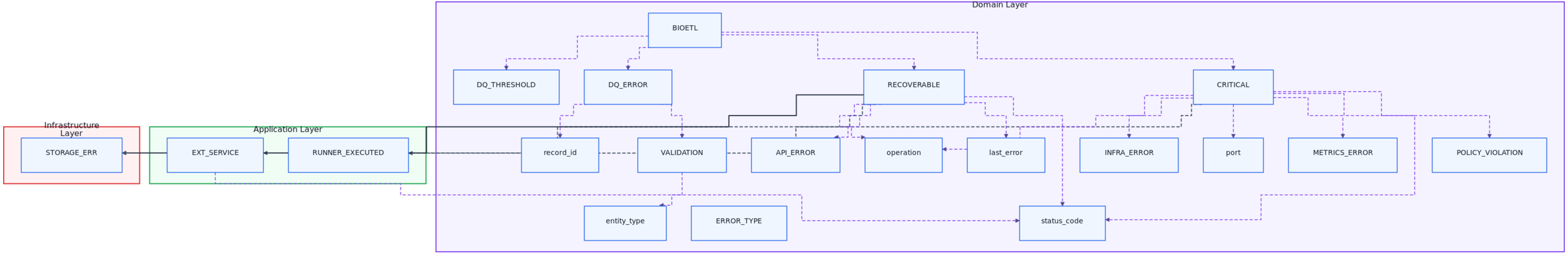

Application Layer
FSMStateHelper.validate_transition()
validates guards
validates guards
Domain Layer
all enrichers failed?
no
yes
ENRICHMENT_COMPLETED
MERGING
COMPLETED
NOT_STARTED
SEED_RUNNING
DEPENDENCIES_RUNNING
DEPENDENCIES_COMPLETED
yes
ENRICHING
has dependencies?
no
dep error
merge error
seed error
FAILED
resume
SEED_COMPLETED

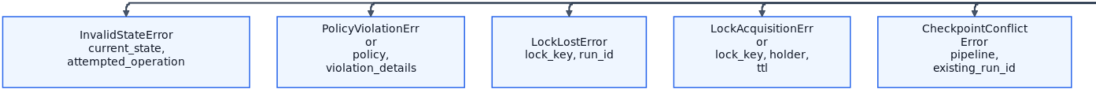

start()
seed success
has dependencies
no dependencies
(skip deps phase)
all deps done
start enrichers
all enrichers done
(COMPLETED or
PARTIAL)
start merge
merge success
merge error
seed error
dep critical error
all enrichers
failed
Resume checkpoint saves
state here.
On resume, seed is skipped
and
keys are re-extracted from
Silver.
FAILED is resumable: seed
/ enrichment
results preserved in
checkpoint.
FSMStateHelper.validate_tr
ansition()
enforces valid
transitions.

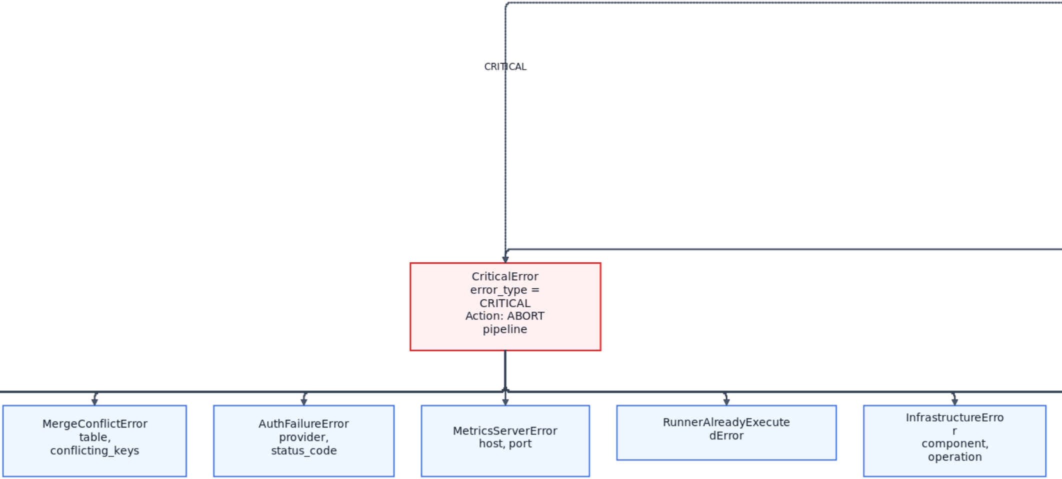

Application Layer
CompositePipelineRunner
PhaseDispatcher
Infrastructure Layer
Metrics / Tracing
MergeService
DependencyCoordinator
PipelineRunner (seed)
EnrichmentCoordinator
CheckpointStore
Domain Layer
CompositePipelineState
emits
emits
updates
updates
updates
updates
recovery state

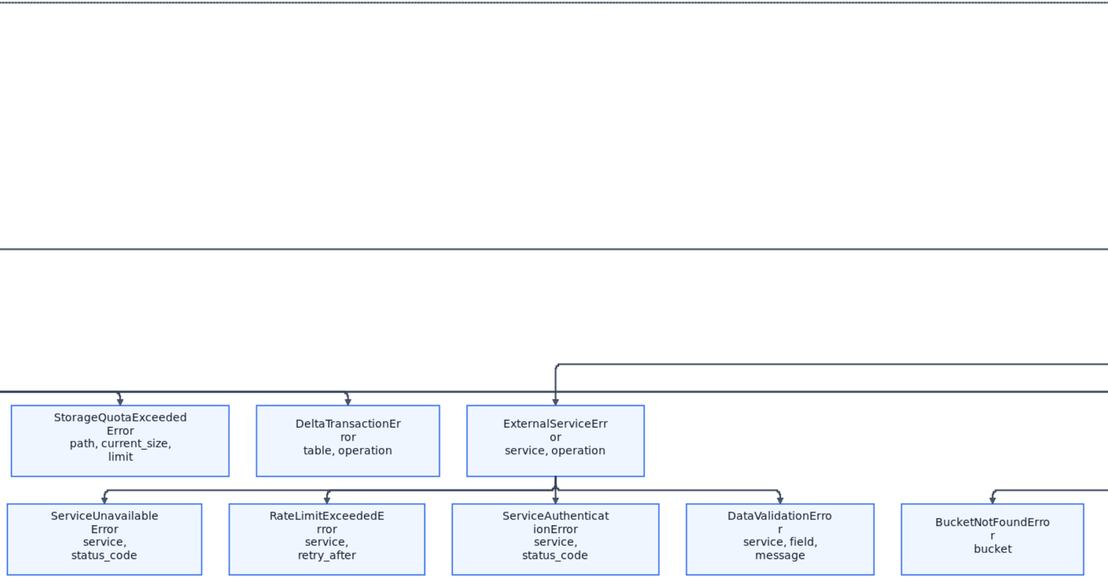

Infrastructure Layer
CheckpointStore
resume
persist / resume
Domain Layer
SEED
skip deps
ENRICHING
MERGING
COMPLETED
End
Start
DEPENDENCIES
all enrichers failed
merge error
dep error
FAILED
seed error
Application Layer
FSMStateHelper.validate_transition()
validates
CompositePipelineOrchestrator
drives phases

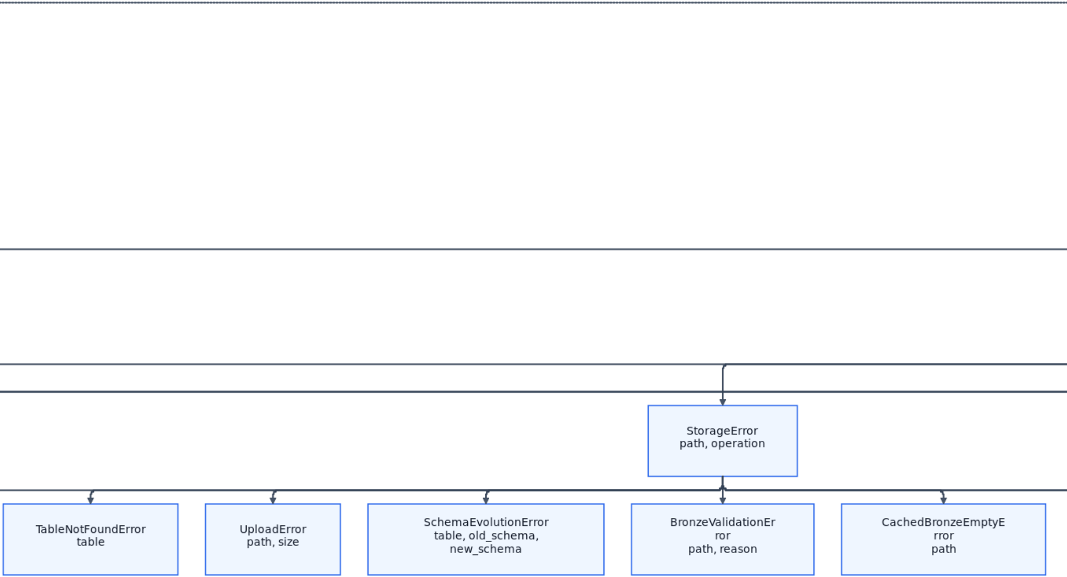

Infrastructure Layer
RECOVERABLE
CRITICAL
STORAGE_ERR
DELTA_TX
DELTA_WRITE
DELTA_OPT
ACTION_QUARANTINE
Application Layer
MERGE_CONFLICT
EXT_SERVICE
Domain Layer
DELTA_SCHEMA_V
BRONZE_VALID
record_id

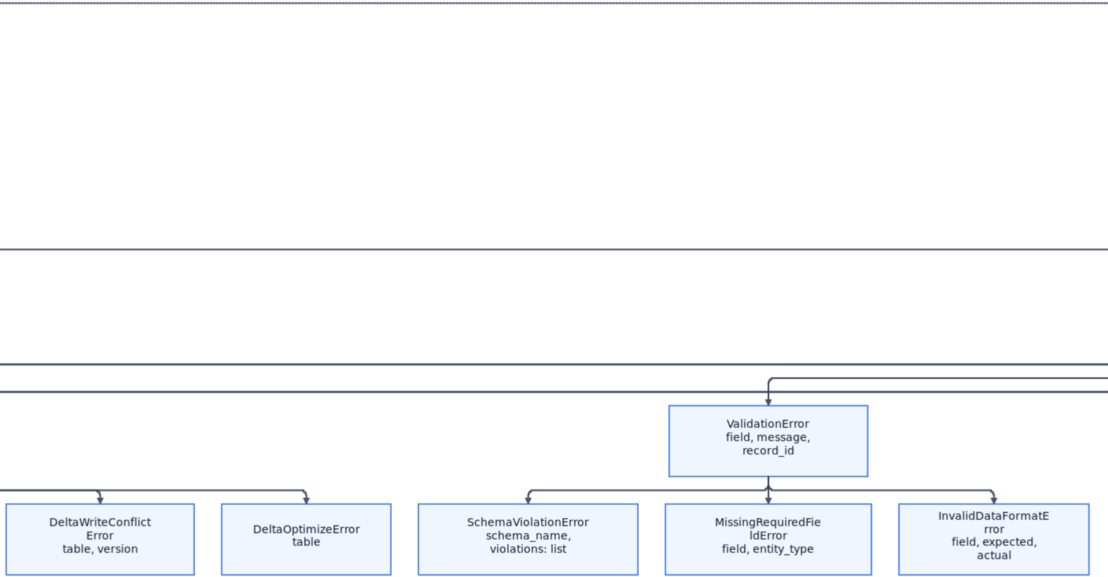

Domain Layer
BIOETL
DQ_THRESHOLD
DQ_ERROR
RECOVERABLE
CRITICAL
record_id
VALIDATION
API_ERROR
operation
last_error
INFRA_ERROR
port
METRICS_ERROR
POLICY_VIOLATION
entity_type
ERROR_TYPE
status_code
Application Layer
EXT_SERVICE
RUNNER_EXECUTED
Infrastructure Layer
STORAGE_ERR

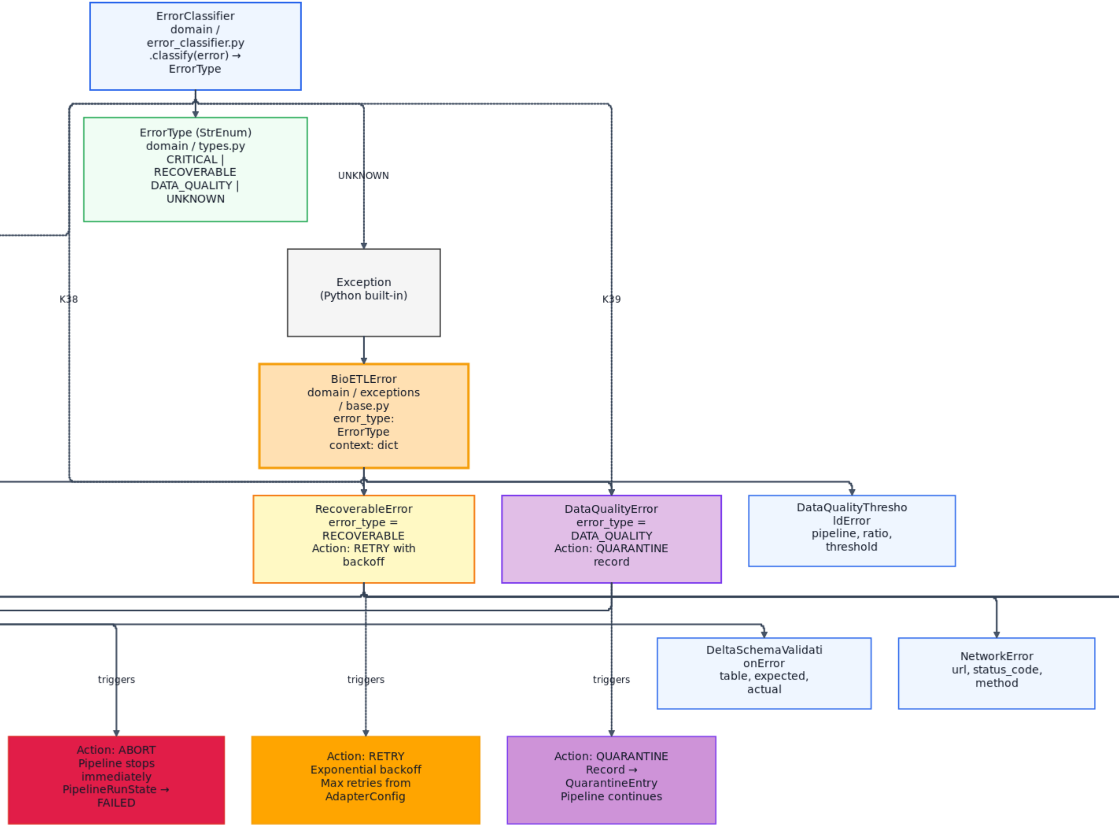

InvalidStateError
current_state,
attempted_operation
PolicyViolationError
policy,
violation_details
LockLostError
lock_key, run_id
LockAcquisitionError
lock_key, holder,
ttl
CheckpointConflictError
pipeline,
existing_run_id

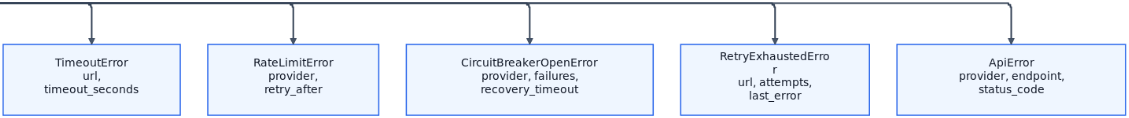

CRITICAL
CriticalError error_type = CRITICAL Action: ABORT pipeline
MergeConflictError table, conflicting_keys
AuthFailureError provider, status_code
MetricsServerError host, port
RunnerAlreadyExecutedError
InfrastructureError component, operation

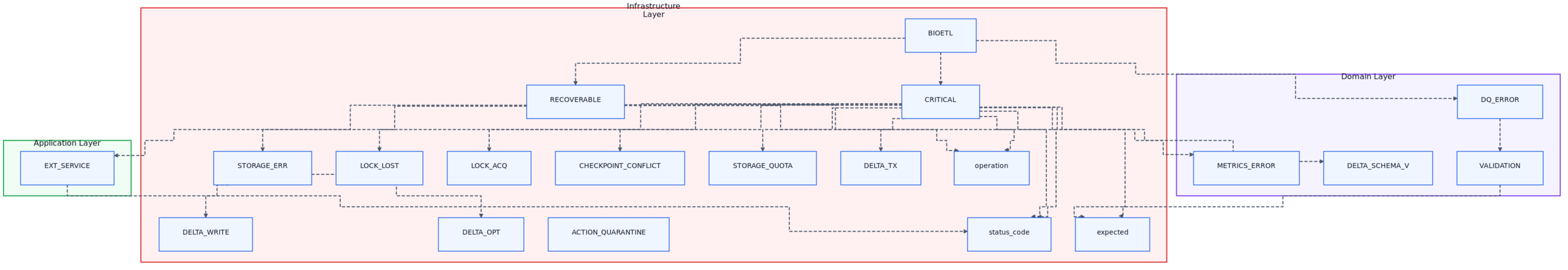

StorageQuotaExceededError
path, current_size, limit
DeltaTransactionError
table, operation
ExternalServiceError
service, operation
ServiceUnavailableError
service, status_code
RateLimitExceededError
service, retry_after
ServiceAuthenticationError
service, status_code
DataValidationError
service, field, message
BucketNotFoundError
bucket

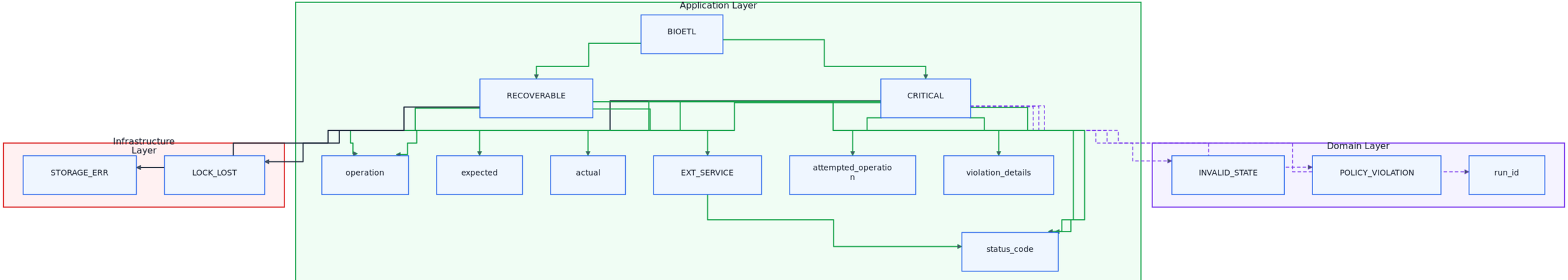

StorageError
path, operation
TableNotFoundError
table
UploadError
path, size
SchemaEvolutionError
table, old_schema, new_schema
BronzeValidationError
path, reason
CachedBronzeEmptyError
path


ValidationError
field, message,
record_id
DeltaWriteConflictError
table, version
DeltaOptimizeError
table
SchemaViolationError
schema_name,
violations: list
MissingRequiredFieldError
field, entity_type
InvalidDataFormatError
field, expected,
actual


ErrorClassifier
domain / error_classifier.py
.classify(error) → ErrorType
ErrorType (StrEnum)
domain / types.py
CRITICAL | RECOVERABLE
DATA_QUALITY | UNKNOWN
UNKNOWN
K38
K39
Exception
(Python built-in)
BioETLError
domain / exceptions / base.py
error_type: ErrorType
context: dict
RecoverableError
error_type = RECOVERABLE
Action: RETRY with backoff
DataQualityError
error_type = DATA_QUALITY
Action: QUARANTINE record
DataQualityThresholdError
pipeline, ratio, threshold
DeltaSchemaValidationError
table, expected, actual
NetworkError
url, status_code, method
triggers
triggers
triggers
Action: ABORT
Pipeline stops immediately
PipelineRunState → FAILED
Action: RETRY
Exponential backoff
Max retries from AdapterConfig
Action: QUARANTINE
Record → QuarantineEntry
Pipeline continues


TimeoutError
url,
timeout_seconds
RateLimitError
provider,
retry_after
CircuitBreakerOpenError
provider, failures,
recovery_timeout
RetryExhaustedErro
r
url, attempts,
last_error
ApiError
provider, endpoint,
status_code


Infrastructure Layer
BIOETL
RECOVERABLE
CRITICAL
Domain Layer
DQ_ERROR
Application Layer
EXT_SERVICE
STORAGE_ERR
LOCK_LOST
LOCK_ACQ
CHECKPOINT_CONFLICT
STORAGE_QUOTA
DELTA_TX
operation
METRICS_ERROR
DELTA_SCHEMA_V
VALIDATION
DELTA_WRITE
DELTA_OPT
ACTION_QUARANTINE
status_code
expected


Application Layer
BIOETL
RECOVERABLE
CRITICAL
Infrastructure Layer
STORAGE_ERR
LOCK_LOST
operation
expected
actual
EXT_SERVICE
attempted_operation
violation_details
status_code
Domain Layer
INVALID_STATE
POLICY_VIOLATION
run_id